Multivariate Data Analysis

Lecture Notes (Ch. 1–11)

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Chapter 1. Multivariate Data and the Geometry of the Sample

Data matrices, descriptive statistics, projections, sphering, Mahalanobis distance, eigendecomposition

1 Multivariate Data and the Goals of Analysis

Multivariate (multidimensional) data are measurements of two or more variables on each individual or experimental unit. The goals of analysis fall into two groups:

(i) **Inference about an unknown population**: understand and infer features of the population from which the data were drawn.

(ii) **Prediction of future observations**: predict the value or category of new observations.

In either case one starts with exploratory data analysis (EDA) to find structure, outliers and relationships among variables. Numerical summaries are descriptive statistics (mean vector, covariance matrix, correlation matrix); graphical summaries are scatterplot matrices, principal component plots, heatmaps, correlation plots and trellis graphs.

Example 1.1: two running examples (slides Examples 1.1, 1.2)

Penguins data. For $n = 333$ penguins from the Palmer archipelago there are 4 numerical variables (bill length, bill depth, flipper length, body mass), 3 categorical variables (species, island, sex) and year. Only the numerical variables go into the data matrix $\mathbf{X}$; categorical variables are kept as separate vectors. This is the typical case $n > p$.
Golub gene expression data (1999). For 72 leukemia patients the expression levels of 7129 genes were measured, together with a categorical label ALL/AML. This is high-dimensional, low-sample-size data ($p = 7129 \gg n = 72$). Scientific question: can the leukemia type be predicted from gene expression alone?

2 The Data Matrix and Descriptive Statistics

Motivation. Multivariate data are an $n \times p$ matrix, and almost every method in the course is a statement about this matrix. Writing the sample mean and covariance as matrix products lets us use linear algebra instead of indices.

2.1 Data matrix

Measurements of p variables on n items form an $n \times p$ matrix

$$\mathbf{X} = \begin{pmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{np} \end{pmatrix} = \begin{pmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix} = [\mathbf{x}_{(1)}, \ldots, \mathbf{x}_{(p)}].$$

The ith row $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})^\top \in \mathbb{R}^p$ is the **observation vector** of item i; the jth column $\mathbf{x}_{(j)} = (x_{1j}, \ldots, x_{nj})^\top \in \mathbb{R}^n$ is the **variable vector** of variable j. All variables in $\mathbf{X}$ are taken to be numerical.

2.2 Sample mean, covariance and correlation

Definition 1.2: descriptive statistics

$$\bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i, \qquad \mathbf{S} = (s_{jk}), \quad s_{jk} = \frac{1}{n-1}\sum_{i=1}^{n}(x_{ij}-\bar{x}_j)(x_{ik}-\bar{x}_k),$$

$$\mathbf{S} = \frac{1}{n-1}\sum_{i=1}^{n}(\mathbf{x}_i-\bar{\mathbf{x}})(\mathbf{x}_i-\bar{\mathbf{x}})^\top, \qquad \mathbf{R} = (r_{jk}), \quad r_{jk} = \frac{s_{jk}}{\sqrt{s_{jj}}\sqrt{s_{kk}}}, \qquad \mathbf{R} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2},$$

with $\mathbf{D} = \operatorname{diag}(s_{11},\dots,s_{pp})$. $\mathbf{R}$ is defined only when all $s_{jj} > 0$. One also uses $\mathbf{S}_n = \frac{n-1}{n}\mathbf{S}$ (divisor n).

Remark 1.3: a Gaussian with covariance S always exists

$\mathbf{S}$ is symmetric and positive semidefinite, so a symmetric square root $\mathbf{S}^{1/2}$ exists (§5). With $\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{I}_p)$ and $\mathbf{X} = \boldsymbol{\mu} + \mathbf{S}^{1/2}\mathbf{Z}$, $\mathrm{E}\mathbf{X} = \boldsymbol{\mu}$ and $\mathrm{Cov}\mathbf{X} = \mathbf{S}^{1/2}\mathbf{I}_p\mathbf{S}^{1/2} = \mathbf{S}$.

2.3 The centering matrix

Let $\mathbf{1}_n = (1,\dots,1)^\top$, $\mathbf{J}_n = \mathbf{1}_n\mathbf{1}_n^\top$ and $\mathbf{H}_n = \mathbf{I}_n - \frac{1}{n}\mathbf{J}_n$.

Theorem 1.4: matrix forms of $\bar{\mathbf{x}}$ and S (slides Exercise 1.1)

$$\bar{\mathbf{x}} = \tfrac{1}{n}\mathbf{X}^\top\mathbf{1}_n, \qquad \mathbf{X}_C := \mathbf{X} - \mathbf{1}_n\bar{\mathbf{x}}^\top = \mathbf{H}_n\mathbf{X}, \qquad \mathbf{S} = \tfrac{1}{n-1}\mathbf{X}^\top\mathbf{H}_n\mathbf{X} = \tfrac{1}{n-1}\mathbf{X}_C^\top\mathbf{X}_C,$$

and $\mathbf{H}_n^\top = \mathbf{H}_n$, $\mathbf{H}_n^2 = \mathbf{H}_n$.

Proof. The jth entry of $\mathbf{X}^\top\mathbf{1}_n$ is $\sum_i x_{ij}$, giving the first identity. Hence $\mathbf{X}_C = \mathbf{X} - \mathbf{1}_n\bar{\mathbf{x}}^\top = \mathbf{X} - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top\mathbf{X} = \mathbf{H}_n\mathbf{X}$. Idempotency follows from $\mathbf{1}_n^\top\mathbf{1}_n = n$:

$$\left(\mathbf{I}_n - \tfrac{1}{n}\mathbf{J}_n\right)^2 = \mathbf{I}_n - \tfrac{2}{n}\mathbf{J}_n + \tfrac{1}{n^2}\mathbf{1}_n(\mathbf{1}_n^\top\mathbf{1}_n)\mathbf{1}_n^\top = \mathbf{I}_n - \tfrac{1}{n}\mathbf{J}_n.$$

Then $\mathbf{X}_C^\top\mathbf{X}_C = \mathbf{X}^\top\mathbf{H}_n^\top\mathbf{H}_n\mathbf{X} = \mathbf{X}^\top\mathbf{H}_n\mathbf{X}$, and $\mathbf{X}_C^\top\mathbf{X}_C = \sum_i(\mathbf{x}_i-\bar{\mathbf{x}})(\mathbf{x}_i-\bar{\mathbf{x}})^\top$. ■

[Supplement] A symmetric idempotent matrix has eigenvalues 0 or 1, so $\operatorname{rank}(\mathbf{H}_n) = \operatorname{tr}(\mathbf{H}_n) = n-1$.

3 Sample Geometry in p-Space

3.1 The data ellipse

Each $\mathbf{x}_i$ is a point in $\mathbb{R}^p$. The mean vector is the center of gravity of the points and the covariance matrix describes their spread. The **data ellipse**

$$\{\mathbf{x} \in \mathbb{R}^p : (\mathbf{x}-\bar{\mathbf{x}})^\top\mathbf{S}^{-1}(\mathbf{x}-\bar{\mathbf{x}}) = c\}$$

summarizes location, shape and size for fixed $c > 0$ (an ellipsoid for $p \geq 3$). The penguin plot uses $c = 5.991 = \chi^2_2(0.05)$. If the cloud is not elliptical, the mean and covariance alone describe it poorly.

Example 1.5: data ellipses

Draw the ellipses for mean $\mathbf{0}$ and $\mathbf{S}_1 = \begin{pmatrix} 9 & 0 \\ 0 & 1 \end{pmatrix}$, $\mathbf{S}_2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$.

[Supplement] For $\mathbf{S}_1$ the axes are the coordinate axes with half-length ratio $3:1$. $\mathbf{S}_2$ has eigenvalues $9, 1$ with eigenvectors $(1,1)^\top/\sqrt{2}$, $(1,-1)^\top/\sqrt{2}$: the same ellipse rotated by $45°$. Half-axis lengths are $\sqrt{c\lambda_j}$.

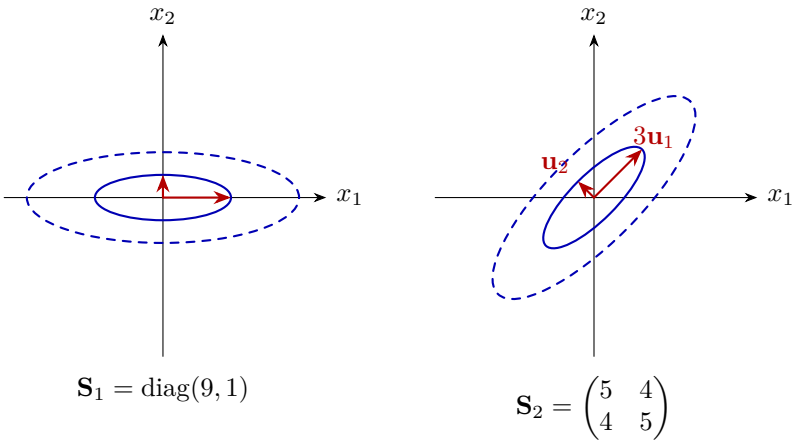


Figure. *Data ellipses for $c = 1$ (solid) and $c = 4$ (dashed). The covariance 4 in $\mathbf{S}_2$ rotates the axes by $45°$.*

3.2 Subspaces, orthogonal projection and projection scores

Definition 1.6: subspace and affine subspace

- **Subspace**: $\mathcal{L}_q = \mathrm{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_q\} = \{\mathbf{V}\mathbf{c} : \mathbf{c} \in \mathbb{R}^q\}$, $\mathbf{V} = [\mathbf{v}_1, \ldots, \mathbf{v}_q]$ (linearly independent). It contains the origin.
- **Affine subspace**: $\mathcal{L}_q(\mathbf{x}) = \mathbf{x} + \mathcal{L}_q$, e.g. a line $\{\mathbf{x} + t\mathbf{v}\}$ or a plane $\{\mathbf{x} + c_1\mathbf{v}_1 + c_2\mathbf{v}_2\}$. It is not a vector space unless it contains the origin, but has the same shape as $\mathcal{L}_q$. Its representation (base point, spanning vectors) is not unique.

Definition 1.7: orthogonal projection and projection score

Projection onto $\mathrm{span}\{\mathbf{v}\}$ and the projection score (signed distance):

$$P_{\mathbf{v}}(\mathbf{x}) = \frac{\mathbf{v}\mathbf{v}^\top}{\mathbf{v}^\top\mathbf{v}}\mathbf{x} = s_{\mathbf{v}}(\mathbf{x})\frac{\mathbf{v}}{\sqrt{\mathbf{v}^\top\mathbf{v}}}, \qquad s_{\mathbf{v}}(\mathbf{x}) = \frac{\mathbf{v}^\top\mathbf{x}}{\sqrt{\mathbf{v}^\top\mathbf{v}}}.$$

The projection matrix onto $\mathcal{L}_q$ is $\mathbf{P}_{\mathbf{V}} = \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top$; for an orthonormal basis ($\mathbf{V}^\top\mathbf{V} = \mathbf{I}_q$)

$$P_{\mathcal{L}_q}(\mathbf{x}) = \mathbf{V}\mathbf{V}^\top\mathbf{x} = s_{\mathbf{v}_1}(\mathbf{x})\mathbf{v}_1 + \cdots + s_{\mathbf{v}_q}(\mathbf{x})\mathbf{v}_q,$$

so $\mathbf{V}^\top\mathbf{x} = (s_{\mathbf{v}_1}(\mathbf{x}), \ldots, s_{\mathbf{v}_q}(\mathbf{x}))^\top$ are the coordinates in the subspace. The projection does not depend on the basis; the scores do. A scatterplot of variables i, j is the projection onto $\mathrm{span}\{\mathbf{e}_i, \mathbf{e}_j\}$.

Derivation: why P_V is the orthogonal projection [Supplement]

$\mathbf{P}_\mathbf{V}^\top = \mathbf{P}_\mathbf{V}$ and $\mathbf{P}_\mathbf{V}^2 = \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top\mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top = \mathbf{P}_\mathbf{V}$. Also $\mathbf{P}_\mathbf{V}\mathbf{x} \in \mathcal{L}_q$, and the residual satisfies $\mathbf{V}^\top(\mathbf{x} - \mathbf{P}_\mathbf{V}\mathbf{x}) = \mathbf{0}$, so it is orthogonal to $\mathcal{L}_q$. By Pythagoras, for any $\mathbf{Vc}$, $\|\mathbf{x} - \mathbf{Vc}\|^2 = \|\mathbf{x} - \mathbf{P}_\mathbf{V}\mathbf{x}\|^2 + \|\mathbf{P}_\mathbf{V}\mathbf{x} - \mathbf{Vc}\|^2 \geq \|\mathbf{x} - \mathbf{P}_\mathbf{V}\mathbf{x}\|^2$: $\mathbf{P}_\mathbf{V}\mathbf{x}$ is the closest point of $\mathcal{L}_q$ to $\mathbf{x}$.

3.3 Projection onto an affine subspace and decomposition of observations

Projection onto $\mathcal{L}_q(\bar{\mathbf{x}})$ is done in two steps:

(1) translate $\mathbf{x} \mapsto \mathbf{x} - \bar{\mathbf{x}}$ and project onto $\mathcal{L}_q$;

(2) translate back by $\bar{\mathbf{x}}$:

$$P_{\mathcal{L}_q(\bar{\mathbf{x}})}(\mathbf{x}) = \bar{\mathbf{x}} + \mathbf{P}_\mathbf{V}(\mathbf{x} - \bar{\mathbf{x}}) = \bar{\mathbf{x}} + s_{\mathbf{v}_1}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_1 + \cdots + s_{\mathbf{v}_q}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_q.$$

Hence an observation splits into projection plus remainder, and by Pythagoras, with $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$,

$$\mathbf{x} = \bar{\mathbf{x}} + \sum_{j=1}^{q} s_{\mathbf{v}_j}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_j + r_{\mathcal{L}_q}(\mathbf{x}), \qquad d(\mathbf{x}, \bar{\mathbf{x}})^2 = \sum_{j=1}^{q} s_{\mathbf{v}_j}(\mathbf{x} - \bar{\mathbf{x}})^2 + \|r_{\mathcal{L}_q}(\mathbf{x})\|^2.$$

This decomposition is the starting point of dimension reduction (PCA).

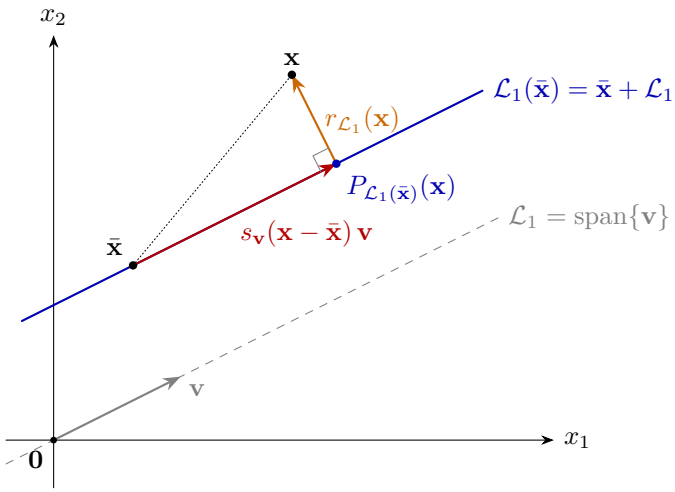


Figure. *Projection onto the affine line $\mathcal{L}_1(\bar{\mathbf{x}})$: the observation splits into $\bar{\mathbf{x}}$, the projected part and an orthogonal remainder.*

Remark 1.8: geometric meaning of variance

Projection scores are tied to conditional variances. If $X_1 \sim N(0, \sigma_1^2)$ and $X_2 = X_1 + \epsilon_2$ with independent $\epsilon_2 \sim N(0, \sigma_2^2)$, then $\mathrm{Var}(X_2 \mid X_1) = \sigma_2^2$ and $\mathrm{Var}(X_1 \mid X_2) = \dfrac{\sigma_1^2\sigma_2^2}{\sigma_1^2 + \sigma_2^2}$.

[Supplement] The latter is $\sigma_1^2 - \sigma_1^4/(\sigma_1^2 + \sigma_2^2)$ (conditional distribution formula, Ch. 3): what remains after removing the projected part is the conditional variance.

3.4 Linear and affine transformations

A $q \times p$ matrix $\mathbf{L}$ defines the linear transformation $\mathbf{x} \mapsto \mathbf{Lx}$; $q > p$ only adds dimensions without information and is rarely used. Examples: the projection $\mathbf{x} \mapsto \mathbf{P}_\mathbf{V}\mathbf{x}$ and the score map $\mathbf{x} \mapsto \mathbf{S}_\mathbf{V}\mathbf{x}$,

$\mathbf{S_V} = (\mathbf{v}_1/\|\mathbf{v}_1\|, \ldots, \mathbf{v}_q/\|\mathbf{v}_q\|)^\top$. Projection onto an affine subspace is not linear (because of the translation) but an **affine transformation** $\mathbf{x} \mapsto \boldsymbol{m} + \mathbf{L}\mathbf{x}$.

Theorem 1.9: affine maps and standardization (slides Exercise 1.2)

1. Orthogonal projection onto $\mathcal{L}_q(\bar{\mathbf{x}})$ is affine with $\boldsymbol{m} = \bar{\mathbf{x}} - \mathbf{P_V}\bar{\mathbf{x}}$, $\mathbf{L} = \mathbf{P_V} = \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top$; so is the score map $\mathbf{x} \mapsto (s_{\mathbf{v}_1}(\mathbf{x}-\bar{\mathbf{x}}), \ldots, s_{\mathbf{v}_q}(\mathbf{x}-\bar{\mathbf{x}}))^\top$.
2. (Standardization) The data $\mathbf{x}_i' = \mathbf{D}^{-1/2}(\mathbf{x}_i - \bar{\mathbf{x}})$ have sample mean $\mathbf{0}$ and unit sample variances, and $\mathbf{x}' = \mathbf{D}^{-1/2}\mathbf{x} - \mathbf{D}^{-1/2}\bar{\mathbf{x}}$ is affine.

Proof. (1) $P_{\mathcal{L}_q(\bar{\mathbf{x}})}(\mathbf{x}) = \bar{\mathbf{x}} + \mathbf{P_V}(\mathbf{x}-\bar{\mathbf{x}}) = (\bar{\mathbf{x}} - \mathbf{P_V}\bar{\mathbf{x}}) + \mathbf{P_V}\mathbf{x}$; the score vector is $\mathbf{S_V}\mathbf{x} - \mathbf{S_V}\bar{\mathbf{x}}$.

(2) $\bar{\mathbf{x}}' = \frac{1}{n}\sum_i \mathbf{D}^{-1/2}(\mathbf{x}_i - \bar{\mathbf{x}}) = \mathbf{0}$; the jth sample variance is $s_{jj}^{-1/2} s_{jj} s_{jj}^{-1/2} = 1$. ■

3.5 Rotation of coordinates

For $p = 2$ and $\mathbf{R}_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, put $\mathbf{v}_1 = \mathbf{R}_\theta\mathbf{e}_1$, $\mathbf{v}_2 = \mathbf{R}_\theta\mathbf{e}_2$; then $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2]$ satisfies $\mathbf{V}^\top\mathbf{V} = \mathbf{V}\mathbf{V}^\top = \mathbf{I}_2$. Since $\mathbf{x} = \mathbf{V}\mathbf{V}^\top\mathbf{x} = [\mathbf{v}_1, \mathbf{v}_2](\mathbf{v}_1^\top\mathbf{x}, \mathbf{v}_2^\top\mathbf{x})^\top$, the coordinates of $\mathbf{x}$ in the new system are the projection scores $(\mathbf{v}_1^\top\mathbf{x}, \mathbf{v}_2^\top\mathbf{x})$. A linear map by an orthonormal basis is a rotation of coordinates.

4 Standardization, Sphering and Mahalanobis Distance

4.1 Univariate standardization and its limitation

For $N(\mu, \sigma^2)$ the distance of x from the center is better measured by the **standardized distance** $|x-\mu|/\sigma$ than by $|x-\mu|$. In the multivariate case, standardizing each variable, $\mathbf{x} \mapsto \mathbf{D}^{-1/2}(\mathbf{x}-\bar{\mathbf{x}})$, removes scale differences but **ignores correlation**, so it does not give a distance that reflects the distribution.

Example 1.10: distances in bivariate data (slides Example 1.3)

$\bar{\mathbf{x}} = \mathbf{0}$, $\mathbf{S} = \begin{pmatrix} 640000 & 9600 \\ 9600 & 225 \end{pmatrix} = \begin{pmatrix} 800^2 & 800\cdot 15\cdot 0.8 \\ 800\cdot 15\cdot 0.8 & 15^2 \end{pmatrix}$ (correlation 0.8). For $\mathbf{y} = (-500, 10)^\top$, $\mathbf{z} = (1500, 15)^\top$: $d(\bar{\mathbf{x}}, \mathbf{y}) = 500.1$, $d(\bar{\mathbf{x}}, \mathbf{z}) = 1500.075$, so $\mathbf{z}$ looks farther. After standardization $\mathbf{y}' = (-0.625, 0.667)^\top$, $\mathbf{z}' = (1.875, 1.000)^\top$ and still $d(\mathbf{0}, \mathbf{y}') = 0.91 < 2.125 = d(\mathbf{0}, \mathbf{z}')$. Yet the scatterplot has many points near $\mathbf{z}$ and almost none near $\mathbf{y}$ ($\mathbf{y}$ goes against the positive correlation). After standardization the SD in the direction of $\mathbf{y}'$ is 0.36 and in the direction of $\mathbf{z}'$ is 1.31, not 1: standardization does not standardize spread in every direction. After sphering (Example 1.14), $\|\mathbf{y}' - \bar{\mathbf{x}}'\| = 2.65 > 2.18 = \|\mathbf{z}' - \bar{\mathbf{x}}'\|$: now $\mathbf{y}$ is correctly farther. [Supplement] These numbers use the covariance of the plotted sample; the rounded $\mathbf{S}$ above gives 2.04 and 2.05.

4.2 Sphering

If the covariance is $\mathbf{I}_p$, the data ellipse is a sphere ($c = \|\mathbf{x}-\bar{\mathbf{x}}\|^2$) and spread is 1 in **every** direction.

Theorem 1.11: isotropy of sphered data (slides homework)

If $\bar{\mathbf{x}} = \mathbf{0}$ and $\mathbf{S} = \mathbf{I}_p$, then for any unit vector $\mathbf{v}$ the scores $S_{\mathbf{v}}(\mathbf{x}_i) = \mathbf{v}^\top \mathbf{x}_i$ have sample variance 1.

Proof. The scores have mean $\mathbf{v}^\top \bar{\mathbf{x}} = 0$ and sample variance $\mathbf{v}^\top \mathbf{S} \mathbf{v} = \mathbf{v}^\top \mathbf{v} = 1$. ■

Example 1.12: sphering a diagonal covariance (slides Example 1.5)

$\bar{\mathbf{x}} = (10, 10)^\top$, $\mathbf{S} = \operatorname{diag}(4, 9)$: the sphering matrix is $\mathbf{D}^{-1/2} = \operatorname{diag}(1/2, 1/3)$ and $\mathbf{S}' = \mathbf{D}^{-1/2} \mathbf{S} \mathbf{D}^{-1/2} = \mathbf{I}_2$.

Definition 1.13: sphering transformation

A transformation after which the covariance is $\mathbf{I}_p$. For general $\mathbf{S}$ it is built in three steps.

1. **Rotation**: choose orthogonal $\mathbf{U}_S = [\mathbf{u}_1, \ldots, \mathbf{u}_p]$ so that $\mathbf{x}'_i = \mathbf{U}_S^\top \mathbf{x}_i$ has diagonal sample covariance $\mathbf{D}_S = \mathbf{U}_S^\top \mathbf{S} \mathbf{U}_S = \operatorname{diag}(d_{11}, \ldots, d_{pp})$ (the eigendecomposition of $\mathbf{S}$).
2. **Standardization**: $\mathbf{x}' \mapsto \mathbf{D}_S^{-1/2} \mathbf{x}'$.
3. **Back-rotation**: $\mathbf{x} \mapsto \mathbf{U}_S \mathbf{x}$.

Altogether $\mathbf{x} \mapsto \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top \mathbf{x} = \mathbf{S}^{-1/2} \mathbf{x}$.

[Supplement] The sphered covariance is $\mathbf{I}_p$.

$$\begin{aligned} \mathbf{S}' &= \tfrac{1}{n-1} \textstyle\sum_i \mathbf{S}^{-1/2} (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{S}^{-1/2} = \mathbf{S}^{-1/2} \mathbf{S} \mathbf{S}^{-1/2} \\ &= \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top \, \mathbf{U}_S \mathbf{D}_S \mathbf{U}_S^\top \, \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top = \mathbf{U}_S \mathbf{U}_S^\top = \mathbf{I}_p. \end{aligned}$$

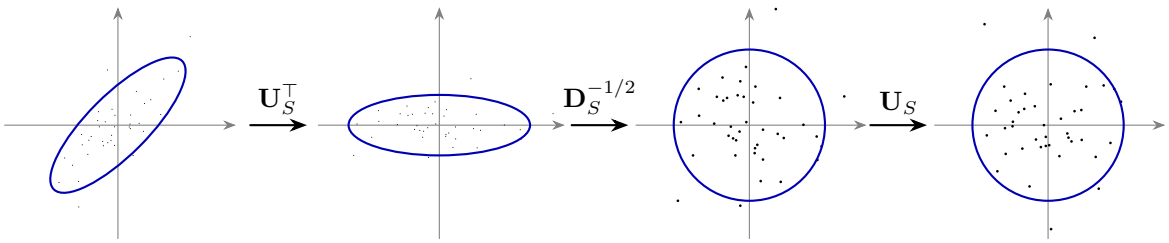
■

Example 1.14: applying the sphering transformation (slides Example 1.6)

$\bar{\mathbf{x}} = \mathbf{0}$, $\mathbf{S} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$. With $\mathbf{U}_S = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ (rotation by $\theta = \pi/4$),

$$\mathbf{U}_S^\top \mathbf{S} \mathbf{U}_S = \tfrac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 9 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{D}_S,$$

so sphering is $\mathbf{x} \mapsto \mathbf{U}_S \operatorname{diag}(1/3, 1) \mathbf{U}_S^\top \mathbf{x}$.



(a) data, cov $\mathbf{S}$ (b) rotated, cov $\operatorname{diag}(9, 1)$ (c) standardized, cov $\mathbf{I}_2$ (d) rotated back, cov $\mathbf{I}_2$

Figure. *The three steps of sphering for $\mathbf{S}$ of Example 1.14 (40 points with this sample covariance). The overall map $\mathbf{S}^{-1/2}$ turns the ellipse into a circle.*

4.3 Mahalanobis distance

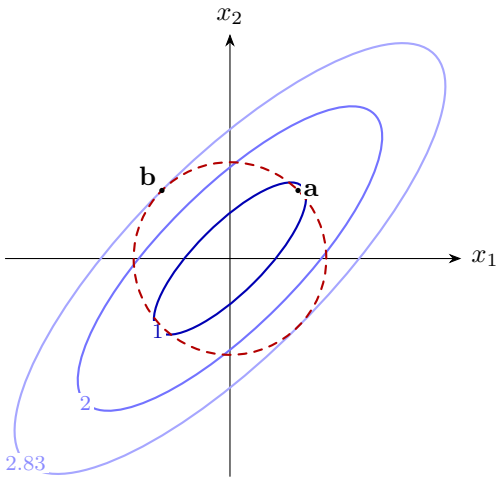
Definition 1.15: Mahalanobis (statistical) distance

For a distribution with mean $\bar{\mathbf{x}}$ and covariance $\mathbf{S}$, the Mahalanobis distance of $\mathbf{y}$ from $\bar{\mathbf{x}}$ is the Euclidean distance after sphering:

$$d_S(\mathbf{y}, \bar{\mathbf{x}}) = \|\mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top (\mathbf{y} - \bar{\mathbf{x}})\|, \qquad d_S(\mathbf{y}, \bar{\mathbf{x}})^2 = (\mathbf{y} - \bar{\mathbf{x}})^\top \mathbf{S}^{-1} (\mathbf{y} - \bar{\mathbf{x}}).$$

Proof. $d_S^2 = (\mathbf{y} - \bar{\mathbf{x}})^\top \mathbf{U}_S \mathbf{D}_S^{-1/2} \underbrace{\mathbf{U}_S^\top \mathbf{U}_S}_{\mathbf{I}} \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top (\mathbf{y} - \bar{\mathbf{x}})$ and $\mathbf{U}_S \mathbf{D}_S^{-1} \mathbf{U}_S^\top = \mathbf{S}^{-1}$. ■

Sphering and Mahalanobis distance are the multivariate counterparts of studentisation and standardized distance.



$\mathbf{a} = (2, 2)$: $d_S(\mathbf{a}, \mathbf{0}) = 0.94$
$\mathbf{b} = (-2, 2)$: $d_S(\mathbf{b}, \mathbf{0}) = 2.83$
dashed: $\|\mathbf{x}\| = 2.83$
solid: $d_S(\mathbf{x}, \mathbf{0}) = 1, 2, 2.83$

Figure. *Euclidean versus Mahalanobis distance for* $\mathbf{S} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$*:* $\mathbf{a}$ *and* $\mathbf{b}$ *are equally far in Euclidean distance, but* $\mathbf{b}$ *lies against the correlation.*

5 Eigendecomposition and Positive Definiteness

Motivation. A covariance matrix is symmetric and positive semidefinite, so it has an orthonormal eigenbasis. The eigenvectors are the natural axes of the data cloud and the eigenvalues the variances along them; PCA, Mahalanobis distance and the normal density all rest on this decomposition.

5.1 Eigendecomposition

For symmetric $\mathbf{A}$, $\mathbf{Au} = \lambda\mathbf{u}$ ($\mathbf{u} \neq \mathbf{0}$) makes $\mathbf{u}$ an eigenvector and λ an eigenvalue: $\mathbf{A}$ only stretches or shrinks $\mathbf{u}$, by the factor λ.

Theorem 1.16: eigendecomposition of a symmetric matrix (slides homework)

A symmetric $\mathbf{A} \in \mathbb{R}^{p \times p}$ satisfies $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top = \sum_{i=1}^{p} \lambda_i \mathbf{u}_i \mathbf{u}_i^\top$ with $\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_p]$ orthogonal and $\mathbf{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_p)$ real.

- If $\mathbf{u}_i$ is an eigenvector so is $c\mathbf{u}_i$; normalized to length 1 it is unique up to sign (for distinct eigenvalues).
- Eigenvectors of distinct eigenvalues are orthogonal.

- With repeated eigenvalues the decomposition exists but $\mathbf{U}$ is not unique, e.g. $\sigma^2\mathbf{I}_p = \mathbf{U}(\sigma^2\mathbf{I})\mathbf{U}^\top$ for every orthogonal $\mathbf{U}$.

Orthogonality. From $\mathbf{A}\mathbf{u}_j = \lambda_j\mathbf{u}_j$, $\mathbf{u}_i^\top\mathbf{A}\mathbf{u}_j = \lambda_j\mathbf{u}_i^\top\mathbf{u}_j$; from $\mathbf{A}^\top = \mathbf{A}$, $\mathbf{u}_i^\top\mathbf{A} = \lambda_i\mathbf{u}_i^\top$, so $\mathbf{u}_i^\top\mathbf{A}\mathbf{u}_j = \lambda_i\mathbf{u}_i^\top\mathbf{u}_j$. Subtracting, $(\lambda_i - \lambda_j)\mathbf{u}_i^\top\mathbf{u}_j = 0$. ■

Example 1.17: eigenstructure of the sphering matrix (slides Example 1.7)

$\mathbf{A} = \mathbf{S}^{-1/2} = \begin{pmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{pmatrix}$ (Example 1.14), $\mathbf{u}_1 = (1,1)^\top/\sqrt{2}$: $\mathbf{A}\mathbf{u}_1 = \frac{1}{\sqrt{2}}(2/3-1/3,\ -1/3+2/3)^\top = \frac{1}{3}\mathbf{u}_1$.
[Supplement] Similarly $\mathbf{A}\mathbf{u}_2 = \mathbf{u}_2$. Sphering rescales each eigen-direction by $1/\sqrt{\lambda_j}$.

5.2 Inverse, square root, definiteness

If no eigenvalue is 0, $\mathbf{A}^{-1} = \mathbf{U}\mathbf{\Lambda}^{-1}\mathbf{U}^\top$.

Definition 1.18: NND and PD

A symmetric $\mathbf{A}$ is **non-negative definite (NND)** if $\mathbf{x}^\top\mathbf{A}\mathbf{x} \geq 0$ for all $\mathbf{x}$, and **positive definite (PD)** if $\mathbf{x}^\top\mathbf{A}\mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.

Theorem 1.19: eigenvalues and definiteness (slides homework)

A symmetric $\mathbf{A}$ with all eigenvalues positive is PD; with all eigenvalues non-negative it is NND.

Proof. With $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ and $\mathbf{y} = \mathbf{U}^\top\mathbf{x}$ (so $\mathbf{y} = \mathbf{0} \iff \mathbf{x} = \mathbf{0}$), $\mathbf{x}^\top\mathbf{A}\mathbf{x} = \mathbf{y}^\top\mathbf{\Lambda}\mathbf{y} = \sum_i \lambda_i y_i^2$. If all $\lambda_i > 0$ and $\mathbf{x} \neq \mathbf{0}$ some $y_i \neq 0$ and the sum is positive; NND is the same. ■

[Supplement] Conversely, if $\mathbf{A}$ is PD (NND), $\lambda_i = \mathbf{u}_i^\top\mathbf{A}\mathbf{u}_i > 0$ (≥ 0): the conditions are equivalent.

Theorem 1.20: the sample covariance is always NND, not necessarily PD (slides homework)

Proof. $\mathbf{z}^\top\mathbf{S}\mathbf{z} = \frac{1}{n-1}\sum_i\{(\mathbf{x}_i - \bar{\mathbf{x}})^\top\mathbf{z}\}^2 \geq 0$. But if the data lie in a lower-dimensional subspace of $\mathbb{R}^p$ (e.g. on a line), any $\mathbf{z} \neq \mathbf{0}$ orthogonal to it gives $\mathbf{z}^\top\mathbf{S}\mathbf{z} = 0$. ■

When n is much larger than p, $\mathbf{S}$ is PD in most cases and $\mathbf{S}^{-1}$ exists.

Definition 1.21: symmetric square root

For NND $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$, $\mathbf{S}^{1/2} = \mathbf{U}\mathbf{\Lambda}^{1/2}\mathbf{U}^\top$; for PD $\mathbf{S}$, $\mathbf{S}^{-1/2} = (\mathbf{S}^{-1})^{1/2} = \mathbf{U}\mathbf{\Lambda}^{-1/2}\mathbf{U}^\top$. $\mathbf{S}^{1/2}$ is symmetric with $\mathbf{S}^{1/2}\mathbf{S}^{1/2} = \mathbf{S}$.

5.3 Covariance matrices and ellipses

With $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ the ellipse $E_S = \{\mathbf{x} : \mathbf{x}^\top \mathbf{S}^{-1}\mathbf{x} = 1\}$ reads

$$\frac{(\mathbf{u}_1^\top \mathbf{x})^2}{\lambda_1} + \cdots + \frac{(\mathbf{u}_p^\top \mathbf{x})^2}{\lambda_p} = 1,$$

so its principal axes are along $\mathbf{u}_i$ with half-lengths $\sqrt{\lambda_i}$. The eigendecomposition of a covariance matrix gives (1) the rotation $\mathbf{x} \mapsto \mathbf{U}^\top \mathbf{x}$ to uncorrelated coordinates and (2) the sphering $\mathbf{x} \mapsto \mathbf{U}\mathbf{\Lambda}^{-1/2}\mathbf{U}^\top \mathbf{x}$.

Theorem 1.22: covariance of rotated data (slides Discussion)

If $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$, the rotated data $\mathbf{x}_i' = \mathbf{U}^\top \mathbf{x}_i$ have sample covariance $\mathbf{\Lambda}$.

[Supplement]. $\frac{1}{n-1}\sum_i \mathbf{U}^\top(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{U} = \mathbf{U}^\top \mathbf{S}\mathbf{U} = \mathbf{\Lambda}$. ■

The eigenvalues of a covariance matrix are the variances of mutually uncorrelated rotated coordinates — this is PCA.

6 Geometry of Principal Component Analysis

With many variables, inspecting all pairwise scatterplots is inefficient. PCA finds, in order, linear combinations of the original variables that explain the most variation, and summarizes the data by a few of them (details in Ch. 7).

6.1 Which direction to project on?

The scores $t_i(\mathbf{a}) = \mathbf{a}^\top(\mathbf{x}_i - \bar{\mathbf{x}})$ along a unit vector $\mathbf{a}$ have sample variance $\frac{1}{n-1}\sum_i t_i(\mathbf{a})^2 = \mathbf{a}^\top \mathbf{S}\mathbf{a}$.

Definition 1.23: principal component directions

First direction $\mathbf{u}_1 \in \arg\max_{\|\mathbf{a}\|=1} \mathbf{a}^\top \mathbf{S}\mathbf{a}$; later directions maximize variance subject to orthogonality to the earlier ones. The solutions are eigenvectors of $\mathbf{S}$.

[Supplement] $\mathbf{u}_1$ is the top eigenvector. With $\mathbf{z} = \mathbf{U}^\top \mathbf{a}$ ($\|\mathbf{z}\| = 1$), $\mathbf{a}^\top \mathbf{S}\mathbf{a} = \sum_j \lambda_j z_j^2 \le \lambda_1$, with equality at $\mathbf{a} = \mathbf{u}_1$. Under $z_1 = 0$ the maximum is λ_2 at $\mathbf{u}_2$, and so on. ■

6.2 Principal component scores

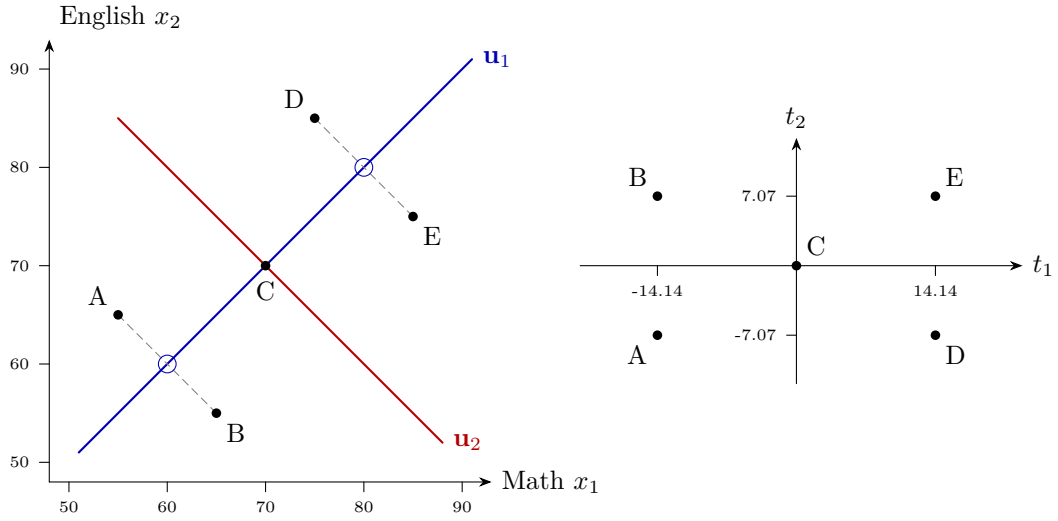
With $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$, $\lambda_1 \ge \cdots \ge \lambda_p \ge 0$, the score matrix is $\mathbf{T} = \mathbf{X}_C\mathbf{U} = (t_{ij})$, $t_{ij} = (\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{u}_j$. Each column of $\mathbf{T}$ has mean 0 and $\frac{1}{n-1}\mathbf{T}^\top\mathbf{T} = \mathbf{U}^\top\mathbf{S}\mathbf{U} = \mathbf{\Lambda}$: the jth PC has sample variance λ_j and different PC scores are uncorrelated.

Example 1.24: scores in two subjects

Student	Math x_1	English x_2	PC1 score	PC2 score
A	55	65	-14.14	-7.07
B	65	55	-14.14	7.07
C	70	70	0	0
D	75	85	14.14	-7.07
E	85	75	14.14	7.07

$\bar{\mathbf{x}} = (70, 70)^\top$, $\mathbf{S} = \begin{pmatrix} 125 & 75 \\ 75 & 125 \end{pmatrix}$, $\lambda_1 = 200$, $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1,1)^\top$, $\lambda_2 = 50$, $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1,-1)^\top$.

$t_{i1} = \{(x_{i1} - 70) + (x_{i2} - 70)\}/\sqrt{2}$ is the overall level; $t_{i2} = \{(x_{i1} - 70) - (x_{i2} - 70)\}/\sqrt{2}$ is the relative difference between the subjects. A and B share the same PC1 score, so PC1 alone cannot distinguish their relative strengths.



(a) original scores (b) principal component scores

Figure. *(a) Principal axes through $\bar{\mathbf{x}}$; A, B (and D, E) project to the same point on the first axis. (b) Score plot: PC2 separates A from B.*

6.3 Dimension reduction and reconstruction error

For student E, $\mathbf{x}_E = (85, 75)^\top$ and $t_{E1} = 10\sqrt{2}$. Reconstructing from PC1 gives $\hat{\mathbf{x}}_E = (70, 70)^\top + 10\sqrt{2} \cdot \frac{1}{\sqrt{2}}(1,1)^\top = (80, 80)^\top$: the average level is kept, the difference between subjects is lost.

Theorem 1.25: reconstruction error

For the reconstruction $\hat{\mathbf{x}}_i = \bar{\mathbf{x}} + \sum_{j \le q} t_{ij}\mathbf{u}_j$ from the first q PCs,

$$\sum_{i=1}^{n} \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2 = (n-1) \sum_{j=q+1}^{p} \lambda_j.$$

Among all q-dimensional planes through $\bar{\mathbf{x}}$, PCA minimizes the reconstruction error, which is equivalent to maximizing the total projected variance.

[Supplement]. $\mathbf{x}_i - \hat{\mathbf{x}}_i = \sum_{j>q} t_{ij}\mathbf{u}_j$, so $\|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2 = \sum_{j>q} t_{ij}^2$ and $\sum_i t_{ij}^2 = (n-1)\lambda_j$. For any

orthonormal $\mathbf{V} \in \mathbb{R}^{p\times q}$, Pythagoras gives $\sum_i \|(\mathbf{x}_i-\bar{\mathbf{x}})-\mathbf{V}\mathbf{V}^\top(\mathbf{x}_i-\bar{\mathbf{x}})\|^2 = (n-1)\{\mathrm{tr}\mathbf{S}-\mathrm{tr}(\mathbf{V}^\top\mathbf{S}\mathbf{V})\}$, so minimizing the error is maximizing $\mathrm{tr}(\mathbf{V}^\top\mathbf{S}\mathbf{V})$, whose maximum $\lambda_1+\cdots+\lambda_q$ follows from von Neumann's inequality (Ch. 2) or Ch. 7. ■

6.4 Proportion of variance, scale and caveats

- Since $\mathrm{tr}\mathbf{S} = \sum_k \lambda_k$, the proportion of variance explained by PC j is $\lambda_j/\sum_k \lambda_k$ and the cumulative proportion is $\sum_{j\le q}\lambda_j/\sum_k\lambda_k$ ($80\% = 200/250$ in the example). Choose q from the cumulative proportion or a scree plot.
- Covariance-based PCA depends on units (rescaling English to 1000 points multiplies its variance by 100). PCA of standardized data $z_{ik} = (x_{ik}-\bar{x}_k)/s_k$ is **correlation PCA**. If differences in variance are meaningful, standardizing is inappropriate.
- High explained variance does not mean good group separation. With $Y \in \{-1,1\}$ equally likely, $X_1 = 10Z$, $X_2 = Y$, $Z \sim N(0,1) \perp\!\!\!\perp Y$: $\mathbf{\Sigma} = \mathrm{diag}(100,1)$ and PC1 explains $100/101 \approx 99\%$, but $X_1 \perp\!\!\!\perp Y$ while X_2 alone separates the groups perfectly. PCA keeps variation, not predictive information.
- Still, for the Golub data ($p = 7129$; slides Example 1.8), the PC1–PC2 plot, built without the labels, separates ALL/AML better than the original variables: PCA can be used to *explore* group differences.

7 Random Vectors and the Geometry of Variable Vectors

Motivation. Viewing each variable as a vector in $\mathbb{R}^n$ gives covariance and correlation a geometric meaning: lengths are standard deviations and angles are correlations. This picture makes projection arguments (regression, partial correlation) transparent.

7.1 Random vectors

The data $\mathbf{X} = [\mathbf{x}_1,\ldots,\mathbf{x}_n]^\top$ are a random sample from a population modeled by the joint distribution of $\mathbf{x} = (X_1,\ldots,X_p)^\top \sim f$; a simple random sample is i.i.d., $\mathbf{x}_1,\ldots,\mathbf{x}_n \overset{\text{iid}}{\sim} f$.

Example 1.26: distributions of random vectors (slides Example 2.1)

1. Independent $X_i \sim \mathrm{Unif}(a_i,b_i)$: $f(\mathbf{x}) = \prod_i \frac{1}{b_i-a_i}\mathbf{1}\{x_i \in (a_i,b_i)\}$, uniform on a hypercube.
2. Independent $X_i \sim N(0,1)$: $f(\mathbf{x}) = (2\pi)^{-p/2}e^{-\|\mathbf{x}\|^2/2}$, the standard multivariate normal.
3. (X_1,X_2) uniform on $0<x_1<x_2<1$: $f = 2\cdot\mathbf{1}\{0<x_1<x_2<1\}$. $X_1 \mid X_2 = x_2 \sim \mathrm{Unif}(0,x_2)$ depends on x_2, so they are not independent.
4. Bivariate normal with correlation ρ: $f = \frac{1}{2\pi\sqrt{1-\rho^2}}\exp\{-\frac{x_1^2-2\rho x_1x_2+x_2^2}{2(1-\rho^2)}\}$. Both marginals are $N(0,1)$ but the joint density is not their product: not independent.

Definition 1.27: moments of a random vector

$$\boldsymbol{\mu} = \mathrm{E}(\mathbf{X}), \qquad \mathrm{Cov}(\mathbf{X}) = \mathrm{E}\{(\mathbf{X}-\boldsymbol{\mu})(\mathbf{X}-\boldsymbol{\mu})^\top\} = \mathrm{E}(\mathbf{X}\mathbf{X}^\top) - \boldsymbol{\mu}\boldsymbol{\mu}^\top,$$
$$\mathrm{Cov}(\mathbf{X},\mathbf{Y}) = \mathrm{E}\{(\mathbf{X}-\mathrm{E}\mathbf{X})(\mathbf{Y}-\mathrm{E}\mathbf{Y})^\top\}.$$

For $\boldsymbol{\Sigma} = \mathrm{Cov}(\mathbf{X}) \succ 0$, $d_{\boldsymbol{\Sigma}}(\mathbf{x}, \boldsymbol{\mu})^2 = (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})$. Generalized variance $|\mathrm{Cov}(\mathbf{X})|$, total variance $\mathrm{tr}\mathrm{Cov}(\mathbf{X})$; sample versions $|\mathbf{S}|$, $\mathrm{tr}\mathbf{S}$.

Theorem 1.28: characterization of covariance matrices

$\boldsymbol{\Sigma}$ is a covariance matrix iff $\boldsymbol{\Sigma}$ is symmetric and NND.

Proof. ($\Rightarrow$) [Supplement] $\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a} = \mathrm{Var}(\mathbf{a}^\top \mathbf{X}) \geq 0$.

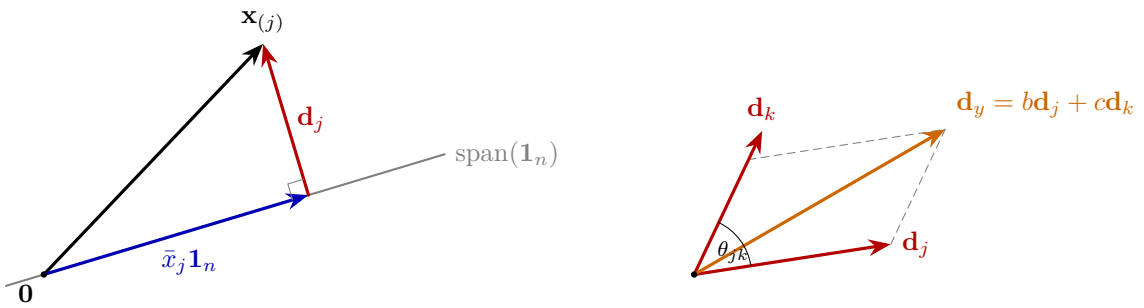
($\Leftarrow$) $\mathbf{X} = \boldsymbol{\Sigma}^{1/2}\mathbf{Z}$, $\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{I}_p)$, has $\mathrm{Cov}(\mathbf{X}) = \boldsymbol{\Sigma}$. ■

7.2 Mean, variance and correlation through variable vectors

Viewing the variable vectors $\mathbf{x}_{(j)} \in \mathbb{R}^n$ (p vectors in n-space) gives a geometric reading of the statistics.

- **Mean**: $\bar{x}_j = \frac{1}{n}\mathbf{1}_n^\top \mathbf{x}_{(j)}$. The projection of $\mathbf{x}_{(j)}$ onto $\mathbf{1}_n$ is $P_{\mathbf{1}_n}(\mathbf{x}_{(j)}) = \mathbf{1}_n(\mathbf{1}_n^\top \mathbf{1}_n)^{-1}\mathbf{1}_n^\top \mathbf{x}_{(j)} = \bar{x}_j \mathbf{1}_n$ with score $\sqrt{n}\,\bar{x}_j$.
- **Deviation vector**: $\mathbf{d}_j = \mathbf{x}_{(j)} - \bar{x}_j \mathbf{1}_n$; $\mathbf{x}_{(j)} = \bar{x}_j \mathbf{1}_n + \mathbf{d}_j$ is an orthogonal decomposition ($\mathbf{1}_n^\top \mathbf{d}_j = 0$).
- $s_j = \|\mathbf{d}_j\|/\sqrt{n-1}$, $s_j^2 = \|\mathbf{d}_j\|^2/(n-1)$, $s_{jk} = \mathbf{d}_j^\top \mathbf{d}_k/(n-1)$.
- **The sample correlation is the cosine of the angle between deviation vectors**: $r_{jk} = \dfrac{\mathbf{d}_j^\top \mathbf{d}_k}{\|\mathbf{d}_j\|\|\mathbf{d}_k\|} = \cos\theta_{jk}$.

A longer deviation vector means more variability. $\mathbf{d}_j = c\mathbf{d}_k$ with $c > 0$ ($\theta = 0$) gives $r = 1$, with $c < 0$ ($\theta = \pi$) gives $r = -1$; orthogonal deviation vectors give zero *linear* correlation.



(a) $\mathbf{x}_{(j)} = \bar{x}_j \mathbf{1}_n + \mathbf{d}_j$ (b) $r_{jk} = \cos\theta_{jk}$, $s_j \propto \|\mathbf{d}_j\|$

Figure. *Variables as vectors in $\mathbb{R}^n$: (a) mean part and deviation vector; (b) lengths are standard deviations, angles give correlations.*

Theorem 1.29: unbiasedness of $\bar{\mathbf{X}}$ and $\mathbf{S}$ (slides Result 2.1)

If $\mathbf{X}_1, \ldots, \mathbf{X}_n$ is a random sample with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$,

$$\mathrm{E}(\bar{\mathbf{X}}) = \boldsymbol{\mu}, \qquad \mathrm{Cov}(\bar{\mathbf{X}}) = \tfrac{1}{n}\boldsymbol{\Sigma}, \qquad \mathrm{E}(\mathbf{S}_n) = \tfrac{n-1}{n}\boldsymbol{\Sigma}, \qquad \mathrm{E}(\mathbf{S}) = \boldsymbol{\Sigma}.$$

Proof. $\mathrm{E}\bar{\mathbf{X}} = \frac{1}{n}\sum \mathrm{E}\mathbf{X}_j = \boldsymbol{\mu}$. $\mathrm{Cov}(\bar{\mathbf{X}}) = \frac{1}{n^2}\sum_j \sum_\ell \mathrm{E}\{(\mathbf{X}_j - \boldsymbol{\mu})(\mathbf{X}_\ell - \boldsymbol{\mu})^\top\}$, and terms with $j \neq \ell$ vanish by independence, giving $\frac{1}{n^2} \cdot n\boldsymbol{\Sigma}$.

[Supplement] Since $\sum_j(\mathbf{X}_j-\bar{\mathbf{X}})(\mathbf{X}_j-\bar{\mathbf{X}})^\top = \sum_j(\mathbf{X}_j-\boldsymbol{\mu})(\mathbf{X}_j-\boldsymbol{\mu})^\top - n(\bar{\mathbf{X}}-\boldsymbol{\mu})(\bar{\mathbf{X}}-\boldsymbol{\mu})^\top$, its expectation is $n\boldsymbol{\Sigma}-\boldsymbol{\Sigma}=(n-1)\boldsymbol{\Sigma}$. ■

7.3 Linear combinations of variables

For the new variable $z = a+\mathbf{b}^\top\mathbf{x} = a+\sum_j b_jx_j$ the variable vector is $\mathbf{z} = a\mathbf{1}_n + \sum_j b_j\mathbf{x}_{(j)}$, with

$$P_{\mathbf{1}_n}(\mathbf{z}) = \Big(a+\sum_j b_j\bar{x}_j\Big)\mathbf{1}_n, \qquad \mathbf{d}_z = \sum_j b_j\mathbf{d}_j = \mathbf{X}_C\mathbf{b}.$$

Theorem 1.30: sample variance and covariance of linear combinations

$$\widehat{\mathrm{Var}}(z) = \tfrac{1}{n-1}\mathbf{d}_z^\top\mathbf{d}_z = \mathbf{b}^\top\mathbf{S}\mathbf{b}, \qquad y=\mathbf{a}^\top\mathbf{x},\ z=\mathbf{b}^\top\mathbf{x} \ \Rightarrow\ \widehat{\mathrm{Cov}}(y,z) = \tfrac{1}{n-1}\mathbf{d}_y^\top\mathbf{d}_z = \mathbf{a}^\top\mathbf{S}\mathbf{b}.$$

For two variables, $y = b\,x_1 + c\,x_2$ has $\widehat{\mathrm{Var}}(y) = \{b^2\|\mathbf{d}_1\|^2 + c^2\|\mathbf{d}_2\|^2 + 2bc\|\mathbf{d}_1\|\|\mathbf{d}_2\|\cos\theta\}/(n-1)$.

8 Dimension and Rank of the Data

Motivation. If $n \le p$, the sample covariance matrix is singular and cannot be inverted. The rank result here explains when classical multivariate procedures break down.

- Linear independence: $\sum c_i\mathbf{x}_i = \mathbf{0} \Rightarrow$ all $c_i = 0$. The rank is the maximal number of linearly independent columns (= rows). $\mathrm{rank}(\mathbf{X}) \le \min(n,p)$.
- $\mathrm{rank}(\mathbf{A}) = \mathrm{rank}(\mathbf{A}^\top) = \mathrm{rank}(\mathbf{A}^\top\mathbf{A})$, hence $\mathrm{rank}(\mathbf{S}) = \mathrm{rank}(\mathbf{X}_C^\top\mathbf{X}_C) = \mathrm{rank}(\mathbf{X}_C)$.
- All deviation vectors are orthogonal to $\mathbf{1}_n$, so they lie in an $(n-1)$-dimensional subspace: $\mathrm{rank}(\mathbf{X}_C) \le \min(n-1,p)$.
- An $m\times k$ matrix is **full-rank** if its rank is $\min(m,k)$ and **rank-deficient** otherwise. If $p>n$, $\mathrm{rank}(\mathbf{S}) \le n-1 < p$: $\mathbf{S}$ is always rank-deficient.

Theorem 1.31: rank-deficiency property (slides Proposition, homework)

For the covariance matrix $\mathbf{S}$ of $\mathbf{X}\in\mathbb{R}^{n\times p}$ the following are equivalent.

1. $\mathbf{S}$ is rank-deficient.
2. There is $\mathbf{a}\neq\mathbf{0}$ with $\mathbf{a}^\top(\mathbf{x}_i-\bar{\mathbf{x}}) = 0$ for all i.
3. All projection scores coincide: $s_{\mathbf{a}}(\mathbf{x}_i) = s_{\mathbf{a}}(\mathbf{x}_j)$ for all i,j.
4. $\mathrm{span}\{\mathbf{d}_1,\dots,\mathbf{d}_p\}$ has dimension less than p.
5. Some deviation vector is a linear combination of the others.
6. At least one eigenvalue of $\mathbf{S}$ is 0.

Proof. (1)⇔(2): $\mathrm{rank}(\mathbf{S}) = \mathrm{rank}(\mathbf{X}_C^\top) = r < p$ iff the columns $\{\mathbf{x}_i-\bar{\mathbf{x}}\}$ of $\mathbf{X}_C^\top$ span a space of dimension $r<p$, iff a nonzero $\mathbf{a}$ in its orthogonal complement exists.

(2)⇔(3): $\mathbf{a}^\top(\mathbf{x}_i-\bar{\mathbf{x}}) = 0 \iff \mathbf{a}^\top\mathbf{x}_i = \mathbf{a}^\top\bar{\mathbf{x}}$ for all i.

(1)⇔(4): $\mathrm{rank}(\mathbf{S}) = \mathrm{rank}(\mathbf{X}_C) = \dim\mathrm{span}\{\mathbf{d}_j\}$.

(4)⇔(5): dimension below p means $\{\mathbf{d}_j\}$ is linearly dependent.

(2)⇒(6): $\mathbf{Sa} = \frac{1}{n-1}\mathbf{X}_C^\top \mathbf{X}_C \mathbf{a} = \mathbf{0} = 0 \cdot \mathbf{a}$.

(6)⇒(2): $\mathbf{Su} = \mathbf{0}$ gives $\mathbf{u}^\top \mathbf{Su} = \frac{1}{n-1}\|\mathbf{X}_C \mathbf{u}\|^2 = 0$, so $\mathbf{X}_C \mathbf{u} = \mathbf{0}$. ■

If $\mathbf{S}$ is full rank, every direction carries independent variation. If it is rank-deficient, some direction carries no information at all (an exact linear relation among the variables), the sphering $\mathbf{x} \mapsto \mathbf{S}^{-1/2}\mathbf{x}$ is undefined, and statistical analysis becomes less reliable. In high dimensions ($p > n$) $\mathbf{S}$ is always rank-deficient: $\operatorname{rank}(\mathbf{S}) \le \min\{n-1, p\} < p$.

Remark 1.32: generalized variance as a squared volume [Supplement]

For $p = 2$, $|\mathbf{S}| = s_{11}s_{22} - s_{12}^2 = (\|\mathbf{d}_1\|\|\mathbf{d}_2\| \sin\theta_{12})^2/(n-1)^2$: the squared area of the parallelogram spanned by $\mathbf{d}_1, \mathbf{d}_2$. In general $|\mathbf{S}| = (n-1)^{-p}(\text{volume})^2$, which is 0 exactly in the rank-deficient case.



weak correlation: $|\mathbf{S}|$ large strong correlation: $|\mathbf{S}|$ small

Figure. [Supplement] *A small angle between* $\mathbf{d}_1$ *and* $\mathbf{d}_2$ *flattens the parallelogram and* $|\mathbf{S}|$ *tends to 0.*

Chapter 2. Matrix, Calculus and Probability Tools

Matrix identities, eigenvalue inequalities, matrix calculus, optimization, convergence

This chapter collects the tools used in later proofs (MLE, Wishart, T^2, large-sample theory). It follows the Lab Session 3 handout "Preliminaries", whose item numbers are quoted in the box titles.

1 Basic Matrix Operations

Motivation. This chapter collects the linear-algebra and probability tools used throughout, so later proofs can cite them. We start with trace, determinant, norms and continuity.

Definition 2.1: basic facts (Lab 3, Recall 1.1)

(i) $(\mathbf{AB})_{ik} = \sum_j a_{ij}b_{jk}$; $(\mathbf{A}^\top)^\top = \mathbf{A}$, $(\mathbf{AB})^\top = \mathbf{B}^\top\mathbf{A}^\top$; a 1×1 matrix equals its transpose.

(ii) $\operatorname{tr}(\mathbf{C}) = \sum_i c_{ii}$ is linear.

(iii) $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$, so $|\mathbf{A}^{-1}| = 1/|\mathbf{A}|$.

(iv) (Leibniz) $|\mathbf{A}| = \sum_\sigma \operatorname{sgn}(\sigma)\prod_i a_{i\sigma(i)}$; hence $|\mathbf{A}|$ is a polynomial in the entries, continuous and infinitely differentiable, and $\{|\mathbf{A}| \neq 0\}$ is open. $|\mathbf{I}_p| = 1$, $|\operatorname{diag}(d_i)| = \prod d_i$, $|c\mathbf{A}| = c^p|\mathbf{A}|$.

(v) (Adjugate) If $|\mathbf{A}| \neq 0$, $\mathbf{A}^{-1} = \operatorname{adj}(\mathbf{A})/|\mathbf{A}|$, so $\mathbf{A} \mapsto \mathbf{A}^{-1}$ is continuous and differentiable on $\{|\mathbf{A}| \neq 0\}$.

(vi) (Cauchy–Schwarz) $|\sum_k a_k b_k| \le (\sum a_k^2)^{1/2}(\sum b_k^2)^{1/2}$.

Definition 2.2: Frobenius norm (Lab 3, Definition 1.2; Lab 2)

$\|\mathbf{A}\|_F = (\sum_{i,j} a_{ij}^2)^{1/2}$ and $\|\mathbf{x}\| = \|\mathbf{x}\|_F$ for vectors. Example: $\mathbf{M} = \begin{pmatrix} -0.1 & 0.3 \\ 0.2 & -0.2 \end{pmatrix}$ has $\|\mathbf{M}\|_F = \sqrt{0.18} \approx 0.424$. Estimation errors of the sample moments can be measured by $E_{\bar u} = \|\bar{\mathbf{u}} - \boldsymbol{\mu}\|_2$, $E_S = \|\mathbf{S} - \boldsymbol{\Sigma}\|_F$, $E_R = \|\mathbf{R} - \mathbf{R}_0\|_F$.

Theorem 2.3: basic identities (Lab 3, Lemma 1.3)

For $\mathbf{x} \in \mathbb{R}^p$, $\mathbf{y} \in \mathbb{R}^q$, $\mathbf{A} \in \mathbb{R}^{p\times q}$, $\mathbf{B} \in \mathbb{R}^{q\times p}$:

(i) $\mathbf{x}^\top\mathbf{A}\mathbf{y} = \sum_{i,j} a_{ij}x_i y_j$; if $\mathbf{A}$ is diagonal, $\mathbf{x}^\top\mathbf{A}\mathbf{x} = \sum_i a_{ii}x_i^2$.

(ii) $\operatorname{tr}(\mathbf{AB}) = \operatorname{tr}(\mathbf{BA})$.

(iii) $\|\mathbf{A}\|_F^2 = \operatorname{tr}(\mathbf{A}^\top\mathbf{A})$ and $\|\mathbf{x}\|^2 = \mathbf{x}^\top\mathbf{x}$.

Proof. (i) $(\mathbf{Ay})_i = \sum_j a_{ij}y_j$, so $\mathbf{x}^\top\mathbf{Ay} = \sum_i x_i(\mathbf{Ay})_i$.

(ii) $\operatorname{tr}(\mathbf{AB}) = \sum_i\sum_j a_{ij}b_{ji} = \sum_j\sum_i b_{ji}a_{ij} = \operatorname{tr}(\mathbf{BA})$.

(iii) $(\mathbf{A}^\top\mathbf{A})_{jj} = \sum_i a_{ij}^2$; sum over j. ■

Theorem 2.4: first-order expansion of the determinant (Lab 3, Lemma 1.4)

$|\mathbf{I}_p + t\mathbf{N}| = 1 + t\,\mathrm{tr}(\mathbf{N}) + O(t^2)$ as $t \to 0$.

Proof. In the Leibniz formula a non-identity σ moves at least two indices, so its term is $O(t^2)$. The identity contributes $\prod_i (1 + t n_{ii}) = 1 + t\sum_i n_{ii} + O(t^2)$. ■

Definition 2.5: open sets and continuity (Lab 3, Definition 1.5)

With distance $\|\mathbf{B} - \mathbf{A}\|_F$: U is open iff every $\mathbf{A} \in U$ has a ball of radius $r > 0$ inside U; f is continuous at $\mathbf{A}$ iff $\forall \epsilon\, \exists \delta$ with $\|\mathbf{B} - \mathbf{A}\|_F < \delta \Rightarrow \|f(\mathbf{B}) - f(\mathbf{A})\|_F < \epsilon$.

Theorem 2.6: entrywise continuity (Lab 3, Lemma 1.6)

If every entry of a map F is continuous at $\mathbf{A}$, so is F.

Proof. With m entries, choose δ so that each $|F_{kl}(\mathbf{B}) - F_{kl}(\mathbf{A})| < \epsilon/\sqrt{m}$; then $\|F(\mathbf{B}) - F(\mathbf{A})\|_F < \epsilon$. ■

1.1 Block-matrix formulas

[Supplement] Used for conditional distributions, the Wishart theorem and likelihood ratio tests.

Theorem 2.7: block determinant and inverse [Supplement]

For $\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}$ with $\mathbf{A}_{22}$ invertible,

$$|\mathbf{A}| = |\mathbf{A}_{22}|\,|\mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}| = |\mathbf{A}_{11}|\,|\mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12}| \quad (\mathbf{A}_{11} \text{ invertible}).$$

With the Schur complement $\mathbf{A}_{11\cdot 2} := \mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}$,

$$\mathbf{A}^{-1} = \begin{pmatrix} \mathbf{A}_{11\cdot 2}^{-1} & -\mathbf{A}_{11\cdot 2}^{-1}\mathbf{A}_{12}\mathbf{A}_{22}^{-1} \\ -\mathbf{A}_{22}^{-1}\mathbf{A}_{21}\mathbf{A}_{11\cdot 2}^{-1} & \mathbf{A}_{22}^{-1} + \mathbf{A}_{22}^{-1}\mathbf{A}_{21}\mathbf{A}_{11\cdot 2}^{-1}\mathbf{A}_{12}\mathbf{A}_{22}^{-1} \end{pmatrix};$$

in particular $(\mathbf{A}^{-1})_{11} = \mathbf{A}_{11\cdot 2}^{-1}$.

Proof. Block Gaussian elimination:

$$\begin{pmatrix} \mathbf{I} & -\mathbf{A}_{12}\mathbf{A}_{22}^{-1} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \mathbf{A} \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{A}_{22}^{-1}\mathbf{A}_{21} & \mathbf{I} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{11\cdot 2} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{22} \end{pmatrix}.$$

The outer factors have determinant 1, giving the determinant formula; inverting both sides gives $\mathbf{A}^{-1}$. ■

Theorem 2.8: matrix determinant lemma and Woodbury identity [Supplement]

$$|\mathbf{A} + \mathbf{u}\mathbf{v}^\top| = |\mathbf{A}|(1 + \mathbf{v}^\top\mathbf{A}^{-1}\mathbf{u}), \qquad |\mathbf{I}_p + \mathbf{u}\mathbf{v}^\top| = 1 + \mathbf{v}^\top\mathbf{u},$$
$$(\mathbf{A} + \mathbf{U}\mathbf{C}\mathbf{V})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}(\mathbf{C}^{-1} + \mathbf{V}\mathbf{A}^{-1}\mathbf{U})^{-1}\mathbf{V}\mathbf{A}^{-1}.$$

Proof. Compute $\begin{vmatrix} \mathbf{A} & -\mathbf{u} \\ \mathbf{v}^\top & 1 \end{vmatrix}$ with both forms of the block formula: $|\mathbf{A}|(1+\mathbf{v}^\top\mathbf{A}^{-1}\mathbf{u}) = 1\cdot|\mathbf{A}+\mathbf{u}\mathbf{v}^\top|$. For Woodbury, multiply the right side by $\mathbf{A}+\mathbf{UCV}$ and check that the product is $\mathbf{I}$. ■

2 Eigenvalues and Positive Definiteness

Motivation. Most multivariate statistics are functions of eigenvalues of covariance matrices; their basic properties are recorded here.

Theorem 2.9: spectral decomposition and consequences (Lab 3, Theorem 2.1, Lemmas 2.2, 2.4)

Symmetric $\mathbf{A} = \mathbf{U}\boldsymbol{\Lambda}(\mathbf{A})\mathbf{U}^\top = \sum_i \lambda_i(\mathbf{A})\mathbf{u}_i\mathbf{u}_i^\top$, $\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_p(\mathbf{A})$ ($\lambda_i(\mathbf{A})$ denotes the ith largest eigenvalue).

$$|\mathbf{A}| = \prod_i \lambda_i(\mathbf{A}), \qquad \operatorname{tr}(\mathbf{A}) = \sum_i \lambda_i(\mathbf{A}), \qquad \mathbf{A} \succ 0 \iff \text{all } \lambda_i(\mathbf{A}) > 0.$$

Proof. $|\mathbf{U}||\mathbf{U}^\top| = |\mathbf{U}\mathbf{U}^\top| = 1$ gives $|\mathbf{A}| = |\boldsymbol{\Lambda}(\mathbf{A})|$; $\operatorname{tr}(\mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top) = \operatorname{tr}(\boldsymbol{\Lambda}\mathbf{U}^\top\mathbf{U}) = \operatorname{tr}\boldsymbol{\Lambda}$. Definiteness follows from $\mathbf{x}^\top\mathbf{A}\mathbf{x} = \sum_i \lambda_i(\mathbf{u}_i^\top\mathbf{x})^2$. ■

Theorem 2.10: smallest eigenvalue (Lab 3, Lemma 2.5)

For symmetric $\mathbf{A}, \mathbf{B}$:
(i) $\lambda_p(\mathbf{A}) = \min_{\|\mathbf{x}\|=1} \mathbf{x}^\top\mathbf{A}\mathbf{x}$;
(ii) $|\lambda_p(\mathbf{A}) - \lambda_p(\mathbf{B})| \leq \|\mathbf{A}-\mathbf{B}\|_F$;
(iii) $\{\mathbf{A} \succ 0\}$ is convex and open in the space of symmetric matrices.

Proof. (i) With $\mathbf{z} = \mathbf{U}^\top\mathbf{x}$, $\|\mathbf{z}\| = 1$ and $\mathbf{x}^\top\mathbf{A}\mathbf{x} = \sum \lambda_i z_i^2 \geq \lambda_p$, equality at $\mathbf{x} = \mathbf{u}_p$.

(ii) For $\|\mathbf{x}\| = 1$, Cauchy–Schwarz gives $|\mathbf{x}^\top\mathbf{A}\mathbf{x} - \mathbf{x}^\top\mathbf{B}\mathbf{x}| = |\sum_{ij}(a_{ij}-b_{ij})x_ix_j| \leq \|\mathbf{A}-\mathbf{B}\|_F(\sum_{ij}x_i^2x_j^2)^{1/2} = \|\mathbf{A}-\mathbf{B}\|_F$; minimize both sides over $\mathbf{x}$.

(iii) Convex: $\mathbf{x}^\top\{(1-s)\mathbf{A}_0 + s\mathbf{A}_1\}\mathbf{x} > 0$. Open: if $\mathbf{A} \succ 0$ and $\|\mathbf{B}-\mathbf{A}\|_F < \lambda_p(\mathbf{A})$, (ii) gives $\lambda_p(\mathbf{B}) \geq \lambda_p(\mathbf{A}) - \|\mathbf{A}-\mathbf{B}\|_F > 0$. ■

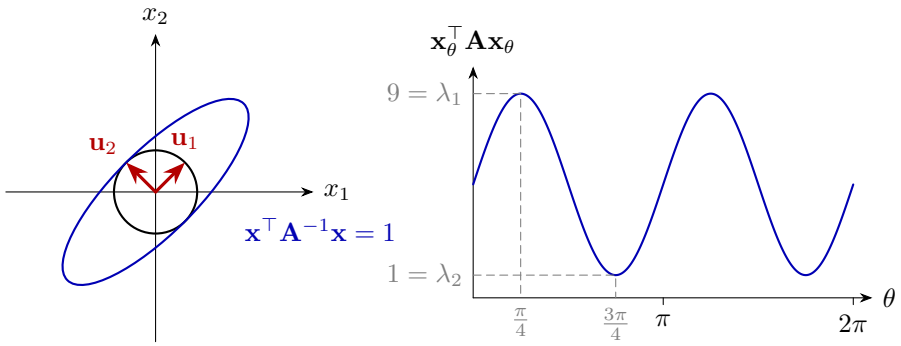


Figure. *For* $\mathbf{A} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$ *the quadratic form on the unit circle is* $5 + 4\sin 2\theta$*, with maximum* $\lambda_1 = 9$ *at* $\pm\mathbf{u}_1$ *and minimum* $\lambda_2 = 1$ *at* $\pm\mathbf{u}_2$.

Theorem 2.11: symmetric square root (Lab 3, Definition 2.6, Lemma 2.7)

For $\mathbf{A} \succ 0$, $\mathbf{A}^{\pm 1/2} = \mathbf{U}\mathbf{\Lambda}^{\pm 1/2}\mathbf{U}^\top$ are symmetric, $\mathbf{A}^{1/2}\mathbf{A}^{1/2} = \mathbf{A}$, $\mathbf{A}^{-1/2}\mathbf{A}^{-1/2} = \mathbf{A}^{-1}$, $\mathbf{A}^{-1/2}\mathbf{A}\mathbf{A}^{-1/2} = \mathbf{I}_p$. $\mathbf{A}^{-1} = \mathbf{U}\mathbf{\Lambda}^{-1}\mathbf{U}^\top \succ 0$ and $\lambda_i(\mathbf{A}^{-1}) = 1/\lambda_{p+1-i}(\mathbf{A})$ with eigenvector $\mathbf{u}_{p+1-i}$.

Proof. Each matrix is $\mathbf{U}\mathbf{D}\mathbf{U}^\top$ with $\mathbf{D}$ diagonal, hence symmetric; products collapse by $\mathbf{U}^\top\mathbf{U} = \mathbf{I}_p$. The entries $1/\lambda_i > 0$ of $\mathbf{\Lambda}^{-1}$, in decreasing order, sit on $\mathbf{u}_p, \dots, \mathbf{u}_1$. ■

3 Eigenvalue Inequalities

Motivation. PCA, canonical correlation and discriminant analysis all maximize a ratio of quadratic forms. The extremal characterization of eigenvalues (Rayleigh quotient, Courant–Fischer) is the common tool that solves them.

Definition 2.12: Birkhoff's theorem and the rearrangement inequality (Lab 3, Recall 3.1)

(i) A non-negative matrix with all row and column sums 1 (doubly stochastic) is a convex combination of permutation matrices.
(ii) If $a_1 \geq \cdots \geq a_p$ and $b_1 \geq \cdots \geq b_p$, then for every permutation σ, $\sum_i a_i b_{p+1-i} \leq \sum_i a_i b_{\sigma(i)} \leq \sum_i a_i b_i$.

Theorem 2.13: von Neumann's trace inequality (Lab 3, Theorem 3.2)

For symmetric $\mathbf{A} = \sum_i \lambda_i(\mathbf{A})\mathbf{u}_i\mathbf{u}_i^\top$, $\mathbf{B} = \sum_j \lambda_j(\mathbf{B})\mathbf{v}_j\mathbf{v}_j^\top$,

$$\sum_{i=1}^{p} \lambda_{p+1-i}(\mathbf{A})\lambda_i(\mathbf{B}) \leq \operatorname{tr}(\mathbf{A}\mathbf{B}) \leq \sum_{i=1}^{p} \lambda_i(\mathbf{A})\lambda_i(\mathbf{B}).$$

The upper bound is attained when $\mathbf{v}_i = \mathbf{u}_i$ and the lower bound when $\mathbf{v}_i = \mathbf{u}_{p+1-i}$.

Proof. Step 1. $\operatorname{tr}(\mathbf{A}\mathbf{B}) = \sum_{i,j} \lambda_i(\mathbf{A})\lambda_j(\mathbf{B})w_{ij}$, $w_{ij} = (\mathbf{u}_i^\top\mathbf{v}_j)^2 \geq 0$. Since $\{\mathbf{u}_i\}, \{\mathbf{v}_j\}$ are orthonormal bases, $\sum_j w_{ij} = \|\mathbf{u}_i\|^2 = 1$ and $\sum_i w_{ij} = \|\mathbf{v}_j\|^2 = 1$: $\mathbf{W}$ is doubly stochastic.

Step 2. By Birkhoff $\mathbf{W}$ is a convex combination of permutation matrices; $\operatorname{tr}(\mathbf{A}\mathbf{B})$ is linear in $\mathbf{W}$, so it lies between the minimum and maximum of $\sum_i \lambda_i(\mathbf{A})\lambda_{\sigma(i)}(\mathbf{B})$.

Step 3. By rearrangement the maximum is at $\sigma(i) = i$, the minimum at $\sigma(i) = p + 1 - i$.

Step 4. If $\mathbf{v}_i = \mathbf{u}_i$ (resp. $\mathbf{u}_{p+1-i}$), $\mathbf{W}$ is the identity (resp. reversal) permutation and the bound is attained. ■

Theorem 2.14: an elementary inequality (Lab 3, Lemma 3.3)

$x - \log x \geq 1$ for $x > 0$, with equality iff $x = 1$.

Proof. $\phi(x) = x - \log x$ has $\phi'(x) = 1 - 1/x$, negative on $(0, 1)$, positive on $(1, \infty)$. For $x \neq 1$ the mean value theorem gives $\phi(x) - \phi(1) = \phi'(\xi)(x - 1) > 0$ with ξ between 1 and x. ■

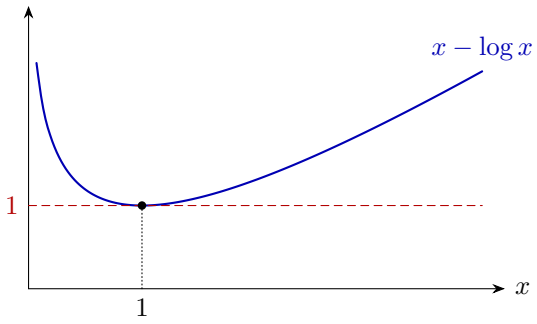


Figure. $x - \log x$ *touches its lower bound 1 only at* $x = 1$.

4 Vector Calculus

Motivation. Maximum likelihood for the multivariate normal requires derivatives with respect to vectors and matrices, including $\log\det\boldsymbol{\Sigma}$. These rules are collected here.

Definition 2.15: calculus recall (Lab 3, Recall 4.1)

(i) $g'' > 0$ on an interval $\Rightarrow$ g' strictly increasing, g strictly convex. (ii) Chain rule $\partial f(\mathbf{g}(\mathbf{x}))/\partial x_k = \sum_i (\partial f/\partial y_i)(\partial g_i/\partial x_k)$. (iii) Product rule.

Definition 2.16: gradient, Jacobian, Hessian (Lab 3, Definition 4.2)

For $f : \mathbb{R}^p \to \mathbb{R}$, $\partial f/\partial \mathbf{x} = (\partial f/\partial x_1, \dots, \partial f/\partial x_p)^\top$. The Jacobian of $\mathbf{g} : \mathbb{R}^p \to \mathbb{R}^q$ is the $q \times p$ matrix $(\partial g_i/\partial x_j)$. The Hessian $\partial^2 f/\partial \mathbf{x}\partial \mathbf{x}^\top$ is the Jacobian of the gradient.

Theorem 2.17: Schwarz's theorem (Lab 3, Theorem 4.3)

If the second partials of f are continuous, $\partial^2 f/\partial x_j \partial x_i = \partial^2 f/\partial x_i \partial x_j$: the Hessian is symmetric.

Proof. Let $\Delta(h) = f(\mathbf{x} + h\mathbf{e}_i + h\mathbf{e}_j) - f(\mathbf{x} + h\mathbf{e}_i) - f(\mathbf{x} + h\mathbf{e}_j) + f(\mathbf{x})$.

Step 1. The mean value theorem in x_i, then x_j, gives $\Delta(h) = h^2 \partial^2 f(\boldsymbol{\xi}_h)/\partial x_j \partial x_i$ with $\boldsymbol{\xi}_h \to \mathbf{x}$.

Step 2. In the other order, $\Delta(h) = h^2 \partial^2 f(\boldsymbol{\eta}_h)/\partial x_i \partial x_j$ with $\boldsymbol{\eta}_h \to \mathbf{x}$.

Step 3. Divide by h^2, let $h \to 0$ and use continuity. ∎

Theorem 2.18: chain rule (Lab 3, Lemma 4.4)

For $\mathbf{y} = \mathbf{g}(\mathbf{x})$ with $q \times p$ Jacobian $\mathbf{J} = \partial \mathbf{g}/\partial \mathbf{x}^\top$,

$$\frac{\partial f(\mathbf{y})}{\partial \mathbf{x}} = \mathbf{J}^\top \frac{\partial f}{\partial \mathbf{y}}, \qquad \frac{\partial^2 f(\mathbf{y})}{\partial \mathbf{x}\partial \mathbf{x}^\top} = \mathbf{J}^\top \frac{\partial^2 f}{\partial \mathbf{y}\partial \mathbf{y}^\top}\mathbf{J} + \sum_{i=1}^{q} \frac{\partial f}{\partial y_i}\frac{\partial^2 g_i}{\partial \mathbf{x}\partial \mathbf{x}^\top}.$$

For an affine map $\mathbf{g}(\mathbf{x}) = \mathbf{B}\mathbf{x} + \mathbf{b}$, $\mathbf{J} = \mathbf{B}$ and the last sum vanishes.

Proof. The first identity is the chain rule entrywise, $\partial f/\partial x_k = \sum_i (\partial f/\partial y_i) J_{ik}$. Differentiating in x_l with the product rule gives the second. ∎

Theorem 2.19: derivatives of quadratic forms and basic maps (Lab 3, Proposition 4.5, Lemma 4.6)

For symmetric $\mathbf{A}$, $\dfrac{\partial\,\mathbf{x}^\top\mathbf{A}\mathbf{x}}{\partial\mathbf{x}} = 2\mathbf{A}\mathbf{x}$ and $\dfrac{\partial^2\,\mathbf{x}^\top\mathbf{A}\mathbf{x}}{\partial\mathbf{x}\partial\mathbf{x}^\top} = 2\mathbf{A}$. Also $\partial\mathbf{x}/\partial\mathbf{x}^\top = \mathbf{I}_p$, $\partial(\mathbf{B}\mathbf{x}+\mathbf{b})/\partial\mathbf{x}^\top = \mathbf{B}$, and $\partial(\mathbf{x}-\boldsymbol{\mu})/\partial\boldsymbol{\mu}^\top = -\mathbf{I}_p$.

Proof. $\mathbf{x}^\top\mathbf{A}\mathbf{x} = \sum_{i,j} a_{ij}x_ix_j$, so $\partial/\partial x_k = \sum_j a_{kj}x_j + \sum_i a_{ik}x_i = 2(\mathbf{A}\mathbf{x})_k$; differentiating in x_l gives $2a_{kl}$. ■

5 Matrix Calculus

To optimize over covariance matrices we differentiate with respect to a matrix: matrix gradients give first-order conditions and directional derivatives give second-order conditions along symmetric directions.

Definition 2.20: directional derivatives and matrix gradient (Lab 3, Definitions 5.1, 5.3)

For f on an open set C, $\boldsymbol{\Theta}\in C$ and a direction $\mathbf{H}$,

$$Df(\boldsymbol{\Theta})[\mathbf{H}] = \frac{d}{dt}f(\boldsymbol{\Theta}+t\mathbf{H})\Big|_{t=0}, \qquad D^2f(\boldsymbol{\Theta})[\mathbf{H}] = \frac{d^2}{dt^2}f(\boldsymbol{\Theta}+t\mathbf{H})\Big|_{t=0}.$$

The matrix gradient $\partial f/\partial\mathbf{X}$ has (i,j) entry $\partial f/\partial x_{ij}$.

Theorem 2.21: directional derivative and gradient (Lab 3, Lemma 5.2, Proposition 5.4)

In $\mathbb{R}^p$: $Df(\boldsymbol{\mu})[\mathbf{h}] = \mathbf{h}^\top\frac{\partial f}{\partial\boldsymbol{\mu}}$, $D^2f(\boldsymbol{\mu})[\mathbf{h}] = \mathbf{h}^\top\frac{\partial^2 f}{\partial\boldsymbol{\mu}\partial\boldsymbol{\mu}^\top}\mathbf{h}$. For matrices: $Df(\mathbf{X})[\mathbf{H}] = \operatorname{tr}\Big\{\Big(\dfrac{\partial f}{\partial\mathbf{X}}\Big)^\top\mathbf{H}\Big\} = \sum_{i,j}\dfrac{\partial f}{\partial x_{ij}}h_{ij}$.

Theorem 2.22: derivatives of basic matrix functions (Lab 3, Propositions 5.5, 5.6)

For $\boldsymbol{\Sigma}\succ 0$, symmetric $\mathbf{H}$ and $p\times p$ $\mathbf{B}$:

$$D\log|\boldsymbol{\Sigma}|[\mathbf{H}] = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H}), \qquad D^2\log|\boldsymbol{\Sigma}|[\mathbf{H}] = -\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H}\boldsymbol{\Sigma}^{-1}\mathbf{H}),$$

$$D\operatorname{tr}(\boldsymbol{\Sigma}\mathbf{B})[\mathbf{H}] = \operatorname{tr}(\mathbf{H}\mathbf{B}), \qquad D^2\operatorname{tr}(\boldsymbol{\Sigma}\mathbf{B})[\mathbf{H}] = 0,$$

$$\frac{\partial\log|\boldsymbol{\Sigma}|}{\partial\boldsymbol{\Sigma}} = \boldsymbol{\Sigma}^{-1}, \qquad \frac{\partial\operatorname{tr}(\boldsymbol{\Sigma}\mathbf{B})}{\partial\boldsymbol{\Sigma}} = \mathbf{B}^\top.$$

D and D^2 are linear.

Proof. Let $\mathbf{M} = \boldsymbol{\Sigma}^{-1/2}\mathbf{H}\boldsymbol{\Sigma}^{-1/2} = \mathbf{V}\boldsymbol{\Lambda}(\mathbf{M})\mathbf{V}^\top$ (symmetric). Then $\boldsymbol{\Sigma}+t\mathbf{H} = \boldsymbol{\Sigma}^{1/2}(\mathbf{I}+t\mathbf{M})\boldsymbol{\Sigma}^{1/2}$ and $\mathbf{I}+t\mathbf{M}$ has eigenvalues $1+t\lambda_k(\mathbf{M})$, so for small $|t|$

$$\log|\boldsymbol{\Sigma}+t\mathbf{H}| = \log|\boldsymbol{\Sigma}| + \sum_k \log\{1+t\lambda_k(\mathbf{M})\}.$$

The first and second derivatives at $t=0$ are $\sum_k\lambda_k(\mathbf{M}) = \operatorname{tr}\mathbf{M} = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H})$ and $-\sum_k\lambda_k(\mathbf{M})^2 = -\operatorname{tr}(\mathbf{M}^2) = -\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H}\boldsymbol{\Sigma}^{-1}\mathbf{H})$. $\operatorname{tr}\{(\boldsymbol{\Sigma}+t\mathbf{H})\mathbf{B}\}$ is linear in t.

Gradients: for any $\mathbf{H}$, $|\boldsymbol{\Sigma}+t\mathbf{H}| = |\boldsymbol{\Sigma}||\mathbf{I}+t\boldsymbol{\Sigma}^{-1}\mathbf{H}| = |\boldsymbol{\Sigma}|\{1+t\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H})+O(t^2)\}$ (Theorem 2.4), so $D\log|\boldsymbol{\Sigma}|[\mathbf{H}] = \operatorname{tr}\{(\boldsymbol{\Sigma}^{-1})^\top\mathbf{H}\}$; compare with Theorem 2.21. Likewise $\operatorname{tr}(\mathbf{HB}) = \operatorname{tr}\{(\mathbf{B}^\top)^\top\mathbf{H}\}$. ■

Theorem 2.23: symmetric matrix variables (Lab 3, Corollary 5.7)

If $\mathbf{X}$ is symmetric and $\mathbf{G} = \partial f/\partial\mathbf{X}$ is symmetric, then $Df(\mathbf{X})[\mathbf{H}] = 0$ for all symmetric $\mathbf{H}$ iff $\mathbf{G} = \mathbf{0}$.

Proof. $\Leftarrow$ is clear. $\Rightarrow$: take $\mathbf{H} = \mathbf{G}$: $0 = \operatorname{tr}(\mathbf{G}^\top\mathbf{G}) = \|\mathbf{G}\|_F^2$. ■

6 Optimization

Motivation. Several estimators are defined as maximizers subject to constraints. Lagrange multipliers and the trace inequality used for the normal MLE are stated here.

Theorem 2.24: convexity and unique minimizer (Lab 3, Proposition 6.1)

Let C be open and convex and suppose $D^2f(\boldsymbol{\Theta})[\mathbf{H}] > 0$ for all $\boldsymbol{\Theta}\in C$, $\mathbf{H}\neq\mathbf{0}$. Then (i) f is strictly convex on C;
(ii) a point $\hat{\boldsymbol{\Theta}}$ with $Df(\hat{\boldsymbol{\Theta}})[\mathbf{H}] = 0$ for all $\mathbf{H}$ is the unique global minimizer;
(iii) for $C = \mathbb{R}^p$ these read: Hessian $\succ 0$ and gradient $= \mathbf{0}$.

Proof. (i) $g(t) = f(\boldsymbol{\Theta}_0 + t(\boldsymbol{\Theta}_1 - \boldsymbol{\Theta}_0))$ has $g''(t) = D^2f(\cdot)[\mathbf{H}] > 0$, so g is strictly convex.

(ii) With $\boldsymbol{\Theta}_0 = \hat{\boldsymbol{\Theta}}$, $g'(0) = 0$ and g' is strictly increasing, so $g'(t) > 0$ on $(0,1]$ and $f(\boldsymbol{\Theta}_1) = g(1) > g(0) = f(\hat{\boldsymbol{\Theta}})$ by the mean value theorem.

(iii) Theorem 2.21 and the symmetry of the Hessian (Theorem 2.17). ■

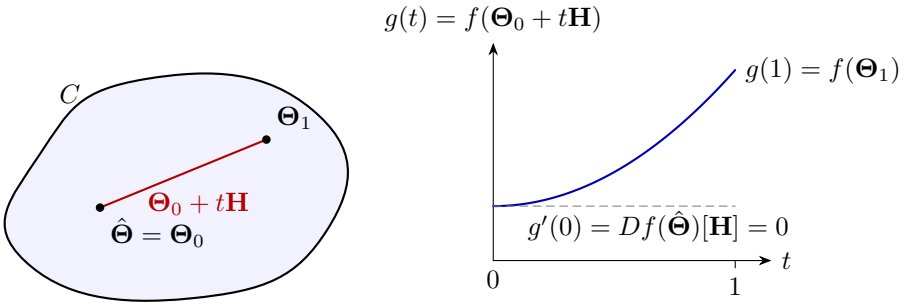


Figure. *Proof of Theorem 2.24: on a segment from the critical point, g is strictly convex with $g'(0) = 0$, so $g(1) > g(0)$.*

Theorem 2.25: transformations that keep the optimizer; profiling (Lab 3, Lemma 6.2, Proposition 6.3)

For $a > 0$ and a constant c: $\arg\min\{af + c\} = \arg\min f$, $\arg\max f = \arg\min(-f)$, $\arg\max L = \arg\max\log L$ for $L > 0$, and for a bijection ϕ, $\arg\min_\theta f(\theta) = \phi^{-1}(\arg\min_\eta f\circ\phi^{-1}(\eta))$. **Profiling**: if $\hat\theta(\eta)$ minimizes $\ell(\cdot,\eta)$ for each η and $\hat\eta$ minimizes the profile $p(\eta) = \ell(\hat\theta(\eta),\eta)$, then $(\hat\theta(\hat\eta),\hat\eta)$ minimizes ℓ jointly: $\ell(\theta,\eta)\geq\ell(\hat\theta(\eta),\eta) = p(\eta)\geq p(\hat\eta)$.

The MLE (Ch. 4) first profiles out $\boldsymbol{\mu}$ and then minimizes over $\boldsymbol{\Sigma}$.

7 Probability and Convergence

Motivation. Large-sample results need convergence in probability and distribution for random vectors; the multivariate CLT and Slutsky's lemma are the tools.

Definition 2.26: multivariate normal facts (Lab 3, Definition 7.1, Proposition 7.2)

$\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\boldsymbol{\Sigma} \succ 0$, has density $|2\pi\boldsymbol{\Sigma}|^{-1/2} \exp\{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\}$. If $\mathbf{Z} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$: $\mathbf{t}^\top \mathbf{Z} \sim N(0, \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t})$ and $\mathbf{Z}^\top \boldsymbol{\Sigma}^{-1} \mathbf{Z} \sim \chi^2_p$ (proofs in Ch. 3).

Definition 2.27: standard results from mathematical statistics (Lab 3, Recall 8.1)

(S0) $\mathbf{Y}_n \xrightarrow{p} \mathbf{c}$: $P(\|\mathbf{Y}_n - \mathbf{c}\| > \epsilon) \to 0$ for all $\epsilon > 0$. $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$: $\mathrm{E} g(\mathbf{Y}_n) \to \mathrm{E} g(\mathbf{Y})$ for all bounded continuous g.

(S1) Markov: for $W \geq 0$, $P(W \geq a) \leq \mathrm{E}W/a$.

(S2) $\mathrm{E}(\mathbf{AY} + \mathbf{b}) = \mathbf{A}\mathrm{E}\mathbf{Y} + \mathbf{b}$, $\mathrm{Cov}(\mathbf{AY} + \mathbf{b}) = \mathbf{A}\mathrm{Cov}(\mathbf{Y})\mathbf{A}^\top$; for independent $\mathbf{Y}_j$, $\mathrm{Cov}(\sum \mathbf{Y}_j) = \sum \mathrm{Cov}(\mathbf{Y}_j)$; $\mathrm{E}\|\mathbf{Y} - \mathrm{E}\mathbf{Y}\|^2 = \mathrm{tr}\mathrm{Cov}(\mathbf{Y})$; $EV^2 = 0 \Rightarrow V = 0$ a.s.

(S3) Weak law of large numbers; (S4) univariate CLT.

(S5) Continuous mapping theorem (CMT): $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$, g continuous $\Rightarrow g(\mathbf{Y}_n) \xrightarrow{d} g(\mathbf{Y})$; $\mathbf{Y}_n \xrightarrow{p} \mathbf{c}$, g continuous at $\mathbf{c} \Rightarrow g(\mathbf{Y}_n) \xrightarrow{p} g(\mathbf{c})$.

(S6) Slutsky: $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$, $\mathbf{W}_n \xrightarrow{p} \mathbf{c} \Rightarrow (\mathbf{Y}_n, \mathbf{W}_n) \xrightarrow{d} (\mathbf{Y}, \mathbf{c})$.

(S7) Componentwise convergence in probability to constants implies joint convergence.

Theorem 2.28: Cauchy–Schwarz for random variables (Lab 3, Lemma 8.2)

$\mathrm{E}|UV| \leq \{\mathrm{E}(U^2)\mathrm{E}(V^2)\}^{1/2}$.

Proof. If $EV^2 = 0$, $V = 0$ a.s. by (S2). Otherwise $0 \leq \mathrm{E}(|U| - t|V|)^2 = \mathrm{E}U^2 - 2t\mathrm{E}|UV| + t^2 EV^2$; take $t = \mathrm{E}|UV|/EV^2$. ∎

Theorem 2.29: vector Markov inequality and matrix convergence (Lab 3, Lemmas 8.3, 8.5)

$P(\|\mathbf{Y}\| > \epsilon) \leq \mathrm{E}\|\mathbf{Y}\|^2/\epsilon^2$. For random matrices, $\mathbf{A}_n \xrightarrow{p} \mathbf{A}$ (entrywise) $\iff \|\mathbf{A}_n - \mathbf{A}\|_F \xrightarrow{p} 0$.

Proof. $\{\|\mathbf{Y}\| > \epsilon\} \subset \{\|\mathbf{Y}\|^2 \geq \epsilon^2\}$; apply Markov. Matrices: $\|\mathbf{A}_n - \mathbf{A}\|_F^2 \leq pq \max_{ij} (\mathbf{A}_n - \mathbf{A})_{ij}^2$, so $\{\|\mathbf{A}_n - \mathbf{A}\|_F > \epsilon\} \subset \bigcup_{ij} \{|(\mathbf{A}_n - \mathbf{A})_{ij}| > \epsilon/\sqrt{pq}\}$ and a union bound gives convergence; conversely $|(\mathbf{A}_n - \mathbf{A})_{ij}| \leq \|\mathbf{A}_n - \mathbf{A}\|_F$. ∎

Theorem 2.30: characteristic functions and the Cramér–Wold device (Lab 3, Theorem 8.6, Corollary 8.7)

$\varphi_{\mathbf{Y}}(\mathbf{t}) = \mathrm{E}\exp(i\mathbf{t}^\top \mathbf{Y})$.

(i) Uniqueness: $\varphi_{\mathbf{Y}} \equiv \varphi_{\mathbf{W}} \iff \mathbf{Y} \overset{d}{=} \mathbf{W}$.

(ii) Lévy continuity: $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y} \iff \varphi_{\mathbf{Y}_n}(\mathbf{t}) \to \varphi_{\mathbf{Y}}(\mathbf{t})$ for all $\mathbf{t}$. **Cramér–Wold**:

(iii) the distribution of $\mathbf{Y}$ is determined by those of all $\mathbf{t}^\top\mathbf{Y}$;
(iv) $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y} \iff \mathbf{t}^\top\mathbf{Y}_n \xrightarrow{d} \mathbf{t}^\top\mathbf{Y}$ for every $\mathbf{t}$.

Proof. The key identity is $\varphi_{\mathbf{t}^\top\mathbf{Y}}(1) = \mathrm{E}e^{i\mathbf{t}^\top\mathbf{Y}} = \varphi_{\mathbf{Y}}(\mathbf{t})$.

(iii) If $\mathbf{t}^\top\mathbf{Y} \overset{d}{=} \mathbf{t}^\top\mathbf{W}$, then $\varphi_{\mathbf{Y}}(\mathbf{t}) = \varphi_{\mathbf{t}^\top\mathbf{Y}}(1) = \varphi_{\mathbf{t}^\top\mathbf{W}}(1) = \varphi_{\mathbf{W}}(\mathbf{t})$.

(iv) $\Leftarrow$: Lévy for scalars gives $\varphi_{\mathbf{t}^\top\mathbf{Y}_n}(s) \to \varphi_{\mathbf{t}^\top\mathbf{Y}}(s)$; take $s = 1$. $\Rightarrow$ is the CMT. ■

"All one-dimensional linear projections determine the multivariate distribution" underlies the general definition of the multivariate normal (Ch. 3) and the multivariate CLT.

8 Selected exercises

Motivation. Three further tools: the min–max characterization of every eigenvalue (which makes sample eigenvalues consistent), Hadamard's determinant inequality (which bounds $|\mathbf{R}|$ in independence tests), and the Kronecker product (the language for the covariance of a whole data matrix).

Exercise 2.1 (Courant–Fischer and Weyl)

[Supplement] Let $\mathbf{A}$ be symmetric $p\times p$ with eigenvalues $\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_p(\mathbf{A})$ and orthonormal eigenvectors $\mathbf{u}_1, \ldots, \mathbf{u}_p$.
(a) Show that for $k = 1, \ldots, p$,

$$\lambda_k(\mathbf{A}) = \max_{\mathcal{V}:\ \dim\mathcal{V}=k}\ \min_{\mathbf{x}\in\mathcal{V},\ \|\mathbf{x}\|=1} \mathbf{x}^\top\mathbf{A}\mathbf{x}.$$

(b) For symmetric $\mathbf{A}, \mathbf{B}$ show $|\lambda_k(\mathbf{A}) - \lambda_k(\mathbf{B})| \leq \|\mathbf{A}-\mathbf{B}\|_F$ for every k, which extends Theorem 2.10(ii) to all eigenvalues.

Solution [Supplement].

(a)

Step 1 ($\geq$). Let $\mathcal{V}_k = \mathrm{span}(\mathbf{u}_1, \ldots, \mathbf{u}_k)$. A unit vector $\mathbf{x} \in \mathcal{V}_k$ is $\mathbf{x} = \sum_{i\leq k} c_i\mathbf{u}_i$ with $\sum_i c_i^2 = 1$, so $\mathbf{x}^\top\mathbf{A}\mathbf{x} = \sum_{i\leq k}\lambda_i c_i^2 \geq \lambda_k$. Hence the minimum over $\mathcal{V}_k$ is at least λ_k (it equals λ_k at $\mathbf{x} = \mathbf{u}_k$), and the maximum over all $\mathcal{V}$ is at least λ_k.

Step 2 ($\leq$). Let $\mathcal{V}$ be any subspace with $\dim\mathcal{V} = k$ and $\mathcal{W} = \mathrm{span}(\mathbf{u}_k, \ldots, \mathbf{u}_p)$, $\dim\mathcal{W} = p-k+1$. Since $\dim\mathcal{V} + \dim\mathcal{W} = p+1 > p$, the intersection $\mathcal{V}\cap\mathcal{W}$ contains a unit vector $\mathbf{x} = \sum_{i\geq k} c_i\mathbf{u}_i$, and $\mathbf{x}^\top\mathbf{A}\mathbf{x} = \sum_{i\geq k}\lambda_i c_i^2 \leq \lambda_k$. So the minimum over $\mathcal{V}$ is at most λ_k for every $\mathcal{V}$.

(b)

Step 1. Let $\epsilon = \|\mathbf{A}-\mathbf{B}\|_F$. As in the proof of Theorem 2.10(ii), $|\mathbf{x}^\top\mathbf{A}\mathbf{x} - \mathbf{x}^\top\mathbf{B}\mathbf{x}| \leq \epsilon$ for every unit $\mathbf{x}$.

Step 2. Fix $\mathcal{V}$ with $\dim\mathcal{V} = k$. The unit sphere of $\mathcal{V}$ is compact, so $\mathbf{x} \mapsto \mathbf{x}^\top\mathbf{B}\mathbf{x}$ attains its minimum on it at some $\mathbf{x}_0$. Then

$$\min_{\mathbf{x}\in\mathcal{V},\|\mathbf{x}\|=1} \mathbf{x}^\top\mathbf{A}\mathbf{x} \leq \mathbf{x}_0^\top\mathbf{A}\mathbf{x}_0 \leq \mathbf{x}_0^\top\mathbf{B}\mathbf{x}_0 + \epsilon = \min_{\mathbf{x}\in\mathcal{V},\|\mathbf{x}\|=1} \mathbf{x}^\top\mathbf{B}\mathbf{x} + \epsilon.$$

Step 3. Taking the maximum over $\mathcal{V}$ and using (a) gives $\lambda_k(\mathbf{A}) \leq \lambda_k(\mathbf{B}) + \epsilon$. Exchanging $\mathbf{A}$ and $\mathbf{B}$ gives $\lambda_k(\mathbf{B}) \leq \lambda_k(\mathbf{A}) + \epsilon$.

Consequently each $\lambda_k(\cdot)$ is continuous, and $\mathbf{S} \xrightarrow{p} \mathbf{\Sigma}$ implies $\lambda_k(\mathbf{S}) \xrightarrow{p} \lambda_k(\mathbf{\Sigma})$ for every k (sample PC variances are consistent). ■

Exercise 2.2 (Hadamard's inequality)

[Supplement] Let $\mathbf{A} \succ 0$ be $p \times p$. Show that $|\mathbf{A}| \le \prod_{i=1}^{p} a_{ii}$, with equality iff $\mathbf{A}$ is diagonal. Deduce that a correlation matrix $\mathbf{R} \succ 0$ satisfies $0 < |\mathbf{R}| \le 1$, with $|\mathbf{R}| = 1$ iff $\mathbf{R} = \mathbf{I}_p$.

Solution [Supplement].

Step 1 (reduction). $a_{ii} = \mathbf{e}_i^\top \mathbf{A}\mathbf{e}_i > 0$. Let $\mathbf{D} = \operatorname{diag}(a_{11}, \dots, a_{pp})$ and $\mathbf{B} = \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$. Then $\mathbf{B} \succ 0$, $b_{ii} = 1$ and $|\mathbf{B}| = |\mathbf{A}| / \prod_i a_{ii}$, so it suffices to show $|\mathbf{B}| \le 1$ with equality iff $\mathbf{B} = \mathbf{I}_p$.

Step 2. Let $\mu_1, \dots, \mu_p > 0$ be the eigenvalues of $\mathbf{B}$. By Theorem 2.9, $\sum_i \mu_i = \operatorname{tr}\mathbf{B} = p$ and $\log|\mathbf{B}| = \sum_i \log \mu_i$. Theorem 2.14 gives $\log \mu_i \le \mu_i - 1$, so

$$\log|\mathbf{B}| \le \sum_{i=1}^{p} (\mu_i - 1) = \operatorname{tr}\mathbf{B} - p = 0.$$

Step 3 (equality). Equality holds iff $\mu_i = 1$ for all i, i.e. $\mathbf{B} = \mathbf{U}\mathbf{I}_p\mathbf{U}^\top = \mathbf{I}_p$, i.e. $\mathbf{A} = \mathbf{D}$ is diagonal. For a correlation matrix $a_{ii} = 1$, and $|\mathbf{R}| > 0$ because $\mathbf{R} \succ 0$. ■

Exercise 2.3 (Kronecker products and vec)

[Supplement] For $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $\mathbf{B} \in \mathbb{R}^{r\times s}$ let $\mathbf{A} \otimes \mathbf{B} = [a_{ij}\mathbf{B}]_{i,j} \in \mathbb{R}^{mr\times ns}$, and let $\operatorname{vec}(\mathbf{X})$ stack the columns of $\mathbf{X}$.
(a) Show $(\mathbf{A}\otimes\mathbf{B})(\mathbf{C}\otimes\mathbf{D}) = \mathbf{AC}\otimes\mathbf{BD}$ whenever the products are defined, $(\mathbf{A}\otimes\mathbf{B})^\top = \mathbf{A}^\top \otimes \mathbf{B}^\top$, and $(\mathbf{A}\otimes\mathbf{B})^{-1} = \mathbf{A}^{-1}\otimes\mathbf{B}^{-1}$ for invertible $\mathbf{A}, \mathbf{B}$.
(b) For symmetric $\mathbf{A}$ $(m\times m)$ and $\mathbf{B}$ $(r\times r)$ with eigenpairs $(\lambda_i, \mathbf{u}_i)$ and $(\mu_j, \mathbf{v}_j)$, show that $\mathbf{A}\otimes\mathbf{B}$ has the orthonormal eigenvectors $\mathbf{u}_i \otimes \mathbf{v}_j$ with eigenvalues $\lambda_i\mu_j$; hence $|\mathbf{A}\otimes\mathbf{B}| = |\mathbf{A}|^r|\mathbf{B}|^m$ and $\operatorname{tr}(\mathbf{A}\otimes\mathbf{B}) = \operatorname{tr}\mathbf{A}\operatorname{tr}\mathbf{B}$.
(c) Show $\operatorname{vec}(\mathbf{AXB}) = (\mathbf{B}^\top \otimes \mathbf{A})\operatorname{vec}(\mathbf{X})$.
(d) If the $n \times p$ data matrix $\mathbf{X}$ has iid rows $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, show $\operatorname{vec}(\mathbf{X}) \sim N_{np}(\boldsymbol{\mu}\otimes\mathbf{1}_n,\ \boldsymbol{\Sigma}\otimes\mathbf{I}_n)$.

Solution [Supplement].

(a) The (i,k) block of the product is $\sum_j (a_{ij}\mathbf{B})(c_{jk}\mathbf{D}) = (\sum_j a_{ij}c_{jk})\mathbf{BD} = (\mathbf{AC})_{ik}\mathbf{BD}$, the (i,k) block of $\mathbf{AC}\otimes\mathbf{BD}$. The (i,j) block of $(\mathbf{A}\otimes\mathbf{B})^\top$ is the transpose of the (j,i) block, $a_{ji}\mathbf{B}^\top$. Finally $(\mathbf{A}\otimes\mathbf{B})(\mathbf{A}^{-1}\otimes\mathbf{B}^{-1}) = \mathbf{I}_m \otimes \mathbf{I}_r = \mathbf{I}_{mr}$.

(b)

Step 1. Treating vectors as one-column matrices, (a) gives $(\mathbf{A}\otimes\mathbf{B})(\mathbf{u}_i\otimes\mathbf{v}_j) = \mathbf{A}\mathbf{u}_i \otimes \mathbf{B}\mathbf{v}_j = \lambda_i\mu_j\, \mathbf{u}_i\otimes\mathbf{v}_j$.

Step 2. $(\mathbf{u}_i\otimes\mathbf{v}_j)^\top(\mathbf{u}_k\otimes\mathbf{v}_l) = (\mathbf{u}_i^\top\mathbf{u}_k)(\mathbf{v}_j^\top\mathbf{v}_l) = \delta_{ik}\delta_{jl}$, so these mr vectors form an orthonormal eigenbasis of the symmetric matrix $\mathbf{A}\otimes\mathbf{B}$.

Step 3. By Theorem 2.9, $|\mathbf{A}\otimes\mathbf{B}| = \prod_{i,j}\lambda_i\mu_j = (\prod_i\lambda_i)^r(\prod_j\mu_j)^m = |\mathbf{A}|^r|\mathbf{B}|^m$ and $\operatorname{tr}(\mathbf{A}\otimes\mathbf{B}) = \sum_{i,j}\lambda_i\mu_j = \operatorname{tr}\mathbf{A}\operatorname{tr}\mathbf{B}$.

(c) Let $\mathbf{X} \in \mathbb{R}^{n\times p}$ have columns $\mathbf{x}_1, \dots, \mathbf{x}_p$ and $\mathbf{B} \in \mathbb{R}^{p\times q}$. Column k of $\mathbf{AXB}$ is $\mathbf{AX}\mathbf{b}_k = \sum_j b_{jk}\mathbf{A}\mathbf{x}_j = [b_{1k}\mathbf{A}, \dots, b_{pk}\mathbf{A}]\operatorname{vec}(\mathbf{X})$, and $[b_{1k}\mathbf{A}, \dots, b_{pk}\mathbf{A}]$ is the kth block row of $\mathbf{B}^\top\otimes\mathbf{A}$.

(d) Column j of $\mathbf{X}$ is $\mathbf{x}_{(j)} = (X_{1j}, \dots, X_{nj})^\top$ with mean $\mu_j\mathbf{1}_n$, so $\mathrm{E}\operatorname{vec}(\mathbf{X}) = \boldsymbol{\mu}\otimes\mathbf{1}_n$. For columns j, k, $\operatorname{Cov}(X_{ij}, X_{lk}) = \sigma_{jk}\delta_{il}$ by independence of the rows, so $\operatorname{Cov}(\mathbf{x}_{(j)}, \mathbf{x}_{(k)}) = \sigma_{jk}\mathbf{I}_n$, the (j,k) block of $\boldsymbol{\Sigma}\otimes\mathbf{I}_n$. Normality: the stacked rows $(\mathbf{X}_1^\top, \dots, \mathbf{X}_n^\top)^\top$ are jointly normal (independent normal blocks), and $\operatorname{vec}(\mathbf{X})$ is a permutation, hence a linear map, of this vector (Theorem 3.7). ■

Chapter 3. The Multivariate Normal Distribution

Definition and properties, conditional distributions, best linear prediction, multiple and partial correlation, quadratic forms

1 Definition

1.1 Non-singular case

The univariate normal density is $f(x) = (2\pi\sigma^2)^{-1/2}\exp[-\frac{1}{2}(x-\mu)\sigma^{-2}(x-\mu)]$. Replacing $(x-\mu)^2/\sigma^2$ by a squared Mahalanobis distance gives the multivariate version.

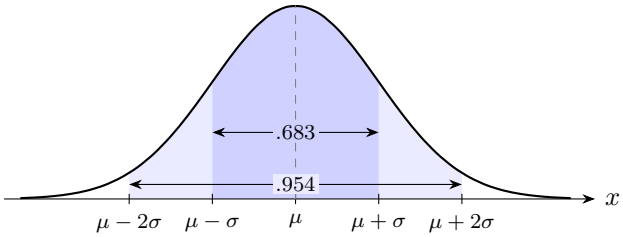


Figure. *The $N(\mu,\sigma^2)$ density; $\mu\pm\sigma$ carries 0.683, $\mu\pm2\sigma$ carries 0.954.*

Definition 3.1: multivariate normal (non-singular)

For $\boldsymbol{\mu}\in\mathbb{R}^p$ and $\boldsymbol{\Sigma}_{(p\times p)}\succ 0$, $\mathbf{X}\sim N_p(\boldsymbol{\mu},\boldsymbol{\Sigma})$ means $\mathbf{X}$ has density

$$f(\mathbf{x}) = |2\pi\boldsymbol{\Sigma}|^{-1/2}\exp\Big[-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\Big],\qquad \mathbf{x}\in\mathbb{R}^p,$$

where $|2\pi\boldsymbol{\Sigma}| = (2\pi)^p|\boldsymbol{\Sigma}|$.

$\int f = 1$. Let $\mathbf{Y} = \boldsymbol{\Sigma}^{-1/2}(\mathbf{X}-\boldsymbol{\mu})$, so $d\mathbf{y}/d\mathbf{x} = |\boldsymbol{\Sigma}|^{-1/2}$. Then

$$\int f\,d\mathbf{x} = \int\cdots\int (2\pi)^{-p/2}|\boldsymbol{\Sigma}|^{-1/2}e^{-\mathbf{y}^\top\mathbf{y}/2}|\boldsymbol{\Sigma}|^{1/2}d\mathbf{y} = \prod_{i=1}^{p}\int(2\pi)^{-1/2}e^{-y_i^2/2}dy_i = 1.$$

Equivalently, with $\boldsymbol{\Sigma} = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^\top$ and $\mathbf{z} = \mathbf{Q}^\top(\mathbf{x}-\boldsymbol{\mu})$, the exponent becomes $\sum_i z_i^2/\lambda_i$ and the integral is a product of univariate Gaussian integrals. ■

The substitution shows $\mathbf{Y} = \boldsymbol{\Sigma}^{-1/2}(\mathbf{X}-\boldsymbol{\mu}) \sim N_p(\mathbf{0},\mathbf{I}_p)$, i.e. $\mathbf{X} \overset{d}{=} \boldsymbol{\Sigma}^{1/2}\mathbf{Y}+\boldsymbol{\mu}$. Hence $\mathrm{E}\mathbf{X} = \boldsymbol{\mu}$, $\mathrm{Var}\mathbf{X} = \boldsymbol{\Sigma}$, and

$$U = (\mathbf{X}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{X}-\boldsymbol{\mu}) = \mathbf{Y}^\top\mathbf{Y} = \textstyle\sum_i Y_i^2 \sim \chi^2(p).$$

1.2 General definition and Cramér–Wold

Definition 3.2: multivariate normal (singular allowed)

$\mathbf{X}\in\mathbb{R}^p$ has a multivariate normal distribution iff $\mathbf{t}^\top\mathbf{X}$ is univariate normal for every $\mathbf{t}\in\mathbb{R}^p$ (including the point mass $N(c,0)$).

This definition does not require $\mathrm{Cov}(\mathbf{X})$ to be non-singular: $\mathbf{X}$ is MVN iff *every* projection onto a one-dimensional subspace is normal. It is well defined because, by Cramér–Wold (Theorem 2.30), the distributions of all $\mathbf{t}^\top\mathbf{X}$ determine that of $\mathbf{X}$.

Example 3.3: normal marginals do not imply joint normality (Example 4.8)

$X_1 \sim N(0,1)$ and $X_2 = -X_1$ if $-1 \le X_1 \le 1$, $X_2 = X_1$ otherwise.
(a) $X_2 \sim N(0,1)$.
(b) (X_1, X_2) is not bivariate normal.

Proof. (a) X_1 is symmetric about 0, so $-X_1 \overset{d}{=} X_1$ on $\{-1 \le X_1 \le 1\}$: $P[X_2 \le x_2] = P[-X_1 \le x_2, |X_1| \le 1] + P[X_1 \le x_2, |X_1| > 1] = P[X_1 \le x_2]$.

(b) $X_1 - X_2 = 2X_1$ if $|X_1| \le 1$ and 0 otherwise, so $P(X_1 - X_2 = 0) = P(|X_1| > 1) \approx 0.317 \in (0,1)$: $X_1 - X_2$ is neither a continuous normal nor a point mass. A linear combination is not normal, so (X_1, X_2) is not jointly normal. ■

1.3 Characteristic function

Theorem 3.4: characteristic function of the MVN

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succeq 0$, then $\varphi_{\mathbf{X}}(\mathbf{t}) = \exp\left(i\mathbf{t}^\top \boldsymbol{\mu} - \frac{1}{2}\mathbf{t}^\top \boldsymbol{\Sigma}\mathbf{t}\right)$. If $\boldsymbol{\Sigma} \succ 0$, the density above exists.

Proof. $\mathbf{t}^\top \mathbf{X} \sim N(\mathbf{t}^\top \boldsymbol{\mu}, \mathbf{t}^\top \boldsymbol{\Sigma}\mathbf{t})$ and $\varphi_{N(m,v)}(s) = \exp(ism - \frac{1}{2}s^2 v)$, so $\varphi_{\mathbf{X}}(\mathbf{t}) = \varphi_{\mathbf{t}^\top \mathbf{X}}(1) = \exp(i\mathbf{t}^\top \boldsymbol{\mu} - \frac{1}{2}\mathbf{t}^\top \boldsymbol{\Sigma}\mathbf{t})$. ■

[Supplement] Since the characteristic function depends only on $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the notation $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is justified under the general definition as well.

2 Geometry: Constant-Density Ellipsoids

Location parameter $\boldsymbol{\mu}$, shape parameter $\boldsymbol{\Sigma} \succ 0$. The contours of constant density are

$$E_c = \{\mathbf{x} \in \mathbb{R}^p : f(\mathbf{x}) = c_0\} = \{\mathbf{x} : (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) = c^2\},$$

where c is the Mahalanobis distance $d_M(\boldsymbol{\mu}, \mathbf{x})$. With $\boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$, $\lambda_1 \ge \cdots \ge \lambda_p > 0$, E_c is an ellipsoid centered at $\boldsymbol{\mu}$ with principal axes along $\mathbf{u}_i$ and half-lengths proportional to $\sqrt{\lambda_i}$. If $\boldsymbol{\Sigma} = \mathbf{I}_p$ it is a sphere of radius c.

Example 3.5: bivariate normal

$\boldsymbol{\Sigma} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = \mathbf{R}_{\pi/4} \mathrm{diag}(3,1) \mathbf{R}_{\pi/4}^\top$: elliptical contours with principal axes at $45°$ and $-45°$ and half-axis ratio $\sqrt{3} : 1$.

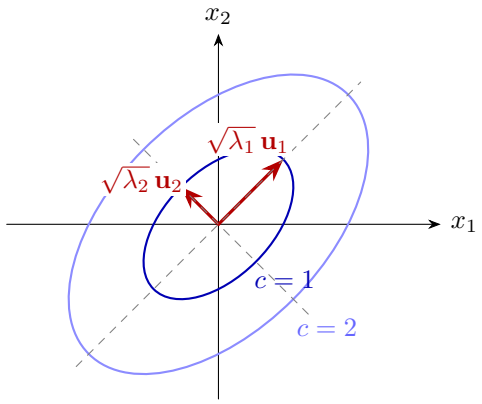


Figure. *Contours of the example ($\lambda_1 = 3$, $\lambda_2 = 1$); half-axes $c\sqrt{\lambda_i}$ along $\mathbf{u}_i$.*

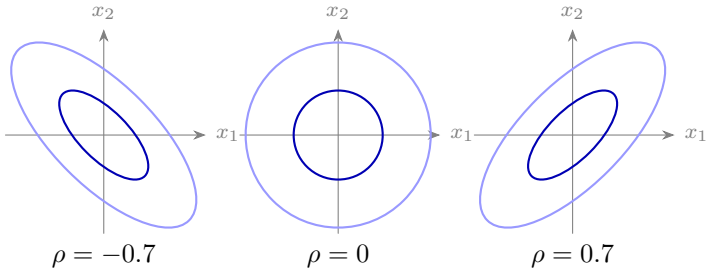


Figure. *[Supplement] 50% and 95% contours for unit variances; eigenvalues $1 \pm \rho$, so ρ sets tilt and eccentricity.*

3 Linear Transformations, Subsets, Independence, Conditionals

Motivation. The multivariate normal is the default model because it is closed under every operation we perform: linear maps, marginals and conditionals are again normal, and zero correlation means independence. The conditional distribution formula is the population version of linear regression.

Theorem 3.6: linear combinations (Result 4.2)

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ then $\mathbf{a}^\top \mathbf{X} = a_1 X_1 + \cdots + a_p X_p \sim N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})$. Conversely, if $\mathbf{a}^\top \mathbf{X} \sim N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})$ for every $\mathbf{a}$, then $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

Proof. $\mathrm{E}(\mathbf{a}^\top \mathbf{X}) = \mathbf{a}^\top \boldsymbol{\mu}$ and $\mathrm{Var}(\mathbf{a}^\top \mathbf{X}) = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$. Normality via the MGF: $M_{\mathbf{a}^\top \mathbf{X}}(t) = \mathrm{E} e^{t \mathbf{a}^\top \mathbf{X}} = M_{\mathbf{X}}(t\mathbf{a}) = \exp(t \mathbf{a}^\top \boldsymbol{\mu} + \tfrac{1}{2} t^2 \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})$, the MGF of $N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})$. The converse is the general definition. ∎

Theorem 3.7: linear transformations (Result 4.3)

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\mathbf{A} \in \mathbb{R}^{q \times p}$ and $\mathbf{d} \in \mathbb{R}^p$, then $\mathbf{A}\mathbf{X} \sim N_q(\mathbf{A}\boldsymbol{\mu}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top)$ and $\mathbf{X} + \mathbf{d} \sim N_p(\boldsymbol{\mu} + \mathbf{d}, \boldsymbol{\Sigma})$.

Proof. $\mathrm{E}(\mathbf{A}\mathbf{X}) = \mathbf{A}\boldsymbol{\mu}$ and $\mathrm{Cov}(\mathbf{A}\mathbf{X}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top$. Any $\mathbf{b}^\top(\mathbf{A}\mathbf{X}) = (\mathbf{A}^\top \mathbf{b})^\top \mathbf{X}$ is a linear combination of $\mathbf{X}$, normal by Theorem 3.6, so $\mathbf{A}\mathbf{X}$ is MVN. $\mathbf{a}^\top(\mathbf{X} + \mathbf{d}) = \mathbf{a}^\top \mathbf{X} + \mathbf{a}^\top \mathbf{d}$ only shifts the mean. ∎

Theorem 3.8: subsets (Result 4.4)

Partition $\mathbf{X} = \begin{pmatrix}\mathbf{X}_1\\ \mathbf{X}_2\end{pmatrix}$, $\boldsymbol{\mu} = \begin{pmatrix}\boldsymbol{\mu}_1\\ \boldsymbol{\mu}_2\end{pmatrix}$, $\boldsymbol{\Sigma} = \begin{pmatrix}\boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12}\\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22}\end{pmatrix}$. Then $\mathbf{X}_1 \sim N_q(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$; all subsets of $\mathbf{X}$ are normal.

Proof. Theorem 3.7 with $\mathbf{A} = [\mathbf{I}_{q\times q}\ \mathbf{0}_{q\times(p-q)}]$. Any subset can be relabelled as $\mathbf{X}_1$. ■

Theorem 3.9: independence and covariance (Result 4.5)

(a) If $\mathbf{X}_1 \perp\!\!\!\perp \mathbf{X}_2$, then $\mathrm{Cov}(\mathbf{X}_1, \mathbf{X}_2) = \mathbf{0}$.

(b) If $\begin{pmatrix}\mathbf{X}_1\\ \mathbf{X}_2\end{pmatrix} \sim N_{q_1+q_2}$, then $\mathbf{X}_1 \perp\!\!\!\perp \mathbf{X}_2 \iff \boldsymbol{\Sigma}_{12} = \mathbf{0}$.

(c) If $\mathbf{X}_1 \sim N_{q_1}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$ and $\mathbf{X}_2 \sim N_{q_2}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$ are independent, then $(\mathbf{X}_1^\top, \mathbf{X}_2^\top)^\top \sim N_{q_1+q_2}\Big(\begin{pmatrix}\boldsymbol{\mu}_1\\ \boldsymbol{\mu}_2\end{pmatrix}, \begin{pmatrix}\boldsymbol{\Sigma}_{11} & \mathbf{0}\\ \mathbf{0} & \boldsymbol{\Sigma}_{22}\end{pmatrix}\Big)$.

Proof.

(a) [Supplement] Each entry $\mathrm{E}[(X_{1i} - \mu_{1i})(X_{2j} - \mu_{2j})]$ factorizes into $0 \cdot 0$.

(b) The joint characteristic function is

$$\varphi_{\mathbf{X}}(\mathbf{t}_1, \mathbf{t}_2) = \exp\Big\{ i(\mathbf{t}_1^\top \boldsymbol{\mu}_1 + \mathbf{t}_2^\top \boldsymbol{\mu}_2) - \tfrac{1}{2} \begin{pmatrix}\mathbf{t}_1\\ \mathbf{t}_2\end{pmatrix}^\top \begin{pmatrix}\boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12}\\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22}\end{pmatrix} \begin{pmatrix}\mathbf{t}_1\\ \mathbf{t}_2\end{pmatrix} \Big\}.$$

If $\boldsymbol{\Sigma}_{12} = \mathbf{0}$ it factorizes as $\varphi_{\mathbf{X}_1}(\mathbf{t}_1)\varphi_{\mathbf{X}_2}(\mathbf{t}_2)$: independence. Conversely, independence forces the factorization, so the cross term $\mathbf{t}_1^\top \boldsymbol{\Sigma}_{12} \mathbf{t}_2$ must vanish for all $\mathbf{t}$, i.e. $\boldsymbol{\Sigma}_{12} = \mathbf{0}$.

(c) [Supplement] Reads the factorization backwards. ■

Remark 3.10: "uncorrelated ⇒ independent" needs joint normality

[Supplement] In Example 3.3 each marginal is normal but the pair is not jointly normal, so (b) does not apply.

Theorem 3.11: conditional distribution (Result 4.6)

If $\boldsymbol{\Sigma}_{22} \succ 0$,

$$\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2 \ \sim\ N_r(\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2),\ \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}).$$

The conditional mean is linear in $\mathbf{x}_2$; the conditional covariance $\boldsymbol{\Sigma}_{11\cdot 2} := \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$ does not depend on $\mathbf{x}_2$.

Proof. Define $\mathbf{X}_1^* = \mathbf{X}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{X}_2$, $\mathbf{X}_2^* = \mathbf{X}_2$, i.e. $\mathbf{X}^* = \mathbf{A}\mathbf{X}$ with $\mathbf{A} = \begin{pmatrix}\mathbf{I}_r & -\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\\ \mathbf{0} & \mathbf{I}_s\end{pmatrix}$. By Theorem 3.7 $\mathbf{X}^*$ is MVN and

$$\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top = \begin{pmatrix}\boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21} & \mathbf{0}\\ \mathbf{0} & \boldsymbol{\Sigma}_{22}\end{pmatrix}.$$

$\mathrm{Cov}(\mathbf{X}_1^*, \mathbf{X}_2^*) = \mathbf{0}$, so $\mathbf{X}_1^* \perp\!\!\!\perp \mathbf{X}_2$ (Theorem 3.9(b)) and $\mathbf{X}_1^* \sim N_r(\boldsymbol{\mu}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{11\cdot 2})$. Since $\mathbf{X}_1 = \mathbf{X}_1^* + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{X}_2$, conditioning on $\mathbf{X}_2 = \mathbf{x}_2$ makes the second term constant without changing the law of $\mathbf{X}_1^*$: $\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2 \overset{d}{=} \mathbf{X}_1^* + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{x}_2$. ■

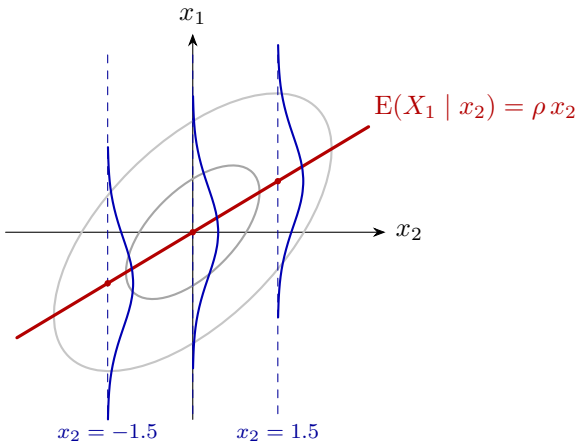


Figure. *Result 4.6 with $\rho = 0.6$: each slice $X_2 = x_2$ is normal, centred on the line $0.6\,x_2$, with the same variance $1 - \rho^2 = 0.64$.*

Remark 3.12: residuals, projections and partial variance

$\mathbf{X}_1^* = \mathbf{X}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{X}_2$ is (up to centring) the residual $\mathbf{X}_1 - \mathrm{BLP}(\mathbf{X}_1 \mid \mathbf{X}_2)$ after projecting $\mathbf{X}_1$ onto the span of $\mathbf{X}_2$. The residual is uncorrelated with $\mathbf{X}_2$, and independent of it in the Gaussian case. $\sigma^2_{11\cdot 2} = \sigma_{11} - \boldsymbol{\sigma}_{21}^\top \boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21}$ (more generally $\boldsymbol{\Sigma}_{11\cdot 2}$) is the **partial variance** of $\mathbf{X}_1$ after removing the effect of $\mathbf{X}_2$, i.e. $\mathrm{Var}(\mathbf{X}_1^*)$.

4 Best Linear Prediction, Multiple and Partial Correlation

Motivation. How well can Y be predicted linearly from $\mathbf{X}$, and how correlated are two variables once a third is held fixed? Both answers come from the conditional covariance of the normal distribution.

Let $\mathbf{X} \in \mathbb{R}^q$ be the predictor and $Y \in \mathbb{R}$ the response, $\mathbf{U} = (\mathbf{X}^\top, Y)^\top \in \mathbb{R}^p$ $(p = q + 1)$,

$$\boldsymbol{\mu} = \mathrm{E}(\mathbf{U}) = \begin{pmatrix} \boldsymbol{\mu}_X \\ \mu_Y \end{pmatrix}, \qquad \boldsymbol{\Sigma} = \mathrm{Var}(\mathbf{U}) = \begin{pmatrix} \boldsymbol{\Sigma}_{XX} & \boldsymbol{\Sigma}_{XY} \\ \boldsymbol{\Sigma}_{YX} & \sigma_Y^2 \end{pmatrix},$$

with sample counterparts $\bar{\mathbf{u}} = (\bar{\mathbf{x}}^\top, \bar{y})^\top$ and $\mathbf{S} = \begin{pmatrix} \mathbf{S}_{XX} & \mathbf{S}_{XY} \\ \mathbf{S}_{YX} & s_Y^2 \end{pmatrix}$.

4.1 Best linear predictor

Theorem 3.13: best linear predictor

$\mathrm{BLP}(Y \mid \mathbf{X}) := \mathbf{A}^*\mathbf{X} + b^*$, $(\mathbf{A}^*, b^*) = \arg\min_{\mathbf{A} \in \mathbb{R}^{1\times q},\, b \in \mathbb{R}} \mathrm{E}\{Y - (\mathbf{A}\mathbf{X} + b)\}^2$. With finite second moments and $\boldsymbol{\Sigma}_{XX} \succ 0$,

$$\mathbf{A}^* = \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}, \qquad b^* = \mu_Y - \mathbf{A}^*\boldsymbol{\mu}_X, \qquad \mathrm{BLP}(Y \mid \mathbf{X}) = \mu_Y + \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}(\mathbf{X} - \boldsymbol{\mu}_X).$$

Proof. Let $W := Y - \mu_Y - \mathbf{A}^*(\mathbf{X} - \boldsymbol{\mu}_X)$; then $\mathrm{E}W = 0$ and $\mathrm{Cov}(W, \mathbf{X}) = \boldsymbol{\Sigma}_{YX} - \mathbf{A}^*\boldsymbol{\Sigma}_{XX} = \mathbf{0}$. For any $(\mathbf{A}, b)$,

$$Y - \mathbf{A}\mathbf{X} - b = W + (\mathbf{A}^* - \mathbf{A})(\mathbf{X} - \boldsymbol{\mu}_X) + \{\mu_Y - \mathbf{A}\boldsymbol{\mu}_X - b\},$$

three mutually uncorrelated terms (the last is constant), so

$$\mathrm{E}(Y - \mathbf{A}\mathbf{X} - b)^2 = \mathrm{E}W^2 + (\mathbf{A}^* - \mathbf{A})\boldsymbol{\Sigma}_{XX}(\mathbf{A}^* - \mathbf{A})^\top + (\mu_Y - \mathbf{A}\boldsymbol{\mu}_X - b)^2 \geq \mathrm{E}W^2,$$

with equality iff $\mathbf{A} = \mathbf{A}^*$ and $b = \mu_Y - \mathbf{A}^*\boldsymbol{\mu}_X$. ■

Remark 3.14: properties of the BLP

$\mathrm{E}[\mathrm{BLP}(Y \mid \mathbf{X})] = \mu_Y$, $\mathrm{Var}[\mathrm{BLP}(Y \mid \mathbf{X})] = \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}$; the residual $W = Y - \mathrm{BLP}(Y \mid \mathbf{X})$ has $\mathrm{E}W = 0$, $\mathrm{Var}W = \sigma_Y^2 - \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}$ and $\mathrm{Cov}(W, \mathrm{BLP}) = 0$. Under joint normality $\mathrm{BLP}(Y \mid \mathbf{X}) = \mathrm{E}(Y \mid \mathbf{X})$ (Theorem 3.11). In block notation $\mathrm{E}[\mathrm{BLP}(\mathbf{X}_2 \mid \mathbf{X}_1)] = \boldsymbol{\mu}_2$, $\mathrm{Var}[\mathrm{BLP}(\mathbf{X}_2 \mid \mathbf{X}_1)] = \boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}$.

4.2 Multiple correlation

Definition 3.15: multiple correlation of Y with $\mathbf{X}$

$$\rho_{\mathrm{mult}} = \mathrm{mcorr}(\mathbf{X}, Y) := \max_{\mathbf{a} \in \mathbb{R}^q \setminus \{\mathbf{0}\}} \mathrm{Corr}(Y, \mathbf{a}^\top \mathbf{X}), \qquad \rho_{\mathrm{mult}}^2 = \frac{\boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}}{\sigma_Y^2}.$$

With $\hat{Y} = \mathrm{BLP}(Y \mid \mathbf{X})$, $\rho_{\mathrm{mult}} = \mathrm{Corr}(Y, \hat{Y})$.

Proof. $\mathrm{Corr}(Y, \mathbf{a}^\top \mathbf{X} + b) = \dfrac{\mathbf{a}^\top \boldsymbol{\Sigma}_{XY}}{\sigma_Y \sqrt{\mathbf{a}^\top \boldsymbol{\Sigma}_{XX} \mathbf{a}}}$. With $\mathbf{b} = \boldsymbol{\Sigma}_{XX}^{1/2}\mathbf{a}$, Cauchy–Schwarz gives

$$\mathbf{a}^\top \boldsymbol{\Sigma}_{XY} = \mathbf{b}^\top \boldsymbol{\Sigma}_{XX}^{-1/2} \boldsymbol{\Sigma}_{XY} \le \sqrt{\mathbf{a}^\top \boldsymbol{\Sigma}_{XX} \mathbf{a}} \sqrt{\boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}},$$

so $\mathrm{Corr} \le \sqrt{\boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}}/\sigma_Y$, with equality iff $\mathbf{a} \propto \boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}$ (the BLP direction). Then $\mathrm{Cov}(Y, \hat{Y}) = \mathrm{Var}(\hat{Y}) = \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}$ and $\mathrm{Corr}(Y, \hat{Y}) = \sqrt{\mathrm{Var}\hat{Y}}/\sigma_Y$. ■

For two vectors, $\rho(\mathbf{X}_1, \mathbf{X}_2) := \max \mathrm{Corr}(\mathbf{a}^\top \mathbf{X}_1, \mathbf{b}^\top \mathbf{X}_2)$ is the largest canonical correlation (Ch. 8).

4.3 Partial covariance and partial correlation

Partition $\mathbf{X} = (X, \mathbf{Z}^\top)^\top$ with X the predictor of interest and $\mathbf{Z} \in \mathbb{R}^{q-1}$ the others ($\mathrm{Var}\mathbf{Z} \succ 0$), and let $\tilde{X} = X - \mathrm{BLP}(X \mid \mathbf{Z})$, $\tilde{Y} = Y - \mathrm{BLP}(Y \mid \mathbf{Z})$.

Definition 3.16: partial covariance and partial correlation

$$\sigma_{YX \cdot Z} := \mathrm{Cov}(\tilde{Y}, \tilde{X}) = \mathrm{Cov}(Y, X) - \mathrm{Cov}(Y, \mathbf{Z})\mathrm{Var}(\mathbf{Z})^{-1}\mathrm{Cov}(\mathbf{Z}, X),$$

$$\rho_{YX \cdot Z} := \frac{\sigma_{YX \cdot Z}}{\sqrt{\mathrm{Var}(\tilde{Y})\mathrm{Var}(\tilde{X})}} = \mathrm{Corr}(\tilde{Y}, \tilde{X}).$$

In block notation: $\mathrm{pcorr}(\mathbf{X}_1, \mathbf{X}_2; \mathbf{X}_3) = \mathrm{Corr}(\mathbf{X}_1 - \mathrm{BLP}(\mathbf{X}_1 \mid \mathbf{X}_3), \mathbf{X}_2 - \mathrm{BLP}(\mathbf{X}_2 \mid \mathbf{X}_3)) = \mathrm{Corr}(\mathbf{X}_1^*, \mathbf{X}_2^*)$ and the partial covariance matrix $\mathrm{pVar}(\mathbf{X}_1; \mathbf{X}_2) = \mathrm{Var}(\mathbf{X}_1 - \mathrm{BLP}(\mathbf{X}_1 \mid \mathbf{X}_2)) = \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$.

Partial covariance formula [Supplement]. $\tilde{Y} = Y - \mu_Y - \boldsymbol{\Sigma}_{YZ}\boldsymbol{\Sigma}_{ZZ}^{-1}(\mathbf{Z} - \boldsymbol{\mu}_Z)$ and similarly for $\tilde{X}$. Expanding, $\mathrm{Cov}(\tilde{Y}, \tilde{X}) = \sigma_{YX} - 2\boldsymbol{\Sigma}_{YZ}\boldsymbol{\Sigma}_{ZZ}^{-1}\boldsymbol{\Sigma}_{ZX} + \boldsymbol{\Sigma}_{YZ}\boldsymbol{\Sigma}_{ZZ}^{-1}\boldsymbol{\Sigma}_{ZZ}\boldsymbol{\Sigma}_{ZZ}^{-1}\boldsymbol{\Sigma}_{ZX} = \sigma_{YX} - \boldsymbol{\Sigma}_{YZ}\boldsymbol{\Sigma}_{ZZ}^{-1}\boldsymbol{\Sigma}_{ZX}$. ■

[Supplement] So the partial covariance matrix of (Y, X) is a Schur complement, and under joint normality it equals the conditional covariance given $\mathbf{Z}$.

Theorem 3.17: Frisch–Waugh–Lovell theorem

Let $\hat{\beta}_X$ be the coefficient of X in the OLS regression of Y on $(X, \mathbf{Z})$, and let $\tilde{y}_i, \tilde{x}_i$ be the residuals from the OLS regressions of Y and X on $\mathbf{Z}$. Then

$$\hat{\beta}_X = \frac{\sum_i \tilde{x}_i \tilde{y}_i}{\sum_i \tilde{x}_i^2},$$

the slope of the residual-on-residual regression (all regressions with intercept; full-column-rank design).

Proof [Supplement]. Let $\mathbf{M}_Z = \mathbf{I} - \mathbf{P}_{[\mathbf{1},\mathbf{Z}]}$. The normal equations give $\mathbf{y} = \hat{\beta}_X \mathbf{x} + [\mathbf{1}, \mathbf{Z}]\hat{\boldsymbol{\gamma}} + \mathbf{e}$ with $\mathbf{e} \perp \mathbf{x}, \mathbf{1}, \mathbf{Z}$. Multiplying by $\mathbf{M}_Z$ ($\mathbf{M}_Z[\mathbf{1}, \mathbf{Z}] = \mathbf{0}$, $\mathbf{M}_Z \mathbf{e} = \mathbf{e}$) gives $\tilde{\mathbf{y}} = \hat{\beta}_X \tilde{\mathbf{x}} + \mathbf{e}$. Multiply by $\tilde{\mathbf{x}}^\top$: $\tilde{\mathbf{x}}^\top \mathbf{e} = \mathbf{x}^\top \mathbf{M}_Z \mathbf{e} = \mathbf{x}^\top \mathbf{e} = 0$, so $\tilde{\mathbf{x}}^\top \tilde{\mathbf{y}} = \hat{\beta}_X \tilde{\mathbf{x}}^\top \tilde{\mathbf{x}}$. ■

Example 3.18: mtcars — marginal versus partial correlation

Y = mpg, X = displacement, $\mathbf{Z}$ = (horsepower, weight). The marginal correlation is $\hat{\rho}_{YX} \approx -0.805$, but after adjusting for horsepower and weight $\hat{\rho}_{YX \cdot Z} \approx -0.044$: little linear association remains. The displacement–mpg relation runs mostly through horsepower and weight.

5 Quadratic Forms in Normal Random Vectors

Motivation. Test statistics are typically quadratic forms such as $(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu})$. Their χ^2 distributions are the basis of all tests in later chapters.

For $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and symmetric $\mathbf{A}$, a quadratic form is $Y = \mathbf{X}^\top \mathbf{A} \mathbf{X} = \sum_{j,k} X_j a_{jk} X_k$. We want its distribution and conditions for independence. Special case: $\mathbf{X} \sim N_p(\mathbf{0}, \mathbf{I}_p)$, $\mathbf{A} = \mathbf{I}_p$ gives $Y = \sum X_i^2 \sim \chi^2(p)$.

Remark 3.19: matrix algebra recall

$\mathbf{A}$ is idempotent if $\mathbf{A}^2 = \mathbf{A}$. If $\mathbf{A}$ is symmetric, $\mathbf{A} = \boldsymbol{\Gamma} \boldsymbol{\Lambda} \boldsymbol{\Gamma}^\top$; if it is also idempotent, its eigenvalues are 0 or 1 and $\operatorname{rank}(\mathbf{A}) = \#\{\text{nonzero eigenvalues}\} = \operatorname{tr}(\mathbf{A})$.

Theorem 3.20: chi-square quadratic forms, $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$ (Result 4.7, Case 001)

Let $\mathbf{X} \sim N_p(\mathbf{0}, \sigma^2 \mathbf{I})$ and $\mathbf{A}$ symmetric. Then $Y = \dfrac{\mathbf{X}^\top \mathbf{A} \mathbf{X}}{\sigma^2} \sim \chi^2(m)$ iff $\mathbf{A}$ is idempotent of rank $m \le p$.

Proof. ($\Leftarrow$) $\mathbf{A} = \boldsymbol{\Gamma} \boldsymbol{\Lambda} \boldsymbol{\Gamma}^\top$ with $\boldsymbol{\Lambda} = \operatorname{diag}(1, \ldots, 1, 0, \ldots, 0)$ (m ones). Let $\mathbf{z} = \frac{1}{\sigma} \boldsymbol{\Gamma}^\top \mathbf{X} \sim N(\mathbf{0}, \boldsymbol{\Gamma}^\top \boldsymbol{\Gamma}) = N(\mathbf{0}, \mathbf{I}_p)$; then $Y = \mathbf{z}^\top \boldsymbol{\Lambda} \mathbf{z} = z_1^2 + \cdots + z_m^2 \sim \chi^2(m)$.

($\Rightarrow$) [Supplement] In general $Y = \sum_i \lambda_i z_i^2$ has MGF $\prod_i (1 - 2t\lambda_i)^{-1/2}$; matching $(1 - 2t)^{-m/2}$ near $t = 0$ forces the nonzero λ_i to be m ones. ■

Simple example: $\mathbf{A} = \operatorname{diag}(1, 0, \ldots, 0)$ gives $\mathbf{X}^\top \mathbf{A} \mathbf{X} / \sigma^2 = X_1^2 / \sigma^2 \sim \chi_1^2$.

Theorem 3.21: chi-square quadratic forms, general Σ (Result 4.7, Case 002)

Let $\mathbf{X} \sim N_p(\mathbf{0}, \mathbf{\Sigma})$ with $\mathbf{\Sigma} \succ 0$ and $\mathbf{A}$ symmetric. Then $Y = \mathbf{X}^\top \mathbf{A}\mathbf{X} \sim \chi^2(m)$ iff either (1) $\mathbf{A}\mathbf{\Sigma}$ is idempotent of rank m, or (2) $\mathbf{\Sigma}\mathbf{A}$ is idempotent of rank m.

Proof. By the representation theorem $\mathbf{X} = \mathbf{\Sigma}^{1/2}\mathbf{z}$, $\mathbf{z} \sim N_p(\mathbf{0}, \mathbf{I}_p)$, so $Y = \mathbf{z}^\top \mathbf{\Sigma}^{1/2}\mathbf{A}\mathbf{\Sigma}^{1/2}\mathbf{z}$. By Theorem 3.20, $Y \sim \chi^2_m$ iff $\mathbf{\Sigma}^{1/2}\mathbf{A}\mathbf{\Sigma}^{1/2}$ is (symmetric) idempotent of rank m. Now

$$(\mathbf{\Sigma}^{1/2}\mathbf{A}\mathbf{\Sigma}^{1/2})^2 = \mathbf{\Sigma}^{1/2}\mathbf{A}\mathbf{\Sigma}^{1/2} \iff \mathbf{\Sigma}^{1/2}\mathbf{A}\mathbf{\Sigma}\mathbf{A}\mathbf{\Sigma}^{1/2} = \mathbf{\Sigma}^{1/2}\mathbf{A}\mathbf{\Sigma}^{1/2} \iff \mathbf{A}\mathbf{\Sigma}\mathbf{A} = \mathbf{A},$$

(multiply by $\mathbf{\Sigma}^{-1/2}$ on both sides), and $\mathbf{A}\mathbf{\Sigma}\mathbf{A} = \mathbf{A} \iff (\mathbf{A}\mathbf{\Sigma})^2 = \mathbf{A}\mathbf{\Sigma}$ (multiply on the right by $\mathbf{\Sigma}$ or $\mathbf{\Sigma}^{-1}$). Ranks: $\operatorname{rank}(\mathbf{\Sigma}^{1/2}\mathbf{A}\mathbf{\Sigma}^{1/2}) = \operatorname{rank}(\mathbf{A}) = \operatorname{rank}(\mathbf{A}\mathbf{\Sigma})$. Since $\mathbf{\Sigma}\mathbf{A} = (\mathbf{A}\mathbf{\Sigma})^\top$, the same holds for $\mathbf{\Sigma}\mathbf{A}$. ■

Example 3.22: distribution of the Mahalanobis distance (Example 4)

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \mathbf{\Sigma})$ with $\mathbf{\Sigma} \succ 0$, then $(\mathbf{X} - \boldsymbol{\mu})^\top \mathbf{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu}) \sim \chi^2(p)$ ($\mathbf{A} = \mathbf{\Sigma}^{-1}$ makes $\mathbf{A}\mathbf{\Sigma} = \mathbf{I}_p$ idempotent of rank p; apply Case 002 to $\mathbf{X} - \boldsymbol{\mu}$). Hence $\{\mathbf{x} : (\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) \le \chi^2_p(\alpha)\}$ is an ellipsoid of probability $1 - \alpha$.

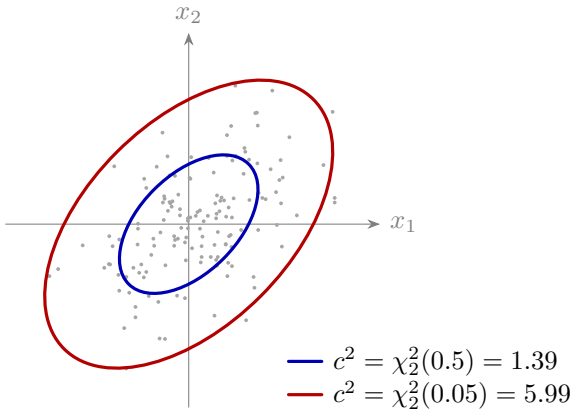


Figure. *Example 4 with* $\mathbf{\Sigma} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$*: 54% and 96% of* 150 *draws fall inside the* 0.5 *and* 0.95 *ellipses.*

Theorem 3.23: independence of linear and quadratic forms (Theorem 9)

Let $\mathbf{X} \sim N_p(\mathbf{0}, \mathbf{I})$, $\mathbf{A}$ a $p \times p$ symmetric matrix and $\mathbf{B}$ a $k \times p$ matrix. If $\mathbf{B}\mathbf{A} = \mathbf{0}$, then $\mathbf{B}\mathbf{X}$ and $\mathbf{X}^\top \mathbf{A}\mathbf{X}$ are independent.

Proof. If $\mathbf{A}$ is full rank, $\mathbf{B}\mathbf{A} = \mathbf{0}$ forces $\mathbf{B} = \mathbf{0}$: trivial. If $\operatorname{rank}(\mathbf{A}) = m < p$, write $\mathbf{A} = \mathbf{\Gamma} \begin{pmatrix} \mathbf{\Lambda}_m & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \mathbf{\Gamma}^\top$ with $\mathbf{\Lambda}_m = \operatorname{diag}(\lambda_1, \ldots, \lambda_m)$, all $\lambda_i \neq 0$. From $\mathbf{B}\mathbf{A} = \mathbf{0}$, $\mathbf{B}\mathbf{\Gamma} \begin{pmatrix} \mathbf{\Lambda}_m & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} = \mathbf{0}$. Writing $\mathbf{B}\mathbf{\Gamma} = [\mathbf{C}_1, \mathbf{C}_2]$ gives $\mathbf{C}_1\mathbf{\Lambda}_m = \mathbf{0}$, so $\mathbf{C}_1 = \mathbf{0}$ and $\mathbf{B}\mathbf{\Gamma} = [\mathbf{0}, \mathbf{C}_2]$. Let $\mathbf{z} = \begin{pmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \end{pmatrix} = \mathbf{\Gamma}^\top \mathbf{X} \sim N_p(\mathbf{0}, \mathbf{I})$, so $\mathbf{z}_1 \perp\!\!\!\perp \mathbf{z}_2$. Then

$$\mathbf{B}\mathbf{X} = \mathbf{B}\mathbf{\Gamma}\mathbf{z} = \mathbf{C}_2\mathbf{z}_2, \qquad \mathbf{X}^\top \mathbf{A}\mathbf{X} = \mathbf{z}^\top \begin{pmatrix} \mathbf{\Lambda}_m & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \mathbf{z} = \mathbf{z}_1^\top \mathbf{\Lambda}_m \mathbf{z}_1,$$

functions of $\mathbf{z}_2$ and $\mathbf{z}_1$ respectively, hence independent. ■

Example 3.24: $\bar{X} \perp\!\!\!\perp S^2$

For $X_1, \dots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ let $\mathbf{Z} = (\mathbf{X} - \mu \mathbf{1}_n)/\sigma \sim N_n(\mathbf{0}, \mathbf{I}_n)$, $\mathbf{B} = \frac{1}{n}\mathbf{1}_n^\top$, $\mathbf{A} = \frac{1}{n-1}\mathbf{H}_n$ with $\mathbf{H}_n = \mathbf{I}_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top$. Since $\mathbf{H}_n\mathbf{1}_n = \mathbf{0}$, $\bar{X} = \mu + \sigma\mathbf{B}\mathbf{Z}$ and $S^2 = \sigma^2\mathbf{Z}^\top\mathbf{A}\mathbf{Z}$, and $\mathbf{B}\mathbf{A} = \mathbf{0}$, so $\bar{X} \perp\!\!\!\perp S^2$ by Theorem 3.23. The multivariate version is in Ch. 4.

Remark 3.25: same argument, two quadratic forms

[Supplement] If $\mathbf{X} \sim N_p(\mathbf{0}, \mathbf{I}_p)$ and symmetric $\mathbf{A}, \mathbf{B}$ satisfy $\mathbf{A}\mathbf{B} = \mathbf{0}$, then $\mathbf{X}^\top\mathbf{A}\mathbf{X} \perp\!\!\!\perp \mathbf{X}^\top\mathbf{B}\mathbf{X}$: $\mathbf{X}^\top\mathbf{B}\mathbf{X} = (\mathbf{B}\mathbf{X})^\top\mathbf{B}^+(\mathbf{B}\mathbf{X})$ is a function of $\mathbf{B}\mathbf{X}$, and $\mathbf{B}\mathbf{A} = \mathbf{0}$.

6 Selected exercises

Motivation. These exercises sharpen two warnings and add one tool: zero correlation with normal marginals still allows dependence, the normal density factorizes into a marginal and a conditional part, and quadratic forms in normal vectors have explicit means and variances.

Exercise 3.1 (Uncorrelated but dependent; J&W 4.9)

Let $X_1 \sim N(0,1)$, $c > 0$, and $X_2 = -X_1$ if $|X_1| \le c$, $X_2 = X_1$ otherwise.
(a) Show that $X_2 \sim N(0,1)$ for every c.
(b) Show that c can be chosen so that $\mathrm{Cov}(X_1, X_2) = 0$.
(c) For this c, show that X_1 and X_2 are not independent, and conclude that (X_1, X_2) is not bivariate normal.

Solution [Supplement].

(a) For a Borel set B, $P(X_2 \in B) = P(X_1 \in B, |X_1| > c) + P(-X_1 \in B, |X_1| \le c)$. Since $-X_1 \overset{d}{=} X_1$ and $\{|X_1| \le c\} = \{|-X_1| \le c\}$, the second term equals $P(X_1 \in B, |X_1| \le c)$, so $P(X_2 \in B) = P(X_1 \in B)$ (Example 3.3 is the case $c = 1$).

(b)

Step 1. With $g(c) = \mathrm{E}[X_1^2 \mathbf{1}\{|X_1| \le c\}] = \int_{-c}^{c} x^2\phi(x)\,dx$,

$$\mathrm{Cov}(X_1, X_2) = \mathrm{E}(X_1X_2) = \mathrm{E}[X_1^2\mathbf{1}\{|X_1| > c\}] - \mathrm{E}[X_1^2\mathbf{1}\{|X_1| \le c\}] = 1 - 2g(c).$$

Step 2. g is continuous, $g(c) \to 0$ as $c \downarrow 0$, and $g(c) \to \mathrm{E}X_1^2 = 1$ as $c \to \infty$ (monotone convergence). So $h(c) = 1 - 2g(c)$ is continuous with limits 1 and -1, and by the intermediate value theorem $h(c^*) = 0$ for some $c^* > 0$ (numerically $c^* \approx 1.54$).

(c)

Step 1. $|X_2| = |X_1|$, so $X_2^2 = X_1^2$. If $X_1 \perp\!\!\!\perp X_2$, then $X_1^2 \perp\!\!\!\perp X_2^2$ and $\mathrm{Var}(X_1^2) = \mathrm{Cov}(X_1^2, X_2^2) = 0$, contradicting $\mathrm{Var}(X_1^2) = \mathrm{E}X_1^4 - 1 = 2$.

Step 2. If (X_1, X_2) were bivariate normal, $\mathrm{Cov}(X_1, X_2) = 0$ would imply independence by Theorem 3.9. Hence it is not jointly normal, although both marginals are $N(0,1)$. ■

Exercise 3.2 (Factorizing the normal density; J&W 4.13)

Let $\mathbf{X} = (\mathbf{X}_1^\top, \mathbf{X}_2^\top)^\top \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$ partitioned as in Theorem 3.11, and $\boldsymbol{m}(\mathbf{x}_2) = \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2)$.
(a) Show $|\boldsymbol{\Sigma}| = |\boldsymbol{\Sigma}_{22}|\,|\boldsymbol{\Sigma}_{11\cdot 2}|$.

(b) Show

$$(\mathbf{x}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) = \{\mathbf{x}_1-\boldsymbol{m}(\mathbf{x}_2)\}^\top\boldsymbol{\Sigma}_{11\cdot2}^{-1}\{\mathbf{x}_1-\boldsymbol{m}(\mathbf{x}_2)\} + (\mathbf{x}_2-\boldsymbol{\mu}_2)^\top\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2-\boldsymbol{\mu}_2).$$

(c) Using only the density, deduce the marginal law of $\mathbf{X}_2$ and the conditional law of $\mathbf{X}_1$ given $\mathbf{X}_2 = \mathbf{x}_2$.

Solution [Supplement].

(a) This is the first determinant formula of Theorem 2.7 with $\mathbf{A} = \boldsymbol{\Sigma}$.

(b)

Step 1. Let $\mathbf{E} = \begin{pmatrix} \mathbf{I} & -\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}$. Since $\boldsymbol{\Sigma}_{21} = \boldsymbol{\Sigma}_{12}^\top$ and $\boldsymbol{\Sigma}_{22}$ is symmetric, the elimination identity in the proof of Theorem 2.7 reads $\mathbf{E}\boldsymbol{\Sigma}\mathbf{E}^\top = \mathbf{D} := \operatorname{diag}(\boldsymbol{\Sigma}_{11\cdot2}, \boldsymbol{\Sigma}_{22})$.

Step 2. Both $\mathbf{E}$ and $\mathbf{D}$ are invertible, so $\boldsymbol{\Sigma}^{-1} = \mathbf{E}^\top\mathbf{D}^{-1}\mathbf{E}$ and the quadratic form equals $\{\mathbf{E}(\mathbf{x}-\boldsymbol{\mu})\}^\top\mathbf{D}^{-1}\{\mathbf{E}(\mathbf{x}-\boldsymbol{\mu})\}$.

Step 3. $\mathbf{E}(\mathbf{x}-\boldsymbol{\mu}) = (\mathbf{x}_1-\boldsymbol{\mu}_1-\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2-\boldsymbol{\mu}_2),\ \mathbf{x}_2-\boldsymbol{\mu}_2) = (\mathbf{x}_1-\boldsymbol{m}(\mathbf{x}_2),\ \mathbf{x}_2-\boldsymbol{\mu}_2)$, and $\mathbf{D}^{-1}$ is block diagonal, which gives the identity.

(c)

Step 1. $\boldsymbol{\Sigma}_{11\cdot2} = \{(\boldsymbol{\Sigma}^{-1})_{11}\}^{-1}$ is the inverse of a principal block of a positive definite matrix, hence positive definite. With $r = \dim\mathbf{X}_1$, $s = \dim\mathbf{X}_2$, (a) and (b) give

$$f(\mathbf{x}_1,\mathbf{x}_2) = g(\mathbf{x}_1 \mid \mathbf{x}_2)\, h(\mathbf{x}_2),$$

where $g(\cdot \mid \mathbf{x}_2)$ is the $N_r(\boldsymbol{m}(\mathbf{x}_2), \boldsymbol{\Sigma}_{11\cdot2})$ density and h the $N_s(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$ density (the constants split as $(2\pi)^{-p/2} = (2\pi)^{-r/2}(2\pi)^{-s/2}$).

Step 2. For each $\mathbf{x}_2$, $\int g(\mathbf{x}_1 \mid \mathbf{x}_2)\, d\mathbf{x}_1 = 1$, so the marginal density of $\mathbf{X}_2$ is $\int f\, d\mathbf{x}_1 = h(\mathbf{x}_2)$: $\mathbf{X}_2 \sim N_s(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$.

Step 3. The conditional density is $f(\mathbf{x}_1,\mathbf{x}_2)/h(\mathbf{x}_2) = g(\mathbf{x}_1 \mid \mathbf{x}_2)$, which is Theorem 3.11 again, now obtained from the density. ■

Exercise 3.3 (Mean and variance of a normal quadratic form)

Let $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ ($\boldsymbol{\Sigma} \succeq 0$) and $\mathbf{A}$ symmetric. Show

$$\mathrm{E}(\mathbf{X}^\top\mathbf{A}\mathbf{X}) = \operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top\mathbf{A}\boldsymbol{\mu}, \qquad \operatorname{Var}(\mathbf{X}^\top\mathbf{A}\mathbf{X}) = 2\operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma}) + 4\boldsymbol{\mu}^\top\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\mu}.$$

Solution [Supplement].

Step 1 (mean). $\mathrm{E}(\mathbf{X}^\top\mathbf{A}\mathbf{X}) = \mathrm{E}\operatorname{tr}(\mathbf{A}\mathbf{X}\mathbf{X}^\top) = \operatorname{tr}\{\mathbf{A}(\boldsymbol{\Sigma}+\boldsymbol{\mu}\boldsymbol{\mu}^\top)\} = \operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top\mathbf{A}\boldsymbol{\mu}$ (normality is not needed here).

Step 2 (representation). Let $\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{I}_p)$ and $\boldsymbol{\Sigma}^{1/2}$ the symmetric square root. By Theorem 3.7, $\boldsymbol{\mu}+\boldsymbol{\Sigma}^{1/2}\mathbf{Z}$ is normal with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, and a normal law is determined by its mean and covariance (Theorem 3.4), so $\mathbf{X} \overset{d}{=} \boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbf{Z}$. Let $\mathbf{B} = \boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}^{1/2} = \mathbf{P}\boldsymbol{\Lambda}\mathbf{P}^\top$ (spectral decomposition), $\mathbf{W} = \mathbf{P}^\top\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{I}_p)$ and $\mathbf{b} = \mathbf{P}^\top\boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\mu}$. Then

$$\mathbf{X}^\top\mathbf{A}\mathbf{X} \overset{d}{=} \boldsymbol{\mu}^\top\mathbf{A}\boldsymbol{\mu} + 2\mathbf{b}^\top\mathbf{W} + \mathbf{W}^\top\boldsymbol{\Lambda}\mathbf{W} = \boldsymbol{\mu}^\top\mathbf{A}\boldsymbol{\mu} + \sum_{i=1}^{p}(2b_iW_i + \lambda_iW_i^2).$$

Step 3 (variance). The summands are independent. With $\mathrm{E}W_i^3 = 0$ and $\operatorname{Var}(W_i^2) = 3-1 = 2$,

$$\operatorname{Var}(2b_iW_i + \lambda_iW_i^2) = 4b_i^2 + 2\lambda_i^2 + 4b_i\lambda_i\operatorname{Cov}(W_i, W_i^2) = 4b_i^2 + 2\lambda_i^2.$$

Step 4. $\sum_i \lambda_i^2 = \mathrm{tr}(\mathbf{B}^2) = \mathrm{tr}(\boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma}^{1/2}) = \mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma})$ and $\sum_i b_i^2 = \mathbf{b}^\top\mathbf{b} = \boldsymbol{\mu}^\top\mathbf{A}\boldsymbol{\Sigma}^{1/2}\mathbf{P}\mathbf{P}^\top\boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\mu} = \boldsymbol{\mu}^\top\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\mu}$, which gives the variance formula.

Check: $\boldsymbol{\mu} = \mathbf{0}$ and $\mathbf{A} = \boldsymbol{\Sigma}^{-1}$ give mean p and variance $2p$, those of χ^2_p (Example 3.22). ■

Chapter 4. Sampling Distributions and Large-Sample Theory

The Wishart distribution, distributions of $\bar{\mathbf{X}}$ and $\mathbf{S}$, MLE, consistency, multivariate CLT

1 The Wishart Distribution

Motivation. The sample covariance matrix is random, and inference about $\boldsymbol{\Sigma}$ needs its distribution. The Wishart distribution is the matrix analogue of χ^2: a sum of outer products of independent normal vectors.

Definition 4.1: Wishart distribution

- The Wishart distribution is a family of distributions for symmetric positive definite matrices.
- The distribution of the $p \times p$ random matrix $\mathbf{M} = \mathbf{X}^\top \mathbf{X} = \sum_{i=1}^{n} \mathbf{X}_i \mathbf{X}_i^\top$ is a Wishart distribution.
- With m degrees of freedom,

$$W_p(m \mid \boldsymbol{\Sigma}) = \text{Wishart with } m \text{ d.f.} = \text{distribution of } \sum_{j=1}^{m} \mathbf{Z}_j \mathbf{Z}_j^\top, \qquad \mathbf{Z}_j \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma}).$$

We also write $W_p(m, \boldsymbol{\Sigma})$.

- It is the multivariate extension of χ^2: if $\mathbf{M} \sim W_1(n, \sigma^2)$ then $\mathbf{M}/\sigma^2 \sim \chi^2_n$. E.g. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(0, \sigma^2)$ gives $\mathbf{M} = \sum X_i^2$, $\mathbf{M}/\sigma^2 \sim \chi^2_n$.
- $W_p(n, \mathbf{I})$ is the *standard Wishart* distribution.
- If $\boldsymbol{\mu} = \mathbf{0}$ is known and $\mathbf{S} = \frac{1}{n}\mathbf{X}^\top \mathbf{X} = \frac{1}{n}\sum \mathbf{X}_i \mathbf{X}_i^\top$, then $n\mathbf{S} \sim W_p(n, \boldsymbol{\Sigma})$.
- [Supplement] If $m \geq p$ and $\boldsymbol{\Sigma} \succ 0$, $\mathbf{M}$ is PD with probability 1.

Theorem 4.2: properties of the Wishart distribution (Proposition 1)

(i) For $\mathbf{M} \sim W_p(n, \boldsymbol{\Sigma})$ and $\mathbf{B}_{(p\times m)}$, $\mathbf{B}^\top \mathbf{M} \mathbf{B} \sim W_m(n, \mathbf{B}^\top \boldsymbol{\Sigma} \mathbf{B})$.

(ii) For $\mathbf{M} \sim W_p(n, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$, $\boldsymbol{\Sigma}^{-1/2} \mathbf{M} \boldsymbol{\Sigma}^{-1/2} \sim W_p(n, \mathbf{I}_p)$.

(iii) If $\mathbf{M}_i \sim W_p(n_i, \boldsymbol{\Sigma})$ $(i = 1, \ldots, k)$ are independent, $\sum_i \mathbf{M}_i \sim W_p(n, \boldsymbol{\Sigma})$ with $n = n_1 + \cdots + n_k$.

(iv) For $\mathbf{M}_n \sim W_p(n, \boldsymbol{\Sigma})$, $\mathrm{E}(\mathbf{M}_n) = n\boldsymbol{\Sigma}$ and $\mathrm{Var}((\mathbf{M}_n)_{ij}) = n(\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj})$. Hence $\hat{\boldsymbol{\Sigma}}_n = \mathbf{M}_n/n$ has $\mathrm{E}(\hat{\boldsymbol{\Sigma}}_n) = \boldsymbol{\Sigma}$ and $\mathrm{Var}[(\hat{\boldsymbol{\Sigma}}_n)_{ij}] = (\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj})/n$.

(v) The law of large numbers and the Cramér–Wold device give $\mathbf{M}_n/n \to \boldsymbol{\Sigma}$ in probability as $n \to \infty$.

Proof. (i) Write $\mathbf{M} = \sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i^\top$ with $\mathbf{x}_i \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma})$. Then $\mathbf{B}^\top \mathbf{M} \mathbf{B} = \sum_i \mathbf{B}^\top \mathbf{x}_i (\mathbf{B}^\top \mathbf{x}_i)^\top$ and $\mathbf{B}^\top \mathbf{x}_i \overset{\text{iid}}{\sim} N_m(\mathbf{0}, \mathbf{B}^\top \boldsymbol{\Sigma} \mathbf{B})$.

(ii) Special case of (i) with $\mathbf{B} = \boldsymbol{\Sigma}^{-1/2}$.

(iii) With $\mathbf{M}_i = \sum_j \mathbf{x}_{ij}\mathbf{x}_{ij}^\top$, $\sum_i \mathbf{M}_i = \sum_i \sum_j \mathbf{x}_{ij}\mathbf{x}_{ij}^\top \sim W_p(\sum n_i, \boldsymbol{\Sigma})$.

(iv) $\mathrm{E}[\mathbf{M}_n] = \mathrm{E}(\sum_k \mathbf{x}_k\mathbf{x}_k^\top) = \sum_k \mathrm{Var}(\mathbf{x}_k) = n\boldsymbol{\Sigma}$. Terms with different k are independent, so $\mathrm{Var}((\mathbf{M}_n)_{ij}) = n\{\mathrm{E}(X_{ki}^2X_{kj}^2) - \Sigma_{ij}^2\}$. Using $\mathrm{E}(X_{ki} \mid X_{kj}) = \frac{\Sigma_{ij}}{\Sigma_{jj}}X_{kj}$ and $X_{ki} - \mathrm{E}(X_{ki} \mid X_{kj}) \perp\!\!\!\perp X_{kj}$,

$$\mathrm{E}[(X_{ki}X_{kj})^2] = \mathrm{E}[X_{kj}^2\mathrm{E}(X_{ki}^2 \mid X_{kj})] = \frac{\Sigma_{ij}^2}{\Sigma_{jj}^2}3\Sigma_{jj}^2 + \Big(\Sigma_{ii} - \frac{\Sigma_{ij}^2}{\Sigma_{jj}}\Big)\Sigma_{jj} = 2\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj},$$

so $\mathrm{Var} = n(\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj})$.

(v) By Cramér–Wold it suffices that $\sum_{ij} a_{ij}M_{ij}/n \xrightarrow{p} \sum_{ij} a_{ij}\Sigma_{ij}$ for all a_{ij}. Since $\mathbf{M}_n/n = \frac{1}{n}\sum_k \mathbf{x}_k\mathbf{x}_k^\top$ is an average of iid random matrices, the LLN gives $\sum_{ij} a_{ij}\frac{M_{ij}}{n} \xrightarrow{p} \sum_{ij} a_{ij}\Sigma_{ij}$. ■

2 Wishart Quadratic Forms and the Distribution of S

Motivation. Projections of a Wishart matrix are again Wishart or χ^2, which yields the distribution of $\mathbf{S}$ and its independence from $\bar{\mathbf{X}}$, just as in the univariate case.

Theorem 4.3: Wishart quadratic forms (Lemma 5)

Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be iid $N_p(\mathbf{0}, \boldsymbol{\Sigma})$, $\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_n)^\top$, and $\mathbf{A}$ an $n \times n$ real symmetric matrix. Then

$$\mathbf{X}^\top\mathbf{A}\mathbf{X} \sim W_p(r, \boldsymbol{\Sigma}) \iff \mathbf{A} \text{ is idempotent with } \mathrm{rank}(\mathbf{A}) = r.$$

Partial proof ($\Leftarrow$). Since $\mathbf{A}$ is symmetric idempotent of rank r, $\mathbf{A} = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top$ with $\boldsymbol{\Lambda} = \mathrm{diag}(1, \dots, 1, 0, \dots, 0)$. Write $\mathbf{A} = \mathbf{P}\begin{pmatrix}\mathbf{I}_r & \mathbf{0}\\ \mathbf{0} & \mathbf{0}\end{pmatrix}\mathbf{P}^\top$ for an $n \times n$ orthogonal $\mathbf{P} = [\mathbf{p}_1, \dots, \mathbf{p}_n]$, so $\mathbf{A} = \sum_{j=1}^r \mathbf{p}_j\mathbf{p}_j^\top$ and

$$\mathbf{X}^\top\mathbf{A}\mathbf{X} = \sum_{j=1}^r \mathbf{X}^\top\mathbf{p}_j\mathbf{p}_j^\top\mathbf{X}.$$

Now $\mathbf{X}^\top\mathbf{p}_j = \sum_{i=1}^n p_{ij}\mathbf{X}_i \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$, since

$$\mathrm{Cov}\Big(\sum_i p_{ij}\mathbf{X}_i\Big) = \boldsymbol{\Sigma}\cdot\mathbf{p}_j^\top\mathbf{p}_j = \boldsymbol{\Sigma}, \qquad \mathrm{Cov}\Big(\sum_i p_{ij}\mathbf{X}_i, \sum_i p_{ik}\mathbf{X}_i\Big) = \mathbf{p}_j^\top\mathbf{p}_k\boldsymbol{\Sigma} = \mathbf{0}\ (j \neq k);$$

being jointly normal, $\mathbf{z}_j = \mathbf{X}^\top\mathbf{p}_j$ are iid $N_p(\mathbf{0}, \boldsymbol{\Sigma})$. Hence $\mathbf{X}^\top\mathbf{A}\mathbf{X} = \sum_{j=1}^r \mathbf{z}_j\mathbf{z}_j^\top \sim W_p(r, \boldsymbol{\Sigma})$. ([Supplement] $\Rightarrow$ reduces to Theorem 3.20 with $p = 1$: $\mathbf{a}^\top\mathbf{X}^\top\mathbf{A}\mathbf{X}\mathbf{a} \sim (\mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a})\chi_r^2$ for every $\mathbf{a}$.) ■

Theorem 4.4: distribution of the sample covariance matrix; $\bar{\mathbf{X}} \perp\!\!\!\perp \mathbf{S}$

If $\mathbf{X}_1, \dots, \mathbf{X}_n \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then

$$\bar{\mathbf{X}}_n \sim N_p(\boldsymbol{\mu}, \tfrac{1}{n}\boldsymbol{\Sigma}), \qquad (n-1)\mathbf{S} = \mathbf{X}^\top\Big(\mathbf{I}_n - \tfrac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top\Big)\mathbf{X} \sim W_p(n-1, \boldsymbol{\Sigma}), \qquad \bar{\mathbf{X}} \perp\!\!\!\perp \mathbf{S}.$$

Proof.

Step 1. $\bar{\mathbf{X}} = \frac{1}{n}\mathbf{X}^\top\mathbf{1}_n$ is linear in the normal rows, with covariance $\frac{1}{n}\boldsymbol{\Sigma}$ (Theorem 3.7).

Step 2. $\mathbf{H}_n = \mathbf{I}_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top$ is symmetric idempotent of rank $n-1$. Since it annihilates $\mathbf{1}_n$, $\mathbf{X}^\top\mathbf{H}_n\mathbf{X} = (\mathbf{X} - \mathbf{1}_n\boldsymbol{\mu}^\top)^\top\mathbf{H}_n(\mathbf{X} - \mathbf{1}_n\boldsymbol{\mu}^\top)$, which reduces to the mean-zero case, and Theorem 4.3 gives $W_p(n-1, \boldsymbol{\Sigma})$.

Step 3 (independence) [Supplement]. Choose an orthogonal $\mathbf{P} = [\mathbf{p}_1, \dots, \mathbf{p}_n]$ with $\mathbf{p}_1 = \mathbf{1}_n/\sqrt{n}$ and let $\mathbf{z}_j = (\mathbf{X} - \mathbf{1}_n\boldsymbol{\mu}^\top)^\top \mathbf{p}_j$. As above $\mathbf{z}_1, \dots, \mathbf{z}_n \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma})$, $\mathbf{z}_1 = \sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu})$, and since $\mathbf{H}_n = \sum_{j=2}^n \mathbf{p}_j\mathbf{p}_j^\top$, $(n-1)\mathbf{S} = \sum_{j=2}^n \mathbf{z}_j\mathbf{z}_j^\top$. The first depends only on $\mathbf{z}_1$, the second only on $\mathbf{z}_2, \dots, \mathbf{z}_n$. ■

3 A Theorem on Inverse Wishart Ratios

Motivation. Hotelling's T^2 involves $\mathbf{S}^{-1}$. This theorem reduces quadratic forms in an inverse Wishart matrix to ratios of χ^2 variables, which is the key step toward the F distribution of T^2.

Theorem 4.5: ratios involving an inverse Wishart matrix (Theorem 3)

Suppose $\mathbf{M} \sim W_p(n, \boldsymbol{\Sigma})$, $\mathbf{a} \in \mathbb{R}^p$, and $n > p - 1$. Then

(i) $(\boldsymbol{\Sigma}^{-1})_{11}/(\mathbf{M}^{-1})_{11} \sim \chi^2_{n-p+1}$;

(ii) for any constant $\mathbf{a} \neq \mathbf{0}$, $\dfrac{\mathbf{a}^\top\boldsymbol{\Sigma}^{-1}\mathbf{a}}{\mathbf{a}^\top\mathbf{M}^{-1}\mathbf{a}} \sim \chi^2_{n-p+1}$;

(iii) if $\mathbf{y} \in \mathbb{R}^p$ is random and independent of $\mathbf{M}$ (with $P(\mathbf{y} = \mathbf{0}) = 0$), then $\dfrac{\mathbf{y}^\top\boldsymbol{\Sigma}^{-1}\mathbf{y}}{\mathbf{y}^\top\mathbf{M}^{-1}\mathbf{y}} \sim \chi^2_{n-p+1}$ and is independent of $\mathbf{y}$.

Proof. (i) Write

$$\mathbf{M} = \begin{pmatrix} m_{11} & \mathbf{M}_{12} \\ \mathbf{M}_{21} & \mathbf{M}_{22} \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} \sigma_{11} & \boldsymbol{\sigma}_{12} \\ \boldsymbol{\sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix}.$$

By the block-inverse formula,

$$\frac{(\boldsymbol{\Sigma}^{-1})_{11}}{(\mathbf{M}^{-1})_{11}} = \frac{m_{11} - \mathbf{M}_{12}\mathbf{M}_{22}^{-1}\mathbf{M}_{21}}{\sigma_{11} - \boldsymbol{\sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21}}.$$

Write $\mathbf{M} = \mathbf{X}^\top\mathbf{X} = \sum_{i=1}^n \mathbf{x}_i\mathbf{x}_i^\top$ with $\mathbf{x}_i \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma})$ and $\mathbf{X} = [\mathbf{x}_1, \mathbf{X}_2]$ ($\mathbf{x}_1$: first column). Then

$$m_{11} - \mathbf{M}_{12}\mathbf{M}_{22}^{-1}\mathbf{M}_{21} = \mathbf{x}_1^\top\mathbf{x}_1 - \mathbf{x}_1^\top\mathbf{X}_2(\mathbf{X}_2^\top\mathbf{X}_2)^{-1}\mathbf{X}_2^\top\mathbf{x}_1 = \mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1,$$

$$\mathbf{A} = \mathbf{A}(\mathbf{X}_2) = \mathbf{I}_n - \mathbf{X}_2(\mathbf{X}_2^\top\mathbf{X}_2)^{-1}\mathbf{X}_2^\top.$$

Given $\mathbf{X}_2$, $\mathbf{x}_1 \mid \mathbf{X}_2 \sim N_n(\mathbf{X}_2\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21},\ \sigma_{11\cdot 2}\mathbf{I}_n)$, $\sigma_{11\cdot 2} = \sigma_{11} - \boldsymbol{\sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21}$. Since $\mathbf{A}\mathbf{X}_2 = \mathbf{0}$, $\mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1 = (\mathbf{x}_1 - \mathbf{X}_2\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21})^\top\mathbf{A}(\mathbf{x}_1 - \mathbf{X}_2\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21})$. For any given $\mathbf{X}_2$, $\mathbf{A}$ is symmetric idempotent with $\operatorname{rank}(\mathbf{A}) = \operatorname{tr}(\mathbf{A}) = n - \operatorname{tr}\{(\mathbf{X}_2^\top\mathbf{X}_2)^{-1}\mathbf{X}_2^\top\mathbf{X}_2\} = n - (p-1)$, so Theorem 3.20 gives

$$\mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1 \mid \mathbf{X}_2 \sim \sigma_{11\cdot 2}\,\chi^2_{n-p+1}.$$

This does not depend on $\mathbf{X}_2$, so marginally $\mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1 \sim \sigma_{11\cdot 2}\chi^2_{n-p+1}$, independent of $\mathbf{X}_2$. Hence the ratio is $\mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1/\sigma_{11\cdot 2} \sim \chi^2_{n-p+1}$.

(i)⇒(ii) Given $\mathbf{a}$, take a $p \times p$ orthogonal $\mathbf{P}$ whose first column is $\mathbf{a}/\|\mathbf{a}\|$, i.e. $\mathbf{a}^\top\mathbf{P} = \mathbf{e}_1^\top$ (w.l.o.g. $\|\mathbf{a}\| = 1$). Then

$$\frac{\mathbf{a}^\top\boldsymbol{\Sigma}^{-1}\mathbf{a}}{\mathbf{a}^\top\mathbf{M}^{-1}\mathbf{a}} = \frac{\mathbf{a}^\top\mathbf{P}\mathbf{P}^\top\boldsymbol{\Sigma}^{-1}\mathbf{P}\mathbf{P}^\top\mathbf{a}}{\mathbf{a}^\top\mathbf{P}\mathbf{P}^\top\mathbf{M}^{-1}\mathbf{P}\mathbf{P}^\top\mathbf{a}} = \frac{\mathbf{e}_1^\top\boldsymbol{\Sigma}_{\mathbf{P}}^{-1}\mathbf{e}_1}{\mathbf{e}_1^\top\mathbf{M}_{\mathbf{P}}^{-1}\mathbf{e}_1}, \quad \boldsymbol{\Sigma}_{\mathbf{P}}^{-1} = \mathbf{P}^\top\boldsymbol{\Sigma}^{-1}\mathbf{P},\ \mathbf{M}_{\mathbf{P}}^{-1} = \mathbf{P}^\top\mathbf{M}^{-1}\mathbf{P}.$$

By Theorem 4.2, $\mathbf{M}_{\mathbf{P}} = \mathbf{P}^\top\mathbf{M}\mathbf{P} \sim W_p(n, \boldsymbol{\Sigma}_{\mathbf{P}})$, so (i) applies: $(\boldsymbol{\Sigma}_{\mathbf{P}}^{-1})_{11}/(\mathbf{M}_{\mathbf{P}}^{-1})_{11} \sim \chi^2_{n-p+1}$.

(ii)⇒(iii) Conditioning on $\mathbf{y} = \mathbf{a}$ leaves the law of $\mathbf{M}$ unchanged (independence), so by (ii) the conditional distribution is χ^2_{n-p+1} for almost all $\mathbf{a} \neq \mathbf{0}$. It does not depend on $\mathbf{a}$, so it is also the marginal law and is independent of $\mathbf{y}$. ■

Remark 4.6: residuals and projections in multivariate analysis

1. The residual of $\mathbf{x}_1$ after projecting onto the span of $\mathbf{X}_2$ is $\mathbf{x}_1^* = \mathbf{x}_1 - \mathbf{X}_2\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21} = \mathbf{x}_1 - \mathrm{BLP}(\mathbf{x}_1 \mid \mathbf{X}_2)$.
2. $\mathbf{x}_1^*$ is orthogonal to (uncorrelated with) $\mathbf{X}_2$; being Gaussian, they are independent.
3. The partial variance $\sigma_{11\cdot 2} = \sigma_{11} - \boldsymbol{\sigma}_{21}^\top\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21} = \mathrm{Var}(\mathbf{x}_1^*)$ is the variance left after removing $\mathbf{X}_2$.

The residual sum of squares has $n - (p-1)$ degrees of freedom. This theorem is the key to the F distribution of Hotelling's T^2 (Ch. 5).

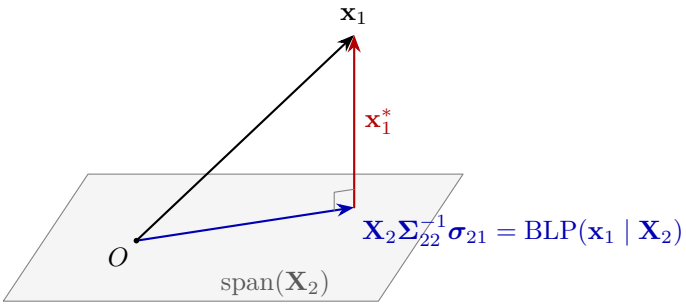


Figure. *The residual* $\mathbf{x}_1^*$ *is orthogonal to* $\mathrm{span}(\mathbf{X}_2)$*;* $\sigma_{11\cdot 2} = \mathrm{Var}(\mathbf{x}_1^*)$.

4 Maximum Likelihood Estimation

Motivation. Under normality the maximum likelihood estimates of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are the sample mean and the (biased) sample covariance. The proof uses a trace inequality rather than differentiation.

Theorem 4.7: MLE of μ and Σ (Result 4.11)

Let $\mathbf{X}_1, \ldots, \mathbf{X}_n$ be iid $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $n > p$. The MLEs are

$$\hat{\boldsymbol{\mu}}_{\mathrm{MLE}} = \bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^n \mathbf{x}_i, \qquad \hat{\boldsymbol{\Sigma}}_{\mathrm{MLE}} = \frac{1}{n}\sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top =: \mathbf{S}_0 = \frac{n-1}{n}\mathbf{S}.$$

The MLE jointly minimizes the negative log-likelihood

$$\begin{aligned}\ell(\boldsymbol{\mu}, \boldsymbol{\Sigma} \mid \mathbf{x}^n) &:= -\log L(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{n}{2}\log|\boldsymbol{\Sigma}| + \frac{1}{2}\sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}_i - \boldsymbol{\mu}) + c \\ &= \frac{n}{2}\log|\boldsymbol{\Sigma}| + \frac{n}{2}(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) + \frac{1}{2}\sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}_i - \bar{\mathbf{x}}) + c.\end{aligned}$$

(Expand $\mathbf{x}_i - \boldsymbol{\mu} = (\mathbf{x}_i - \bar{\mathbf{x}}) + (\bar{\mathbf{x}} - \boldsymbol{\mu})$; the cross term vanishes.)

Step 1 (profile out μ). ℓ depends on $\boldsymbol{\mu}$ only through $(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) \geq 0$, minimized by $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$ for every $\boldsymbol{\Sigma}$. With $\sum_i (\mathbf{x}_i - \bar{\mathbf{x}})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}_i - \bar{\mathbf{x}}) = \mathrm{tr}\{\boldsymbol{\Sigma}^{-1}\sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top\} = n\mathrm{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0)$,

$$2\ell(\hat{\boldsymbol{\mu}}_{\mathrm{MLE}}, \boldsymbol{\Sigma}) = n\log|\boldsymbol{\Sigma}| + n\mathrm{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0) + c,$$

so it remains to minimize $f(\boldsymbol{\Sigma}) = \log|\boldsymbol{\Sigma}| + \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0)$ over $\boldsymbol{\Sigma} \succ 0$.

First proof: von Neumann's inequality. Decompose $\boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$ and $\mathbf{S}_0 = \mathbf{V}\mathbf{D}\mathbf{V}^\top$ (λ_i, d_i in decreasing order; $d_i > 0$ a.s. when $n > p$). The eigenvalues of $\boldsymbol{\Sigma}^{-1}$ in increasing order are $1/\lambda_1 \le \cdots \le 1/\lambda_p$, so von Neumann's lower bound gives

$$\log|\boldsymbol{\Sigma}| + \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0) \ge \log(\lambda_1 \cdots \lambda_p) + \sum_{i=1}^{p} \frac{d_i}{\lambda_i} = \sum_{i=1}^{p} \Big(\log\lambda_i + \frac{d_i}{\lambda_i}\Big),$$

with equality iff $\mathbf{U} = \mathbf{V}$. Let $a_i = d_i/\lambda_i$. Then

$$\sum_{i=1}^{p} \Big(\log\lambda_i + \frac{d_i}{\lambda_i}\Big) = \sum_{i=1}^{p} (\log d_i + a_i - \log a_i) = \log|\mathbf{S}_0| + \sum_{i=1}^{p} (a_i - \log a_i) \ge \log|\mathbf{S}_0| + p,$$

since $x - \log x \ge 1$. The bound is attained iff (1) $\mathbf{U} = \mathbf{V}$ and (2) $a_i = 1$ for all i, i.e. $\boldsymbol{\Lambda} = \mathbf{D}$: $\hat{\boldsymbol{\Sigma}}_{\text{MLE}} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top = \mathbf{V}\mathbf{D}\mathbf{V}^\top = \mathbf{S}_0$. ■

Second proof: matrix calculus. Reparametrise by the precision matrix $\mathbf{K} = \boldsymbol{\Sigma}^{-1}$ ($\boldsymbol{\Sigma} \mapsto \boldsymbol{\Sigma}^{-1}$ is a bijection of the PD cone, so by the change-of-variables part of Theorem 2.25 the optimizers correspond). $g(\mathbf{K}) = -\log|\mathbf{K}| + \operatorname{tr}(\mathbf{K}\mathbf{S}_0)$ is defined on the open convex set $\{\mathbf{K} \succ 0\}$, and for symmetric $\mathbf{H} \neq \mathbf{0}$

$$D^2 g(\mathbf{K})[\mathbf{H}] = \operatorname{tr}(\mathbf{K}^{-1}\mathbf{H}\mathbf{K}^{-1}\mathbf{H}) = \|\mathbf{K}^{-1/2}\mathbf{H}\mathbf{K}^{-1/2}\|_F^2 > 0,$$

so g is strictly convex (Theorem 2.24). The first-order condition

$$Dg(\mathbf{K})[\mathbf{H}] = -\operatorname{tr}(\mathbf{K}^{-1}\mathbf{H}) + \operatorname{tr}(\mathbf{S}_0\mathbf{H}) = \operatorname{tr}\{(\mathbf{S}_0 - \mathbf{K}^{-1})\mathbf{H}\} = 0 \quad \forall \mathbf{H}$$

and Theorem 2.23 give $\mathbf{K}^{-1} = \mathbf{S}_0$, the unique minimizer. By profiling (Theorem 2.25), $(\bar{\mathbf{x}}, \mathbf{S}_0)$ is the joint minimizer. ■

Remark 4.8: invariance and the maximized likelihood

[Supplement] A function of the MLE is the MLE of the function: e.g. the eigenvalues of $\mathbf{S}_0$ estimate those of $\boldsymbol{\Sigma}$, and r_{ij} is the MLE of ρ_{ij}. The maximum is $L(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Sigma}}) = (2\pi)^{-np/2}|\mathbf{S}_0|^{-n/2}e^{-np/2}$, which depends on the data only through the generalized variance $|\mathbf{S}_0|$; this is used for likelihood ratio tests (Ch. 5).

5 Large-Sample Behavior of $\bar{\mathbf{X}}$ and $\mathbf{S}$

Motivation. Without normality, the CLT still makes $\bar{\mathbf{X}}$ approximately normal and $\mathbf{S}$ consistent, so normal-theory procedures remain valid for large samples.

Theorem 4.9: law of large numbers (Result 4.12)

Let $\mathbf{X}_1, \ldots, \mathbf{X}_n \in \mathbb{R}^p$ be iid with $\mathrm{E}(\mathbf{X}_j) = \boldsymbol{\mu}$ and finite $\mathrm{Cov}(\mathbf{X}_j) = \boldsymbol{\Sigma}$. Then, as $n \to \infty$,

$$\text{(a) } \bar{\mathbf{X}}_n \xrightarrow{p} \boldsymbol{\mu}, \qquad \text{(b) } \mathbf{S} \xrightarrow{p} \boldsymbol{\Sigma}, \qquad \text{(c) } \mathbf{S}_0 \xrightarrow{p} \boldsymbol{\Sigma}.$$

Moreover $\bar{\mathbf{X}}_n$ and $\mathbf{S}$ are unbiased.

Proof. (a) By Markov's inequality,

$$P(\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\| > \epsilon) \le \frac{1}{\epsilon^2}\mathrm{E}\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|^2 = \frac{1}{\epsilon^2}\mathrm{E}(\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) = \frac{1}{\epsilon^2}\mathrm{E}[\operatorname{tr}\{(\bar{\mathbf{X}}_n - \boldsymbol{\mu})(\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top\}]$$

$$= \frac{1}{\epsilon^2}\mathrm{tr}(\mathrm{Var}(\bar{\mathbf{X}}_n)) = \frac{\mathrm{tr}(\boldsymbol{\Sigma})}{n\epsilon^2} \xrightarrow[n\to\infty]{} 0.$$

(b) Adding and subtracting $\boldsymbol{\mu}$ inside each factor and using $\sum_i(\mathbf{X}_i - \boldsymbol{\mu}) = n(\bar{\mathbf{X}} - \boldsymbol{\mu})$,

$$(n-1)\mathbf{S} = \sum_{i=1}^{n}(\mathbf{X}_i - \bar{\mathbf{X}}_n)(\mathbf{X}_i - \bar{\mathbf{X}}_n)^\top = \sum_{i=1}^{n}(\mathbf{X}_i - \boldsymbol{\mu})(\mathbf{X}_i - \boldsymbol{\mu})^\top - n(\bar{\mathbf{X}} - \boldsymbol{\mu})(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top.$$

Hence

$$\frac{n-1}{n}\mathbf{S} = \frac{1}{n}\sum_{i=1}^{n}(\mathbf{X}_i - \boldsymbol{\mu})(\mathbf{X}_i - \boldsymbol{\mu})^\top - (\bar{\mathbf{X}} - \boldsymbol{\mu})(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \xrightarrow{p} \boldsymbol{\Sigma} - \mathbf{0} = \boldsymbol{\Sigma}.$$

The first term is an average of iid matrices with mean $\mathrm{E}(\mathbf{X}-\boldsymbol{\mu})(\mathbf{X}-\boldsymbol{\mu})^\top = \boldsymbol{\Sigma}$, so it converges to $\boldsymbol{\Sigma}$ by the weak LLN applied entrywise (matrix convergence is entrywise, Theorem 2.29). The second term tends to $\mathbf{0}$ by (a) and the CMT ($\mathbf{x} \mapsto \mathbf{x}\mathbf{x}^\top$ is continuous). Since $n/(n-1) \to 1$, $\mathbf{S} \xrightarrow{p} \boldsymbol{\Sigma}$.

(c) $\mathbf{S}_0 = \frac{n-1}{n}\mathbf{S}$, which was shown to converge to $\boldsymbol{\Sigma}$ in (b).

Unbiasedness. $\mathrm{E}\bar{\mathbf{X}}_n = \boldsymbol{\mu}$, and taking expectations in the identity of (b), $\mathrm{E}\{(n-1)\mathbf{S}\} = n\boldsymbol{\Sigma} - n\mathrm{Var}(\bar{\mathbf{X}}_n) = (n-1)\boldsymbol{\Sigma}$. [Supplement] Under normality, $(n-1)\mathbf{S} \sim W_p(n-1, \boldsymbol{\Sigma})$ and Theorem 4.2(iv) also gives the rate $\mathrm{E}\|\mathbf{S} - \boldsymbol{\Sigma}\|_F^2 = \sum_{ij}(\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj})/(n-1) \to 0$. ■

Remark 4.10: rate of the estimation error

[Supplement] $\mathrm{E}\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|_2^2 = \mathrm{tr}\boldsymbol{\Sigma}/n$, so $E_{\bar{u}} = \|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|_2$ decays at rate $n^{-1/2}$; so does $E_S = \|\mathbf{S} - \boldsymbol{\Sigma}\|_F$ under normality. On a log–log plot of $E_{\bar{u}}, E_S, E_R = \|\mathbf{R} - \mathbf{R}_0\|_F$ against n the slope is about $-1/2$.

Theorem 4.11: multivariate central limit theorem (Result 4.13(i))

Let $\mathbf{X}_1, \mathbf{X}_2, \cdots \in \mathbb{R}^p$ be iid with $\mathrm{E}(\mathbf{X}_i) = \boldsymbol{\mu}$ and $\mathrm{Cov}(\mathbf{X}_i) = \boldsymbol{\Sigma}$. Then

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}(\mathbf{X}_i - \boldsymbol{\mu}) \xrightarrow{d} N_p(\mathbf{0}, \boldsymbol{\Sigma}), \qquad \text{equivalently} \qquad \sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \xrightarrow{d} N_p(\mathbf{0}, \boldsymbol{\Sigma}).$$

Proof. By Cramér–Wold it suffices to show $\mathbf{t}^\top\sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \xrightarrow{d} \mathbf{t}^\top\mathbf{Y}$, $\mathbf{Y} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$, for every $\mathbf{t}$. $W_i = \mathbf{t}^\top(\mathbf{X}_i - \boldsymbol{\mu})$ are iid scalars with mean 0 and variance $\mathbf{t}^\top\boldsymbol{\Sigma}\mathbf{t}$, so the univariate CLT gives $\sqrt{n}\bar{W}_n \xrightarrow{d} N(0, \mathbf{t}^\top\boldsymbol{\Sigma}\mathbf{t})$, the law of $\mathbf{t}^\top\mathbf{Y}$ (if $\mathbf{t}^\top\boldsymbol{\Sigma}\mathbf{t} = 0$, $W_i = 0$ a.s. and the claim is trivial). ■

Theorem 4.12: large-sample chi-square (Result 4.13(ii))

Under the same assumptions with $\boldsymbol{\Sigma} \succ 0$, $T_n := n(\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top\mathbf{S}^{-1}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \xrightarrow{d} \chi_p^2$. (Where $\mathbf{S}$ is singular, read $\mathbf{S}^{-1}$ as $r(\mathbf{S})$ defined in the proof.)

Proof. Let $\mathbf{Z}_n := \sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu})$ and $\mathbf{Z} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$.

(i) $\mathbf{Z}_n \xrightarrow{d} \mathbf{Z}$. For every $\mathbf{t} \in \mathbb{R}^p$, $\mathbf{t}^\top\mathbf{Z}_n = \sqrt{n}(\mathbf{t}^\top\bar{\mathbf{X}}_n - \mathbf{t}^\top\boldsymbol{\mu}) \xrightarrow{d} N(0, \mathbf{t}^\top\boldsymbol{\Sigma}\mathbf{t})$ by the univariate CLT, so $\mathbf{Z}_n \xrightarrow{d} \mathbf{Z}$ by Cramér–Wold (this is Theorem 4.11).

(ii) Handling a possibly singular S. Let $A_n := \{\mathbf{S} \text{ is invertible}\}$. Since $\mathbf{S} \xrightarrow{p} \boldsymbol{\Sigma}$ (Theorem 4.9(b)) and $\mathbf{A} \mapsto |\mathbf{A}|$ is continuous, $|\mathbf{S}| \xrightarrow{p} |\boldsymbol{\Sigma}| > 0$, hence

$$P(A_n) \geq P(|\mathbf{S}| \geq \tfrac{1}{2}|\boldsymbol{\Sigma}|) \longrightarrow 1.$$

Define $r(\mathbf{A}) = \mathbf{A}^{-1}$ if $|\mathbf{A}| \neq 0$ and $r(\mathbf{A}) = \mathbf{I}_p$ otherwise. By the definition of r,

$$T_n = n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{X}} - \boldsymbol{\mu})\, I_{A_n} + n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top (\bar{\mathbf{X}} - \boldsymbol{\mu})\, I_{A_n^c} = \mathbf{Z}_n^\top r(\mathbf{S}) \mathbf{Z}_n,$$

and T_n coincides with the statistic of the theorem on A_n, an event of probability tending to one.

(iii) $r(\mathbf{S}) \xrightarrow{p} \boldsymbol{\Sigma}^{-1}$**.** Since $\mathbf{S} \xrightarrow{p} \boldsymbol{\Sigma}$, by the CMT it suffices that r is continuous at the point $\boldsymbol{\Sigma}$. Since $|\boldsymbol{\Sigma}| > 0$ and $\{\mathbf{A} : |\mathbf{A}| \neq 0\}$ is open, $r(\mathbf{A}) = \mathbf{A}^{-1} = \operatorname{adj}(\mathbf{A})/|\mathbf{A}|$ on a neighborhood of $\boldsymbol{\Sigma}$ (Section 1 of Ch. 2), whose entries are polynomials divided by a nonvanishing polynomial; so r is continuous at $\boldsymbol{\Sigma}$ and $r(\mathbf{S}) \xrightarrow{p} \boldsymbol{\Sigma}^{-1}$.

(iv) Conclusion. Let $Q(\mathbf{z}, \mathbf{A}) = \mathbf{z}^\top \mathbf{A} \mathbf{z}$, so $T_n = Q(\mathbf{Z}_n, r(\mathbf{S}))$. By Slutsky, $(\mathbf{Z}_n, r(\mathbf{S})) \xrightarrow{d} (\mathbf{Z}, \boldsymbol{\Sigma}^{-1})$, and Q is continuous, so by the CMT

$$T_n = Q(\mathbf{Z}_n, r(\mathbf{S})) \xrightarrow{d} Q(\mathbf{Z}, \boldsymbol{\Sigma}^{-1}) = \mathbf{Z}^\top \boldsymbol{\Sigma}^{-1} \mathbf{Z} \sim \chi^2_p$$

(Example 3.22). ■

This is the basis of large-sample tests and confidence regions for the mean without normality (Theorems 5.13 and 5.14).

Theorem 4.13: asymptotic normality of $h(\bar{\mathbf{X}}_n)$ (delta method)

For $h : \mathbb{R}^p \to \mathbb{R}$ differentiable at $\boldsymbol{\mu}$ with gradient $\nabla h(\boldsymbol{\mu})$,

$$\sqrt{n}\{h(\bar{\mathbf{X}}_n) - h(\boldsymbol{\mu})\} \xrightarrow{d} N\big(0,\ (\nabla h(\boldsymbol{\mu}))^\top \boldsymbol{\Sigma} (\nabla h(\boldsymbol{\mu}))\big).$$

Proof. Taylor: $h(\bar{\mathbf{X}}_n) = h(\boldsymbol{\mu}) + (\nabla h(\boldsymbol{\mu}))^\top (\bar{\mathbf{X}}_n - \boldsymbol{\mu}) + O_p(\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|_2^2)$. Multiply by $\sqrt{n}$: $\sqrt{n}(h(\bar{\mathbf{X}}_n) - h(\boldsymbol{\mu})) = (\nabla h(\boldsymbol{\mu}))^\top \sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) + O_p(\sqrt{n}\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|^2)$, and the remainder is $o_p(1)$. Slutsky gives $\xrightarrow{d} (\nabla h(\boldsymbol{\mu}))^\top N_p(\mathbf{0}, \boldsymbol{\Sigma})$. ■

6 Assessing the Assumption of Normality

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then (necessary conditions only):

- each marginal X_i is normal (Result 4.2): about 68.3% of x_{ji} within $\bar{x}_i \pm \sqrt{s_{ii}}$ and 95.4% within $\pm 2\sqrt{s_{ii}}$ (flag if off by more than $1.396/\sqrt{n}$ or $0.628/\sqrt{n}$); Q–Q plot and r_Q;
- each pair is elliptical: about half the points satisfy $(\mathbf{x}_j - \bar{\mathbf{x}})^\top \mathbf{S}^{-1} (\mathbf{x}_j - \bar{\mathbf{x}}) \le \chi^2_2(0.5) = 1.39$;
- $(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) \sim \chi^2_p$ (Result 4.7): chi-square plot.

6.1 Q–Q plot and r_Q

Order $x_{(1)} \le \cdots \le x_{(n)}$ and plot $(q_{(j)}, x_{(j)})$ with $q_{(j)} = \Phi^{-1}((j - \tfrac{1}{2})/n)$; under normality $x_{(j)} \approx \mu + \sigma q_{(j)}$ (a straight line). Reject normality if

$$r_Q := \frac{\sum_j (x_{(j)} - \bar{x})(q_{(j)} - \bar{q})}{\sqrt{\sum_j (x_{(j)} - \bar{x})^2}\, \sqrt{\sum_j (q_{(j)} - \bar{q})^2}}$$

is below the critical point (J&W Table 4.2):

n	$\alpha = .01$	.05	.10
10	.8801	.9198	.9351
20	.9269	.9508	.9604
30	.9479	.9652	.9715
50	.9671	.9768	.9809
100	.9822	.9873	.9895

Example 4.14: Q–Q plot and r_Q (Examples 4.9 and 4.11)

$n = 10$, $\bar{x} = 0.770$, $\sqrt{s_{11}} = 0.970$: $r_Q = 0.994 > 0.9351$, normality not rejected at $\alpha = 0.10$.

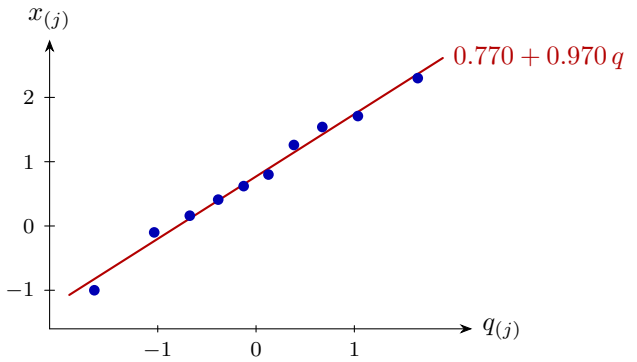


Figure. *Q–Q plot for Example 4.9: points near the line $\bar{x} + \sqrt{s_{11}}\,q$.*

6.2 Chi-square plot

Plot the ordered $d^2_{(j)}$, $d^2_j = (\mathbf{x}_j - \bar{\mathbf{x}})^\top \mathbf{S}^{-1}(\mathbf{x}_j - \bar{\mathbf{x}})$, against $q_{c,p}\big((j - \tfrac{1}{2})/n\big)$, the χ^2_p quantiles. A line of slope 1 supports normality; curvature indicates non-normality; points far above the line are outliers. Use as a guide when n, $n - p \geq 25$–30.

Example 4.15: 50% contour and chi-square plots (Examples 4.12–4.14)

(a) Sales–profits ($p = 2$, $n = 10$): 4 of 10 $d^2_j \leq 1.39$; the plot is reasonably straight.
(b) Board stiffness ($p = 4$, $n = 30$): 17 of 30 $d^2_j \leq q_{c,4}(0.5) = 3.36$, but boards 9 and 16 break away from the line.

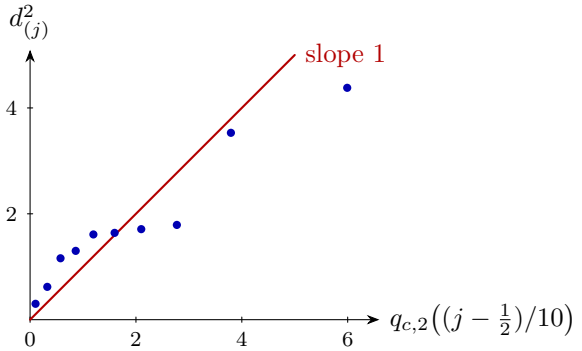


Figure. *Chi-square plot for the sales–profits data (Example 4.13).*

7 Summary: Distributional Facts Used Later

[Supplement] The facts of Chapters 3 and 4 that the inference chapters rely on:

Fact	Source
$\sqrt{n}(\bar{\mathbf{X}}-\boldsymbol{\mu}) \sim N_p(\mathbf{0},\boldsymbol{\Sigma})$	Theorem 3.7
$(n-1)\mathbf{S} \sim W_p(n-1,\boldsymbol{\Sigma})$, $\bar{\mathbf{X}} \perp\!\!\!\perp \mathbf{S}$	Theorem 4.3
$(\mathbf{X}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{X}-\boldsymbol{\mu}) \sim \chi^2_p$	Theorem 3.21, Example 3.22
$\mathbf{a}^\top\boldsymbol{\Sigma}^{-1}\mathbf{a}/\mathbf{a}^\top\mathbf{M}^{-1}\mathbf{a} \sim \chi^2_{m-p+1}$, also for independent random $\mathbf{y}$	Theorem 4.5
$n(\bar{\mathbf{X}}-\boldsymbol{\mu})^\top\mathbf{S}^{-1}(\bar{\mathbf{X}}-\boldsymbol{\mu}) \xrightarrow{d} \chi^2_p$ (no normality)	Theorem 4.11 + Slutsky

8 Selected exercises

Motivation. Two classical consequences of this chapter: the exact law of the generalized variance $|\mathbf{S}|$, obtained from the argument behind Theorem 4.5, and the large-sample law of the sample correlation, obtained from the CLT and the delta method.

Exercise 4.1 (The generalized variance)

Let $\mathbf{M} \sim W_p(n,\boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$ and $n \geq p$.

(a) Show that $|\mathbf{M}|/|\boldsymbol{\Sigma}| \overset{d}{=} \prod_{i=1}^{p} V_i$ with $V_1,\dots,V_p$ independent and $V_i \sim \chi^2_{n-i+1}$.

(b) For a sample of size $n > p$ from $N_p(\boldsymbol{\mu},\boldsymbol{\Sigma})$, deduce $\mathrm{E}|\mathbf{S}| = |\boldsymbol{\Sigma}| \prod_{i=1}^{p} \dfrac{n-i}{n-1}$, so $|\mathbf{S}|$ underestimates $|\boldsymbol{\Sigma}|$ on average when $p \geq 2$.

Solution [Supplement].

(a) Induction on p. For $p=1$, $M/\sigma \sim \chi^2_n$ by definition.

Step 1 (set-up). Let $p \geq 2$ and write $\mathbf{M} = \mathbf{X}^\top\mathbf{X}$, where the $n \times p$ matrix $\mathbf{X} = [\mathbf{x}_1, \mathbf{X}_2]$ has iid rows $N_p(\mathbf{0},\boldsymbol{\Sigma})$ and $\mathbf{x}_1$ is its first column. The rows of $\mathbf{X}_2$ are iid $N_{p-1}(\mathbf{0},\boldsymbol{\Sigma}_{22})$, so $\mathbf{M}_{22} = \mathbf{X}_2^\top\mathbf{X}_2 \sim W_{p-1}(n,\boldsymbol{\Sigma}_{22})$, and $\mathbf{M}_{22} \succ 0$ a.s. since $n \geq p-1$.

Step 2 (block determinants). By Theorem 2.7, $|\mathbf{M}| = |\mathbf{M}_{22}|\, m_{11\cdot 2}$ and $|\boldsymbol{\Sigma}| = |\boldsymbol{\Sigma}_{22}|\, \sigma_{11\cdot 2}$, so

$$\frac{|\mathbf{M}|}{|\boldsymbol{\Sigma}|} = \frac{|\mathbf{M}_{22}|}{|\boldsymbol{\Sigma}_{22}|} \cdot \frac{m_{11\cdot 2}}{\sigma_{11\cdot 2}}.$$

Step 3 (the last factor). The proof of Theorem 4.5(i) shows that, conditionally on $\mathbf{X}_2$, $U := m_{11\cdot 2}/\sigma_{11\cdot 2} \sim \chi^2_{n-p+1}$. This conditional law does not depend on $\mathbf{X}_2$, so $U \sim \chi^2_{n-p+1}$ and U is independent of $\mathbf{X}_2$, hence of $|\mathbf{M}_{22}|/|\boldsymbol{\Sigma}_{22}|$.

Step 4 (induction). By the induction hypothesis for $\mathbf{M}_{22}$ (dimension $p-1$, $n \geq p-1$), $|\mathbf{M}_{22}|/|\boldsymbol{\Sigma}_{22}| \overset{d}{=} \prod_{i=1}^{p-1} V_i$ with independent $V_i \sim \chi^2_{n-i+1}$. Put $V_p = U \sim \chi^2_{n-p+1}$; by Step 3 it is independent of $(V_1,\dots,V_{p-1})$.

(b) By Theorem 4.4, $(n-1)\mathbf{S} \sim W_p(n-1,\boldsymbol{\Sigma})$ with $n-1 \geq p$, so by (a) $(n-1)^p|\mathbf{S}|/|\boldsymbol{\Sigma}|$ is a product of independent χ^2_{n-i} variables, $i=1,\dots,p$. Independence gives $\mathrm{E}\{(n-1)^p|\mathbf{S}|/|\boldsymbol{\Sigma}|\} = \prod_{i=1}^{p}(n-i)$. For $p \geq 2$ the factor $(n-2)/(n-1)$ is < 1 and the others are ≤ 1. ∎

Exercise 4.2 (Asymptotics of the sample correlation)

Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be iid $N_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\rho = \sigma_{12}/\sqrt{\sigma_{11}\sigma_{22}}$, $|\rho| < 1$, and $r = s_{12}/\sqrt{s_{11}s_{22}}$.
(a) Show $\sqrt{n}(r - \rho) \xrightarrow{d} N\big(0, (1-\rho^2)^2\big)$.
(b) Show that Fisher's $z = \frac{1}{2}\log\frac{1+r}{1-r}$ satisfies $\sqrt{n}(z - \zeta) \xrightarrow{d} N(0,1)$, $\zeta = \frac{1}{2}\log\frac{1+\rho}{1-\rho}$, and give an approximate $1-\alpha$ confidence interval for ρ.

Solution [Supplement].

(a)

Step 1 (reduction). r is unchanged by $X_{ij} \mapsto (X_{ij} - \mu_j)/\sqrt{\sigma_{jj}}$, so assume $\boldsymbol{\mu} = \mathbf{0}$, $\sigma_{11} = \sigma_{22} = 1$, $\sigma_{12} = \rho$.

Step 2 (moments about 0). Let $\mathbf{Y}_i = (X_{i1}^2, X_{i2}^2, X_{i1}X_{i2})^\top$, iid with mean $\boldsymbol{m} = (1, 1, \rho)^\top$, and

$$\mathbf{W}_n = \bar{\mathbf{Y}}_n - (\bar{X}_1^2,\ \bar{X}_2^2,\ \bar{X}_1\bar{X}_2)^\top = \tfrac{n-1}{n}(s_{11}, s_{22}, s_{12})^\top.$$

Since $\sqrt{n}\,\bar{X}_j\bar{X}_k = (\sqrt{n}\bar{X}_j)\bar{X}_k = O_p(1)\,o_p(1) \xrightarrow{p} 0$ (CLT and LLN), Theorem 4.11 and Slutsky give $\sqrt{n}(\mathbf{W}_n - \boldsymbol{m}) \xrightarrow{d} N_3(\mathbf{0}, \boldsymbol{\Gamma})$ with $\boldsymbol{\Gamma} = \mathrm{Cov}(\mathbf{Y}_1)$. The factor $\frac{n-1}{n}$ cancels in r, so $r = g(\mathbf{W}_n)$ with $g(a,b,c) = c/\sqrt{ab}$.

Step 3 ($\boldsymbol{\Gamma}$). Write $X_2 = \rho X_1 + \sqrt{1-\rho^2}\,E$ with $E \sim N(0,1)$ independent of X_1 (Theorem 3.11). Using $\mathrm{E}X_1^4 = 3$ and $\mathrm{E}X_1^3 = 0$,

$$\mathrm{E}X_1^2X_2^2 = 3\rho^2 + (1-\rho^2) = 1 + 2\rho^2, \qquad \mathrm{E}X_1^3X_2 = 3\rho, \qquad \mathrm{E}X_1X_2^3 = 3\rho,$$

and subtracting the products of the means,

$$\boldsymbol{\Gamma} = \begin{pmatrix} 2 & 2\rho^2 & 2\rho \\ 2\rho^2 & 2 & 2\rho \\ 2\rho & 2\rho & 1+\rho^2 \end{pmatrix}.$$

Step 4 (delta method). g is continuously differentiable near $\boldsymbol{m}$ with $\nabla g(\boldsymbol{m}) = (-\rho/2, -\rho/2, 1)^\top$. The proof of Theorem 4.13 uses only $\sqrt{n}(\mathbf{W}_n - \boldsymbol{m}) \xrightarrow{d} N_3(\mathbf{0}, \boldsymbol{\Gamma})$ and a Taylor expansion, so $\sqrt{n}(r - \rho) \xrightarrow{d} N(0, \nabla g^\top \boldsymbol{\Gamma} \nabla g)$ with

$$\begin{aligned} \nabla g^\top \boldsymbol{\Gamma} \nabla g &= \frac{\rho^2}{4}(\Gamma_{11} + \Gamma_{22} + 2\Gamma_{12}) + \Gamma_{33} - \rho(\Gamma_{13} + \Gamma_{23}) \\ &= \rho^2(1+\rho^2) + 1 + \rho^2 - 4\rho^2 = (1-\rho^2)^2. \end{aligned}$$

(b) $h(x) = \frac{1}{2}\log\frac{1+x}{1-x}$ has $h'(\rho) = 1/(1-\rho^2)$. The one-dimensional delta method applied to (a) gives $\sqrt{n}(z - \zeta) \xrightarrow{d} N\big(0, (1-\rho^2)^2/(1-\rho^2)^2\big) = N(0,1)$: the transformation stabilizes the variance. Hence $P(|z - \zeta| \le z_{\alpha/2}/\sqrt{n}) \to 1 - \alpha$, and since $\tanh = h^{-1}$ is increasing,

$$\big[\tanh(z - z_{\alpha/2}/\sqrt{n}),\ \tanh(z + z_{\alpha/2}/\sqrt{n})\big]$$

is an approximate $1-\alpha$ interval for ρ (in practice $\sqrt{n-3}$ is used, which has the same limit). ■

Chapter 5. Inference about a Mean Vector

Hotelling's T^2, likelihood ratio tests, confidence regions and simultaneous intervals, large-sample inference

1 Starting Point: the Univariate t and Known Σ

In one dimension $H_0 : \mu = \mu_0$ is tested with $t = \sqrt{n}(\bar{X} - \mu_0)/s$, and

$$t^2_{n-1} = \sqrt{n}(\bar{X} - \mu_0)\,(s^2)^{-1}\sqrt{n}(\bar{X} - \mu_0) = \begin{pmatrix}\text{normal}\\ \text{r.v.}\end{pmatrix}\begin{pmatrix}\text{(scaled) chi-square}\\ \text{r.v./d.f.}\end{pmatrix}^{-1}\begin{pmatrix}\text{normal}\\ \text{r.v.}\end{pmatrix}.$$

If $\boldsymbol{\Sigma}$ is known, $\sqrt{n}\boldsymbol{\Sigma}^{-1/2}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \sim N_p(\mathbf{0}, \mathbf{I}_p)$, so

$$n(\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \sim \chi^2_p, \qquad C_\alpha = \{\boldsymbol{\mu} \in \mathbb{R}^p : (\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top (\tfrac{1}{n}\boldsymbol{\Sigma})^{-1}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \le \chi^2_{p,\alpha}\}.$$

Replacing $\boldsymbol{\Sigma}$ by $\mathbf{S}$ gives Hotelling's T^2.

2 Hotelling's T^2 Statistic

Motivation. Testing $\boldsymbol{\mu} = \boldsymbol{\mu}_0$ with p variables by p separate t-tests ignores correlation and inflates error. Hotelling's T^2 is the multivariate t-statistic: it combines all coordinates using $\mathbf{S}^{-1}$ and has an exact F distribution.

Definition 5.1: Hotelling's T^2

Suppose $\mathbf{X}$ and $\mathbf{S}$ are independent with $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $m\mathbf{S} \sim W_p(m, \boldsymbol{\Sigma})$. Then

$$T^2_p(m) = (\mathbf{X} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\mathbf{X} - \boldsymbol{\mu})$$

is Hotelling's T^2 statistic. It plays the role of Student's t when $\boldsymbol{\Sigma}$ is *unknown*:

$$T^2_p(n-1) = \begin{pmatrix}\text{multivariate normal}\\ \text{random vector}\end{pmatrix}^\top \begin{pmatrix}\text{Wishart random}\\ \text{matrix/d.f.}\end{pmatrix}^{-1} \begin{pmatrix}\text{multivariate normal}\\ \text{random vector}\end{pmatrix}$$

$$= N_p(\mathbf{0}, \boldsymbol{\Sigma})^\top \left[\tfrac{1}{n-1} W_p(n-1, \boldsymbol{\Sigma})\right]^{-1} N_p(\mathbf{0}, \boldsymbol{\Sigma}).$$

Theorem 5.2 (slides: Theorem 7): T^2 and the F distribution

If $m > p - 1$, then $\dfrac{m - p + 1}{mp} T^2_p(m) \sim F_{p, m-p+1}$.

Proof. Multiplying and dividing by $(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu})$,

$$\frac{m - p + 1}{mp} T^2_p(m) = \frac{(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu})/p}{\dfrac{(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu})}{(\mathbf{X} - \boldsymbol{\mu})^\top (m\mathbf{S})^{-1}(\mathbf{X} - \boldsymbol{\mu})} \Big/ (m - p + 1)} = \frac{Y/p}{W/(m - p + 1)},$$

where $Y = (\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu}) \sim \chi^2_p$ (Example 3.22) and $W = \dfrac{\mathbf{a}^\top \boldsymbol{\Sigma}^{-1}\mathbf{a}}{\mathbf{a}^\top (m\mathbf{S})^{-1}\mathbf{a}}$ with the random vector $\mathbf{a} = \mathbf{X} - \boldsymbol{\mu}$ independent of $m\mathbf{S} \sim W_p(m, \boldsymbol{\Sigma})$; by Theorem 4.5(iii) $W \sim \chi^2_{m-p+1}$ and is independent of $\mathbf{a}$, hence of Y. By definition the ratio is $F_{p, m-p+1}$. ■

Theorem 5.3: one-sample T^2

If $\mathbf{X}_1, \ldots, \mathbf{X}_n \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\bar{\mathbf{X}} = \frac{1}{n}\sum \mathbf{X}_j$, $\mathbf{S} = \frac{1}{n-1}\sum(\mathbf{X}_j - \bar{\mathbf{X}})(\mathbf{X}_j - \bar{\mathbf{X}})^\top$, then $\sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$ and $(n-1)\mathbf{S} \sim W_p(n-1, \boldsymbol{\Sigma})$ independently, so

$$n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \sim T^2_p(n-1), \qquad n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \sim \frac{(n-1)p}{n-p} F_{p,n-p}.$$

($m = n-1$ gives $\frac{m-p+1}{mp} = \frac{n-p}{(n-1)p}$.) Hence $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ is rejected at level α if

$$T^2 = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)^\top \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0) > \frac{(n-1)p}{n-p} F_{p,n-p}(\alpha),$$

with p-value $P(F_{p,n-p} \geq \frac{n-p}{(n-1)p} T^2_{\text{obs}})$.

Theorem 5.4: invariance of T^2 under linear transformations

The T^2 statistic is unchanged by changes of units of the form $\mathbf{Y}_{(p\times 1)} = \mathbf{C}_{(p\times p)}\mathbf{X}_{(p\times 1)} + \mathbf{d}_{(p\times 1)}$ with $\mathbf{C}$ non-singular: $\bar{\mathbf{y}} = \mathbf{C}\bar{\mathbf{x}} + \mathbf{d}$, $\mathbf{S}_y = \mathbf{C}\mathbf{S}\mathbf{C}^\top$, $\boldsymbol{\mu}_Y = \mathbf{C}\boldsymbol{\mu}_X + \mathbf{d}$, and

$$n(\bar{\mathbf{y}} - \boldsymbol{\mu}_Y)^\top \mathbf{S}_y^{-1}(\bar{\mathbf{y}} - \boldsymbol{\mu}_Y) = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_x)^\top \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_x) \sim T^2_p(n-1).$$

Proof. $n(\mathbf{C}(\bar{\mathbf{x}} - \boldsymbol{\mu}_x))^\top(\mathbf{C}\mathbf{S}\mathbf{C}^\top)^{-1}\mathbf{C}(\bar{\mathbf{x}} - \boldsymbol{\mu}_x) = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_x)^\top \mathbf{C}^\top \mathbf{C}^{-\top}\mathbf{S}^{-1}\mathbf{C}^{-1}\mathbf{C}(\bar{\mathbf{x}} - \boldsymbol{\mu}_x)$. ■

Example 5.5 (Example 5.2): testing a mean vector with T^2 (sweat data)

Perspiration from $n = 20$ healthy females: X_1 sweat rate, X_2 sodium, X_3 potassium. Test $H_0 : \boldsymbol{\mu}^\top = [4, 50, 10]$ at $\alpha = 0.10$.

$$\bar{\mathbf{x}} = \begin{pmatrix} 4.640 \\ 45.400 \\ 9.965 \end{pmatrix}, \quad \mathbf{S} = \begin{pmatrix} 2.879 & 10.010 & -1.810 \\ 10.010 & 199.788 & -5.640 \\ -1.810 & -5.640 & 3.628 \end{pmatrix}, \quad \mathbf{S}^{-1} = \begin{pmatrix} .586 & -.022 & .258 \\ -.022 & .006 & -.002 \\ .258 & -.002 & .402 \end{pmatrix}.$$

$\bar{\mathbf{x}} - \boldsymbol{\mu}_0 = (.640, -4.600, -.035)^\top$ and $\mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0) = (.467, -.042, .160)^\top$, so $T^2 = 20(.640 \cdot .467 + 4.600 \cdot .042 - .035 \cdot .160) = 9.74$. Critical value $\frac{(n-1)p}{n-p} F_{p,n-p}(.10) = \frac{19(3)}{17} F_{3,17}(.10) = 3.353(2.44) = 8.18$. Since $9.74 > 8.18$, reject H_0 at the 10% level.

3 Likelihood Ratio Tests

3.1 $\boldsymbol{\Sigma}$ unknown

Under $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ the likelihood is

$$L(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{np/2}|\boldsymbol{\Sigma}|^{n/2}} \exp\Big(-\frac{1}{2}\sum_{j=1}^{n}(\mathbf{x}_j - \boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}_j - \boldsymbol{\mu}_0)\Big),$$

and the exponent equals $-\frac{1}{2}\text{tr}[\boldsymbol{\Sigma}^{-1}\sum_j(\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)^\top]$. Maximizing over $\boldsymbol{\Sigma}$ as in the proof of Theorem 4.7 (with $\sum_j(\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)^\top$ in place of $\sum_j(\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})^\top$),

$$\max_{\boldsymbol{\Sigma}} L(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{np/2}|\hat{\boldsymbol{\Sigma}}_0|^{n/2}} e^{-np/2}, \qquad \hat{\boldsymbol{\Sigma}}_0 = \frac{1}{n}\sum_{j=1}^{n}(\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)^\top.$$

The unrestricted maximum has the same form with $\hat{\boldsymbol{\Sigma}} = \frac{1}{n}\sum(\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})^\top$, so the *likelihood ratio statistic* is

$$\Lambda = \frac{\max_{\boldsymbol{\Sigma}} L(\boldsymbol{\mu}_0, \boldsymbol{\Sigma})}{\max_{\boldsymbol{\mu},\boldsymbol{\Sigma}} L(\boldsymbol{\mu}, \boldsymbol{\Sigma})} = \Big(\frac{|\hat{\boldsymbol{\Sigma}}|}{|\hat{\boldsymbol{\Sigma}}_0|}\Big)^{n/2}, \qquad \Lambda^{2/n} = \frac{|\hat{\boldsymbol{\Sigma}}|}{|\hat{\boldsymbol{\Sigma}}_0|} = \frac{|\sum_j (\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})^\top|}{|\sum_j (\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)^\top|}.$$

$\Lambda^{2/n}$ is *Wilks' lambda*. Reject H_0 if $\Lambda < c_\alpha$, the lower 100αth percentile of its distribution; fortunately the distribution of $\Lambda^{2/n}$ is available.

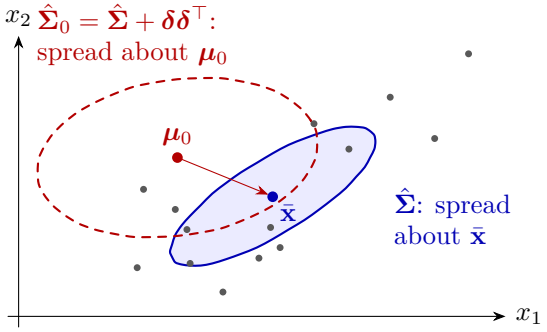


Figure. *Wilks' lambda $|\hat{\boldsymbol{\Sigma}}|/|\hat{\boldsymbol{\Sigma}}_0|$ is the squared ratio of the ellipse areas (here 0.16); the further $\boldsymbol{\mu}_0$ is from $\bar{\mathbf{x}}$, the smaller Λ.*

Theorem 5.6 (Result 5.1): likelihood ratio test and T^2

For a random sample from $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the T^2 test is equivalent to the likelihood ratio test of $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ versus $H_1 : \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$ because

$$\Lambda^{2/n} = \Big(1 + \frac{T^2}{n-1}\Big)^{-1}.$$

H_0 is rejected for small $\Lambda^{2/n}$, equivalently large T^2.

First proof: block determinant. Let $\mathbf{Q} = \sum_j (\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})^\top = (n-1)\mathbf{S}$, $\boldsymbol{\delta} = \bar{\mathbf{x}} - \boldsymbol{\mu}_0$, and consider the $(p+1) \times (p+1)$ matrix

$$\mathbf{A} = \begin{pmatrix} \mathbf{Q} & \sqrt{n}\,\boldsymbol{\delta} \\ \sqrt{n}\,\boldsymbol{\delta}^\top & -1 \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}.$$

From $|\mathbf{A}| = |\mathbf{A}_{22}||\mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}| = |\mathbf{A}_{11}||\mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12}|$,

$$(-1)|\mathbf{Q} + n\boldsymbol{\delta}\boldsymbol{\delta}^\top| = |\mathbf{Q}|(-1 - n\boldsymbol{\delta}^\top\mathbf{Q}^{-1}\boldsymbol{\delta}) = -|\mathbf{Q}|\Big(1 + \frac{T^2}{n-1}\Big).$$

Also $\sum_j (\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)^\top = \sum_j (\mathbf{x}_j - \bar{\mathbf{x}} + \bar{\mathbf{x}} - \boldsymbol{\mu}_0)(\cdot)^\top = \mathbf{Q} + n\boldsymbol{\delta}\boldsymbol{\delta}^\top$. Hence $|n\hat{\boldsymbol{\Sigma}}_0| = |n\hat{\boldsymbol{\Sigma}}|(1 + \frac{T^2}{n-1})$ and $\Lambda^{2/n} = |\hat{\boldsymbol{\Sigma}}|/|\hat{\boldsymbol{\Sigma}}_0| = (1 + \frac{T^2}{n-1})^{-1}$. ■

Second proof: determinant lemma. With $\boldsymbol{\delta} = \bar{\mathbf{x}} - \boldsymbol{\mu}_0$ and $\mathbf{S}_0 = \hat{\boldsymbol{\Sigma}}$, $\hat{\boldsymbol{\Sigma}}_0 = \mathbf{S}_0 + \boldsymbol{\delta}\boldsymbol{\delta}^\top$ and $|\mathbf{S}_0 + \boldsymbol{\delta}\boldsymbol{\delta}^\top| = |\mathbf{S}_0^{1/2}||\mathbf{I}_p + \mathbf{S}_0^{-1/2}\boldsymbol{\delta}\boldsymbol{\delta}^\top\mathbf{S}_0^{-1/2}||\mathbf{S}_0^{1/2}| = |\mathbf{S}_0|(1 + \boldsymbol{\delta}^\top\mathbf{S}_0^{-1}\boldsymbol{\delta})$ using $|\mathbf{I} + \mathbf{u}\mathbf{v}^\top| = 1 + \mathbf{v}^\top\mathbf{u}$. Since $\mathbf{S}_0 = \frac{n-1}{n}\mathbf{S}$, $\boldsymbol{\delta}^\top\mathbf{S}_0^{-1}\boldsymbol{\delta} = \frac{n}{n-1}\boldsymbol{\delta}^\top\mathbf{S}^{-1}\boldsymbol{\delta} = \frac{T^2}{n-1} =: \frac{T_0^2}{n-1}$. ■

Remark 5.7: approximation of $W = -2\log\Lambda$

$W = -2\log(L_0/L_1) = n\log\left(1 + \frac{T^2}{n-1}\right)$. Under H_0, $T^2 = O_p(1)$, so $W \approx \frac{n}{n-1}T^2 \approx T^2$, and for large n, $\frac{np}{n-p}F_{p,n-p} \approx \chi^2_p$: $W \dot{\sim} \chi^2_p$, in agreement with general likelihood ratio theory. Exact test: reject if $T_0^2 > \frac{(n-1)p}{n-p}F_{p,n-p,\alpha}$; approximate test: reject if $W > \chi^2_{p,\alpha}$.

3.2 $\boldsymbol{\Sigma}$ known

With $L_0 = L(\boldsymbol{\mu}_0, \boldsymbol{\Sigma})$ and $L_1 = L(\bar{\mathbf{x}}, \boldsymbol{\Sigma})$, $W = -2\log L_0 - (-2\log L_1) = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)$, which is exactly χ^2_p under H_0.

4 Confidence Regions

Motivation. Inverting the T^2 test gives a confidence ellipsoid for $\boldsymbol{\mu}$, whose axes are the eigenvectors of $\mathbf{S}$.

Theorem 5.8: $100(1-\alpha)\%$ confidence region for $\boldsymbol{\mu}$

$$R(\mathbf{X}) = \left\{\boldsymbol{\mu} \in \mathbb{R}^p : \frac{n}{n-1}\frac{n-p}{p}(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \mathbf{S}_0^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) \le F_{p,n-p,\alpha}\right\}$$
$$= \left\{\boldsymbol{\mu} : n(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) \le \frac{(n-1)p}{n-p}F_{p,n-p}(\alpha)\right\}.$$

[Supplement] With $\mathbf{S} = \sum \lambda_i \mathbf{e}_i \mathbf{e}_i^\top$ it is an ellipsoid centered at $\bar{\mathbf{x}}$ with axes along $\mathbf{e}_i$ and half-lengths $\sqrt{\lambda_i}\sqrt{\frac{(n-1)p}{n(n-p)}F_{p,n-p}(\alpha)}$. Checking whether $\boldsymbol{\mu}_0 \in R$ is the T^2 test. For $p \ge 3$ the ellipsoid cannot be drawn, so componentwise intervals are needed.

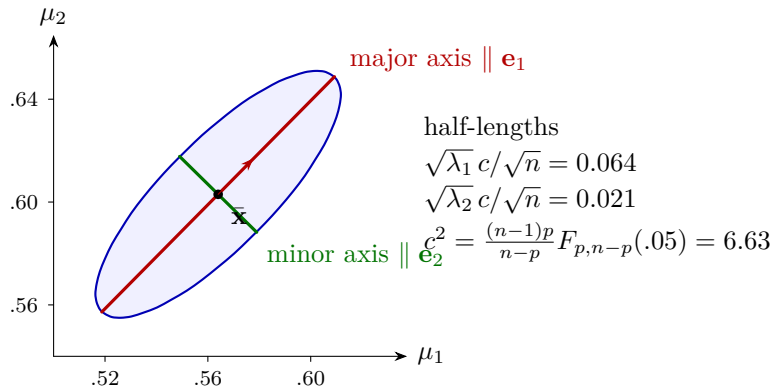


Figure. *[Supplement] 95% confidence ellipse for the microwave data (Example 5.10); axes along the eigenvectors of* $\mathbf{S}$.

5 Simultaneous Confidence Intervals

5.1 A single linear combination

For fixed $\mathbf{a}$, $\mathbf{a}^\top(\bar{\mathbf{X}}-\boldsymbol{\mu}) \sim N_1(0, \frac{\mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a}}{n})$ and $(n-1)\mathbf{a}^\top\mathbf{S}\mathbf{a} \sim W_1(n-1, \mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a}) \iff (n-1)\frac{\mathbf{a}^\top\mathbf{S}\mathbf{a}}{\mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a}} \sim \chi^2_{n-1}$, independent, so

$$t(\mathbf{a}) = \frac{\sqrt{n}\,\mathbf{a}^\top(\bar{\mathbf{X}}-\boldsymbol{\mu})/\sqrt{\mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a}}}{\sqrt{(n-1)\frac{\mathbf{a}^\top\mathbf{S}\mathbf{a}}{\mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a}}/(n-1)}} = \frac{\mathbf{a}^\top(\bar{\mathbf{X}}-\boldsymbol{\mu})}{\sqrt{\mathbf{a}^\top\mathbf{S}\mathbf{a}/n}} \sim t_{n-1}, \qquad C_\alpha = \Big\{\theta : \frac{|\mathbf{a}^\top\bar{\mathbf{x}}-\theta|}{\sqrt{\mathbf{a}^\top\mathbf{S}\mathbf{a}/n}} \le t_{n-1,\alpha/2}\Big\}.$$

Using these intervals for several $\mathbf{a}$ at once lowers the overall confidence below $1-\alpha$.

5.2 T^2 simultaneous intervals

Theorem 5.9 (Result 5.3): simultaneous confidence intervals for $\mathbf{a}^\top\boldsymbol{\mu}$

Let $\mathbf{X}_1,\dots,\mathbf{X}_n$ be a random sample from $N_p(\boldsymbol{\mu},\boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$. Then, *simultaneously for all* $\mathbf{a}$, the interval

$$\Big(\mathbf{a}^\top\bar{\mathbf{X}} - \sqrt{\frac{p(n-1)}{n(n-p)}F_{p,n-p}(\alpha)\mathbf{a}^\top\mathbf{S}\mathbf{a}},\ \ \mathbf{a}^\top\bar{\mathbf{X}} + \sqrt{\frac{p(n-1)}{n(n-p)}F_{p,n-p}(\alpha)\mathbf{a}^\top\mathbf{S}\mathbf{a}}\Big)$$

contains $\mathbf{a}^\top\boldsymbol{\mu}$ with probability $1-\alpha$.

Proof. Simple algebra gives, for every $\mathbf{a}$,

$$\mathbf{a}^\top\bar{\mathbf{x}} - c\sqrt{\frac{\mathbf{a}^\top\mathbf{S}\mathbf{a}}{n}} \le \mathbf{a}^\top\boldsymbol{\mu} \le \mathbf{a}^\top\bar{\mathbf{x}} + c\sqrt{\frac{\mathbf{a}^\top\mathbf{S}\mathbf{a}}{n}} \iff t(\mathbf{a})^2 = \frac{n(\mathbf{a}^\top\bar{\mathbf{x}}-\mathbf{a}^\top\boldsymbol{\mu})^2}{\mathbf{a}^\top\mathbf{S}\mathbf{a}} \le c^2.$$

The key is $\max_{\mathbf{a}} t(\mathbf{a})^2 = T^2$: with $\mathbf{b} = \mathbf{S}^{1/2}\mathbf{a}$ and $\mathbf{v} = \sqrt{n}\mathbf{S}^{-1/2}(\bar{\mathbf{x}}-\boldsymbol{\mu})$,

$$\max_{\mathbf{a}} t(\mathbf{a})^2 = \max_{\mathbf{b}} \frac{(\mathbf{b}^\top\mathbf{v})^2}{\mathbf{b}^\top\mathbf{b}} \le \mathbf{v}^\top\mathbf{v} = n(\bar{\mathbf{x}}-\boldsymbol{\mu})^\top\mathbf{S}^{-1}(\bar{\mathbf{x}}-\boldsymbol{\mu}) = T^2(n-1)$$

by Cauchy–Schwarz, with equality at $\mathbf{a} \propto \mathbf{S}^{-1}(\bar{\mathbf{x}}-\boldsymbol{\mu})$. By Theorem 5.2, choosing $c^2 = \frac{p(n-1)}{n-p}F_{p,n-p}(\alpha)$ gives $P(\bigcap_{\mathbf{a}}\{t(\mathbf{a})^2 \le c^2\}) = P(\max_{\mathbf{a}} t(\mathbf{a})^2 \le c^2) = P(T^2 \le c^2) = 1-\alpha$. ∎

- These are called T^2-intervals, since the coverage is determined by the distribution of T^2. With $\mathbf{a} = \mathbf{e}_j$: $\bar{x}_j \pm \sqrt{\frac{p(n-1)}{n-p}F_{p,n-p}(\alpha)}\sqrt{\frac{s_{jj}}{n}}$, holding simultaneously with confidence $1-\alpha$.
- With $\mathbf{a}^\top = [0,\dots,0,a_j,0,\dots,0,a_k,0,\dots]$, $a_j = 1$, $a_k = -1$: $\mathbf{a}^\top\mathbf{S}\mathbf{a} = s_{jj} - 2s_{jk} + s_{kk}$, giving intervals for $\mu_j - \mu_k$.
- The coefficient $1-\alpha$ holds for any choice of $\mathbf{a}$, so these intervals are ideal for "data snooping".
- The rectangle formed by the componentwise T^2 intervals is the shadow of the confidence ellipsoid on the coordinate axes.

Example 5.10 (Example 5.4): simultaneous intervals (microwave radiation data)

$n = 42$, $p = 2$, $\bar{\mathbf{x}} = (0.564, 0.603)^\top$, $s_{11} = 0.0144$, $s_{22} = 0.0146$ ($s_{12} = 0.0117$), $F_{2,40}(.05) =$

3.23. The 95% simultaneous T^2 intervals are

$$0.564 \pm \sqrt{\tfrac{2(41)}{40}3.23}\sqrt{\tfrac{0.0144}{42}} = (0.516, 0.612), \qquad 0.603 \pm \sqrt{\tfrac{2(41)}{40}3.23}\sqrt{\tfrac{0.0146}{42}} = (0.555, 0.651).$$

5.3 The Bonferroni method

When attention is restricted to a small number m of pre-specified linear combinations, shorter (more precise) simultaneous intervals are possible.

Theorem 5.11: Bonferroni inequality

Let C_i be a confidence statement about $\mathbf{a}_i^\top \boldsymbol{\mu}$ with $P(C_i \text{ true}) = 1 - \alpha_i$. Then

$$P[\text{all } C_i \text{ true}] = 1 - P[\text{at least one } C_i \text{ false}] \geq 1 - \sum_{i=1}^{m} P(C_i \text{ false}) = 1 - (\alpha_1 + \cdots + \alpha_m).$$

With $\alpha_i = \alpha/m$, the individual intervals $\bar{x}_i \pm t_{n-1}(\frac{\alpha}{2m})\sqrt{s_{ii}/n}$, $i = 1, \ldots, m$, hold simultaneously with confidence at least $1 - \alpha$. (Equivalently, with $C_i(\alpha) = \{\mu_i : |\bar{x}_i - \mu_i|/(s_i/\sqrt{n}) < t_{n-1,\alpha/2}\}$; setting $\mu_i \in C_i(\alpha/m)$ gives $P(\mu_i \in C_i(\alpha/m),\ i = 1, \ldots, m) \geq 1 - \sum_i P(\mu_i \notin C_i(\alpha/m)) = 1 - \alpha$.)

Remark 5.12: comparison of Bonferroni and T^2 interval lengths

Both have the form $\mathbf{a}^\top \bar{\mathbf{x}} \pm (\text{critical value})\sqrt{\mathbf{a}^\top \mathbf{S}\mathbf{a}/n}$, so with $\alpha_i = \alpha/m$

$$\frac{\text{length of Bonferroni interval}}{\text{length of } T^2 \text{ interval}} = \frac{t_{n-1}(\alpha/2m)}{\sqrt{\frac{p(n-1)}{n-p}F_{p,n-p}(\alpha)}},$$

which does not depend on $\bar{\mathbf{X}}$ or $\mathbf{S}$. For $1 - \alpha = .95$, $\alpha_i = .05/m$, $m = p$:

n	$m = 2$	$m = 4$	$m = 10$
15	.88	.69	.29
25	.90	.75	.48
50	.91	.78	.58
100	.91	.80	.62
∞	.91	.81	.66

For a small number of specified functions $\mathbf{a}^\top \boldsymbol{\mu}$ the Bonferroni intervals are shorter; being easy to apply, they are used often. In Example 5.10 the Bonferroni intervals are $(0.521, 0.607)$ and $(0.560, 0.646)$, inside the T^2 intervals.

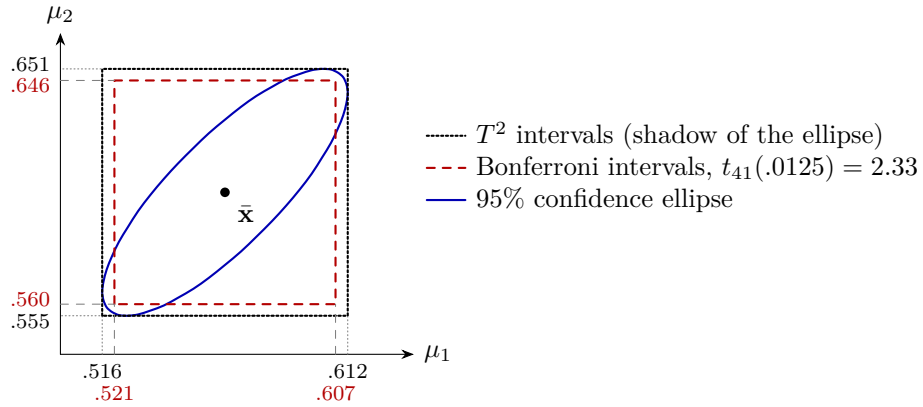


Figure. *T^2 intervals (shadow of the ellipse) and the slightly smaller Bonferroni rectangle, microwave data.*

6 Large-Sample Inference about a Mean Vector

Motivation. When normality is doubtful but n is large, T^2 is approximately χ^2_p, and the same procedures apply with χ^2 critical values.

Theorem 5.13 (Result 5.4): hypothesis testing for $\boldsymbol{\mu}$ with large $n-p$

Let $\mathbf{X}_1, \ldots, \mathbf{X}_n$ be a random sample from a population with mean $\boldsymbol{\mu}$ and positive definite covariance $\boldsymbol{\Sigma}$ (not necessarily normal). When $n-p$ is large, $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ is rejected in favor of $H_1 : \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$, at level approximately α, if

$$n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)^\top \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0) > \chi^2_p(\alpha).$$

The basis is $n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \xrightarrow{d} \chi^2_p$ (Theorem 4.12). The statistic is the same as in normal theory; only the critical values differ, since for n large relative to p, $\frac{(n-1)pF_{p,n-p}(\alpha)}{n-p} \approx \chi^2_p(\alpha)$ ([Supplement] $pF_{p,\nu} \xrightarrow{d} \chi^2_p$ as $\nu \to \infty$).

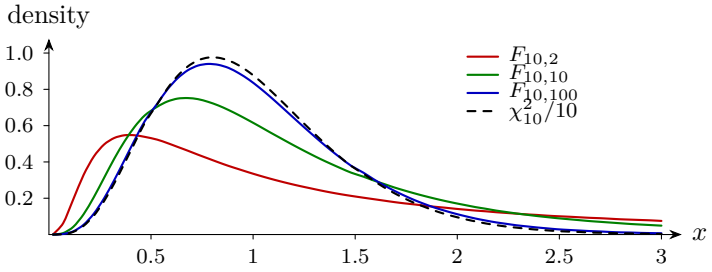


Figure. *$F_{10,\nu}$ approaches $\chi^2_{10}/10$ as ν grows.*

Theorem 5.14 (Result 5.5): large-sample simultaneous intervals

If $n-p$ is large, $\mathbf{a}^\top \bar{\mathbf{X}} \pm \sqrt{\chi^2_p(\alpha)}\sqrt{\mathbf{a}^\top \mathbf{S}\mathbf{a}/n}$ contains $\mathbf{a}^\top \boldsymbol{\mu}$ for every $\mathbf{a}$ with probability approximately $1-\alpha$. In particular $\bar{x}_j \pm \sqrt{\chi^2_p(\alpha)}\sqrt{s_{jj}/n}$ contains μ_j, $j = 1, \ldots, p$.

Proof. [Supplement] As in Theorem 5.9, coverage for every $\mathbf{a} \iff \max_{\mathbf{a}} t(\mathbf{a})^2 = n(\bar{\mathbf{X}}-\boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{X}}-\boldsymbol{\mu}) \leq \chi^2_p(\alpha)$, which has probability $\to 1-\alpha$ by Theorem 4.12. ■

Example (Golub data, two genes, 11 AML and 27 ALL): the 95% and 99% regions and intervals show clearly different group means.

Chapter 6. Comparing Mean Vectors and MANOVA

Paired comparisons, contrasts, two populations, one- and two-way MANOVA, simultaneous intervals

1 Paired Comparisons and Repeated Measures

1.1 Paired comparisons

[Supplement] When two treatments are applied to the same unit and p responses are measured, analyze the differences $\mathbf{D}_j = \mathbf{X}_{1j} - \mathbf{X}_{2j}$, $j = 1, \dots, n$; unit effects cancel, increasing precision.

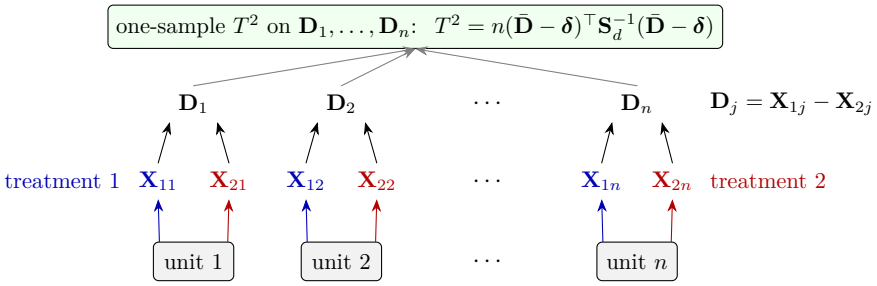


Figure. *[Supplement] Paired design: differences within a unit turn the comparison into a one-sample problem for $\boldsymbol{\delta} = \mathrm{E}\mathbf{D}_j$.*

Theorem 6.1 (Result 6.1): paired T^2

Let $\mathbf{D}_1, \dots, \mathbf{D}_n$ be a random sample from $N_p(\boldsymbol{\delta}, \boldsymbol{\Sigma}_d)$. Then

$$T^2 = n(\bar{\mathbf{D}} - \boldsymbol{\delta})^\top \mathbf{S}_d^{-1}(\bar{\mathbf{D}} - \boldsymbol{\delta})$$

is distributed as $\frac{(n-1)p}{n-p} F_{p,n-p}$, whatever the true $\boldsymbol{\delta}$ and $\boldsymbol{\Sigma}_d$. If n and $n-p$ are both large, T^2 is approximately χ^2_p regardless of the population of differences.

Proof. [Supplement] This is the one-sample T^2 (Theorem 5.3) applied to $\mathbf{D}_j$: $(n-1)\mathbf{S}_d \sim W_p(n-1, \boldsymbol{\Sigma}_d)$, independent of $\bar{\mathbf{D}}$ (the slide prints $(n-p)\mathbf{S}_d$; the degrees of freedom are $n-1$). Without normality use Theorem 4.12. ■

[Supplement] The test of $H_0 : \boldsymbol{\delta} = \mathbf{0}$, the confidence region, simultaneous intervals $\bar{d}_i \pm \sqrt{\frac{(n-1)p}{n-p} F_{p,n-p}(\alpha)} \sqrt{s^2_{d_i}/n}$ and Bonferroni intervals $\bar{d}_i \pm t_{n-1}(\alpha/2p)\sqrt{s^2_{d_i}/n}$ are as in Ch. 5.

1.2 Contrasts and repeated measures designs

[Supplement] With q repeated measurements $\mathbf{X}_j \in \mathbb{R}^q$ on each unit, "all treatment means are equal" is a hypothesis $\mathbf{C}\boldsymbol{\mu} = \mathbf{0}$, e.g. $\mathbf{C} = \begin{pmatrix} -1 & 1 & 0 & \cdots \\ -1 & 0 & 1 & \cdots \\ & & \ddots & \end{pmatrix}$ ($(q-1) \times q$, rows are contrasts).

Theorem 6.2: test for contrasts

If $\mathbf{X}_j \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\mathbf{C} \in \mathbb{R}^{q\times p}$ has rank q, then under $H_0 : \mathbf{C}\boldsymbol{\mu} = \mathbf{b}$

$$T_c^2 = n(\mathbf{C}\bar{\mathbf{x}} - \mathbf{b})^\top (\mathbf{C}\mathbf{S}\mathbf{C}^\top)^{-1}(\mathbf{C}\bar{\mathbf{x}} - \mathbf{b}) \sim \frac{(n-1)q}{n-q} F_{q,n-q} = \frac{q(n-1)}{(n-1)-(q-1)} F_{q,n-q}.$$

Proof. (i) $\sqrt{n}(\mathbf{C}\bar{\mathbf{X}} - \mathbf{C}\boldsymbol{\mu}) \sim N_q(\mathbf{0}, \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}^\top)$;

(ii) $(n-1)\mathbf{C}\mathbf{S}\mathbf{C}^\top \sim W_q(n-1, \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}^\top)$ (Theorem 4.2(i)), independent of $\mathbf{C}\bar{\mathbf{X}}$ since $\bar{\mathbf{X}} \perp\!\!\!\perp \mathbf{S}$. Apply Theorem 5.2 in dimension q with $m = n-1$. ■

[Supplement] By invariance (Theorem 5.4), T_c^2 does not depend on which contrast matrix with the same row space is used.

2 Comparing Mean Vectors from Two Populations

Motivation. Two-group comparisons generalize the two-sample t-test. With a common covariance matrix, the pooled $\mathbf{S}$ gives an exact T^2 test; with unequal covariances, large-sample approximations are used.

Remark 6.3: comparing two distributions versus two means

The general question is $H_0 : F_1 = F_2$ versus $H_1 : F_1 \neq F_2$ for the two population distributions. If both populations are normal with a common covariance, $F_\ell = N_p(\boldsymbol{\mu}_\ell, \boldsymbol{\Sigma})$, then $F_1 = F_2 \iff \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$, so the problem reduces to $H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$ versus $H_1 : \boldsymbol{\mu}_1 \neq \boldsymbol{\mu}_2$. Without these assumptions, equal means does not imply equal distributions.

Remark 6.4: the univariate case $p = 1$

With samples $X_1, \ldots, X_m$ and $Y_1, \ldots, Y_n$, the pooled two-sample t statistic is

$$T = \left(\frac{1}{m} + \frac{1}{n}\right)^{-1/2} \frac{\bar{X}_m - \bar{Y}_n}{S_p}, \qquad S_p^2 = \frac{\sum_{i=1}^m (X_i - \bar{X}_m)^2 + \sum_{i=1}^n (Y_i - \bar{Y}_n)^2}{m+n-2},$$

and $T \sim t_{m+n-2}$ under H_0. For $p \geq 1$ the pooled covariance $\mathbf{S}_p = (s_{ij})$ has entries

$$s_{ij} = \frac{\sum_{k=1}^m (X_{ki} - \bar{X}_{\cdot i})(X_{kj} - \bar{X}_{\cdot j}) + \sum_{k=1}^n (Y_{ki} - \bar{Y}_{\cdot i})(Y_{kj} - \bar{Y}_{\cdot j})}{m+n-2},$$

and the multivariate statistic is the squared norm generalizing T^2:

$$T^2 = \left(\frac{1}{m} + \frac{1}{n}\right)^{-1} \left\|\mathbf{S}_p^{-1/2}(\bar{\mathbf{X}}_m - \bar{\mathbf{Y}}_n)\right\|_2^2 \sim \frac{(m+n-2)p}{m+n-p-1} F_{p,m+n-p-1}.$$

For $p = 1$ this is exactly the square of the t statistic, since $t_\nu^2 = F_{1,\nu}$. The p-value is $P(T^2 > T^2_{\text{obs}} \mid H_0) = P(F_{p,m+n-p-1} \geq F_{\text{obs}})$ with $F_{\text{obs}} = \frac{m+n-p-1}{(m+n-2)p} T^2_{\text{obs}}$. This follows from the general form $T^2 = \mathbf{z}^\top (\mathbf{W}/\nu)^{-1}\mathbf{z} \sim \frac{p\nu}{\nu-(p-1)} F_{p,\nu-(p-1)}$ for $\mathbf{z} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$ independent of $\mathbf{W} \sim W_p(\nu, \boldsymbol{\Sigma})$ (Theorem 5.2 with $m = \nu$), here with $\nu = m+n-2$.

2.1 Common covariance: the pooled T^2

$\mathbf{X}_{11},\dots,\mathbf{X}_{1n_1} \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}_1,\boldsymbol{\Sigma})$ and an independent sample $\mathbf{X}_{21},\dots,\mathbf{X}_{2n_2} \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}_2,\boldsymbol{\Sigma})$. Pooled covariance:

$$\mathbf{S}_{\text{pooled}} = \frac{(n_1-1)\mathbf{S}_1 + (n_2-1)\mathbf{S}_2}{n_1+n_2-2}.$$

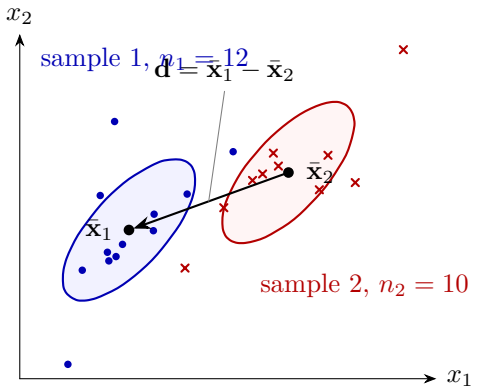


Figure. *Two samples with a common* $\boldsymbol{\Sigma}$*: the same contour of* $\mathbf{S}_{pooled}$ *about* $\bar{\mathbf{x}}_1$ *and* $\bar{\mathbf{x}}_2$ *(here* $T^2 = 57.1$*).*

Theorem 6.5 (Result 6.2): two-sample T^2

$$T^2 = (\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 - (\boldsymbol{\mu}_1-\boldsymbol{\mu}_2))^\top \Big[\Big(\tfrac{1}{n_1}+\tfrac{1}{n_2}\Big)\mathbf{S}_{\text{pooled}}\Big]^{-1}(\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 - (\boldsymbol{\mu}_1-\boldsymbol{\mu}_2))$$
$$\sim \frac{(n_1+n_2-2)p}{n_1+n_2-p-1}F_{p,n_1+n_2-p-1}.$$

Consequently $P(T^2 \le c_\alpha^2) = 1-\alpha$ with $c_\alpha^2 = \frac{(n_1+n_2-2)p}{n_1+n_2-p-1}F_{p,n_1+n_2-p-1}(\alpha)$.

Proof. (i) $\bar{\mathbf{X}}_1-\bar{\mathbf{X}}_2 \sim N_p(\boldsymbol{\mu}_1-\boldsymbol{\mu}_2, (\frac{1}{n_1}+\frac{1}{n_2})\boldsymbol{\Sigma})$, so $\mathbf{z} = (\frac{1}{n_1}+\frac{1}{n_2})^{-1/2}\{\bar{\mathbf{X}}_1-\bar{\mathbf{X}}_2-(\boldsymbol{\mu}_1-\boldsymbol{\mu}_2)\} \sim N_p(\mathbf{0},\boldsymbol{\Sigma})$. (ii) $(n_i-1)\mathbf{S}_i \sim W_p(n_i-1,\boldsymbol{\Sigma})$ independently, so $(n_1+n_2-2)\mathbf{S}_{\text{pooled}} \sim W_p(n_1+n_2-2,\boldsymbol{\Sigma})$ by Theorem 4.2(iii).

(iii) $\bar{\mathbf{X}}_i \perp\!\!\!\perp \mathbf{S}_i$ within each sample and the samples are independent, so $\mathbf{z} \perp\!\!\!\perp \mathbf{S}_{\text{pooled}}$. Hence $T^2 = \mathbf{z}^\top\mathbf{S}_{\text{pooled}}^{-1}\mathbf{z} = (\frac{1}{n_1}+\frac{1}{n_2})^{-1}\|\mathbf{S}_p^{-1/2}(\bar{\mathbf{x}}-\bar{\mathbf{y}}-(\boldsymbol{\mu}_1-\boldsymbol{\mu}_2))\|_2^2 \sim T_p^2(n_1+n_2-2)$, and Theorem 5.2 with $m = n_1+n_2-2$ gives the F form. ■

The p-value is $P(F_{p,n_1+n_2-p-1} \ge F_{\text{obs}})$ with $F_{\text{obs}} = \frac{n_1+n_2-p-1}{(n_1+n_2-2)p}T^2_{\text{obs}}$. For contrasts, $H_0 : \mathbf{C}\boldsymbol{\mu}_1 = \mathbf{C}\boldsymbol{\mu}_2$ ($\mathbf{C}$: $q\times p$):

$$\Big(\tfrac{1}{n_1}+\tfrac{1}{n_2}\Big)^{-1}\|(\mathbf{C}\mathbf{S}_p\mathbf{C}^\top)^{-1/2}\mathbf{C}(\bar{\mathbf{x}}-\bar{\mathbf{y}})\|_2^2 \sim \frac{q(n_1+n_2-2)}{n_1+n_2-q-1}F_{q,n_1+n_2-q-1}.$$

Theorem 6.6 (Result 6.3): simultaneous intervals with a pooled covariance

Let c_α^2 be as above. With probability $1-\alpha$,

$$\mathbf{a}^\top(\bar{\mathbf{X}}_1-\bar{\mathbf{X}}_2) \pm c_\alpha\sqrt{\mathbf{a}^\top\Big(\tfrac{1}{n_1}+\tfrac{1}{n_2}\Big)\mathbf{S}_{\text{pooled}}\mathbf{a}}$$

covers $\mathbf{a}^\top(\boldsymbol{\mu}_1-\boldsymbol{\mu}_2)$ for all $\mathbf{a}$. In particular $\mu_{1j}-\mu_{2j}$ is covered by $(\bar{X}_{1j}-\bar{X}_{2j}) \pm c_\alpha\sqrt{(\frac{1}{n_1}+\frac{1}{n_2})[\mathbf{S}_{\text{pooled}}]_{jj}}$, $j=1,\dots,p$.

The proof is as for Theorem 5.9, using $\max_{\mathbf{a}} t(\mathbf{a})^2 = T^2$.

2.2 Unequal covariances: large samples

Theorem 6.7 (Result 6.4): simultaneous intervals without a pooled covariance

If $n_1 - p$ and $n_2 - p$ are large, an approximate $100(1-\alpha)\%$ confidence ellipsoid for $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$ is

$$[\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)]^\top \left(\tfrac{1}{n_1}\mathbf{S}_1 + \tfrac{1}{n_2}\mathbf{S}_2\right)^{-1} [\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)] \le \chi^2_p(\alpha),$$

and $\mathbf{a}^\top(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$ belongs to $\mathbf{a}^\top(\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2) \pm \sqrt{\chi^2_p(\alpha)}\sqrt{\mathbf{a}^\top(\tfrac{1}{n_1}\mathbf{S}_1 + \tfrac{1}{n_2}\mathbf{S}_2)\mathbf{a}}$ simultaneously for all $\mathbf{a}$.

Sketch. (i) $\boldsymbol{\delta}_n := (\bar{\mathbf{x}} - \bar{\mathbf{y}}) - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) \sim N_p(\mathbf{0}, \tfrac{1}{n_1}\boldsymbol{\Sigma}_1 + \tfrac{1}{n_2}\boldsymbol{\Sigma}_2)$ (exactly under normality, approximately by the CLT), so $\boldsymbol{\delta}_n^\top(\tfrac{1}{n_1}\boldsymbol{\Sigma}_1 + \tfrac{1}{n_2}\boldsymbol{\Sigma}_2)^{-1}\boldsymbol{\delta}_n \sim \chi^2_p$.

(ii) $(n_1 - 1)\mathbf{S}_1 \sim W_p(n_1 - 1, \boldsymbol{\Sigma}_1)$, $(n_2 - 1)\mathbf{S}_2 \sim W_p(n_2 - 1, \boldsymbol{\Sigma}_2)$, and $\mathbf{S}_i \xrightarrow{p} \boldsymbol{\Sigma}_i$.

(iii) For $n_1 = n_2 = n$: $\mathbf{S}_1 + \mathbf{S}_2 \to \boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2$, so $(\mathbf{S}_1 + \mathbf{S}_2)^{-1} \to (\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)^{-1}$ and

$$T^2 = n\boldsymbol{\delta}_n^\top(\mathbf{S}_1 + \mathbf{S}_2)^{-1}\boldsymbol{\delta}_n = n\boldsymbol{\delta}_n^\top(\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)^{-1}\boldsymbol{\delta}_n + n\boldsymbol{\delta}_n^\top\left[(\mathbf{S}_1 + \mathbf{S}_2)^{-1} - (\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)^{-1}\right]\boldsymbol{\delta}_n \xrightarrow{d} \chi^2_p,$$

the second term being $O_p(1) \cdot o_p(1)$. ■

Proof for general n_1, n_2 [Supplement]. Let $n_1, n_2 \to \infty$ (no condition on n_1/n_2), $\boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2 \succ 0$, and

$$\mathbf{V}_n = \tfrac{1}{n_1}\boldsymbol{\Sigma}_1 + \tfrac{1}{n_2}\boldsymbol{\Sigma}_2, \qquad \hat{\mathbf{V}}_n = \tfrac{1}{n_1}\mathbf{S}_1 + \tfrac{1}{n_2}\mathbf{S}_2, \qquad T^2 = \boldsymbol{\delta}_n^\top \hat{\mathbf{V}}_n^{-1}\boldsymbol{\delta}_n.$$

Step 1. $\mathbf{Z}_n := \mathbf{V}_n^{-1/2}\boldsymbol{\delta}_n \xrightarrow{d} N_p(\mathbf{0}, \mathbf{I})$ by (i).

Step 2. $\mathbf{M}_n := \mathbf{V}_n^{-1/2}\hat{\mathbf{V}}_n\mathbf{V}_n^{-1/2} \xrightarrow{p} \mathbf{I}$. For a unit $\mathbf{x}$ put $\mathbf{y} = \mathbf{V}_n^{-1/2}\mathbf{x}$; since $\mathbf{V}_n \succeq \boldsymbol{\Sigma}_1/n_1$, $\mathbf{y}^\top\boldsymbol{\Sigma}_1\mathbf{y}/n_1 \le \mathbf{y}^\top\mathbf{V}_n\mathbf{y} = 1$, so

$$\tfrac{1}{n_1}|\mathbf{y}^\top(\mathbf{S}_1 - \boldsymbol{\Sigma}_1)\mathbf{y}| \le \|\boldsymbol{\Sigma}_1^{-1/2}\mathbf{S}_1\boldsymbol{\Sigma}_1^{-1/2} - \mathbf{I}\| \, \tfrac{\mathbf{y}^\top\boldsymbol{\Sigma}_1\mathbf{y}}{n_1} \le \|\boldsymbol{\Sigma}_1^{-1/2}\mathbf{S}_1\boldsymbol{\Sigma}_1^{-1/2} - \mathbf{I}\| \xrightarrow{p} 0$$

by (ii); likewise for sample 2. Hence $\|\mathbf{M}_n - \mathbf{I}\| \le \|\boldsymbol{\Sigma}_1^{-1/2}\mathbf{S}_1\boldsymbol{\Sigma}_1^{-1/2} - \mathbf{I}\| + \|\boldsymbol{\Sigma}_2^{-1/2}\mathbf{S}_2\boldsymbol{\Sigma}_2^{-1/2} - \mathbf{I}\| \xrightarrow{p} 0$.

Step 3. $T^2 = \mathbf{Z}_n^\top\mathbf{M}_n^{-1}\mathbf{Z}_n$ and $\mathbf{M}_n^{-1} \xrightarrow{p} \mathbf{I}$, so $T^2 \xrightarrow{d} \mathbf{Z}^\top\mathbf{Z} \sim \chi^2_p$ (Slutsky).

Step 4. By Cauchy–Schwarz, $\max_{\mathbf{a}}(\mathbf{a}^\top\boldsymbol{\delta}_n)^2/\mathbf{a}^\top\hat{\mathbf{V}}_n\mathbf{a} = T^2$, so all intervals hold together with probability $P(T^2 \le \chi^2_p(\alpha)) \to 1 - \alpha$. ■

3 One-Way MANOVA

3.1 Data and assumptions

Random samples from g populations: population ℓ: $\mathbf{X}_{\ell 1}, \mathbf{X}_{\ell 2}, \ldots, \mathbf{X}_{\ell n_\ell}$, $\ell = 1, \ldots, g$. MANOVA (multivariate analysis of variance) asks whether the mean vectors are equal and, if not, which components differ.

1. $\mathbf{X}_{\ell 1}, \ldots, \mathbf{X}_{\ell n_\ell}$ is a random sample of size n_ℓ from a population with mean $\boldsymbol{\mu}_\ell$; samples from different populations are **independent**.
2. All populations have a **common** covariance matrix $\boldsymbol{\Sigma}$.
3. Each population is **multivariate normal** (can be relaxed by the CLT, Theorem 4.11).

3.2 Review: univariate ANOVA

$X_{\ell j} = \mu + \tau_\ell + e_{\ell j}$, $e_{\ell j}$ independent $N(0, \sigma^2)$, with the constraint $\sum_\ell n_\ell \tau_\ell = 0$ and $H_0 : \tau_1 = \cdots = \tau_g = 0$ ($\mu_\ell = \mu + \tau_\ell$, $\tau_\ell = \mu_\ell - \mu$). The decomposition $x_{\ell j} = \bar{x} + (\bar{x}_\ell - \bar{x}) + (x_{\ell j} - \bar{x}_\ell)$ ($\bar{x}$ estimates μ, $\hat{\tau}_\ell = \bar{x}_\ell - \bar{x}$ estimates τ_ℓ, $x_{\ell j} - \bar{x}_\ell$ estimates $e_{\ell j}$) gives

$$\underbrace{\sum_\ell \sum_j (x_{\ell j} - \bar{x})^2}_{SS_{\text{cor}}\ \text{(total)}} = \underbrace{\sum_\ell n_\ell (\bar{x}_\ell - \bar{x})^2}_{SS_{\text{tr}}\ \text{(between)}} + \underbrace{\sum_\ell \sum_j (x_{\ell j} - \bar{x}_\ell)^2}_{SS_{\text{res}}\ \text{(within)}},$$

since the cross term is $\sum_\ell (\bar{x}_\ell - \bar{x}) \sum_j (x_{\ell j} - \bar{x}_\ell) = 0$. Equivalently $SS_{\text{obs}} = SS_{\text{mean}} + SS_{\text{tr}} + SS_{\text{res}}$. Reject H_0 if $F = \dfrac{SS_{\text{tr}}/(g-1)}{SS_{\text{res}}/(\sum n_\ell - g)} > F_{g-1, \sum n_\ell - g}(\alpha)$.

Source	SS	d.f.
Treatments	$SS_{\text{tr}} = \sum_\ell n_\ell (\bar{x}_\ell - \bar{x})^2$	$g - 1$
Residual	$SS_{\text{res}} = \sum_\ell \sum_j (x_{\ell j} - \bar{x}_\ell)^2$	$\sum n_\ell - g$
Total (corrected)	$SS_{\text{cor}} = \sum_\ell \sum_j (x_{\ell j} - \bar{x})^2$	$\sum n_\ell - 1$

3.3 The MANOVA model and the SSP decomposition

Definition 6.8: one-way MANOVA model

$\mathbf{X}_{\ell j} = \boldsymbol{\mu} + \boldsymbol{\tau}_\ell + \mathbf{e}_{\ell j}$, $j = 1, \ldots, n_\ell$, $\ell = 1, \ldots, g$, with $\mathbf{e}_{\ell j}$ independent $N_p(\mathbf{0}, \boldsymbol{\Sigma})$ and $\sum_\ell n_\ell \boldsymbol{\tau}_\ell = \mathbf{0}$. (Indeed, $\boldsymbol{\mu} = \mathrm{E}[\bar{\mathbf{X}}] \iff \sum_\ell \sum_j \mathrm{E}(\mathbf{X}_{\ell j}) = n\boldsymbol{\mu} \iff n\boldsymbol{\mu} + \sum n_\ell \boldsymbol{\tau}_\ell = n\boldsymbol{\mu} \iff \sum n_\ell \boldsymbol{\tau}_\ell = \mathbf{0}$.)

Theorem 6.9: SSP decomposition

From $\mathbf{x}_{\ell j} = \bar{\mathbf{x}} + (\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}}) + (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)$ (overall mean $\hat{\boldsymbol{\mu}}$ + estimated treatment effect $\hat{\boldsymbol{\tau}}_\ell$ + residual $\hat{\mathbf{e}}_{\ell j}$),

$$\underbrace{\sum_{\ell=1}^{g} \sum_{j=1}^{n_\ell} (\mathbf{x}_{\ell j} - \bar{\mathbf{x}})(\mathbf{x}_{\ell j} - \bar{\mathbf{x}})^\top}_{\mathbf{T}=\mathbf{B}+\mathbf{W}} = \underbrace{\sum_{\ell=1}^{g} n_\ell (\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}})(\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}})^\top}_{\mathbf{B}\text{: between}} + \underbrace{\sum_{\ell=1}^{g} \sum_{j=1}^{n_\ell} (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)(\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)^\top}_{\mathbf{W}\text{: within}}.$$

$\mathbf{W} = (n_1 - 1)\mathbf{S}_1 + \cdots + (n_g - 1)\mathbf{S}_g$ generalizes $(n_1 + n_2 - 2)\mathbf{S}_{\text{pooled}}$.

Proof. [Supplement] Expanding, the cross terms are $\sum_\ell (\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}}) \sum_j (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)^\top = \mathbf{0}$. ■

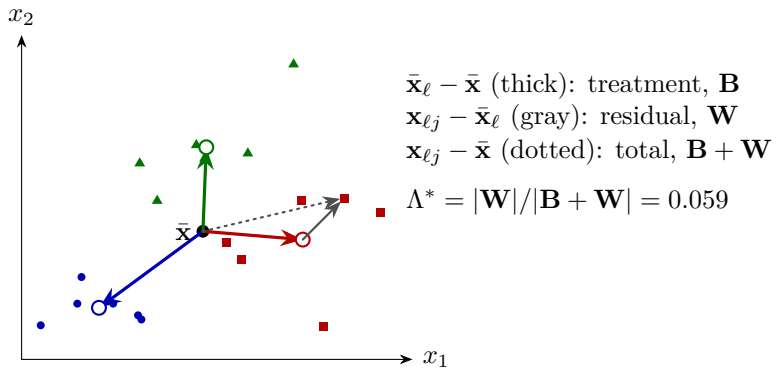


Figure. *One-way MANOVA, $g = 3$, $p = 2$: $\mathbf{x}_{\ell j} - \bar{\mathbf{x}} = (\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}}) + (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)$; groups far apart relative to their spread give a small Λ^*.*

Source	Matrix of SSP	d.f.
Treatment	$\mathbf{B} = \sum_\ell n_\ell(\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}})(\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}})^\top$	$g-1$
Residual (error)	$\mathbf{W} = \sum_\ell \sum_j (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)(\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)^\top$	$\sum n_\ell - g$
Total	$\mathbf{B}+\mathbf{W} = \sum_\ell \sum_j (\mathbf{x}_{\ell j} - \bar{\mathbf{x}})(\mathbf{x}_{\ell j} - \bar{\mathbf{x}})^\top$	$\sum n_\ell - 1$

Remark 6.10: $\mathbf{T} = \mathbf{B} + \mathbf{W}$ as a mixture

Observe $(\mathbf{X}, G)$ with $G \in \{1, \dots, g\}$, $P(G = i) = \pi_i$, $\mathbf{X} \mid G = i \sim N_p(\boldsymbol{\mu}_i, \boldsymbol{\Sigma})$, so $f(\mathbf{x}) = \sum_i f(\mathbf{x} \mid G = i) P(G = i) = \sum_i \phi(\mathbf{x}; \boldsymbol{\mu}_i, \boldsymbol{\Sigma})\pi_i$. Then $\mathrm{E}(\mathbf{X}) = \int \mathbf{x} \sum_i \phi(\mathbf{x}; \boldsymbol{\mu}_i, \boldsymbol{\Sigma})\pi_i \, d\mathbf{x} = \sum_i \pi_i \boldsymbol{\mu}_i = \boldsymbol{\mu}$ and

$$\mathrm{Var}(\mathbf{X}) = \sum_i \pi_i \mathrm{E}\big[(\mathbf{x} - \boldsymbol{\mu}_i)(\mathbf{x} - \boldsymbol{\mu}_i)^\top \mid G = i\big] + \sum_i \pi_i (\boldsymbol{\mu}_i - \boldsymbol{\mu})(\boldsymbol{\mu}_i - \boldsymbol{\mu})^\top$$

$$= \boldsymbol{\Sigma} + \sum_i \pi_i (\boldsymbol{\mu}_i - \boldsymbol{\mu})(\boldsymbol{\mu}_i - \boldsymbol{\mu})^\top.$$

The first term is the population version of $\mathbf{W}/n$, the second of $\mathbf{B}/n$.

3.4 Wilks' lambda

Definition 6.11: Wilks' test

Reject $H_0 : \boldsymbol{\tau}_1 = \cdots = \boldsymbol{\tau}_g = \mathbf{0}$ if the ratio of generalized variances

$$\Lambda^* = \frac{|\mathbf{W}|}{|\mathbf{B}+\mathbf{W}|} = \frac{\big|\sum_\ell \sum_j (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)(\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)^\top\big|}{\big|\sum_\ell \sum_j (\mathbf{x}_{\ell j} - \bar{\mathbf{x}})(\mathbf{x}_{\ell j} - \bar{\mathbf{x}})^\top\big|}$$

is **too small**. Proposed by Wilks, it corresponds to the univariate F test and is related to the likelihood ratio criterion.

Theorem 6.12: Wilks' lambda is the likelihood ratio statistic

With $n = \sum n_\ell$, $\Lambda^{2/n} = \Lambda^*$ and $-2\log\Lambda = n \log \dfrac{|\mathbf{B}+\mathbf{W}|}{|\mathbf{W}|} = n \log \dfrac{|\mathbf{T}|}{|\mathbf{W}|} \dot{\sim} \chi^2_{p(g-1)}$.

Proof. Under H_1: $\hat{\boldsymbol{\mu}}_\ell^{\mathrm{MLE}} = \bar{\mathbf{x}}_\ell$, $\hat{\boldsymbol{\Sigma}}^{\mathrm{MLE}} = \mathbf{S}_p = \dfrac{\sum_\ell (n_\ell - 1)\mathbf{S}_\ell}{n} = \dfrac{\mathbf{W}}{n}$. Under H_0: $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$, $\hat{\boldsymbol{\Sigma}} = \dfrac{\mathbf{T}}{n}$. The maximized likelihoods are $(2\pi)^{-np/2}|\hat{\boldsymbol{\Sigma}}|^{-n/2} e^{-np/2}$, so $\Lambda = (|\mathbf{W}|/|\mathbf{T}|)^{n/2}$ and $-2\log(L_0/L_1) = n \log \frac{|\mathbf{T}|}{|\mathbf{W}|}$. Reject H_0 if $n \log \frac{|\mathbf{T}|}{|\mathbf{W}|} > \chi^2_{p(g-1),\alpha}$. ■

Theorem 6.13: distribution of Wilks' lambda for multivariate normal data

No. of variables	No. of groups	Sampling distribution
$p = 1$	$g \geq 2$	$\left(\frac{\sum n_\ell - g}{g-1}\right)\left(\frac{1-\Lambda^*}{\Lambda^*}\right) \sim F_{g-1,\sum n_\ell - g}$
$p = 2$	$g \geq 2$	$\left(\frac{\sum n_\ell - g - 1}{g-1}\right)\left(\frac{1-\sqrt{\Lambda^*}}{\sqrt{\Lambda^*}}\right) \sim F_{2(g-1),2(\sum n_\ell - g - 1)}$
$p \geq 1$	$g = 2$	$\left(\frac{\sum n_\ell - p - 1}{p}\right)\left(\frac{1-\Lambda^*}{\Lambda^*}\right) \sim F_{p,\sum n_\ell - p - 1}$
$p \geq 1$	$g = 3$	$\left(\frac{\sum n_\ell - p - 2}{p}\right)\left(\frac{1-\sqrt{\Lambda^*}}{\sqrt{\Lambda^*}}\right) \sim F_{2p,2(\sum n_\ell - p - 2)}$

The proof is omitted.

Derivation: the $g = 2$ row — relation to the two-sample T^2

[Supplement] For $g = 2$, $\bar{\mathbf{x}} = (n_1\bar{\mathbf{x}}_1 + n_2\bar{\mathbf{x}}_2)/n$, so $\bar{\mathbf{x}}_1 - \bar{\mathbf{x}} = \frac{n_2}{n}\mathbf{d}$ and $\bar{\mathbf{x}}_2 - \bar{\mathbf{x}} = -\frac{n_1}{n}\mathbf{d}$ with $\mathbf{d} = \bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2$, and

$$\mathbf{B} = n_1\frac{n_2^2}{n^2}\mathbf{d}\mathbf{d}^\top + n_2\frac{n_1^2}{n^2}\mathbf{d}\mathbf{d}^\top = \frac{n_1 n_2}{n}\mathbf{d}\mathbf{d}^\top.$$

By the determinant lemma $|\mathbf{W}+\mathbf{B}| = |\mathbf{W}|(1+\frac{n_1 n_2}{n}\mathbf{d}^\top\mathbf{W}^{-1}\mathbf{d})$, and $\mathbf{W} = (n-2)\mathbf{S}_{\text{pooled}}$ gives $\frac{n_1 n_2}{n}\mathbf{d}^\top\mathbf{W}^{-1}\mathbf{d} = \frac{T^2}{n-2}$. Hence

$$\Lambda^* = \left(1 + \frac{T^2}{n-2}\right)^{-1}, \qquad \frac{1-\Lambda^*}{\Lambda^*} = \frac{T^2}{n-2},$$

and Theorem 6.5 ($\frac{n-p-1}{(n-2)p}T^2 \sim F_{p,n-p-1}$) gives the third row. Wilks' test with $g = 2$ is the two-sample T^2 test.

Theorem 6.14: Bartlett's test for Wilks' lambda

If H_0 is true and $\sum n_\ell = n$ is large,

$$-\left(n - 1 - \frac{p+g}{2}\right)\ln\Lambda^* = -\left(n - 1 - \frac{p+g}{2}\right)\ln\left(\frac{|\mathbf{W}|}{|\mathbf{B}+\mathbf{W}|}\right) \;\dot{\sim}\; \chi^2_{p(g-1)},$$

so reject H_0 at level α if the left side exceeds $\chi^2_{p(g-1)}(\alpha)$.

[Supplement] The degrees of freedom $p(g-1)$ are the difference in the number of mean parameters (g mean vectors versus one); the factor $n - 1 - (p+g)/2$ improves the chi-square approximation.

Remark 6.15: other test statistics

With the eigenvalues $\eta_1 \geq \cdots \geq \eta_s$ of $\mathbf{W}^{-1}\mathbf{B}$ ($s = \min(p, g-1)$),

$$\frac{|\mathbf{T}|}{|\mathbf{W}|} = |\mathbf{I}_p + \mathbf{W}^{-1/2}\mathbf{B}\mathbf{W}^{-1/2}| = \prod_i(1+\eta_i) = \frac{1}{\Lambda^*}, \qquad \text{Lawley–Hotelling } \operatorname{tr}(\mathbf{W}^{-1}\mathbf{B}) = \sum\eta_i,$$

$$\text{Pillai } \operatorname{tr}\{\mathbf{B}(\mathbf{B}+\mathbf{W})^{-1}\} = \sum\frac{\eta_i}{1+\eta_i}, \qquad \text{Roy } \eta_1 = \|\mathbf{W}^{-1}\mathbf{B}\|_2.$$

[Supplement] All summarize "$\mathbf{B}$ is large relative to $\mathbf{W}$". The η_i are also the eigenvalues of Fisher's discriminant analysis (Ch. 10).

Example 6.16 (Example 6.10): Wisconsin nursing home data

Costs per patient-day: X_1 nursing labour, X_2 dietary labour, X_3 plant operation and maintenance labour, X_4 housekeeping and laundry labour. Ownership: private ($n_1 = 271$, $\bar{\mathbf{x}}_1^\top = (2.066, .480, .082, .360)$), non-profit ($n_2 = 138$, $\bar{\mathbf{x}}_2^\top = (2.167, .596, .124, .418)$), government ($n_3 = 107$, $\bar{\mathbf{x}}_3^\top = (2.273, .521, .125, .383)$), $\sum n_\ell = 516$. The $\mathbf{S}_\ell$ look compatible and are pooled:

$$\mathbf{W} = \begin{pmatrix} 182.962 & & & \\ 4.408 & 8.200 & & \\ 1.695 & 0.633 & 1.484 & \\ 9.581 & 2.428 & 0.934 & 6.538 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 3.475 & & & \\ 1.111 & 1.225 & & \\ 0.821 & 0.453 & 0.235 & \\ 0.584 & 0.610 & 0.230 & 0.304 \end{pmatrix}$$

(symmetric; lower triangles shown), $\bar{\mathbf{x}} = (2.136, 0.519, 0.102, 0.380)^\top$. With $\Lambda^* = 0.7714$, $p = 4$, $g = 3$:

$$\Big(\frac{\sum n_\ell - p - 2}{p}\Big)\Big(\frac{1 - \sqrt{\Lambda^*}}{\sqrt{\Lambda^*}}\Big) = \Big(\frac{516 - 4 - 2}{4}\Big)\Big(\frac{1 - \sqrt{.7714}}{\sqrt{.7714}}\Big) = 17.67 > F_{8,1020}(.01) \approx 2.51.$$

Reject H_0 at the 1% level: average costs differ by type of ownership. ([Supplement] Bartlett: $-(516 - 1 - 3.5)\ln 0.7714 = 132.8 > \chi^2_8(.01) = 20.09$.)

4 Simultaneous Intervals for Treatment Effects

Let τ_{kj} be the jth component of $\boldsymbol{\tau}_k$. Since $\hat{\boldsymbol{\tau}}_k = \bar{\mathbf{X}}_k - \bar{\mathbf{X}}$, $\hat{\tau}_{kj} - \hat{\tau}_{\ell j} = \bar{X}_{kj} - \bar{X}_{\ell j}$ and a two-sample t-based interval is valid with an appropriately corrected α:

$$\mathrm{Var}(\hat{\tau}_{kj} - \hat{\tau}_{\ell j}) = \Big(\frac{1}{n_k} + \frac{1}{n_\ell}\Big)\sigma_{jj}, \qquad \widehat{\mathrm{Var}}(\bar{X}_{kj} - \bar{X}_{\ell j}) = \Big(\frac{1}{n_k} + \frac{1}{n_\ell}\Big)\frac{\mathbf{W}_{jj}}{n - g}.$$

Apportioning the error rate: for p variables and $g(g-1)/2$ pairwise differences there are $m = p\,g(g-1)/2$ statements, so each interval uses $t_{n-g}(\alpha/2m)$.

Theorem 6.17 (Result 6.5): simultaneous intervals for differences in treatment effects

Let $n = \sum_k n_k$. With confidence at least $1 - \alpha$,

$$\tau_{kj} - \tau_{\ell j} \;\text{ belongs to }\; \bar{x}_{kj} - \bar{x}_{\ell j} \pm t_{n-g}\Big(\frac{\alpha}{pg(g-1)}\Big)\sqrt{\frac{\mathbf{W}_{jj}}{n-g}\Big(\frac{1}{n_k} + \frac{1}{n_\ell}\Big)}$$

for all components $j = 1, \ldots, p$ and all differences $\ell < k = 1, \ldots, g$.

Proof. [Supplement] $W_{jj}/\sigma_{jj} \sim \chi^2_{n-g}$ independent of the group means, so each studentized difference is t_{n-g} and each interval has coverage $1 - \alpha/m$; Bonferroni (Theorem 5.11) over the m statements gives at least $1 - \alpha$. ■

Example 6.18 (Example 6.11): nursing home cost comparison

Compare plant operation and maintenance (X_3) between private and government homes, i.e. estimate $\tau_{13} - \tau_{33}$: $\hat{\boldsymbol{\tau}}_1 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}) = (-.070, -.039, -.020, -.020)^\top$, $\hat{\boldsymbol{\tau}}_3 = (\bar{\mathbf{x}}_3 - \bar{\mathbf{x}}) = (.137, .002, .023, .003)^\top$, so $\hat{\tau}_{13} - \hat{\tau}_{33} = -.020 - .023 = -.043$. With $n = 271 + 138 + 107 = 516$: $\sqrt{(\frac{1}{271} + \frac{1}{107})\frac{1.484}{516-3}} = 0.00614$; for $p = 4$, $g = 3$ the 95% interval uses $t_{513}(.05/4(3)2) =$

$t_{513}(.00208) \approx 2.87$. Hence $\tau_{13}-\tau_{33} \in -0.043 \pm 2.87 \times 0.00614 = (-0.061, -0.025)$: government homes have higher plant operation costs.

5 Two-Way MANOVA

5.1 Review: univariate two-way ANOVA

Factor 1 with g levels, factor 2 with b levels, n independent observations at each of the gb combinations; $X_{\ell kr}$ is the rth observation at level ℓ of factor 1 and level k of factor 2:

	$k=1$	$k=2$	$\cdots$	$k=b$	
$\ell=1$	$\{\mathbf{x}_{\ell kr}\}_{r=1}^n$	$\{\mathbf{x}_{\ell kr}\}_{r=1}^n$	$\cdots$	$\{\mathbf{x}_{\ell kr}\}_{r=1}^n$	$\bar{\mathbf{x}}_{\ell\cdot}$
$\ell=2$	$\{\mathbf{x}_{\ell kr}\}_{r=1}^n$	$\{\mathbf{x}_{\ell kr}\}_{r=1}^n$	$\cdots$	$\{\mathbf{x}_{\ell kr}\}_{r=1}^n$	$\bar{\mathbf{x}}_{\ell\cdot}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	
$\ell=g$	$\{\mathbf{x}_{\ell kr}\}_{r=1}^n$	$\{\mathbf{x}_{\ell kr}\}_{r=1}^n$	$\cdots$	$\{\mathbf{x}_{\ell kr}\}_{r=1}^n$	$\bar{\mathbf{x}}_{\ell\cdot}$
	$\bar{\mathbf{x}}_{\cdot k}$	$\bar{\mathbf{x}}_{\cdot k}$		$\bar{\mathbf{x}}_{\cdot k}$	$\bar{\mathbf{x}}$

factor 1 margin (mean of bn obs.)

factor 2 margin (mean of gn obs.)

Figure. *Two-way layout: $g \times b$ cells with n replicates each; cell, row, column and grand means.*

$$X_{\ell kr} = \underbrace{\mu}_{\text{overall mean}} + \underbrace{\tau_\ell}_{\text{factor 1 effect}} + \underbrace{\beta_k}_{\text{factor 2 effect}} + \underbrace{\gamma_{\ell k}}_{\text{interaction}} + \underbrace{e_{\ell kr}}_{\text{normal error}},$$

$\sum_\ell \tau_\ell = \sum_k \beta_k = \sum_\ell \gamma_{\ell k} = \sum_k \gamma_{\ell k} = 0$, and $SS_{\text{cor}} = SS_{\text{fac1}} + SS_{\text{fac2}} + SS_{\text{int}} + SS_{\text{res}}$:

Source	Sum of squares	d.f.
Factor 1	$SS_{\text{fac1}} = \sum_\ell bn(\bar{x}_{\ell\cdot} - \bar{x})^2$	$g-1$
Factor 2	$SS_{\text{fac2}} = \sum_k gn(\bar{x}_{\cdot k} - \bar{x})^2$	$b-1$
Interaction	$SS_{\text{int}} = \sum_\ell \sum_k n(\bar{x}_{\ell k} - \bar{x}_{\ell\cdot} - \bar{x}_{\cdot k} + \bar{x})^2$	$(g-1)(b-1)$
Residual	$SS_{\text{res}} = \sum_\ell \sum_k \sum_r (x_{\ell kr} - \bar{x}_{\ell k})^2$	$gb(n-1)$
Total	$SS_{\text{cor}} = \sum\sum\sum (x_{\ell kr} - \bar{x})^2$	$gbn-1$

F-ratios: $F_{\text{fac1}} = \frac{SS_{\text{fac1}}/(g-1)}{SS_{\text{res}}/(gb(n-1))}$, $F_{\text{fac2}} = \frac{SS_{\text{fac2}}/(b-1)}{SS_{\text{res}}/(gb(n-1))}$, $F_{\text{int}} = \frac{SS_{\text{int}}/((g-1)(b-1))}{SS_{\text{res}}/(gb(n-1))}$.

5.2 Multivariate two-way fixed-effects model with interaction

Definition 6.19: two-way MANOVA

$\mathbf{X}_{\ell kr} = \boldsymbol{\mu} + \boldsymbol{\tau}_\ell + \boldsymbol{\beta}_k + \boldsymbol{\gamma}_{\ell k} + \mathbf{e}_{\ell kr}$ with $\mathbf{e}_{\ell kr}$ independent $N_p(\mathbf{0}, \boldsymbol{\Sigma})$ and $\sum_\ell \boldsymbol{\tau}_\ell = \sum_k \boldsymbol{\beta}_k = \sum_\ell \boldsymbol{\gamma}_{\ell k} = \sum_k \boldsymbol{\gamma}_{\ell k} = \mathbf{0}$; all vectors are $p \times 1$.

The SSP decomposition $\text{SSP}_{\text{cor}} = \text{SSP}_{\text{fac1}} + \text{SSP}_{\text{fac2}} + \text{SSP}_{\text{int}} + \text{SSP}_{\text{res}}$ replaces each $(\cdot)^2$ by $(\cdot)(\cdot)^\top$, e.g. $\text{SSP}_{\text{int}} = \sum_\ell \sum_k n(\bar{\mathbf{x}}_{\ell k} - \bar{\mathbf{x}}_{\ell\cdot} - \bar{\mathbf{x}}_{\cdot k} + \bar{\mathbf{x}})(\bar{\mathbf{x}}_{\ell k} - \bar{\mathbf{x}}_{\ell\cdot} - \bar{\mathbf{x}}_{\cdot k} + \bar{\mathbf{x}})^\top$, with the same degrees of freedom.

Theorem 6.20: two-way MANOVA tests (Bartlett-corrected)

Interaction, $H_0 : \boldsymbol{\gamma}_{11} = \boldsymbol{\gamma}_{12} = \cdots = \boldsymbol{\gamma}_{gb} = \mathbf{0}$: with $\Lambda^* = \dfrac{|\mathrm{SSP}_{\mathrm{res}}|}{|\mathrm{SSP}_{\mathrm{int}} + \mathrm{SSP}_{\mathrm{res}}|}$, reject if

$$-\Big[gb(n-1) - \frac{p+1-(g-1)(b-1)}{2}\Big] \ln \Lambda^* > \chi^2_{(g-1)(b-1)p}(\alpha).$$

Main effects of factor 1, $H_0 : \boldsymbol{\tau}_1 = \cdots = \boldsymbol{\tau}_g = \mathbf{0}$: with $\Lambda^* = \dfrac{|\mathrm{SSP}_{\mathrm{res}}|}{|\mathrm{SSP}_{\mathrm{fac1}} + \mathrm{SSP}_{\mathrm{res}}|}$, reject if

$$-\Big[gb(n-1) - \frac{p+1-(g-1)}{2}\Big] \ln \Lambda^* > \chi^2_{(g-1)p}(\alpha).$$

[Supplement] Factor 2 is the same with $g-1$ replaced by $b-1$. Test the interaction first; if present, interpret main effects with care.

Theorem 6.21: simultaneous intervals in two-way MANOVA

With $\nu = gb(n-1)$ and E_{jj} the jth diagonal element of $\mathbf{E} = \mathrm{SSP}_{\mathrm{res}}$,

$$\tau_{\ell j} - \tau_{mj} \quad \text{belongs to} \quad (\bar{x}_{\ell\cdot j} - \bar{x}_{m\cdot j}) \pm t_\nu\Big(\frac{\alpha}{pg(g-1)}\Big)\sqrt{\frac{E_{jj}}{\nu}\cdot\frac{2}{bn}}.$$

Proof. [Supplement] $\bar{X}_{\ell\cdot j} - \bar{X}_{m\cdot j} \sim N(\tau_{\ell j} - \tau_{mj}, \frac{2}{bn}\sigma_{jj})$ and $E_{jj}/\sigma_{jj} \sim \chi^2_\nu$ independently; Bonferroni over the $m = p\,g(g-1)/2$ statements. ■

Chapter 7. Principal Component Analysis

Population and sample principal components, SVD, choosing the number of components

Ch. 1, §6, introduced PCA through sample geometry. Here it is formulated at the population level, with the Lagrange-multiplier derivation, the SVD and the choice of the number of components.

1 Population Principal Components

Motivation. With many correlated variables, most of the variation often lies in a few directions. Principal components are the orthogonal directions of maximal variance, the eigenvectors of $\boldsymbol{\Sigma}$, and they allow us to summarize the data in fewer dimensions.

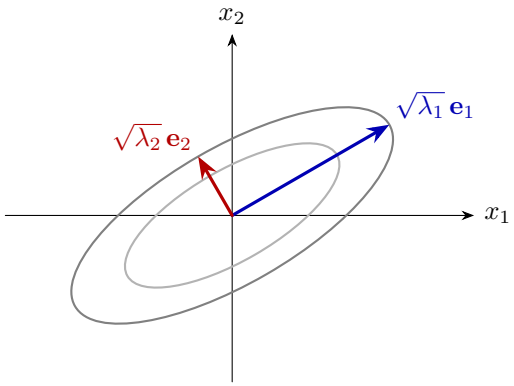


Figure. *Contours of a bivariate normal density are ellipses whose axes are the eigenvectors $\mathbf{e}_1, \mathbf{e}_2$ of $\boldsymbol{\Sigma}$, with half-lengths proportional to $\sqrt{\lambda_1}, \sqrt{\lambda_2}$. The first principal component is the projection on the long axis, the direction of maximal variance.*

Let $\mathbf{X} \in \mathbb{R}^p$ with $\mathrm{Cov}(\mathbf{X}) = \boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$, $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_p]$, $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \dots, \lambda_p)$, $\lambda_1 \geq \cdots \geq \lambda_p \geq 0$. A linear combination $Z(\mathbf{a}) = \mathbf{a}^\top \mathbf{X}$ has variance $\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$. (For $d = 1$: $\mathbf{a}_1 = \arg\max_{\|\mathbf{a}\|=1} \mathrm{Var}(Z(\mathbf{a}))$.)

Definition 7.1: principal components

First PC direction vector: $\mathbf{u}_1 = \arg\max_{\|\mathbf{u}\|=1} \mathrm{Var}(Z(\mathbf{u}))$, with $\lambda_1 = \mathrm{Var}(Z_1)$. Second: $\mathbf{u}_2 = \arg\max_{\|\mathbf{u}\|=1,\ \mathbf{u}^\top \mathbf{u}_1 = 0} \mathrm{Var}(\mathbf{u}^\top \mathbf{X})$, with $\lambda_2 = \mathrm{Var}(Z_2)$; and so on:

$$\mathbf{u}_k = \arg \max_{\|\mathbf{u}\|=1,\ \mathbf{u}^\top \mathbf{u}_j = 0\ (j<k)} \mathrm{Var}(\mathbf{u}^\top \mathbf{X}), \qquad Z_k = \mathbf{u}_k^\top \mathbf{X}.$$

Then $\lambda_1 \geq \lambda_2 \geq \cdots$ and $\mathrm{Cov}(Z_j, Z_k) = 0$ for $j \neq k$.

Theorem 7.2: solution

$\mathbf{u}_k$ is the kth eigenvector of $\boldsymbol{\Sigma}$, $\mathrm{Var}(Z_k) = \lambda_k$, and $\mathrm{Cov}(Z_j, Z_k) = 0$ for $j \neq k$.

Lagrange multipliers. **1.** $\Phi(\mathbf{u}, \lambda) = \mathrm{Var}(\mathbf{u}^\top \mathbf{X}) - \lambda(\mathbf{u}^\top \mathbf{u} - 1) = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{u} - 1)$. $\frac{1}{2} \frac{\partial}{\partial \mathbf{u}} \Phi(\mathbf{u}, \lambda) = \boldsymbol{\Sigma}\mathbf{u} - \lambda \mathbf{u} = \mathbf{0}$, so $\boldsymbol{\Sigma}\mathbf{u} = \lambda \mathbf{u}$ and $\lambda = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u}$: the objective equals the eigenvalue, so take the largest, λ_1.

2. With $\boldsymbol{\gamma}^k = (\gamma_1, \dots, \gamma_{k-1})$,

$$\Phi(\mathbf{u}, \lambda, \boldsymbol{\gamma}^k) = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{u} - 1) - \sum_{j=1}^{k-1} 2\gamma_j \mathbf{u}_j^\top \mathbf{u},$$

$$\frac{1}{2}\frac{\partial \Phi}{\partial \mathbf{u}} = \boldsymbol{\Sigma}\mathbf{u} - \lambda\mathbf{u} - \sum_{j=1}^{k-1}\gamma_j\mathbf{u}_j = \mathbf{0}, \quad \frac{\partial \Phi}{\partial \gamma_j} = \mathbf{u}_j^\top \mathbf{u} = 0.$$

Premultiplying by $\mathbf{u}_i^\top$ $(i<k)$: $\mathbf{u}_i^\top\boldsymbol{\Sigma}\mathbf{u} = \gamma_i$, and $\mathbf{u}_i^\top\boldsymbol{\Sigma}\mathbf{u} = \lambda_i\mathbf{u}_i^\top\mathbf{u} = 0$, so $\gamma_i = 0$. Hence $\boldsymbol{\Sigma}\mathbf{u} = \lambda\mathbf{u}$ again, and among eigenvectors orthogonal to $\mathbf{u}_1, \ldots, \mathbf{u}_{k-1}$ the largest eigenvalue λ_k is chosen: $\mathbf{u}_k, \lambda_k$. Finally $\mathrm{Cov}(Z_j, Z_k) = \mathbf{u}_j^\top\boldsymbol{\Sigma}\mathbf{u}_k = \lambda_k\mathbf{u}_j^\top\mathbf{u}_k = 0$. ■

Theorem 7.3: total variance and correlations

1. $\sum_{j=1}^p \mathrm{Var}(X_j) = \mathrm{tr}\boldsymbol{\Sigma} = \sum_{k=1}^p \lambda_k = \sum_k \mathrm{Var}(Z_k)$; the proportion of variance explained by PC k is $\lambda_k / \sum_j \lambda_j$.
2. $\mathrm{Corr}(Z_k, X_j) = \dfrac{u_{jk}\sqrt{\lambda_k}}{\sqrt{\sigma_{jj}}}$, where u_{jk} is the jth entry of $\mathbf{u}_k$.

Proof. (1) $\mathrm{tr}(\mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top) = \mathrm{tr}(\boldsymbol{\Lambda}\mathbf{U}^\top\mathbf{U}) = \mathrm{tr}\boldsymbol{\Lambda}$, and $\mathrm{Var}(Z_k) = \lambda_k$ by Theorem 7.2.
(2) $\mathrm{Cov}(Z_k, X_j) = \mathbf{u}_k^\top\boldsymbol{\Sigma}\mathbf{e}_j = \lambda_k\mathbf{u}_k^\top\mathbf{e}_j = \lambda_k u_{jk}$, so $\mathrm{Corr} = \lambda_k u_{jk}/(\sqrt{\lambda_k}\sqrt{\sigma_{jj}})$. ■

Theorem 7.4: equicorrelation structure

If $\boldsymbol{\Sigma} = \sigma^2\{(1-\rho)\mathbf{I}_p + \rho\mathbf{1}_p\mathbf{1}_p^\top\}$ with $0 < \rho < 1$, then $\lambda_1 = \sigma^2\{1+(p-1)\rho\}$ with $\mathbf{u}_1 = \mathbf{1}_p/\sqrt{p}$, and $\lambda_2 = \cdots = \lambda_p = \sigma^2(1-\rho)$ with any orthonormal basis of $\mathbf{1}_p^\perp$ as $\mathbf{u}_2, \ldots, \mathbf{u}_p$. The first PC $Z_1 = \frac{1}{\sqrt{p}}\sum_j X_j$ is a scaled total and explains the proportion $\rho + (1-\rho)/p$ of the total variance.

Proof. $\boldsymbol{\Sigma}\mathbf{1}_p = \sigma^2\{(1-\rho)+p\rho\}\mathbf{1}_p$, and for $\mathbf{v} \perp \mathbf{1}_p$, $\boldsymbol{\Sigma}\mathbf{v} = \sigma^2(1-\rho)\mathbf{v}$ because $\mathbf{1}_p^\top\mathbf{v} = 0$. Since $\mathbb{R}^p = \mathrm{span}(\mathbf{1}_p) \oplus \mathbf{1}_p^\perp$, these are all the eigenpairs, and $1+(p-1)\rho > 1-\rho$ for $\rho > 0$. Finally $\lambda_1/\mathrm{tr}\boldsymbol{\Sigma} = \{1+(p-1)\rho\}/p = \rho + (1-\rho)/p$. ■

With ρ close to 1, a single component summarizes all p variables; with $\rho = 0$ every direction is a principal component and PCA reduces nothing.

Theorem 7.5: Ky Fan maximum principle — q-dimensional subspaces

[Supplement]

$$\max_{\mathbf{V}\in\mathbb{R}^{p\times q},\ \mathbf{V}^\top\mathbf{V}=\mathbf{I}_q} \mathrm{tr}(\mathbf{V}^\top\boldsymbol{\Sigma}\mathbf{V}) = \lambda_1 + \cdots + \lambda_q,$$

attained at $\mathbf{V} = [\mathbf{u}_1, \ldots, \mathbf{u}_q]$ (and its rotations).

Proof [Supplement]. $\mathbf{P} = \mathbf{V}\mathbf{V}^\top$ is a rank-q projection with eigenvalues 1 (q times) and 0. By von Neumann's upper bound, $\mathrm{tr}(\mathbf{V}^\top\boldsymbol{\Sigma}\mathbf{V}) = \mathrm{tr}(\boldsymbol{\Sigma}\mathbf{P}) \le \sum_i \lambda_i(\boldsymbol{\Sigma})\lambda_i(\mathbf{P}) = \lambda_1 + \cdots + \lambda_q$. ■

Remark 7.6: PCA as optimal approximation

[Supplement] Since $\mathbf{I} - \mathbf{V}\mathbf{V}^\top$ is symmetric idempotent, $\mathrm{E}\|(\mathbf{X}-\boldsymbol{\mu}) - \mathbf{V}\mathbf{V}^\top(\mathbf{X}-\boldsymbol{\mu})\|^2 = \mathrm{tr}\{(\mathbf{I}-\mathbf{V}\mathbf{V}^\top)\boldsymbol{\Sigma}\} = \mathrm{tr}\boldsymbol{\Sigma} - \mathrm{tr}(\mathbf{V}^\top\boldsymbol{\Sigma}\mathbf{V})$, so among q-dimensional subspaces through the mean, the first q PCs minimize the expected squared reconstruction error, the minimum being $\lambda_{q+1} + \cdots + \lambda_p$.

2 Sample Principal Components and the SVD

2.1 Sample PCs

Let $\tilde{\mathbf{X}} = \mathbf{H}_n\mathbf{X} = \mathbf{X} - \mathbf{1}_n\bar{\mathbf{x}}^\top$ be the centered data matrix, $\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \hat{\mathbf{U}}\hat{\mathbf{\Lambda}}\hat{\mathbf{U}}^\top$, and

$$\hat{\mathbf{u}}_k = \arg\max_{\|\mathbf{u}\|=1,\ \mathbf{u}^\top\hat{\mathbf{u}}_j=0,\ j=1,\dots,k-1} \mathbf{u}^\top\mathbf{S}\mathbf{u}, \qquad \text{scores } t_{ik} = \hat{\mathbf{u}}_k^\top(\mathbf{x}_i - \bar{\mathbf{x}}).$$

The sample PC scores have sample variance $\hat{\lambda}_k$ and are mutually uncorrelated.

Proof [Supplement]. The score vectors are the columns of $\tilde{\mathbf{X}}\hat{\mathbf{U}}$, which are centered because $\mathbf{1}_n^\top\tilde{\mathbf{X}} = \mathbf{0}^\top$. Their sample covariance matrix is $\frac{1}{n-1}\hat{\mathbf{U}}^\top\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}\hat{\mathbf{U}} = \hat{\mathbf{U}}^\top\mathbf{S}\hat{\mathbf{U}} = \hat{\mathbf{\Lambda}}$, which is diagonal with entries $\hat{\lambda}_k$. ■

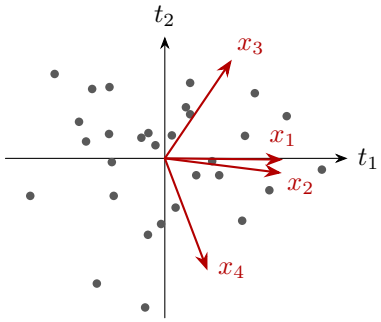


Figure. *Biplot of synthetic standardized data ($n = 30$, $p = 4$; $\hat{\lambda}_1 = 2.29$, $\hat{\lambda}_2 = 1.52$, together 95% of $\mathrm{tr}\mathbf{R} = 4$). Dots are the scores (t_{i1}, t_{i2}); arrow j has coordinates $(\mathrm{Corr}(Z_1, X_j), \mathrm{Corr}(Z_2, X_j))$, scaled by 2.6. x_1, x_2 load only on the first PC; x_3 and x_4 point to opposite sides of the second, which is a contrast between them.*

2.2 Relation to the singular value decomposition

Write the SVD of the $p \times n$ matrix $\tilde{\mathbf{X}}^\top$ $(n > p)$ as

$$\tilde{\mathbf{X}}^\top = \mathbf{U}\mathbf{D}\mathbf{V}^\top = \underbrace{[\mathbf{u}_1 \cdots \mathbf{u}_p]}_{p\times p} \underbrace{\begin{pmatrix} d_1 & & \\ & \ddots & \\ & & d_p \end{pmatrix}}_{p\times p} \underbrace{\begin{pmatrix} \mathbf{v}_1^\top \\ \vdots \\ \mathbf{v}_p^\top \end{pmatrix}}_{p\times n}, \qquad \mathbf{U}^\top\mathbf{U} = \mathbf{I}_p,\ \mathbf{V}^\top\mathbf{V} = \mathbf{I}_p,\ d_1 \geq \cdots \geq d_p \geq 0.$$

Then

$$\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \mathbf{U}\mathrm{diag}\Big(\frac{d_1^2}{n-1}, \dots, \frac{d_p^2}{n-1}\Big)\mathbf{U}^\top, \qquad \begin{pmatrix} \mathbf{z}_{(1)}^\top \\ \vdots \\ \mathbf{z}_{(p)}^\top \end{pmatrix} = \mathbf{Z} = \mathbf{U}^\top\tilde{\mathbf{X}}^\top = \mathbf{D}\mathbf{V}^\top = \begin{pmatrix} d_1\mathbf{v}_1^\top \\ \vdots \\ d_p\mathbf{v}_p^\top \end{pmatrix}.$$

The PC directions are the left singular vectors, the eigenvalues are $d_k^2/(n-1)$, and the kth PC score vector is $\mathbf{z}_{(k)} = \hat{\mathbf{u}}_k^\top\tilde{\mathbf{X}}^\top = d_k\mathbf{v}_k^\top$. Numerically the SVD is preferred to forming $\mathbf{S}$ (computation of PCA).

3 PCA and the Best Approximating Affine Subspace

Motivation. A second view of PCA: the first r components span the r-dimensional affine subspace closest to the data in squared distance. Maximizing variance and minimizing reconstruction error are the same problem.

Theorem 7.7: claim

PCA with r components is equivalent to finding the best r-dimensional affine subspace (hyperplane) approximating the data.

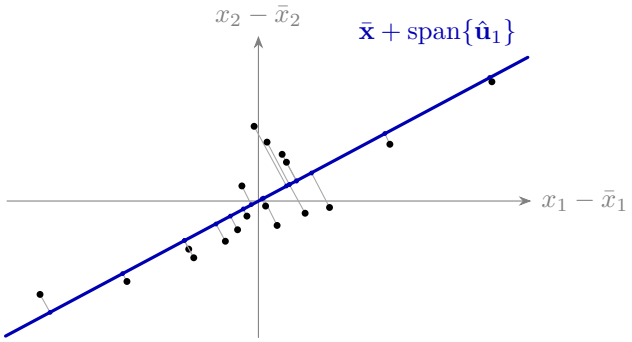


Figure. *Synthetic centered data ($n = 18$, $p = 2$, $r = 1$). Each point $\mathbf{x}_i$ (black) is replaced by its orthogonal projection $\bar{\mathbf{x}} + (\hat{\mathbf{u}}_1^\top \tilde{\mathbf{x}}_i)\hat{\mathbf{u}}_1$ (blue) on the first-PC line, $\hat{\mathbf{u}}_1 = (0.88, 0.47)^\top$. The sum of the squared gray segments is $\text{SSE} = (n-1)\hat{\lambda}_2$, the smallest over all lines; the spread of the blue points along the line is the retained variance $\hat{\lambda}_1$.*

3.1 The problem

Let the data be $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^p$ and $r \in \{1, \ldots, p\}$. We wish to approximate $\mathbf{x}_1, \ldots, \mathbf{x}_n$ by $\bar{\mathbf{x}} + \mathbf{a}_1, \ldots, \bar{\mathbf{x}} + \mathbf{a}_n$ where $\text{rank}([\mathbf{a}_1, \ldots, \mathbf{a}_n]) = r$. The approximation of $\mathbf{x}_i$ by $\bar{\mathbf{x}} + \mathbf{a}_i$ is better when the error $\|\mathbf{x}_i - (\bar{\mathbf{x}} + \mathbf{a}_i)\|_2^2$ is smaller, and the approximation of $\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ by $\{\bar{\mathbf{x}} + \mathbf{a}_i\}$ is better when $\text{SSE} = \sum_{i=1}^n \|\mathbf{x}_i - (\bar{\mathbf{x}} + \mathbf{a}_i)\|_2^2$ is smaller. That is, we wish to find $\mathbf{A}_r = [\mathbf{a}_1, \ldots, \mathbf{a}_n]$ $(p \times n)$ solving

$$(\star) \qquad \min_{\mathbf{A}_r} \sum_{i=1}^n \|(\mathbf{x}_i - \bar{\mathbf{x}}) - \mathbf{a}_i\|_2^2 \quad \text{subject to} \quad \text{rank}(\mathbf{A}_r) = r.$$

Since every column of a rank-r matrix lies in an r-dimensional subspace, the points $\bar{\mathbf{x}} + \mathbf{a}_i$ lie on an r-dimensional affine subspace through $\bar{\mathbf{x}}$; conversely any such subspace gives feasible $\mathbf{A}_r$.

3.2 Observations

1. With $\tilde{\mathbf{X}}^\top = [\mathbf{x}_1 - \bar{\mathbf{x}}, \ldots, \mathbf{x}_n - \bar{\mathbf{x}}]$ $(p \times n)$,
$$\sum_{i=1}^n \|(\mathbf{x}_i - \bar{\mathbf{x}}) - \mathbf{a}_i\|_2^2 = \|\tilde{\mathbf{X}}^\top - \mathbf{A}_r\|_F^2,$$
because for a $p \times n$ matrix $\mathbf{B} = [\mathbf{b}_1, \ldots, \mathbf{b}_n]$ the Frobenius norm is $\|\mathbf{B}\|_F^2 = \sum_{i=1}^p \sum_{j=1}^n b_{ij}^2 = \sum_{j=1}^n \|\mathbf{b}_j\|_2^2 = \text{tr}(\mathbf{B}\mathbf{B}^\top) = \text{tr}(\mathbf{B}^\top\mathbf{B})$.
2. The sample PCA of $(\mathbf{x}_1, \ldots, \mathbf{x}_n)$ is given by
   (1) the eigendecomposition $\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \hat{\mathbf{U}}\hat{\mathbf{\Lambda}}\hat{\mathbf{U}}^\top = \sum_{i=1}^p \hat{\lambda}_i \hat{\mathbf{u}}_i \hat{\mathbf{u}}_i^\top$, where $\hat{\mathbf{u}}_i$ is the ith PC direction vector and $\hat{\lambda}_i$ the ith PC variance, $\hat{\lambda}_1 \geq \cdots \geq \hat{\lambda}_r > \hat{\lambda}_{r+1} \geq \cdots \geq \hat{\lambda}_p$;
   (2) the SVD $\tilde{\mathbf{X}}_{(p\times n)}^\top = \hat{\mathbf{U}}\hat{\mathbf{D}}\hat{\mathbf{V}}^\top = \sum_{i=1}^p \hat{d}_i \hat{\mathbf{u}}_i \hat{\mathbf{v}}_i^\top$, where $\hat{\mathbf{u}}_i$ is the ith PC direction vector and $\hat{d}_i\hat{\mathbf{v}}_i$ the ith PC score vector;
   (3) $\hat{\mathbf{u}}_i^\top \tilde{\mathbf{X}}^\top = \hat{d}_i \hat{\mathbf{v}}_i^\top$, i.e.
$$\begin{pmatrix} \hat{\mathbf{u}}_1^\top \\ \vdots \\ \hat{\mathbf{u}}_r^\top \end{pmatrix} \tilde{\mathbf{X}}^\top = \begin{pmatrix} \hat{d}_1 & & \\ & \ddots & \\ & & \hat{d}_r \end{pmatrix} \begin{pmatrix} \hat{\mathbf{v}}_1^\top \\ \vdots \\ \hat{\mathbf{v}}_r^\top \end{pmatrix}.$$

Theorem 7.8: solution (Eckart–Young–Mirsky)

The $\mathbf{A}_r$ that minimizes $(\star)$ is

$$\mathbf{A}_r = \sum_{i=1}^{r} \hat{\mathbf{u}}_i \hat{d}_i \hat{\mathbf{v}}_i^\top = [\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r][\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r]^\top \tilde{\mathbf{X}}^\top,$$

the projection of $\tilde{\mathbf{X}}^\top$ onto the first r PC subspace, and the minimum is $\sum_{i>r} \hat{d}_i^2 = (n-1)\sum_{i=r+1}^{p} \hat{\lambda}_i$.

Proof. Step 1 (parametrize rank-r matrices). Since $\operatorname{rank}(\mathbf{A}_r) = r$, we can write $\mathbf{A}_r = \mathbf{E}\mathbf{C}^\top$ with $\mathbf{E} = [\mathbf{e}_1, \dots, \mathbf{e}_r]$ $(p \times r)$ satisfying $\mathbf{E}^\top\mathbf{E} = \mathbf{I}_r$ (an orthonormal basis of the column space of $\mathbf{A}_r$) and $\mathbf{C} = \begin{pmatrix} \mathbf{c}_1^\top \\ \vdots \\ \mathbf{c}_n^\top \end{pmatrix}$ an $n \times r$ matrix. Writing $\tilde{\mathbf{x}}_i = \mathbf{x}_i - \bar{\mathbf{x}}$, $(\star)$ is equivalent to $\min_{\mathbf{E},\mathbf{C}} \|\tilde{\mathbf{X}}^\top - \mathbf{E}\mathbf{C}^\top\|_F^2$, where

$$(\star\star) \qquad \|\tilde{\mathbf{X}}^\top - \mathbf{E}\mathbf{C}^\top\|_F^2 = \big\|(\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_n) - \mathbf{E}[\mathbf{c}_1, \dots, \mathbf{c}_n]\big\|_F^2 = \sum_{i=1}^{n} \|\tilde{\mathbf{x}}_i - \mathbf{E}\mathbf{c}_i\|_2^2.$$

Step 2 (optimal $\mathbf{C}$ for fixed $\mathbf{E}$). For any given $\mathbf{E}$, $(\star\star)$ separates over i and its minimizer is

$$\hat{\mathbf{c}}_i = \arg\min_{\mathbf{c}_i} \|\tilde{\mathbf{x}}_i - \mathbf{E}\mathbf{c}_i\|_2^2 = (\mathbf{E}^\top\mathbf{E})^{-1}\mathbf{E}^\top\tilde{\mathbf{x}}_i = \mathbf{E}^\top\tilde{\mathbf{x}}_i,$$

the projection scores of $\tilde{\mathbf{x}}_i$ onto $\operatorname{span}(\mathbf{E})$ (least squares). Stacking, $\hat{\mathbf{C}} = \begin{pmatrix} \tilde{\mathbf{x}}_1^\top\mathbf{E} \\ \vdots \\ \tilde{\mathbf{x}}_n^\top\mathbf{E} \end{pmatrix} = \tilde{\mathbf{X}}\mathbf{E}$, i.e. $\hat{\mathbf{C}}^\top = \mathbf{E}^\top\tilde{\mathbf{X}}^\top$.

Step 3 (profile out $\mathbf{C}$). Plugging $\hat{\mathbf{C}}$ into $(\star\star)$ (profiling, Theorem 2.25), and using that $\mathbf{I} - \mathbf{E}\mathbf{E}^\top$ is symmetric idempotent, $(\mathbf{I} - \mathbf{E}\mathbf{E}^\top)^2 = \mathbf{I} - \mathbf{E}\mathbf{E}^\top$,

$$\begin{aligned}
\min_{\mathbf{E}} \|\tilde{\mathbf{X}}^\top - \mathbf{E}\mathbf{E}^\top\tilde{\mathbf{X}}^\top\|_F^2 &= \min_{\mathbf{E}} \|(\mathbf{I} - \mathbf{E}\mathbf{E}^\top)\tilde{\mathbf{X}}^\top\|_F^2 = \min_{\mathbf{E}} \operatorname{tr}(\tilde{\mathbf{X}}(\mathbf{I} - \mathbf{E}\mathbf{E}^\top)\tilde{\mathbf{X}}^\top) \\
&= \min_{\mathbf{E}} \Big\{ \operatorname{tr}(\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top) - (n-1)\operatorname{tr}\Big(\mathbf{E}^\top \frac{\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}}{n-1}\mathbf{E}\Big) \Big\} \\
&= \operatorname{tr}(\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top) - (n-1)\max_{\mathbf{E}} \operatorname{tr}(\mathbf{S}\mathbf{E}\mathbf{E}^\top).
\end{aligned}$$

Step 4 (von Neumann). Recall $\mathbf{S} = \hat{\mathbf{U}}\hat{\mathbf{\Lambda}}\hat{\mathbf{U}}^\top$. $\mathbf{E}\mathbf{E}^\top$ is symmetric with eigenvalues 1 (r times) and 0 ($p - r$ times), so von Neumann's trace inequality (Theorem 2.13) gives

$$\operatorname{tr}(\mathbf{S}\mathbf{E}\mathbf{E}^\top) \le \hat{\lambda}_1 + \cdots + \hat{\lambda}_r,$$

with equality when $\mathbf{E} = [\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r]$ (or any $\mathbf{E}$ spanning the same space).

Step 5 (the minimizer). Therefore

$$\mathbf{A}_r = \mathbf{E}\hat{\mathbf{C}}^\top = [\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r] \begin{pmatrix} \hat{\mathbf{u}}_1^\top \\ \vdots \\ \hat{\mathbf{u}}_r^\top \end{pmatrix} \tilde{\mathbf{X}}^\top = [\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r] \begin{pmatrix} \hat{d}_1 & & \\ & \ddots & \\ & & \hat{d}_r \end{pmatrix} \begin{pmatrix} \hat{\mathbf{v}}_1^\top \\ \vdots \\ \hat{\mathbf{v}}_r^\top \end{pmatrix} = \sum_{i=1}^{r} \hat{\mathbf{u}}_i \hat{d}_i \hat{\mathbf{v}}_i^\top$$

by Observation 2(3): the projection of $\tilde{\mathbf{X}}^\top$ onto the first r PC subspace. Since $\operatorname{tr}(\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top) = (n-1)\operatorname{tr}\mathbf{S} = (n-1)\sum_i \hat{\lambda}_i$, the minimum is $(n-1)\sum_{i>r} \hat{\lambda}_i = \sum_{i>r} \hat{d}_i^2$. ∎

Remark 7.9: interpretation

[Supplement]

- The best approximating hyperplane is $\bar{\mathbf{x}}+\text{span}\{\hat{\mathbf{u}}_1,\dots,\hat{\mathbf{u}}_r\}$ and each $\mathbf{x}_i$ is approximated by its orthogonal projection $\bar{\mathbf{x}}+\sum_{k\le r}(\hat{\mathbf{u}}_k^\top\tilde{\mathbf{x}}_i)\hat{\mathbf{u}}_k$: the PC scores are the coordinates on the hyperplane.
- The fitted values satisfy $\hat{\mathbf{c}}_i=\mathbf{E}^\top\tilde{\mathbf{x}}_i$, i.e. reconstruction uses projection scores, exactly the decomposition $\mathbf{x}=\bar{\mathbf{x}}+\sum s_{\mathbf{v}_j}(\mathbf{x}-\bar{\mathbf{x}})\mathbf{v}_j+r_{\mathcal{L}_q}(\mathbf{x})$ of Ch. 1.
- Minimizing squared error and maximizing retained variance (the $\arg\max\mathbf{u}^\top\mathbf{S}\mathbf{u}$ definition) are the same problem, since $\text{SSE}=(n-1)\{\text{tr}\mathbf{S}-\text{tr}(\mathbf{S}\mathbf{E}\mathbf{E}^\top)\}$.
- If $\hat{\lambda}_r=\hat{\lambda}_{r+1}$ the optimal subspace is not unique; the strict gap $\hat{\lambda}_r>\hat{\lambda}_{r+1}$ makes it unique.

3.3 Correlation PCA

If the units differ, use PCA of standardized data: $\mathbf{R}=\mathbf{D}_{\boldsymbol{\Sigma}}^{-1/2}\boldsymbol{\Sigma}\mathbf{D}_{\boldsymbol{\Sigma}}^{-1/2}=\mathbf{U}_R\boldsymbol{\Lambda}_R\mathbf{U}_R^\top$ (sample: $\mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2}$). Since $\text{tr}\mathbf{R}=p$, the proportion explained is λ_k^R/p. Covariance and correlation PCs are in general not simple transformations of each other (scale dependence).

4 How Many Components to Keep?

- **Scree plot / cumulative scree plot**: plot $\hat{\lambda}_k$ against k and look for the elbow, or take the q at which the cumulative proportion reaches a target (e.g. 80–90%).
- **Kaiser's rule**: keep components with $\hat{\lambda}_k>\bar{\lambda}=\frac{1}{p}\sum_j\hat{\lambda}_j$; for correlation PCA, $\hat{\lambda}_k>1$.
- **Likelihood ratio testing**: under normality test $H_0(k):\lambda_{k+1}=\cdots=\lambda_p$ (the remaining space is spherical). With arithmetic mean $\bar{a}$ and geometric mean $\bar{g}$ of the last $p-k$ sample eigenvalues,

$$-2\log\Lambda=n(p-k)\log\frac{\bar{a}}{\bar{g}}\ \dot{\sim}\ \chi^2_{(p-k+2)(p-k-1)/2};$$

since $\bar{a}\ge\bar{g}$ (AM–GM), the statistic is small when the eigenvalues are nearly equal.

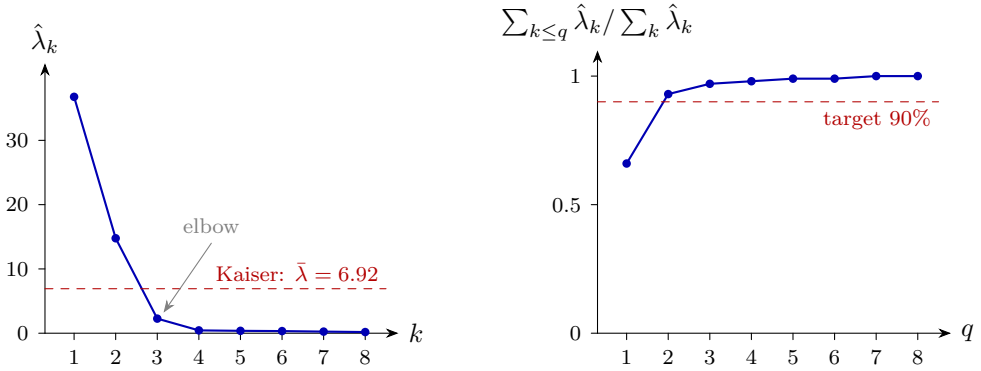


Figure. *Scree plot (left) and cumulative scree plot (right) for synthetic data ($n=200$, $p=8$, generated with three strong directions). The elbow is at $k=3$, Kaiser's rule keeps $q=2$ ($\hat{\lambda}_1,\hat{\lambda}_2>\bar{\lambda}$), and the 90% target is reached at $q=2$ (93%).*

Remark 7.10: cautions in interpretation

- The sign of an eigenvector is arbitrary; what matters is the sign pattern (same signs: "size"; mixed signs: "contrast"). In the scores example $\mathbf{u}_1 \propto (1,1)$ was overall level and $\mathbf{u}_2 \propto (1,-1)$ the contrast between subjects.
- With tied eigenvalues the directions within the eigenspace are not unique.
- Sample eigenvalues are biased: large ones are overestimated and small ones underestimated. Under normality, for distinct λ_k, $\sqrt{n}(\hat{\lambda}_k - \lambda_k) \xrightarrow{d} N(0, 2\lambda_k^2)$.
- High explained variance does not mean usefulness for classification or prediction (the $X_1 = 10Z$, $X_2 = Y$ example of Ch. 1).

5 Example: Scores in Two Subjects Revisited

For $\mathbf{S} = \begin{pmatrix} 125 & 75 \\ 75 & 125 \end{pmatrix}$ the characteristic equation $(125-\lambda)^2 - 75^2 = 0$ gives $\lambda = 200, 50$; the null space of $\mathbf{S} - 200\mathbf{I} = \begin{pmatrix} -75 & 75 \\ 75 & -75 \end{pmatrix}$ is spanned by $(1,1)^\top$. $\mathrm{Corr}(Z_1, X_1) = u_{11}\sqrt{\lambda_1}/\sqrt{s_{11}} = \frac{1}{\sqrt{2}}\sqrt{200}/\sqrt{125} = 0.894$, $\mathrm{Corr}(Z_2, X_1) = \frac{1}{\sqrt{2}}\sqrt{50}/\sqrt{125} = 0.447$, and $0.894^2 + 0.447^2 = 1$: each variable's variance is fully decomposed over the PCs.

6 Selected exercises

Motivation. PCA is worthwhile only if the variables are correlated, and the eigenvalue tests above compare a covariance matrix with a spherical one. The first two exercises derive the corresponding likelihood ratio tests; the third is a covariance whose components can be found by hand.

Exercise 7.1 (Testing that all variables are independent; J&W 8.9(a))

Let $\mathbf{x}_1, \ldots, \mathbf{x}_n$ be a sample from $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $n > p$. Show that the likelihood ratio statistic for $H_0 : \boldsymbol{\Sigma} = \mathrm{diag}(\sigma_{11}, \ldots, \sigma_{pp})$ is

$$\Lambda = \frac{|\mathbf{S}|^{n/2}}{\prod_{j=1}^{p} s_{jj}^{n/2}} = |\mathbf{R}|^{n/2},$$

that $0 < \Lambda \le 1$, and that $-2\log\Lambda$ is approximately $\chi^2_{p(p-1)/2}$ for large n.

Solution.

Step 1 (unrestricted maximum). By Theorem 4.7 and the remark after it, $\max_{\boldsymbol{\mu},\boldsymbol{\Sigma}} L = (2\pi)^{-np/2}|\mathbf{S}_0|^{-n/2}e^{-np/2}$ with $\mathbf{S}_0 = \frac{n-1}{n}\mathbf{S}$.

Step 2 (maximum under H_0). If $\boldsymbol{\Sigma}$ is diagonal, the density of $\mathbf{x}_i$ is the product of univariate $N(\mu_j, \sigma_{jj})$ densities, so $L(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \prod_{j=1}^{p} L_j(\mu_j, \sigma_{jj})$ with L_j the likelihood of the jth coordinate. The factors have separate parameters, so each is maximized separately; Theorem 4.7 with $p = 1$ gives $\hat{\mu}_j = \bar{x}_j$, $\hat{\sigma}_{jj} = (\mathbf{S}_0)_{jj}$ and

$$\max_{H_0} L = \prod_{j=1}^{p} (2\pi)^{-n/2}(\mathbf{S}_0)_{jj}^{-n/2}e^{-n/2} = (2\pi)^{-np/2}\Big\{\prod_{j}(\mathbf{S}_0)_{jj}\Big\}^{-n/2}e^{-np/2}.$$

Step 3 (the ratio). Dividing,

$$\Lambda = \Big(\frac{|\mathbf{S}_0|}{\prod_j (\mathbf{S}_0)_{jj}}\Big)^{n/2} = \Big(\frac{|\mathbf{S}|}{\prod_j s_{jj}}\Big)^{n/2},$$

since $|\mathbf{S}_0| = (\frac{n-1}{n})^p |\mathbf{S}|$ and $\prod_j (\mathbf{S}_0)_{jj} = (\frac{n-1}{n})^p \prod_j s_{jj}$. With $\mathbf{D} = \operatorname{diag}(s_{11}, \dots, s_{pp})$, $\mathbf{R} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2}$ and $|\mathbf{R}| = |\mathbf{S}| / \prod_j s_{jj}$.

Step 4. $\mathbf{S} \succ 0$ a.s. (as $n > p$), so $\mathbf{R} \succ 0$ and Exercise 2.2 gives $0 < |\mathbf{R}| \le 1$, with $\Lambda = 1$ iff $\mathbf{R} = \mathbf{I}_p$.

Step 5 (degrees of freedom). The unrestricted model has $p + p(p+1)/2$ free parameters and H_0 has $2p$; the large-sample likelihood ratio theory used in Ch. 5 gives $-2\log\Lambda \overset{\cdot}{\sim} \chi^2_\nu$ with $\nu = p(p+1)/2 - p = p(p-1)/2$, the number of off-diagonal correlations. ■

Exercise 7.2 (Sphericity test; J&W 8.9(b))

In the setting of Exercise 7.1, show that the likelihood ratio statistic for $H_0 : \boldsymbol{\Sigma} = \sigma^2 \mathbf{I}_p$ ($\sigma^2 > 0$ unknown) is

$$\Lambda = \Big(\frac{|\mathbf{S}|}{(\operatorname{tr}\mathbf{S}/p)^p}\Big)^{n/2} = \Big(\frac{\bar g}{\bar a}\Big)^{np/2},$$

where $\bar a$ and $\bar g$ are the arithmetic and geometric means of the eigenvalues $\hat\lambda_1, \dots, \hat\lambda_p$ of $\mathbf{S}$. Show $-2\log\Lambda \ge 0$ and find its degrees of freedom; compare with the test of $H_0(k)$ in the last section.

Solution.

Step 1 (log-likelihood under H_0). With $Q(\boldsymbol{\mu}) = \sum_i \|\mathbf{x}_i - \boldsymbol{\mu}\|^2$,

$$\log L(\boldsymbol{\mu}, \sigma^2 \mathbf{I}) = -\frac{np}{2}\log(2\pi\sigma^2) - \frac{Q(\boldsymbol{\mu})}{2\sigma^2}.$$

$Q(\boldsymbol{\mu}) = Q(\bar{\mathbf{x}}) + n\|\bar{\mathbf{x}} - \boldsymbol{\mu}\|^2$ is minimized at $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$ for every σ^2, and $Q(\bar{\mathbf{x}}) = n\operatorname{tr}\mathbf{S}_0$.

Step 2 ($\hat\sigma^2$). Let $c = Q(\bar{\mathbf{x}})/(np) = \operatorname{tr}\mathbf{S}_0/p$ and $x = c/\sigma^2$. Then

$$\log\sigma^2 + \frac{Q(\bar{\mathbf{x}})}{np\,\sigma^2} = \log c - \log x + x \;\ge\; \log c + 1$$

by Theorem 2.14, with equality iff $x = 1$. Hence $\hat\sigma^2 = \operatorname{tr}\mathbf{S}_0/p$ and $\max_{H_0} L = (2\pi\hat\sigma^2)^{-np/2} e^{-np/2}$.

Step 3 (the ratio). With Step 1 of Exercise 7.1,

$$\Lambda = \frac{|\mathbf{S}_0|^{n/2}}{(\operatorname{tr}\mathbf{S}_0/p)^{np/2}} = \Big(\frac{|\mathbf{S}|}{(\operatorname{tr}\mathbf{S}/p)^p}\Big)^{n/2},$$

because numerator and denominator are both homogeneous of degree $np/2$ in $\mathbf{S}_0$. Since $|\mathbf{S}| = \prod_i \hat\lambda_i = \bar g^p$ and $\operatorname{tr}\mathbf{S}/p = \bar a$, $\Lambda = (\bar g/\bar a)^{np/2}$.

Step 4. $-2\log\Lambda = np\log(\bar a/\bar g) \ge 0$ by the AM–GM inequality, with equality iff all $\hat\lambda_i$ are equal.

Step 5 (degrees of freedom). The free parameters are $p + p(p+1)/2$ versus $p+1$ under H_0, so $\nu = p(p+1)/2 - 1 = (p+2)(p-1)/2$. This is the statistic $n(p-k)\log(\bar a/\bar g)$ with $\chi^2_{(p-k+2)(p-k-1)/2}$ of the last section for $k = 0$; for $k \ge 1$ the same comparison is applied to the last $p-k$ eigenvalues. ■

Exercise 7.3 (A tridiagonal covariance; J&W 8.4)

Find the principal components and the proportion of total variance explained by each for

$$\boldsymbol{\Sigma} = \sigma^2 \begin{pmatrix} 1 & \rho & 0 \\ \rho & 1 & \rho \\ 0 & \rho & 1 \end{pmatrix}, \qquad 0 < \rho < \frac{1}{\sqrt{2}}.$$

Solution.

Step 1 (eigenvalues). Let $\mathbf{C} = \boldsymbol{\Sigma}/\sigma^2$. Expanding along the first row,

$$|\mathbf{C} - \lambda \mathbf{I}| = (1-\lambda)\{(1-\lambda)^2 - \rho^2\} - \rho \cdot \rho(1-\lambda) = (1-\lambda)\{(1-\lambda)^2 - 2\rho^2\},$$

so the eigenvalues of $\boldsymbol{\Sigma}$ are $\sigma^2(1+\sqrt{2}\rho) > \sigma^2 > \sigma^2(1-\sqrt{2}\rho)$, all positive because $\rho < 1/\sqrt{2}$.

Step 2 (eigenvectors). For $\lambda = 1 \pm \sqrt{2}\rho$ the first and third rows of $(\mathbf{C} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$ read $\mp\sqrt{2}\rho x_1 + \rho x_2 = 0$ and $\rho x_2 \mp \sqrt{2}\rho x_3 = 0$, so $x_2 = \pm\sqrt{2}x_1$ and $x_3 = x_1$. For $\lambda = 1$ they give $x_2 = 0$ and the middle row gives $x_1 = -x_3$. Normalizing,

$$\mathbf{u}_1 = \tfrac{1}{2}(1, \sqrt{2}, 1)^\top, \qquad \mathbf{u}_2 = \tfrac{1}{\sqrt{2}}(1, 0, -1)^\top, \qquad \mathbf{u}_3 = \tfrac{1}{2}(1, -\sqrt{2}, 1)^\top.$$

Step 3 (components). $Z_1 = \frac{1}{2}(X_1 + \sqrt{2}X_2 + X_3)$, $Z_2 = \frac{1}{\sqrt{2}}(X_1 - X_3)$, $Z_3 = \frac{1}{2}(X_1 - \sqrt{2}X_2 + X_3)$. Since $\operatorname{tr}\boldsymbol{\Sigma} = 3\sigma^2$, the proportions are $(1+\sqrt{2}\rho)/3$, $1/3$ and $(1-\sqrt{2}\rho)/3$.

Z_1 is a weighted level with the middle variable counted most; Z_2 contrasts the two ends and does not depend on ρ. As $\rho \uparrow 1/\sqrt{2}$, $\operatorname{Var}(Z_3) \to 0$ and $\boldsymbol{\Sigma}$ becomes singular. ■

Chapter 8. Canonical Correlation Analysis

Maximal correlation between two sets of variables

1 Setting

We want to summarize the association between two sets of variables $\mathbf{X} \in \mathbb{R}^p$ and $\mathbf{Y} \in \mathbb{R}^q$ with $\operatorname{Cov}\begin{pmatrix}\mathbf{X}\\ \mathbf{Y}\end{pmatrix} = \begin{pmatrix}\boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12}\\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22}\end{pmatrix}$, $\boldsymbol{\Sigma}_{11}, \boldsymbol{\Sigma}_{22} \succ 0$. Instead of inspecting pq correlations, maximize the correlation of linear combinations

$$\xi = \mathbf{g}^\top \mathbf{X}, \qquad \omega = \mathbf{h}^\top \mathbf{Y}.$$

For $q = 1$ this is the multiple correlation (Ch. 3). PCA maximizes variance *within* one set; CCA maximizes correlation *between* two sets.

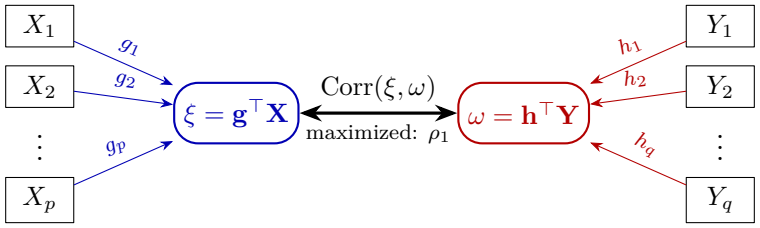


Figure. *CCA projects each set onto one direction, $\mathbf{X} \mapsto \xi = \mathbf{g}^\top \mathbf{X}$ and $\mathbf{Y} \mapsto \omega = \mathbf{h}^\top \mathbf{Y}$, and chooses $(\mathbf{g}, \mathbf{h})$ so that the two projections are as correlated as possible. Later pairs repeat this subject to being uncorrelated with the earlier variates of their own set.*

Definition 8.1: canonical variates and canonical correlations

$$(\mathbf{g}_1, \mathbf{h}_1) = \arg \max_{\mathbf{g} \in \mathbb{R}^p,\, \mathbf{h} \in \mathbb{R}^q} \operatorname{Corr}(\mathbf{g}^\top \mathbf{X}, \mathbf{h}^\top \mathbf{Y}),$$

$$(\mathbf{g}_k, \mathbf{h}_k) = \arg \max_{\substack{\mathbf{g} \in \mathbb{R}^p,\, \mathbf{h} \in \mathbb{R}^q \\ \mathbf{g}^\top \boldsymbol{\Sigma}_{11} \mathbf{g}_j = 0,\ \mathbf{h}^\top \boldsymbol{\Sigma}_{22} \mathbf{h}_j = 0,\ j = 1, \dots, k-1}} \operatorname{Corr}(\mathbf{g}^\top \mathbf{X}, \mathbf{h}^\top \mathbf{Y}).$$

$\xi_k = \mathbf{g}_k^\top \mathbf{X}$, $\omega_k = \mathbf{h}_k^\top \mathbf{Y}$ form the kth pair of canonical variates and $\rho_k = \operatorname{Corr}(\xi_k, \omega_k)$ is the kth canonical correlation. Correlation is scale-free, so normalize $\operatorname{Var}(\xi_k) = \operatorname{Var}(\omega_k) = 1$. The constraints mean $\operatorname{Cov}(\xi_k, \xi_j) = \operatorname{Cov}(\omega_k, \omega_j) = 0$, and $\rho_k \le \rho_j$ for $j < k$.

$\rho(\mathbf{X}, \mathbf{Y}) := \max \operatorname{Corr}(\mathbf{a}^\top \mathbf{X}, \mathbf{b}^\top \mathbf{Y}) = \rho_1$ is the **largest canonical coefficient**.

2 Solution via the Singular Value Decomposition

Motivation. After each set is whitened to identity covariance, correlation becomes a bilinear form $\mathbf{a}^\top \mathbf{M} \mathbf{b}$ in two unit vectors, and maximizing a bilinear form over unit vectors is exactly what the singular value decomposition does.

Theorem 8.2: solution of CCA

Let $\mathbf{M} = \boldsymbol{\Sigma}_{11}^{-1/2} \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1/2}$ $(p \times q)$ have SVD $\mathbf{M} = \sum_{k=1}^{r} d_k \mathbf{a}_k \mathbf{b}_k^\top$, $d_1 \ge \cdots \ge d_r > 0$, $r = \operatorname{rank} \boldsymbol{\Sigma}_{12}$. Then

$$\rho_k = d_k, \qquad \mathbf{g}_k = \boldsymbol{\Sigma}_{11}^{-1/2} \mathbf{a}_k, \qquad \mathbf{h}_k = \boldsymbol{\Sigma}_{22}^{-1/2} \mathbf{b}_k.$$

Equivalently $\mathbf{a}_k$ is the kth eigenvector of $\mathbf{M}\mathbf{M}^\top = \boldsymbol{\Sigma}_{11}^{-1/2}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1/2}$, $\mathbf{b}_k$ that of $\mathbf{M}^\top\mathbf{M} = \boldsymbol{\Sigma}_{22}^{-1/2}\boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1/2}$, and ρ_k^2 is the eigenvalue.

Proof. Substituting $\mathbf{a} = \boldsymbol{\Sigma}_{11}^{1/2}\mathbf{g}$, $\mathbf{b} = \boldsymbol{\Sigma}_{22}^{1/2}\mathbf{h}$,

$$\operatorname{Corr}(\mathbf{g}^\top\mathbf{X}, \mathbf{h}^\top\mathbf{Y}) = \frac{\mathbf{g}^\top\boldsymbol{\Sigma}_{12}\mathbf{h}}{\sqrt{\mathbf{g}^\top\boldsymbol{\Sigma}_{11}\mathbf{g}}\sqrt{\mathbf{h}^\top\boldsymbol{\Sigma}_{22}\mathbf{h}}} = \frac{\mathbf{a}^\top\mathbf{M}\mathbf{b}}{\sqrt{\mathbf{a}^\top\mathbf{a}}\sqrt{\mathbf{b}^\top\mathbf{b}}},$$

so we maximize $\mathbf{a}^\top\mathbf{M}\mathbf{b}$ over $\|\mathbf{a}\| = \|\mathbf{b}\| = 1$. With $\mathbf{M} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$, $\boldsymbol{\alpha} = \mathbf{U}^\top\mathbf{a}$, $\boldsymbol{\beta} = \mathbf{V}^\top\mathbf{b}$, Cauchy–Schwarz gives

$$\mathbf{a}^\top\mathbf{M}\mathbf{b} = \mathbf{a}^\top\mathbf{U}\mathbf{D}\mathbf{V}^\top\mathbf{b} = \sum_k d_k\alpha_k\beta_k \le d_1\sum_k|\alpha_k\beta_k| \le d_1\|\boldsymbol{\alpha}\|\|\boldsymbol{\beta}\| \le d_1,$$

with equality at $\mathbf{a} = \mathbf{a}_1$, $\mathbf{b} = \mathbf{b}_1$. Under $\mathbf{g}^\top\boldsymbol{\Sigma}_{11}\mathbf{g}_j = \mathbf{a}^\top\mathbf{a}_j = 0$ and $\mathbf{b}^\top\mathbf{b}_j = 0$ $(j < k)$ we have $\alpha_j = \beta_j = 0$ for $j < k$, and the same argument gives maximum d_k at $(\mathbf{a}_k, \mathbf{b}_k)$. ■

Theorem 8.3: properties of canonical variates

1. $\operatorname{Var}(\xi_k) = \operatorname{Var}(\omega_k) = 1$ and $\operatorname{Corr}(\xi_k, \xi_l) = \operatorname{Corr}(\omega_k, \omega_l) = 0$ for $k \ne l$.
2. $\operatorname{Corr}(\xi_k, \omega_l) = 0$ for $k \ne l$, $\operatorname{Corr}(\xi_k, \omega_k) = \rho_k$.
3. Invariance: canonical correlations are unchanged by $\mathbf{X} \mapsto \mathbf{D}_1\mathbf{X} + \mathbf{c}_1$, $\mathbf{Y} \mapsto \mathbf{D}_2\mathbf{Y} + \mathbf{c}_2$ with $\mathbf{D}_i$ non-singular; covariance and correlation matrices give the same ρ_k.
4. For $q = 1$, $\rho_1^2 = \boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}/\sigma_{22}$ is the squared multiple correlation.

Proof. (1) $\operatorname{Cov}(\xi_k, \xi_l) = \mathbf{g}_k^\top\boldsymbol{\Sigma}_{11}\mathbf{g}_l = \mathbf{a}_k^\top\mathbf{a}_l = \delta_{kl}$.

(2) $\operatorname{Cov}(\xi_k, \omega_l) = \mathbf{g}_k^\top\boldsymbol{\Sigma}_{12}\mathbf{h}_l = \mathbf{a}_k^\top\mathbf{M}\mathbf{b}_l = d_l\mathbf{a}_k^\top\mathbf{a}_l = d_l\delta_{kl}$.

(3) $\operatorname{Corr}(\mathbf{g}^\top\mathbf{D}_1\mathbf{X}, \mathbf{h}^\top\mathbf{D}_2\mathbf{Y}) = \operatorname{Corr}((\mathbf{D}_1^\top\mathbf{g})^\top\mathbf{X}, (\mathbf{D}_2^\top\mathbf{h})^\top\mathbf{Y})$ and $\mathbf{g} \mapsto \mathbf{D}_1^\top\mathbf{g}$ is a bijection, so the maxima agree.

(4) For $q = 1$ the only nonzero eigenvalue of $\mathbf{M}\mathbf{M}^\top$ is $\mathbf{M}^\top\mathbf{M} = \boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}/\sigma_{22}$. ■

Theorem 8.4: interpreting the canonical variates

1. With $\mathbf{G} = [\mathbf{g}_1, \dots, \mathbf{g}_r]$, $\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_r]$, $\boldsymbol{\xi} = \mathbf{G}^\top\mathbf{X}$ and $\boldsymbol{\omega} = \mathbf{H}^\top\mathbf{Y}$,

$$\begin{aligned} \operatorname{Cov}(\mathbf{X}, \boldsymbol{\xi}) &= \boldsymbol{\Sigma}_{11}\mathbf{G}, & \operatorname{Cov}(\mathbf{Y}, \boldsymbol{\omega}) &= \boldsymbol{\Sigma}_{22}\mathbf{H}, \\ \operatorname{Cov}(\mathbf{X}, \boldsymbol{\omega}) &= \boldsymbol{\Sigma}_{12}\mathbf{H}, & \operatorname{Cov}(\mathbf{Y}, \boldsymbol{\xi}) &= \boldsymbol{\Sigma}_{21}\mathbf{G}, \end{aligned}$$

   and the correlations follow by dividing row i by $\sqrt{\sigma_{ii}}$ (the variates have unit variance). For standardized variables these are the **canonical loadings** used to name the variates.
2. $\rho_1 \ge \max_{i,j}|\operatorname{Corr}(X_i, Y_j)|$, and more generally $\rho_1 \ge$ the multiple correlation of any single X_i with $\mathbf{Y}$ (and of any Y_j with $\mathbf{X}$).

Proof. (1) $\operatorname{Cov}(\mathbf{X}, \mathbf{G}^\top\mathbf{X}) = \operatorname{Cov}(\mathbf{X})\mathbf{G} = \boldsymbol{\Sigma}_{11}\mathbf{G}$ and $\operatorname{Cov}(\mathbf{X}, \mathbf{H}^\top\mathbf{Y}) = \operatorname{Cov}(\mathbf{X}, \mathbf{Y})\mathbf{H} = \boldsymbol{\Sigma}_{12}\mathbf{H}$; the other two are the same computation.

(2) ρ_1 is the maximum of $\operatorname{Corr}(\mathbf{g}^\top\mathbf{X}, \mathbf{h}^\top\mathbf{Y})$ over all $(\mathbf{g}, \mathbf{h})$. Taking $\mathbf{g} = \mathbf{e}_i$ and maximizing over $\mathbf{h}$ only gives the multiple correlation of X_i with $\mathbf{Y}$, a value attained in a smaller set; taking also $\mathbf{h} = \pm\mathbf{e}_j$ gives $|\operatorname{Corr}(X_i, Y_j)|$. ■

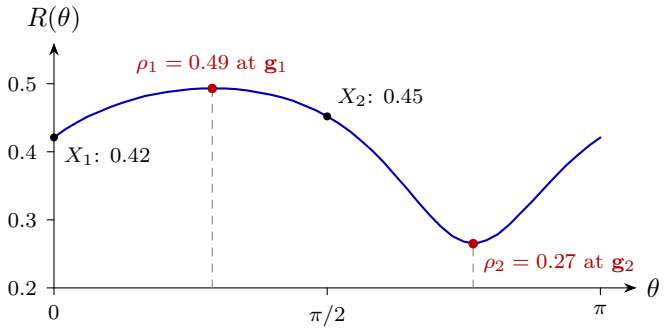


Figure. *A synthetic population with $p = q = 2$ (unit variances, within-set correlations 0.4 in $\mathbf{\Sigma}_{11}$ and 0.2 in $\mathbf{\Sigma}_{22}$, $\mathbf{\Sigma}_{12} = \left(\begin{smallmatrix} 0.30 & 0.35 \\ 0.05 & 0.45 \end{smallmatrix}\right)$). For $\mathbf{g}(\theta) = (\cos\theta, \sin\theta)^\top$, $R(\theta) = \max_{\mathbf{h}} \mathrm{Corr}(\mathbf{g}(\theta)^\top \mathbf{X}, \mathbf{h}^\top \mathbf{Y})$ is the multiple correlation of $\mathbf{g}(\theta)^\top \mathbf{X}$ with $\mathbf{Y}$. Its maximum is ρ_1, attained in the direction of $\mathbf{g}_1$, and for $p = 2$ its minimum is ρ_2, attained at $\mathbf{g}_2$. The single variables X_1 ($\theta = 0$) and X_2 ($\theta = \pi/2$) reach only 0.42 and 0.45, as Theorem 8.4(2) requires.*

3 Sample Canonical Correlations

Replace the covariances by $\mathbf{S}_{11} = \widehat{\mathrm{Cov}}(\mathbf{X})$, $\mathbf{S}_{22} = \widehat{\mathrm{Cov}}(\mathbf{Y})$, $\mathbf{S}_{12} = \widehat{\mathrm{Cov}}(\mathbf{X}, \mathbf{Y})$:

$$(\hat{\mathbf{g}}_k, \hat{\mathbf{h}}_k) = \arg \max_{\substack{\mathbf{g} \in \mathbb{R}^p,\, \mathbf{h} \in \mathbb{R}^q \\ \mathbf{g}^\top \mathbf{S}_{11} \hat{\mathbf{g}}_j = 0,\ \mathbf{h}^\top \mathbf{S}_{22} \hat{\mathbf{h}}_j = 0,\ j = 1, \dots, k-1}} \frac{\mathbf{g}^\top \mathbf{S}_{12} \mathbf{h}}{\sqrt{\mathbf{g}^\top \mathbf{S}_{11} \mathbf{g}\, \mathbf{h}^\top \mathbf{S}_{22} \mathbf{h}}}.$$

With $\mathbf{a} = \mathbf{S}_{11}^{1/2} \mathbf{g}$, $\mathbf{b} = \mathbf{S}_{22}^{1/2} \mathbf{h}$ this is maximizing $\dfrac{\mathbf{a}^\top \mathbf{S}_{11}^{-1/2} \mathbf{S}_{12} \mathbf{S}_{22}^{-1/2} \mathbf{b}}{\sqrt{\mathbf{a}^\top \mathbf{a}\, \mathbf{b}^\top \mathbf{b}}} = \max_{\|\mathbf{a}\| = \|\mathbf{b}\| = 1} \mathbf{a}^\top \mathbf{M} \mathbf{b} = \mathbf{a}^\top \mathbf{U} \mathbf{D} \mathbf{V}^\top \mathbf{b}$, so

$$\mathbf{a}_i = \mathbf{v}_i(\mathbf{S}_{11}^{-1/2} \mathbf{S}_{12} \mathbf{S}_{22}^{-1} \mathbf{S}_{21} \mathbf{S}_{11}^{-1/2}), \qquad \mathbf{b}_i = \mathbf{v}_i(\mathbf{S}_{22}^{-1/2} \mathbf{S}_{21} \mathbf{S}_{11}^{-1} \mathbf{S}_{12} \mathbf{S}_{22}^{-1/2}),$$

where $\mathbf{v}_i(\mathbf{M})$ is the ith eigenvector of $\mathbf{M}$.

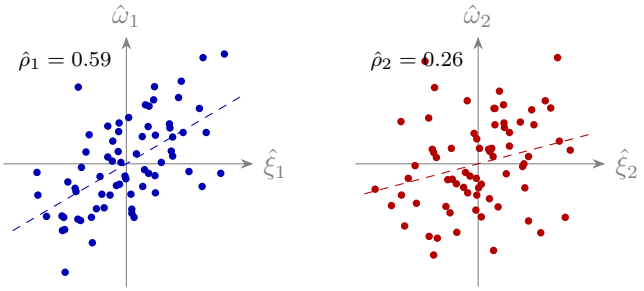


Figure. *Sample canonical variates for $n = 70$ draws from the population of the previous figure ($p = q = 2$). Each variate has sample variance 1, so the dashed least-squares line has slope $\hat{\rho}_k$. The first pair is clearly related; the second, uncorrelated with the first pair, carries the remaining weaker association.*

3.1 Regularized CCA

If p or q is large relative to n, $\mathbf{S}_{11}, \mathbf{S}_{22}$ are singular or unstable and sample canonical correlations are biased towards 1. **Regularized CCA** adds a ridge term,

$$\mathbf{S}_{11} \mapsto \mathbf{S}_{11} + \lambda_1 \mathbf{I}, \qquad \mathbf{S}_{22} \mapsto \mathbf{S}_{22} + \lambda_2 \mathbf{I},$$

with λ_1, λ_2 chosen by cross-validation.

4 Testing Canonical Correlations

Under multivariate normality the likelihood ratio statistic for $H_0 : \mathbf{\Sigma}_{12} = \mathbf{0}$ (all $\rho_k = 0$) is

$$\Lambda^{2/n} = \frac{|\mathbf{S}|}{|\mathbf{S}_{11}||\mathbf{S}_{22}|} = \prod_{k=1}^{\min(p,q)} (1 - \hat{\rho}_k^2),$$

and Bartlett's approximation gives $-(n-1-\frac{p+q+1}{2}) \sum_k \ln(1-\hat{\rho}_k^2) \dot{\sim} \chi^2_{pq}$. The hypothesis that only the first k are nonzero, $H_0^{(k)} : \rho_{k+1} = \cdots = 0$, is tested sequentially with

$$-\Big(n - 1 - \frac{p+q+1}{2}\Big) \sum_{i=k+1}^{\min(p,q)} \ln(1-\hat{\rho}_i^2) \;\dot{\sim}\; \chi^2_{(p-k)(q-k)}.$$

Derivation: the determinant identity

[Supplement] By the block determinant formula, $|\mathbf{S}| = |\mathbf{S}_{22}||\mathbf{S}_{11} - \mathbf{S}_{12}\mathbf{S}_{22}^{-1}\mathbf{S}_{21}| = |\mathbf{S}_{22}||\mathbf{S}_{11}|\,|\mathbf{I} - \hat{\mathbf{M}}\hat{\mathbf{M}}^\top|$, and the eigenvalues of $\hat{\mathbf{M}}\hat{\mathbf{M}}^\top$ are $\hat{\rho}_k^2$, so $|\mathbf{S}|/(|\mathbf{S}_{11}||\mathbf{S}_{22}|) = \prod(1-\hat{\rho}_k^2)$.

5 Relations to PCA, Multiple Correlation and Discriminant Analysis

- **Multiple correlation**: CCA with $q = 1$.
- [Supplement] **Discriminant analysis**: with $\mathbf{Y}$ a set of group indicator (dummy) variables, the canonical variates of $\mathbf{X}$ give the same directions as Fisher's discriminants (Ch. 10).
- **PCA**: variance maximization within one set (eigendecomposition of $\mathbf{\Sigma}$) versus correlation maximization between two sets (SVD of $\mathbf{\Sigma}_{11}^{-1/2}\mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1/2}$).

6 Selected exercises

Motivation. The exercises give a square-root-free way to compute canonical variates, two structures where the canonical correlation is found in closed form, and the likelihood ratio derivation of the test of $\mathbf{\Sigma}_{12} = \mathbf{0}$ quoted above.

Exercise 8.1 (Computing without square roots; J&W 10.4)

Let $\mathbf{M} = \mathbf{\Sigma}_{11}^{-1/2}\mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1/2}$ as in Theorem 8.2 and $\mathbf{K} = \mathbf{\Sigma}_{11}^{-1}\mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1}\mathbf{\Sigma}_{21}$.
(a) Show that if $\mathbf{M}\mathbf{M}^\top\mathbf{e} = \lambda\mathbf{e}$ with $\mathbf{e} \neq \mathbf{0}$, then $\mathbf{K}(\mathbf{\Sigma}_{11}^{-1/2}\mathbf{e}) = \lambda\,\mathbf{\Sigma}_{11}^{-1/2}\mathbf{e}$, and that $\mathbf{K}$ and $\mathbf{M}\mathbf{M}^\top$ have the same characteristic polynomial.
(b) Deduce that $\mathbf{K}\mathbf{g}_k = \rho_k^2\mathbf{g}_k$ and that $\mathbf{h}_k = \rho_k^{-1}\mathbf{\Sigma}_{22}^{-1}\mathbf{\Sigma}_{21}\mathbf{g}_k$ for $k \le r$.

Solution.

(a) $\mathbf{K} = \mathbf{\Sigma}_{11}^{-1/2}(\mathbf{M}\mathbf{M}^\top)\mathbf{\Sigma}_{11}^{1/2}$, because $\mathbf{M}\mathbf{M}^\top = \mathbf{\Sigma}_{11}^{-1/2}\mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1}\mathbf{\Sigma}_{21}\mathbf{\Sigma}_{11}^{-1/2}$. Hence $\mathbf{K}\mathbf{\Sigma}_{11}^{-1/2}\mathbf{e} = \mathbf{\Sigma}_{11}^{-1/2}\mathbf{M}\mathbf{M}^\top\mathbf{e} = \lambda\mathbf{\Sigma}_{11}^{-1/2}\mathbf{e}$, and $\mathbf{\Sigma}_{11}^{-1/2}\mathbf{e} \neq \mathbf{0}$ since $\mathbf{\Sigma}_{11}^{-1/2}$ is invertible. Moreover

$$|\mathbf{K} - \lambda\mathbf{I}| = |\mathbf{\Sigma}_{11}^{-1/2}(\mathbf{M}\mathbf{M}^\top - \lambda\mathbf{I})\mathbf{\Sigma}_{11}^{1/2}| = |\mathbf{M}\mathbf{M}^\top - \lambda\mathbf{I}|,$$

so the eigenvalues of the non-symmetric $\mathbf{K}$ are real and equal to $\rho_1^2, \ldots, \rho_r^2$ and zeros.

(b)

Step 1. With $\mathbf{e} = \mathbf{a}_k$ and $\lambda = d_k^2 = \rho_k^2$ in (a), $\mathbf{K}\mathbf{g}_k = \rho_k^2 \mathbf{g}_k$ for $\mathbf{g}_k = \boldsymbol{\Sigma}_{11}^{-1/2}\mathbf{a}_k$.

Step 2. From $\mathbf{M} = \sum_j d_j \mathbf{a}_j \mathbf{b}_j^\top$ and $\mathbf{a}_j^\top \mathbf{a}_k = \delta_{jk}$, $\mathbf{M}^\top \mathbf{a}_k = d_k \mathbf{b}_k$. Hence for $k \le r$

$$\mathbf{h}_k = \boldsymbol{\Sigma}_{22}^{-1/2}\mathbf{b}_k = \frac{1}{\rho_k}\boldsymbol{\Sigma}_{22}^{-1/2}\boldsymbol{\Sigma}_{22}^{-1/2}\boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1/2}\mathbf{a}_k = \frac{1}{\rho_k}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\mathbf{g}_k.$$

So one solves the eigenproblem of $\mathbf{K}$, normalizes $\mathbf{g}_k^\top \boldsymbol{\Sigma}_{11}\mathbf{g}_k = 1$, and obtains $\mathbf{h}_k$ by regression of the $\mathbf{Y}$-set on ξ_k. ■

Exercise 8.2 (Equicorrelation between and within sets; J&W 10.7)

Let $\mathbf{X} \in \mathbb{R}^p$ and $\mathbf{Y} \in \mathbb{R}^q$ be standardized with every correlation (within and between the sets) equal to $\rho \neq 0$:

$$\mathbf{R}_{11} = (1-\rho)\mathbf{I}_p + \rho \mathbf{1}_p \mathbf{1}_p^\top, \quad \mathbf{R}_{22} = (1-\rho)\mathbf{I}_q + \rho \mathbf{1}_q \mathbf{1}_q^\top, \quad \mathbf{R}_{12} = \rho \mathbf{1}_p \mathbf{1}_q^\top,$$

with the joint matrix positive definite. Find the canonical correlations and the first pair of canonical variates. Specialize to $p = q = 2$.

Solution.

Step 1 (action on 1). Let $\alpha_p = 1 + (p-1)\rho$. Then $\mathbf{R}_{11}\mathbf{1}_p = \alpha_p \mathbf{1}_p$ with $\alpha_p > 0$ (positive definiteness), so $\mathbf{1}_p$ is an eigenvector of $\mathbf{R}_{11}$ and, by Theorem 2.11, $\mathbf{R}_{11}^{-1/2}\mathbf{1}_p = \alpha_p^{-1/2}\mathbf{1}_p$. Likewise $\mathbf{R}_{22}^{-1/2}\mathbf{1}_q = \alpha_q^{-1/2}\mathbf{1}_q$.

Step 2 (the matrix M). Using symmetry of $\mathbf{R}_{22}^{-1/2}$,

$$\mathbf{M} = \mathbf{R}_{11}^{-1/2}\rho \mathbf{1}_p \mathbf{1}_q^\top \mathbf{R}_{22}^{-1/2} = \frac{\rho}{\sqrt{\alpha_p \alpha_q}} \mathbf{1}_p \mathbf{1}_q^\top = \frac{\rho\sqrt{pq}}{\sqrt{\alpha_p \alpha_q}}\, \mathbf{a}\mathbf{b}^\top, \qquad \mathbf{a} = \frac{\mathbf{1}_p}{\sqrt{p}},\ \mathbf{b} = \frac{\mathbf{1}_q}{\sqrt{q}}.$$

This is a rank-one SVD (replace $\mathbf{b}$ by $-\mathbf{b}$ if $\rho < 0$), so by Theorem 8.2

$$\rho_1 = |\rho| \sqrt{\frac{pq}{\{1+(p-1)\rho\}\{1+(q-1)\rho\}}}, \qquad \rho_2 = \cdots = 0.$$

Step 3 (variates). $\mathbf{g}_1 = \mathbf{R}_{11}^{-1/2}\mathbf{a} = \mathbf{1}_p/\sqrt{p\alpha_p}$, so $\xi_1 = \sum_i X_i/\sqrt{p\alpha_p}$, the standardized sum (indeed $\operatorname{Var}(\mathbf{1}_p^\top \mathbf{X}) = \mathbf{1}_p^\top \mathbf{R}_{11}\mathbf{1}_p = p\alpha_p$); similarly $\omega_1 = \operatorname{sign}(\rho)\sum_j Y_j/\sqrt{q\alpha_q}$. Check: $\operatorname{Corr}(\mathbf{1}^\top\mathbf{X}, \mathbf{1}^\top\mathbf{Y}) = \rho pq/\sqrt{p\alpha_p\, q\alpha_q}$, which is ρ_1 up to sign.

For $p = q = 2$: $\rho_1 = 2|\rho|/(1+\rho)$, $\xi_1 = (X_1 + X_2)/\sqrt{2(1+\rho)}$ and $\omega_1 = \operatorname{sign}(\rho)(Y_1 + Y_2)/\sqrt{2(1+\rho)}$. ■

Exercise 8.3 (Correlation for angular data; J&W 10.8)

An angle θ_2 is recorded as $\mathbf{Y} = (\cos\theta_2, \sin\theta_2)^\top$; assume $\operatorname{Cov}(\mathbf{Y}) \succ 0$.
(a) Show $\mathbf{b}^\top\mathbf{Y} = \sqrt{b_1^2 + b_2^2}\,\cos(\theta_2 - \beta)$, where $\cos\beta = b_1/\sqrt{b_1^2+b_2^2}$ and $\sin\beta = b_2/\sqrt{b_1^2+b_2^2}$.
(b) For a scalar X, show that the canonical correlation is $\rho_1 = \max_\beta \operatorname{Corr}\{X, \cos(\theta_2 - \beta)\}$.
(c) If also $\mathbf{X} = (\cos\theta_1, \sin\theta_1)^\top$, show $\rho_1 = \max_{\alpha,\beta} \operatorname{Corr}\{\cos(\theta_1 - \alpha), \cos(\theta_2 - \beta)\}$.

Solution.

(a) With $c = \sqrt{b_1^2 + b_2^2} > 0$, $\mathbf{b}^\top\mathbf{Y} = c(\cos\beta\cos\theta_2 + \sin\beta\sin\theta_2) = c\cos(\theta_2 - \beta)$.

(b)

Step 1. For scalar X and $a \neq 0$, $\mathrm{Corr}(aX, \mathbf{b}^\top \mathbf{Y}) = \mathrm{sign}(a)\mathrm{Corr}(X, \mathbf{b}^\top \mathbf{Y}) = \mathrm{Corr}(X, \mathrm{sign}(a)\mathbf{b}^\top \mathbf{Y})$, so $\rho_1 = \max_{\mathbf{b}\neq\mathbf{0}} \mathrm{Corr}(X, \mathbf{b}^\top \mathbf{Y})$.

Step 2. By (a) and scale invariance of correlation, $\mathrm{Corr}(X, \mathbf{b}^\top \mathbf{Y}) = \mathrm{Corr}\{X, \cos(\theta_2 - \beta)\}$; as $\mathbf{b}$ ranges over $\mathbb{R}^2 \setminus \{\mathbf{0}\}$, β ranges over $[0, 2\pi)$. Both variances are positive because $\mathrm{Cov}(\mathbf{Y}) \succ 0$.

Choosing the canonical variate therefore means choosing a new origin β for the angle.

(c) Apply (a) to both $\mathbf{a}^\top \mathbf{X}$ and $\mathbf{b}^\top \mathbf{Y}$ and use scale invariance; the maximum over $(\mathbf{a}, \mathbf{b})$ becomes a maximum over the two origins (α, β). ■

Exercise 8.4 (The likelihood ratio test of $\boldsymbol{\Sigma}_{12} = \mathbf{0}$)

For a sample of size $n > p + q$ from $N_{p+q}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, derive the likelihood ratio statistic for $H_0 : \boldsymbol{\Sigma}_{12} = \mathbf{0}$ and show $\Lambda^{2/n} = |\mathbf{S}|/(|\mathbf{S}_{11}||\mathbf{S}_{22}|) = \prod_k (1 - \hat{\rho}_k^2)$, with pq degrees of freedom.

Solution [Supplement].

Step 1 (unrestricted). By Theorem 4.7, $\max L = (2\pi)^{-n(p+q)/2} |\mathbf{S}_0|^{-n/2} e^{-n(p+q)/2}$.

Step 2 (under H_0). If $\boldsymbol{\Sigma}_{12} = \mathbf{0}$, then $|\boldsymbol{\Sigma}| = |\boldsymbol{\Sigma}_{11}||\boldsymbol{\Sigma}_{22}|$ and $\boldsymbol{\Sigma}^{-1} = \mathrm{diag}(\boldsymbol{\Sigma}_{11}^{-1}, \boldsymbol{\Sigma}_{22}^{-1})$, so the density of each observation factorizes and $L(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = L_1(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11}) L_2(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$, the likelihoods of the $\mathbf{X}$- and $\mathbf{Y}$-samples. The two factors have separate parameters; Theorem 4.7 applied to each gives

$$\max_{H_0} L = (2\pi)^{-n(p+q)/2} |\mathbf{S}_{0,11}|^{-n/2} |\mathbf{S}_{0,22}|^{-n/2} e^{-n(p+q)/2}.$$

Step 3 (the ratio). $\Lambda^{2/n} = |\mathbf{S}_0|/(|\mathbf{S}_{0,11}||\mathbf{S}_{0,22}|)$, and the factors $(\frac{n-1}{n})^{p+q}$ cancel, so $\Lambda^{2/n} = |\mathbf{S}|/(|\mathbf{S}_{11}||\mathbf{S}_{22}|)$. The determinant identity above turns this into $\prod_k (1 - \hat{\rho}_k^2)$.

Step 4 (degrees of freedom). H_0 removes exactly the pq free entries of $\boldsymbol{\Sigma}_{12}$, so $-2 \log \Lambda \dot{\sim} \chi^2_{pq}$. For $q = 1$ the test is the test of zero multiple correlation; with blocks of size one, Exercise 7.1 is the analogous test for p blocks. ■

Chapter 9. Factor Analysis

Model, estimation (principal components, principal factors, MLE), rotation, factor scores

1 The Orthogonal Factor Model

Motivation. Factor analysis explains correlations by a few unobserved common factors: $\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^\top + \boldsymbol{\Psi}$. Unlike PCA, it is a model with specific error variances, and the loadings are determined only up to rotation.

The covariance of p observed variables is explained by a few, k, unobserved common factors.

Definition 9.1: factor model

$$X_i = \mu_i + \sum_{\ell=1}^{k} \lambda_{i\ell} F_\ell + U_i \quad (i = 1, \dots, p), \qquad \mathbf{X} = \boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{F} + \mathbf{U},$$

where $\boldsymbol{\Lambda} = (\lambda_{i\ell})$ is the $p \times k$ **loading matrix**, $\mathbf{F} \in \mathbb{R}^k$ the common factors and $\mathbf{U} \in \mathbb{R}^p$ the specific factors, with

$$\mathrm{E}(\mathbf{F}) = \mathbf{0}, \quad \mathrm{Cov}(\mathbf{F}) = \mathbf{I}_k, \quad \mathrm{E}(\mathbf{U}) = \mathbf{0}, \quad \mathrm{Cov}(\mathbf{U}) = \boldsymbol{\Psi} = \mathrm{diag}(\psi_1, \dots, \psi_p), \quad \mathrm{Cov}(\mathbf{F}, \mathbf{U}) = \mathbf{0}.$$

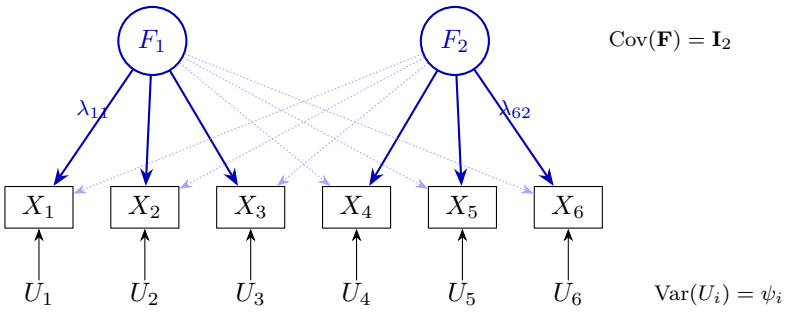


Figure. *Path diagram of a two-factor model for six variables. Every factor may load on every variable (dotted arrows carry small loadings in a simple structure), each variable has its own specific factor U_i, and the U_i are uncorrelated with each other and with $\mathbf{F}$. So any correlation between two X's must pass through the common factors:* $\mathrm{Cov}(X_i, X_j) = \sum_\ell \lambda_{i\ell}\lambda_{j\ell}$.

Theorem 9.2: covariance structure

$$\mathrm{E}(\mathbf{X}) = \boldsymbol{\mu}, \qquad \mathrm{Cov}(\mathbf{X}) = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}, \qquad \mathrm{Cov}(\mathbf{X}, \mathbf{F}) = \boldsymbol{\Lambda},$$

$$\mathrm{Var}(X_i) = \underbrace{\sum_{\ell=1}^{k} \lambda_{i\ell}^2}_{h_i^2:\ \text{communality}} + \underbrace{\psi_i}_{\text{specific variance}}, \qquad \mathrm{Cov}(X_i, X_j) = \sum_{\ell} \lambda_{i\ell}\lambda_{j\ell} \ (i \neq j).$$

Proof. $\mathrm{Cov}(\boldsymbol{\Lambda}\mathbf{F} + \mathbf{U}) = \boldsymbol{\Lambda}\mathrm{Cov}(\mathbf{F})\boldsymbol{\Lambda}^\top + \mathrm{Cov}(\mathbf{U}) + \boldsymbol{\Lambda}\mathrm{Cov}(\mathbf{F}, \mathbf{U}) + \mathrm{Cov}(\mathbf{U}, \mathbf{F})\boldsymbol{\Lambda}^\top = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}$, and $\mathrm{Cov}(\mathbf{X}, \mathbf{F}) = \mathrm{Cov}(\boldsymbol{\Lambda}\mathbf{F} + \mathbf{U}, \mathbf{F}) = \boldsymbol{\Lambda}$. ■

Because $\boldsymbol{\Psi}$ is diagonal, covariances between variables come *entirely* from the common factors. The model explains the $p(p+1)/2$ entries of $\boldsymbol{\Sigma}$ with $pk + p$ parameters.

1.1 Non-identifiability and rotation

For orthogonal $\mathbf{T}$ $(k \times k)$, $\mathbf{\Lambda}^* = \mathbf{\Lambda}\mathbf{T}$ and $\mathbf{F}^* = \mathbf{T}^\top \mathbf{F}$ give $\mathbf{\Lambda}^*\mathbf{F}^* = \mathbf{\Lambda}\mathbf{F}$, $\mathrm{Cov}\mathbf{F}^* = \mathbf{I}_k$ and $\mathbf{\Lambda}^*\mathbf{\Lambda}^{*\top} = \mathbf{\Lambda}\mathbf{\Lambda}^\top$: loadings are determined only up to rotation. If $\hat{\mathbf{\Lambda}}$ is an MLE, so is $\hat{\mathbf{\Lambda}}\mathbf{R}$ for any orthogonal $\mathbf{R}$. Removing the $k(k-1)/2$ rotational degrees of freedom leaves $pk + p - k(k-1)/2$ parameters and

$$\nu = \frac{p(p+1)}{2} - \left\{pk + p - \frac{k(k-1)}{2}\right\} = \frac{(p-k)^2 - p - k}{2},$$

which must be positive for the model to impose a real restriction.

1.2 Invariance to scale changes

If $Y_i = c_i X_i$, i.e. $\mathbf{Y} = \mathrm{diag}(\mathbf{c})\mathbf{X}$, then $\mathbf{Y} = \mathrm{diag}(\mathbf{c})\boldsymbol{\mu} + \mathrm{diag}(\mathbf{c})\mathbf{\Lambda}\mathbf{F} + \mathrm{diag}(\mathbf{c})\mathbf{U}$ is again a factor model with $\mathbf{\Lambda}_Y = \mathrm{diag}(\mathbf{c})\mathbf{\Lambda}$ and $\mathbf{\Psi}_Y = \mathrm{diag}(\mathbf{c})^2\mathbf{\Psi}$. For standardized variables ($\mathrm{Var}X_i = 1$),

$$\lambda_{i\ell} = \mathrm{Cov}(X_i, F_\ell) = \mathrm{Corr}(X_i, F_\ell), \qquad \mathrm{Corr}(Y_i, F_\ell) = \lambda_{i\ell},$$

so loadings are read as variable–factor correlations. In practice factor analysis is often run on the correlation matrix $\mathbf{R}$.

2 Estimation

2.1 Principal component method

From $\widehat{\mathrm{Corr}}(\mathbf{X}) = \mathbf{R} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2} = \sum_{j=1}^{p} l_j \mathbf{u}_j \mathbf{u}_j^\top \approx \sum_{j=1}^{k} l_j \mathbf{u}_j \mathbf{u}_j^\top$ (largest k eigenvalues),

$$\hat{\mathbf{\Lambda}} = [\sqrt{l_1}\mathbf{u}_1, \ldots, \sqrt{l_k}\mathbf{u}_k], \qquad \hat{\mathbf{\Psi}} = \mathrm{diag}(\mathbf{R} - \hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top).$$

The diagonal of $\mathbf{R}$ is reproduced exactly, and the sum of squared off-diagonal residuals of $\mathbf{R} - \hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top - \hat{\mathbf{\Psi}}$ is at most $l_{k+1}^2 + \cdots + l_p^2$. Factor ℓ explains a proportion l_ℓ/p of the total variance. Since eigenvalues do not change, increasing k leaves earlier loadings unchanged.

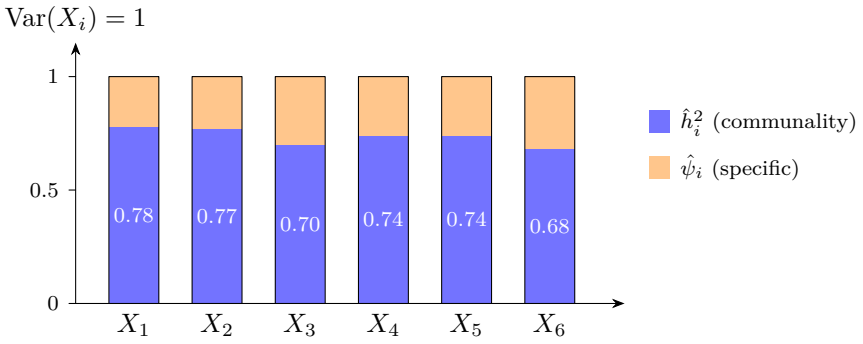


Figure. *Principal component solution with $k = 2$ for a synthetic correlation matrix ($n = 300$, $p = 6$; $l_1 = 2.43$, $l_2 = 1.96$, $l_3 = 0.48$). Each unit bar is $1 = \hat{h}_i^2 + \hat{\psi}_i$: the diagonal of $\mathbf{R}$ is reproduced exactly, and the two factors explain $(l_1 + l_2)/p = 73\%$ of the total standardized variance.*

Remark 9.3: number of factors with the principal component method

Heuristics are used when PCA is used to estimate the factor loadings:

1. use a **scree plot** or a **cumulative scree plot** of $l_1 \geq l_2 \geq \cdots \geq l_p$;

2. or use **Kaiser's rule**: retain factors $1, \dots, k$ satisfying

$$l_k > \bar{l} = \frac{1}{p}\sum_{i=1}^{p} l_i.$$

When correlation PCA is used, $\sum_i l_i = \mathrm{tr}\mathbf{R} = p$, so $\bar{l} = 1$ and the rule keeps factors with $l_k > 1$ (a factor must explain more than one standardized variable's worth of variance).

These are the same rules as for choosing the number of principal components (Ch. 7); the likelihood ratio test of §2.4 is available only under the normal MLE approach.

2.2 Principal factor method

Starting from initial communality estimates $\hat{h}_i^2$ (e.g. $1 - 1/r^{ii}$, r^{ii} the diagonal of $\mathbf{R}^{-1}$), form the reduced correlation matrix $\mathbf{R} - \hat{\boldsymbol{\Psi}}$ and iterate

1. $\mathbf{R} - \hat{\boldsymbol{\Psi}} \approx \hat{\boldsymbol{\Lambda}}_{(k)}\hat{\boldsymbol{\Lambda}}_{(k)}^{\top}$ (top-k eigendecomposition of the reduced matrix);
2. $\hat{\psi}_{(k)i} = 1 - \sum_{\ell=1}^{k} \hat{\lambda}_{(k)i\ell}^2$

until convergence.

2.3 Maximum likelihood

Assume $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Lambda}\boldsymbol{\Lambda}^{\top} + \boldsymbol{\Psi})$. Profiling $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$, twice the negative log-likelihood is, up to constants and a factor n,

$$\ell(\hat{\boldsymbol{\mu}}, \boldsymbol{\Lambda}, \boldsymbol{\Psi}) = \log|\boldsymbol{\Lambda}\boldsymbol{\Lambda}^{\top} + \boldsymbol{\Psi}| + \mathrm{tr}\{\mathbf{S}_0(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^{\top} + \boldsymbol{\Psi})^{-1}\} + c.$$

For identifiability impose that $\boldsymbol{\Delta} = \boldsymbol{\Lambda}^{\top}\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}$ is diagonal (if $\hat{\boldsymbol{\Lambda}}$ is an MLE, so is $\hat{\boldsymbol{\Lambda}}\mathbf{R}$).

Theorem 9.4: ($*$) $\boldsymbol{\Lambda}$ given $\boldsymbol{\Psi}$

Given $\boldsymbol{\Psi} = \mathrm{diag}(\psi_1, \dots, \psi_p)$, let $\mathbf{A} = \boldsymbol{\Psi}^{-1/2}\mathbf{S}_0\boldsymbol{\Psi}^{-1/2} = \sum_{i=1}^{p} d_i \mathbf{e}_i\mathbf{e}_i^{\top}$ ($d_1 \geq \cdots \geq d_p$) and $\delta_i = d_i - 1 > 0$ ($i \leq k$). Then

$$\hat{\boldsymbol{\Lambda}} = \boldsymbol{\Psi}^{1/2}[\mathbf{e}_1, \dots, \mathbf{e}_k]\mathrm{diag}(\delta_1^{1/2}, \dots, \delta_k^{1/2}).$$

Proof [Supplement]. Let $\boldsymbol{\Lambda}^* = \boldsymbol{\Psi}^{-1/2}\boldsymbol{\Lambda}$ and $\boldsymbol{\Sigma}^* = \boldsymbol{\Psi}^{-1/2}\boldsymbol{\Sigma}\boldsymbol{\Psi}^{-1/2} = \boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top} + \mathbf{I}_p$. Then $\ell = \log|\boldsymbol{\Psi}| + \log|\boldsymbol{\Sigma}^*| + \mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma}^{*-1})$. If $\boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top}$ has eigenvalues $\gamma_1 \geq \cdots \geq \gamma_k \geq 0 = \cdots = 0$, $\boldsymbol{\Sigma}^*$ has eigenvalues $1 + \gamma_j$ and 1. Von Neumann's lower bound (pair the small eigenvalues of $\boldsymbol{\Sigma}^{*-1}$ with the large ones of $\mathbf{A}$) gives

$$\log|\boldsymbol{\Sigma}^*| + \mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma}^{*-1}) \geq \sum_{j=1}^{k}\left\{\log(1+\gamma_j) + \frac{d_j}{1+\gamma_j}\right\} + \sum_{j>k} d_j,$$

with equality when the eigenvectors of $\boldsymbol{\Sigma}^*$ coincide with the $\mathbf{e}_j$ in the same order. Each term $\log x + d/x = \log d + \{d/x - \log(d/x)\} \geq \log d + 1$ (Theorem 2.14) is minimized at $x = d$, so $1 + \gamma_j = d_j$, i.e. $\gamma_j = \delta_j$. Hence $\boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top} = \sum_{j \leq k} \delta_j \mathbf{e}_j\mathbf{e}_j^{\top}$ and $\boldsymbol{\Lambda}^* = [\mathbf{e}_1, \dots, \mathbf{e}_k]\mathrm{diag}(\delta_j^{1/2})$ up to rotation; this choice makes $\boldsymbol{\Lambda}^{*\top}\boldsymbol{\Lambda}^* = \boldsymbol{\Delta}$ diagonal. ■

Theorem 9.5: (**) $\boldsymbol{\Psi}$ given $\boldsymbol{\Lambda}$

The MLE satisfies $\operatorname{diag}(\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}}) = \operatorname{diag}(\mathbf{S}_0)$ (the model reproduces the sample variances). Accordingly, given $\boldsymbol{\Lambda}$, update

$$\hat{\boldsymbol{\Psi}} = \operatorname{diag}(\hat{\psi}_1, \dots, \hat{\psi}_p), \qquad \hat{\psi}_i = (\mathbf{S}_0 - \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top)_{ii}.$$

Alternate the two steps. If some $\hat{\psi}_i \le 0$ (a Heywood case), bound it below by a small positive number.

Logic for (*) and (**): the likelihood equations

We evaluate $\dfrac{\partial \ell(\boldsymbol{\Lambda}, \boldsymbol{\Psi})}{\partial \boldsymbol{\Psi}} = 0$ and $\dfrac{\partial \ell(\boldsymbol{\Lambda}, \boldsymbol{\Psi})}{\partial \boldsymbol{\Lambda}} = 0$ for

$$\ell(\boldsymbol{\Lambda}, \boldsymbol{\Psi}) = \log|\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}| + \operatorname{tr}\{\mathbf{S}_0(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})^{-1}\}, \qquad \boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi},$$

using the facts that for a matrix $\mathbf{A} = \mathbf{A}(x)$,

$$\frac{\partial \mathbf{A}^{-1}}{\partial x} = -\mathbf{A}^{-1}\frac{\partial \mathbf{A}}{\partial x}\mathbf{A}^{-1}, \qquad \frac{\partial \log|\mathbf{A}|}{\partial x} = \operatorname{tr}\Big(\mathbf{A}^{-1}\frac{\partial \mathbf{A}}{\partial x}\Big)$$

(the second is the scalar form of Theorem 2.22; the first follows from differentiating $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$).

Derivative in ψ_i. With $\mathbf{e}_i$ the ith unit vector, $\frac{\partial}{\partial \psi_i}\operatorname{diag}(\psi_1, \dots, \psi_p) = \mathbf{e}_i\mathbf{e}_i^\top$, so $\frac{\partial \boldsymbol{\Sigma}}{\partial \psi_i} = \mathbf{e}_i\mathbf{e}_i^\top$ and

$$\begin{aligned}\frac{\partial \ell}{\partial \psi_i} &= \underbrace{\operatorname{tr}\Big(\boldsymbol{\Sigma}^{-1}\frac{\partial \boldsymbol{\Sigma}}{\partial \psi_i}\Big)}_{\partial \log|\boldsymbol{\Sigma}|/\partial\psi_i} + \underbrace{\operatorname{tr}\Big(\mathbf{S}_0\frac{\partial \boldsymbol{\Sigma}^{-1}}{\partial \psi_i}\Big)}_{\partial \operatorname{tr}(\mathbf{S}_0\boldsymbol{\Sigma}^{-1})/\partial\psi_i} = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{e}_i\mathbf{e}_i^\top) - \operatorname{tr}(\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\mathbf{e}_i\mathbf{e}_i^\top\boldsymbol{\Sigma}^{-1}) \\ &= \mathbf{e}_i^\top\boldsymbol{\Sigma}^{-1}\mathbf{e}_i - \mathbf{e}_i^\top\boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\mathbf{e}_i = \mathbf{e}_i^\top(\boldsymbol{\Sigma}^{-1} - \boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1})\mathbf{e}_i = 0, \qquad i = 1, \dots, p.\end{aligned}$$

$$\textcircled{1} \qquad \operatorname{diag}(\boldsymbol{\Sigma}^{-1} - \boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1}) = \mathbf{0}.$$

Derivative in λ_{ij}. With $\boldsymbol{\Lambda} = [\mathbf{l}_1, \dots, \mathbf{l}_k]$ (columns), $\dfrac{\partial \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top}{\partial \lambda_{ij}} = \mathbf{l}_j\mathbf{e}_i^\top + \mathbf{e}_i\mathbf{l}_j^\top$. Then

$$\begin{aligned}\frac{\partial \ell}{\partial \lambda_{ij}} &= \operatorname{tr}\Big(\boldsymbol{\Sigma}^{-1}\frac{\partial \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top}{\partial \lambda_{ij}}\Big) - \operatorname{tr}\Big(\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\frac{\partial \boldsymbol{\Sigma}}{\partial \lambda_{ij}}\boldsymbol{\Sigma}^{-1}\Big) \\ &= \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{l}_j\mathbf{e}_i^\top) + \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{e}_i\mathbf{l}_j^\top) - \operatorname{tr}(\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\mathbf{l}_j\mathbf{e}_i^\top\boldsymbol{\Sigma}^{-1}) - \operatorname{tr}(\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\mathbf{e}_i\mathbf{l}_j^\top\boldsymbol{\Sigma}^{-1}) \\ &= 2\mathbf{e}_i^\top\boldsymbol{\Sigma}^{-1}\mathbf{l}_j - 2\mathbf{e}_i^\top\boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\mathbf{l}_j = 0 \qquad (i = 1, \dots, p,\ j = 1, \dots, k),\end{aligned}$$

using symmetry of $\boldsymbol{\Sigma}^{-1}$ and $\boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1}$ (each pair of traces is equal). Collecting over i, j:

$$\boldsymbol{\Sigma}^{-1}\boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\boldsymbol{\Lambda} \overset{\text{premultiply } \boldsymbol{\Sigma}}{\Longrightarrow} \boldsymbol{\Lambda} = \mathbf{S}_0\boldsymbol{\Sigma}^{-1}\boldsymbol{\Lambda} = \mathbf{S}_0(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})^{-1}\boldsymbol{\Lambda} = \mathbf{S}_0\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}(\mathbf{I} + \boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda})^{-1},$$

where the last step uses $\boldsymbol{\Lambda}(\mathbf{I} + \boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}) = (\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}$, i.e. $(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})^{-1}\boldsymbol{\Lambda} = \boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}(\mathbf{I} + \boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda})^{-1}$.

Write $\boldsymbol{\Delta} \equiv \boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}$ (and require $\boldsymbol{\Delta}_{k\times k}$ to be diagonal). Then

$$\textcircled{2}' \qquad \boldsymbol{\Lambda}(\mathbf{I} + \boldsymbol{\Delta}) = \mathbf{S}_0\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}.$$

Premultiplying by $\boldsymbol{\Psi}^{-1/2}$:

$$\textcircled{2} \qquad \underbrace{\boldsymbol{\Psi}^{-1/2}\mathbf{S}_0\boldsymbol{\Psi}^{-1/2}}_{=:\mathbf{A}}\ \underbrace{\boldsymbol{\Psi}^{-1/2}\boldsymbol{\Lambda}}_{=:\tilde{\mathbf{E}}} = \underbrace{\boldsymbol{\Psi}^{-1/2}\boldsymbol{\Lambda}}_{\tilde{\mathbf{E}}}\ \underbrace{(\mathbf{I} + \boldsymbol{\Delta})}_{=:\mathbf{D}=\operatorname{diag}(d_1,\dots,d_k)}, \qquad \tilde{\mathbf{E}} = [\tilde{\mathbf{e}}_1, \dots, \tilde{\mathbf{e}}_k].$$

Hence $\mathbf{A}\tilde{\mathbf{e}}_i = \tilde{\mathbf{e}}_i d_i$, $i = 1, \dots, k$: $(\tilde{\mathbf{e}}_i, d_i)$ is an eigenpair of $\mathbf{A}$, although $\tilde{\mathbf{e}}_i^\top \tilde{\mathbf{e}}_i \neq 1$ in general. Since $\tilde{\mathbf{e}}_i^\top \tilde{\mathbf{e}}_i = (\tilde{\mathbf{E}}^\top \tilde{\mathbf{E}})_{ii} = (\mathbf{\Lambda}^\top \mathbf{\Psi}^{-1} \mathbf{\Lambda})_{ii} = \Delta_{ii}$, let $\mathbf{e}_i \equiv \tilde{\mathbf{e}}_i / \Delta_{ii}^{1/2}$ so that $\mathbf{e}_i^\top \mathbf{e}_i = 1$ and, for $\mathbf{E} = [\mathbf{e}_1, \dots, \mathbf{e}_k]$, $\mathbf{E}^\top \mathbf{E} = \mathbf{I}_k$ ($\mathbf{\Delta}$ diagonal). Then $(\mathbf{e}_i, d_i)$ is a unit eigenvector–eigenvalue pair of $\mathbf{A}$, and

$$\mathbf{D} = \mathbf{I} + \mathbf{\Delta}, \qquad \tilde{\mathbf{E}} = \mathbf{\Psi}^{-1/2} \mathbf{\Lambda} = \mathbf{E} \mathbf{\Delta}^{1/2}.$$

$(*)$ **$\mathbf{\Psi}$ is fixed.** From ②, $\mathbf{A} = \mathbf{\Psi}^{-1/2} \mathbf{S}_0 \mathbf{\Psi}^{-1/2}$ is computable, with eigenpairs $(\mathbf{e}_i(\mathbf{A}), d_i(\mathbf{A}))$. Use $\mathbf{D} = \mathbf{I} + \mathbf{\Delta}$ to set $\hat{\mathbf{\Delta}} = \mathrm{diag}(d_i(\mathbf{A})) - \mathbf{I}$, and $\mathbf{E}\mathbf{\Delta}^{1/2} = \mathbf{\Psi}^{-1/2}\mathbf{\Lambda}$ to set

$$\hat{\mathbf{\Lambda}} = \mathbf{\Psi}^{1/2} \mathbf{E}(\mathbf{A}) \hat{\mathbf{\Delta}}^{1/2}, \qquad \mathbf{E}(\mathbf{A}) = [\mathbf{e}_1(\mathbf{A}), \dots, \mathbf{e}_k(\mathbf{A})],$$

taking the k largest eigenvalues (the von Neumann argument above shows these give the minimum).

$(**)$ **$\mathbf{\Lambda}$ is fixed.** From ①, $\mathrm{diag}(\mathbf{\Sigma}^{-1} - \mathbf{\Sigma}^{-1}\mathbf{S}_0\mathbf{\Sigma}^{-1}) = \mathbf{0}$.

Claim: if ②′ holds, then

$$\mathbf{\Sigma}^{-1} - \mathbf{\Sigma}^{-1}\mathbf{S}_0\mathbf{\Sigma}^{-1} = \mathbf{\Sigma}^{-1}(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Sigma}^{-1} = \mathbf{\Psi}^{-1}(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Psi}^{-1} = \mathbf{\Psi}^{-1}(\mathbf{\Lambda}\mathbf{\Lambda}^\top - \mathbf{S}_0 + \mathbf{\Psi})\mathbf{\Psi}^{-1}.$$

Proof of claim [Supplement]. $\mathbf{\Sigma}\mathbf{\Psi}^{-1}\mathbf{\Lambda} = (\mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi})\mathbf{\Psi}^{-1}\mathbf{\Lambda} = \mathbf{\Lambda}(\mathbf{I} + \mathbf{\Delta})$, and by ②′ also $\mathbf{S}_0\mathbf{\Psi}^{-1}\mathbf{\Lambda} = \mathbf{\Lambda}(\mathbf{I} + \mathbf{\Delta})$, so $(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Psi}^{-1}\mathbf{\Lambda} = \mathbf{0}$. By Woodbury, $\mathbf{\Sigma}^{-1} = \mathbf{\Psi}^{-1} - \mathbf{\Psi}^{-1}\mathbf{\Lambda}(\mathbf{I} + \mathbf{\Delta})^{-1}\mathbf{\Lambda}^\top\mathbf{\Psi}^{-1}$, hence

$$(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Sigma}^{-1} = (\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Psi}^{-1} - \underbrace{(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Psi}^{-1}\mathbf{\Lambda}}_{=\mathbf{0}}(\mathbf{I} + \mathbf{\Delta})^{-1}\mathbf{\Lambda}^\top\mathbf{\Psi}^{-1} = (\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Psi}^{-1},$$

and transposing, $\mathbf{\Sigma}^{-1}(\mathbf{\Sigma} - \mathbf{S}_0) = \mathbf{\Psi}^{-1}(\mathbf{\Sigma} - \mathbf{S}_0)$. Therefore $\mathbf{\Sigma}^{-1}(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Sigma}^{-1} = \mathbf{\Sigma}^{-1}(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Psi}^{-1} = \mathbf{\Psi}^{-1}(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Psi}^{-1}$.

Since $\mathbf{\Psi}^{-1}$ is diagonal, the ith diagonal entry of $\mathbf{\Psi}^{-1}(\mathbf{\Lambda}\mathbf{\Lambda}^\top - \mathbf{S}_0 + \mathbf{\Psi})\mathbf{\Psi}^{-1}$ is $(\mathbf{\Lambda}\mathbf{\Lambda}^\top - \mathbf{S}_0 + \mathbf{\Psi})_{ii}/\psi_i^2$, so ① becomes

$$\mathrm{diag}(\mathbf{\Lambda}\mathbf{\Lambda}^\top - \mathbf{S}_0 + \mathbf{\Psi}) = \mathbf{0} \quad \Longrightarrow \quad \psi_i = (\mathbf{S}_0 - \mathbf{\Lambda}\mathbf{\Lambda}^\top)_{ii},$$

which is the update $(**)$ and also shows $\mathrm{diag}(\hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top + \hat{\mathbf{\Psi}}) = \mathrm{diag}(\mathbf{S}_0)$ at the MLE.

2.4 Likelihood ratio test for the number of factors

$H_0 : \mathbf{\Sigma} = \mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi}$ (k factors) versus H_1: unrestricted $\mathbf{\Sigma}$. With $L_1 = L(\bar{\mathbf{x}}, \mathbf{S}_0)$ and $L_0 = L(\bar{\mathbf{x}}, \hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top + \hat{\mathbf{\Psi}})$,

$$W = -2 \log \frac{L_0}{L_1} = n \log\Big(\frac{|\hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top + \hat{\mathbf{\Psi}}|}{|\mathbf{S}_0|}\Big) \;\dot{\sim}\; \chi^2_\nu,$$

$$\nu = \frac{p(p+1)}{2} - \Big\{pk + p - \frac{k(k-1)}{2}\Big\} = \frac{(p-k)^2 - p - k}{2}.$$

(At the MLE $\mathrm{tr}\{\mathbf{S}_0\hat{\mathbf{\Sigma}}^{-1}\} = p$, so the trace terms cancel.) With Bartlett's correction, reject the k-factor model if $(n - 1 - \frac{2p+4k+5}{6}) \ln \frac{|\hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top + \hat{\mathbf{\Psi}}|}{|\mathbf{S}_0|} > \chi^2_\nu(\alpha)$.

3 Factor Rotation

Non-identifiability is used to find interpretable loadings, ideally a simple structure in which each variable loads heavily on only one factor. Rotate $\mathbf{\Lambda}^* = \mathbf{\Lambda}\mathbf{J}$ with $\mathbf{J}$ orthogonal.

Definition 9.6: varimax and quartimax

Let $\tilde{\lambda}^*_{i\ell} = \lambda^*_{i\ell}/h_i$ (normalized by the communality).

$$\text{Varimax: } \mathbf{J}^\dagger = \arg\max_{\mathbf{J}} \sum_{\ell=1}^{k} \widehat{\text{Var}}\big((\tilde{\lambda}^*_{1\ell})^2, \dots, (\tilde{\lambda}^*_{p\ell})^2\big)$$

— spread the squared loadings within each *column* (factor), so each factor has some large and some near-zero loadings.

$$\text{Quartimax: } \mathbf{J}^\dagger = \arg\max_{\mathbf{J}} \sum_{i=1}^{p} \widehat{\text{Var}}\big((\lambda^*_{i1})^2, \dots, (\lambda^*_{ik})^2\big)$$

— spread the squared loadings within each *row* (variable), so each variable concentrates on one factor.

Rotation leaves the communalities h_i^2 and $\hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top$ unchanged.

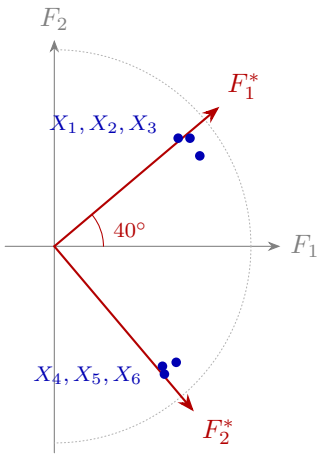


Figure. *Loadings of the six variables of the previous figure on the unrotated principal component factors (blue). Every variable loads about* 0.6 *on* F_1*, so* F_1 *looks like a general factor. Varimax rotates the axes by* 40° *(red); in the new coordinates the loadings are about* $(0.86, 0)$ *for variables 1–3 and* $(0, 0.85)$ *for variables 4–6, a simple structure. The dotted unit circle bounds the communalities, which rotation does not change.*

4 Factor Score Prediction

4.1 Weighted least squares (Bartlett) scores

(1) Choose $\hat{\mathbf{F}}$ to minimize the distance between $\mathbf{X}$ and $\boldsymbol{\mu} + \mathbf{\Lambda F}$: with the estimates fixed, regard $\mathbf{x}_i - \bar{\mathbf{x}} = \hat{\mathbf{\Lambda}}\mathbf{F}_i + \mathbf{U}_i$ as a heteroscedastic regression,

$$\hat{\mathbf{F}}_i^{\text{wls}} = \arg\min_{\mathbf{F}_i} (\mathbf{x}_i - \bar{\mathbf{x}} - \hat{\mathbf{\Lambda}}\mathbf{F}_i)^\top \hat{\mathbf{\Psi}}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}} - \hat{\mathbf{\Lambda}}\mathbf{F}_i) = (\hat{\mathbf{\Lambda}}^\top \hat{\mathbf{\Psi}}^{-1} \hat{\mathbf{\Lambda}})^{-1} \hat{\mathbf{\Lambda}}^\top \hat{\mathbf{\Psi}}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}})$$

(set the gradient of the quadratic form to zero).

4.2 Regression scores

(2) Since $\begin{pmatrix} \mathbf{X} \\ \mathbf{F} \end{pmatrix} \sim N_{p+k}\Big(\begin{pmatrix} \boldsymbol{\mu} \\ \mathbf{0} \end{pmatrix}, \begin{pmatrix} \mathbf{\Lambda\Lambda}^\top + \mathbf{\Psi} & \mathbf{\Lambda} \\ \mathbf{\Lambda}^\top & \mathbf{I}_k \end{pmatrix} \Big)$, the conditional expectation (Theorem 3.11) gives

$$\hat{\mathbf{F}}_i^{\text{reg}} = \text{E}(\mathbf{F} \mid \mathbf{X} = \mathbf{x}_i) = \hat{\mathbf{\Lambda}}^\top (\hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top + \hat{\mathbf{\Psi}})^{-1} (\mathbf{x}_i - \bar{\mathbf{x}}).$$

Theorem 9.7: relation between the two scores

With $\boldsymbol{\Delta} = \boldsymbol{\Lambda}^\top \boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}$, $\hat{\mathbf{F}}^{\text{reg}} = (\mathbf{I}+\boldsymbol{\Delta})^{-1}\boldsymbol{\Delta}\,\hat{\mathbf{F}}^{\text{wls}} = (\mathbf{I}+\boldsymbol{\Delta}^{-1})^{-1}\hat{\mathbf{F}}^{\text{wls}}$. If $\boldsymbol{\Delta}$ is diagonal each component is shrunk by $\delta_\ell/(1+\delta_\ell) < 1$, so $|\hat{F}_\ell^{\text{wls}}| \geq |\hat{F}_\ell^{\text{reg}}|$.

Proof [Supplement]. By Woodbury, $(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})^{-1} = \boldsymbol{\Psi}^{-1} - \boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}(\mathbf{I}+\boldsymbol{\Delta})^{-1}\boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}$, so

$$\boldsymbol{\Lambda}^\top(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})^{-1} = \{\mathbf{I} - \boldsymbol{\Delta}(\mathbf{I}+\boldsymbol{\Delta})^{-1}\}\boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1} = (\mathbf{I}+\boldsymbol{\Delta})^{-1}\boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}$$

(as $\mathbf{I} - \boldsymbol{\Delta}(\mathbf{I}+\boldsymbol{\Delta})^{-1} = (\mathbf{I}+\boldsymbol{\Delta}-\boldsymbol{\Delta})(\mathbf{I}+\boldsymbol{\Delta})^{-1}$), while $\boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}(\mathbf{x}-\bar{\mathbf{x}}) = \boldsymbol{\Delta}\hat{\mathbf{F}}^{\text{wls}}$. ■

Regression scores shrink towards 0, reflecting the prior $\mathbf{F} \sim N(\mathbf{0}, \mathbf{I})$ (a Bayes estimate); WLS scores treat $\mathbf{F}$ as a fixed parameter and are conditionally unbiased.

5 PCA versus Factor Analysis

[Supplement] The table collects the differences met in Ch. 7 and in this chapter.

	PCA	Factor analysis
Goal	linear combinations explaining total variance $\text{tr}\boldsymbol{\Sigma}$	latent factors explaining covariances (off-diagonal)
Model	none (descriptive transformation)	$\boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}$
Direction	variables $\to$ components ($\mathbf{Z} = \mathbf{U}^\top\mathbf{X}$)	factors $\to$ variables ($\mathbf{X} = \boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{F} + \mathbf{U}$)
Uniqueness	unique if eigenvalues distinct	only up to rotation
Increasing k	earlier components unchanged	(MLE) earlier loadings change

6 Selected exercises

Motivation. The exercises prove two claims made in passing above, the residual bound of the principal component method and the simplification of the likelihood ratio statistic, and show by the smallest example why a factor model need not be identified.

Exercise 9.1 (Residuals of the principal component method; J&W 9.5)

Let $\mathbf{R} = \sum_{j=1}^{p} l_j \mathbf{u}_j\mathbf{u}_j^\top$ ($l_1 \geq \cdots \geq l_p$), $\hat{\boldsymbol{\Lambda}} = [\sqrt{l_1}\mathbf{u}_1, \ldots, \sqrt{l_k}\mathbf{u}_k]$, $\hat{\boldsymbol{\Psi}} = \text{diag}(\mathbf{R} - \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top)$ and $\mathbf{E} = \mathbf{R} - \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top - \hat{\boldsymbol{\Psi}}$. Show

$$\sum_{i,j} E_{ij}^2 \leq l_{k+1}^2 + \cdots + l_p^2.$$

Solution.

Step 1 (zero diagonal). By the definition of $\hat{\boldsymbol{\Psi}}$, $E_{ii} = 0$, and $E_{ij} = (\mathbf{R} - \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top)_{ij}$ for $i \neq j$. Hence

$$\|\mathbf{E}\|_F^2 = \sum_{i\neq j}(\mathbf{R} - \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top)_{ij}^2 \leq \|\mathbf{R} - \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top\|_F^2.$$

Step 2 (the discarded part). $\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top = \sum_{j\leq k} l_j\mathbf{u}_j\mathbf{u}_j^\top$, so $\mathbf{R} - \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top = \mathbf{P}_2\mathbf{L}_2\mathbf{P}_2^\top$ with $\mathbf{P}_2 = [\mathbf{u}_{k+1}, \ldots, \mathbf{u}_p]$, $\mathbf{P}_2^\top\mathbf{P}_2 = \mathbf{I}$ and $\mathbf{L}_2 = \text{diag}(l_{k+1}, \ldots, l_p)$.

Step 3. By Theorem 2.3(ii)–(iii), $\|\mathbf{P}_2\mathbf{L}_2\mathbf{P}_2^\top\|_F^2 = \text{tr}(\mathbf{P}_2\mathbf{L}_2\mathbf{P}_2^\top\mathbf{P}_2\mathbf{L}_2\mathbf{P}_2^\top) = \text{tr}(\mathbf{L}_2^2\mathbf{P}_2^\top\mathbf{P}_2) = \sum_{j>k} l_j^2$.

The same argument applies with $\mathbf{S}$ in place of $\mathbf{R}$. ■

Exercise 9.2 (Non-uniqueness for $p = 2$, $k = 1$; J&W 9.7)

For $p = 2$ and $k = 1$ the model reads $\sigma_{11} = \lambda_1^2 + \psi_1$, $\sigma_{12} = \lambda_1\lambda_2$, $\sigma_{22} = \lambda_2^2 + \psi_2$. Given $\mathbf{\Sigma} \succ 0$ with $\sigma_{12} \neq 0$, find all solutions with $\psi_1, \psi_2 \geq 0$ and show that there are infinitely many. Relate this to the sign of ν.

Solution.

Step 1. $\sigma_{12} \neq 0$ forces $\lambda_1 = t \neq 0$ and $\lambda_2 = \sigma_{12}/t$; then $\psi_1 = \sigma_{11} - t^2$ and $\psi_2 = \sigma_{22} - \sigma_{12}^2/t^2$ are determined.

Step 2. $\psi_1 \geq 0$ and $\psi_2 \geq 0$ iff $\sigma_{12}^2/\sigma_{22} \leq t^2 \leq \sigma_{11}$. Since $\mathbf{\Sigma} \succ 0$, $\sigma_{12}^2 < \sigma_{11}\sigma_{22}$, so this is a non-degenerate interval for t^2, and every t in it (with either sign) gives a valid $(\mathbf{\Lambda}, \mathbf{\Psi})$.

Step 3. For $k = 1$ the only orthogonal $\mathbf{T}$ is ± 1, so rotation explains just the sign of t; the continuum of solutions is a genuine lack of identification. This matches $\nu = \{(p-k)^2 - p - k\}/2 = (1-3)/2 = -1 < 0$: three equations for four parameters. ■

Exercise 9.3 (Simplifying the likelihood ratio statistic)

Let $(\hat{\mathbf{\Lambda}}, \hat{\mathbf{\Psi}})$, $\hat{\mathbf{\Psi}} \succ 0$, solve the likelihood equations ① and ②′, and $\hat{\mathbf{\Sigma}} = \hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top + \hat{\mathbf{\Psi}}$. Show $\operatorname{tr}(\mathbf{S}_0\hat{\mathbf{\Sigma}}^{-1}) = p$, and deduce $W = -2\log(L_0/L_1) = n\log(|\hat{\mathbf{\Sigma}}|/|\mathbf{S}_0|)$.

Solution [Supplement].

Step 1. The proof of the Claim above shows that ②′ implies $\hat{\mathbf{\Sigma}}^{-1}(\hat{\mathbf{\Sigma}} - \mathbf{S}_0) = \hat{\mathbf{\Psi}}^{-1}(\hat{\mathbf{\Sigma}} - \mathbf{S}_0)$, and ① then gives $\operatorname{diag}(\hat{\mathbf{\Sigma}}) = \operatorname{diag}(\mathbf{S}_0)$.

Step 2. Since $\hat{\mathbf{\Psi}}^{-1}$ is diagonal,

$$p - \operatorname{tr}(\hat{\mathbf{\Sigma}}^{-1}\mathbf{S}_0) = \operatorname{tr}\{\hat{\mathbf{\Sigma}}^{-1}(\hat{\mathbf{\Sigma}} - \mathbf{S}_0)\} = \operatorname{tr}\{\hat{\mathbf{\Psi}}^{-1}(\hat{\mathbf{\Sigma}} - \mathbf{S}_0)\} = \sum_{i=1}^{p} \frac{(\hat{\mathbf{\Sigma}} - \mathbf{S}_0)_{ii}}{\hat{\psi}_i} = 0.$$

Step 3. With $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$, $L(\bar{\mathbf{x}}, \mathbf{\Sigma}) = (2\pi)^{-np/2}|\mathbf{\Sigma}|^{-n/2}\exp\{-\tfrac{n}{2}\operatorname{tr}(\mathbf{\Sigma}^{-1}\mathbf{S}_0)\}$, so

$$-2\log\frac{L_0}{L_1} = n\log\frac{|\hat{\mathbf{\Sigma}}|}{|\mathbf{S}_0|} + n\{\operatorname{tr}(\hat{\mathbf{\Sigma}}^{-1}\mathbf{S}_0) - \operatorname{tr}(\mathbf{S}_0^{-1}\mathbf{S}_0)\} = n\log\frac{|\hat{\mathbf{\Sigma}}|}{|\mathbf{S}_0|},$$

because both traces equal p. As $L_1 \geq L_0$, also $|\hat{\mathbf{\Sigma}}| \geq |\mathbf{S}_0|$. ■

Chapter 10. Discrimination and Classification

Bayes rule, LDA/QDA, Fisher's discriminant, logistic regression, assessing classifiers

1 The Bayes Classification Rule

Motivation. Classification is a decision problem: assign a new observation to a group so as to minimize expected misclassification cost. The Bayes rule compares prior-weighted densities and is optimal; every practical classifier estimates it.

1.1 Setting

Let $Y \in \{1, \dots, K\}$ be the group and $\mathbf{X} \in \mathbb{R}^p$ the features, with priors $P(Y = k) = \pi_k$ and $\mathbf{X} \mid Y = k \sim f_k$. For $K = 2$: $\mathbf{X} \mid Y = 1 \sim f_1(\mathbf{x})$, $\mathbf{X} \mid Y = 2 \sim f_2(\mathbf{x})$, $P(Y = 1) = \pi_1$, and

$$\eta(\mathbf{x}) := P(Y = 1 \mid \mathbf{X} = \mathbf{x}) = \frac{f_1(\mathbf{x})\pi_1}{f_1(\mathbf{x})\pi_1 + f_2(\mathbf{x})\pi_2}.$$

The misclassification probability of a classifier δ (or ϕ) : $\mathbb{R}^p \to \{1, \dots, K\}$ is $P(\phi(\mathbf{X}) \neq Y) = \sum_k P(\phi(\mathbf{X}) \neq k \mid Y = k)P(Y = k)$.

Theorem 10.1: optimality of the Bayes rule

$\phi_{\text{Bayes}}(\mathbf{x}) = \arg\max_k P(Y = k \mid \mathbf{X} = \mathbf{x}) = \arg\max_k \pi_k f_k(\mathbf{x})$ minimizes the misclassification probability: $P(\phi_{\text{Bayes}}(\mathbf{X}) \neq Y) \leq P(\phi(\mathbf{X}) \neq Y)$ for every ϕ.

$K = 2$.

$$\begin{aligned}
\text{Risk}(\delta) &= \mathrm{E}\big[\mathbf{1}_{\{Y \neq \delta(\mathbf{X})\}}\big] = \mathrm{E}_Y\Big[\mathrm{E}_{\mathbf{X}|Y}\big[\mathbf{1}_{\{Y \neq \delta(\mathbf{X})\}} \mid Y\big]\Big] \\
&= \pi_1 \mathrm{E}_{\mathbf{X}|Y=1}\big[\mathbf{1}_{\{\delta(\mathbf{X}) \neq 1\}} \mid Y = 1\big] + \pi_2 \mathrm{E}_{\mathbf{X}|Y=2}\big[\mathbf{1}_{\{\delta(\mathbf{X}) \neq 2\}} \mid Y = 2\big] \\
&= \int \underbrace{\Big[\pi_1 \mathbf{1}_{\{\delta(\mathbf{x}) \neq 1\}} f_1(\mathbf{x}) + \pi_2 \mathbf{1}_{\{\delta(\mathbf{x}) \neq 2\}} f_2(\mathbf{x})\Big]}_{=: h(\mathbf{x})} d\mathbf{x}.
\end{aligned}$$

At each $\mathbf{x}$, $\delta(\mathbf{x}) = 1$ gives $h(\mathbf{x}) = \pi_2 f_2(\mathbf{x})$ and $\delta(\mathbf{x}) = 2$ gives $h(\mathbf{x}) = \pi_1 f_1(\mathbf{x})$. To minimize the integral choose the smaller pointwise: if $\pi_2 f_2(\mathbf{x}) < \pi_1 f_1(\mathbf{x})$ choose $\delta(\mathbf{x}) = 1$; if $>$, choose $\delta(\mathbf{x}) = 2$. Thus $\phi_{\text{Bayes}}(\mathbf{x}) = \arg\max_k P(Y = k \mid \mathbf{X} = \mathbf{x})$. For general K the same argument minimizes $1 - P(Y = \phi(\mathbf{x}) \mid \mathbf{x})$ pointwise. ∎

Remark 10.2: misclassification costs

If $c(2 \mid 1)$ is the cost of classifying a member of group 1 into group 2, the rule minimizing expected cost classifies into group 1 when $\dfrac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq \dfrac{c(1 \mid 2)}{c(2 \mid 1)} \dfrac{\pi_2}{\pi_1}$.

2 Normal Populations: LDA and QDA

2.1 Common covariance: linear discriminant analysis

If $f_k = N_p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma})$ ($\boldsymbol{\Sigma}_k = \boldsymbol{\Sigma}$), then $\log\{\pi_k f_k(\mathbf{x})\} = -\frac{1}{2}\mathbf{x}^\top \boldsymbol{\Sigma}^{-1}\mathbf{x} + \boldsymbol{\mu}_k^\top \boldsymbol{\Sigma}^{-1}\mathbf{x} - \frac{1}{2}\boldsymbol{\mu}_k^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k + \log \pi_k +$ const, and the first term does not depend on k, so

$$\phi(\mathbf{x}) = \arg\min_k \delta_k(\mathbf{x}), \qquad \delta_k(\mathbf{x}) = b_{0k} - \mathbf{b}_k^\top \mathbf{x}, \quad b_{0k} = \tfrac{1}{2}\boldsymbol{\mu}_k^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k - \log(\pi_k), \quad \mathbf{b}_k = \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k.$$

The discriminant function is linear in $\mathbf{x}$: **linear discriminant analysis (LDA)**. For $K = 2$, $\delta_1 < \delta_2$ rearranges to

$$\phi(\mathbf{x}) = 1 \;\text{ if }\; \mathbf{b}^\top\Big(\mathbf{x} - \frac{\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2}{2}\Big) < \log\Big(\frac{\pi_1}{\pi_2}\Big), \qquad \mathbf{b} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1).$$

$K = 2$. $\delta_1 < \delta_2 \iff (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top\boldsymbol{\Sigma}^{-1}\mathbf{x} - \frac{1}{2}(\boldsymbol{\mu}_1^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_2) > \log(\pi_2/\pi_1)$. Since $\boldsymbol{\mu}_1^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_2 = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2)$, this is $-\mathbf{b}^\top(\mathbf{x} - \frac{\boldsymbol{\mu}_1+\boldsymbol{\mu}_2}{2}) > \log(\pi_2/\pi_1)$. ■

The sample version plugs in $\boldsymbol{\mu}_k \to \bar{\mathbf{x}}_k$, $\boldsymbol{\Sigma} \to \mathbf{S}_p$ (pooled covariance), $\pi_k \to n_k/n$: $\hat{\phi}(\mathbf{x}) = \arg\min_k(\hat{b}_{0k} - \hat{\mathbf{b}}_k^\top\mathbf{x})$ with $\hat{\mathbf{b}}_k = \mathbf{S}_p^{-1}\bar{\mathbf{x}}_k$, $\hat{b}_{0k} = \frac{1}{2}\bar{\mathbf{x}}_k^\top\mathbf{S}_p^{-1}\bar{\mathbf{x}}_k - \log(n_k/n)$, and for $K = 2$, $\hat{\mathbf{b}} = \mathbf{S}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)$ with the rule "$\hat{\mathbf{b}}^\top(\mathbf{x} - \frac{\bar{\mathbf{x}}_1+\bar{\mathbf{x}}_2}{2}) < \log(n_1/n_2) \Rightarrow 1$".

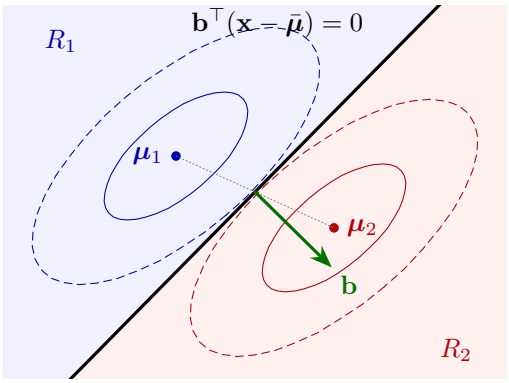


Figure. *LDA for two synthetic normal populations with common* $\boldsymbol{\Sigma} = \left(\begin{smallmatrix} 1 & 0.6 \\ 0.6 & 0.8 \end{smallmatrix}\right)$ *and equal priors; solid and dashed ellipses contain* 39% *and* 86% *of each population. The boundary is the line through* $\bar{\boldsymbol{\mu}} = (\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2)/2$ *orthogonal to Fisher's direction* $\mathbf{b} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)$ *(green), which is* not *the direction of* $\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1$ *(dotted) because* $\boldsymbol{\Sigma}$ *tilts the populations.*

Theorem 10.3: error rates of LDA

Let $K = 2$, $f_k = N_p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma})$, $\Delta^2 = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$ and $c = \log(\pi_2/\pi_1)$. The Bayes rule has

$$P(2 \mid 1) = \Phi\Big(\frac{c - \Delta^2/2}{\Delta}\Big), \qquad P(1 \mid 2) = \Phi\Big(\frac{-c - \Delta^2/2}{\Delta}\Big),$$

so with equal priors the optimal error rate is $\mathrm{OER} = \Phi(-\Delta/2)$, a decreasing function of the Mahalanobis distance between the populations.

Proof. By the proof above the rule assigns $\mathbf{x}$ to population 1 iff $W(\mathbf{x}) := (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x} - \frac{\boldsymbol{\mu}_1+\boldsymbol{\mu}_2}{2}) > c$. W is linear in $\mathbf{X}$, so under population k it is normal with variance $(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) = \Delta^2$ and mean

$$\mathrm{E}_1 W = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top\boldsymbol{\Sigma}^{-1}\tfrac{\boldsymbol{\mu}_1-\boldsymbol{\mu}_2}{2} = \tfrac{\Delta^2}{2}, \qquad \mathrm{E}_2 W = -\tfrac{\Delta^2}{2}.$$

Hence $P(2 \mid 1) = P_1(W \le c) = \Phi\{(c - \Delta^2/2)/\Delta\}$ and $P(1 \mid 2) = P_2(W > c) = 1 - \Phi\{(c + \Delta^2/2)/\Delta\} = \Phi\{(-c - \Delta^2/2)/\Delta\}$. For $c = 0$ both equal $\Phi(-\Delta/2)$, and so does $\frac{1}{2}P(2 \mid 1) + \frac{1}{2}P(1 \mid 2)$. ■

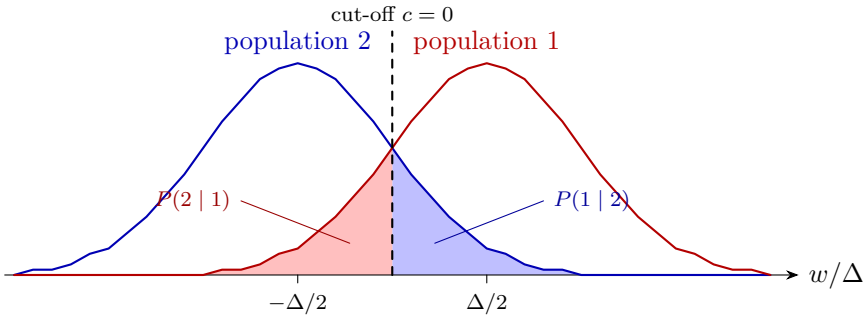


Figure. *Densities of the standardized score W/Δ under the two populations, $N(\pm\Delta/2, 1)$, drawn for $\Delta = 2$ and equal priors. The shaded tails are the two misclassification probabilities, each $\Phi(-\Delta/2) = 0.159$. A prior or cost ratio moves the cut-off c and trades one tail against the other; larger Δ pulls the curves apart and shrinks both.*

2.2 Different covariances: quadratic discriminant analysis

If $f_k = N_p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$,

$$\phi(\mathbf{x}) = \arg\max_k \Big\{ -\tfrac{1}{2}\log|\boldsymbol{\Sigma}_k| - \tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1}(\mathbf{x}-\boldsymbol{\mu}_k) + \log\pi_k \Big\} = \arg\max_k \left(\mathbf{x}^\top \mathbf{C}_k \mathbf{x} + \mathbf{b}_k^\top \mathbf{x} - b_{0k}\right)$$

with $\mathbf{C}_k = -\frac{1}{2}\boldsymbol{\Sigma}_k^{-1}$, $\mathbf{b}_k = \boldsymbol{\Sigma}_k^{-1}\boldsymbol{\mu}_k$ and $b_{0k} = \frac{1}{2}\boldsymbol{\mu}_k^\top \boldsymbol{\Sigma}_k^{-1}\boldsymbol{\mu}_k + \frac{1}{2}\log|\boldsymbol{\Sigma}_k| - \log\pi_k$ (expand the quadratic form). The quadratic terms no longer cancel, so the boundaries are quadratic surfaces: **QDA**. It needs $(K-1)\,p(p+1)/2$ more covariance parameters than LDA, so with small samples LDA is often more stable (bias–variance trade-off).

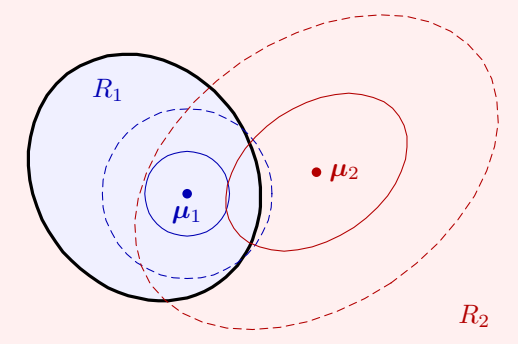


Figure. *QDA for two synthetic normal populations with equal priors, a tight spherical $\boldsymbol{\Sigma}_1 = 0.35\,\mathbf{I}$ (blue) and a wide $\boldsymbol{\Sigma}_2 = \left(\begin{smallmatrix} 1.6 & 0.5 \\ 0.5 & 1.2 \end{smallmatrix}\right)$ (red); ellipses as in the previous figure. The boundary $\{\mathbf{x} : \pi_1 f_1(\mathbf{x}) = \pi_2 f_2(\mathbf{x})\}$ is a closed conic: R_1 is a bounded region around $\boldsymbol{\mu}_1$, and points far away in* every *direction, even far to the left of $\boldsymbol{\mu}_1$, go to the more spread-out population 2.*

3 Fisher's Linear Discriminant

Motivation. Fisher's approach needs no distributional assumption: find the projection that best separates the groups relative to the within-group spread. Under normality with equal covariances it coincides with the Bayes rule.

Without distributional assumptions, find the projection that best separates the groups.

Definition 10.4: Fisher's LDA direction ($K = 2$)

$$\mathbf{u}_0 = \arg\max_{\mathbf{u}\in\mathbb{R}^p} \frac{(\mathbf{u}^\top \bar{\mathbf{x}}_1 - \mathbf{u}^\top \bar{\mathbf{x}}_2)^2}{\mathbf{u}^\top \mathbf{S}_p \mathbf{u}} :$$

squared difference of projected means over projected within-group variance.

Theorem 10.5: solution

$\mathbf{u}_0 \propto \mathbf{S}_p^{-1}(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)$ (equivalently $\mathbf{S}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)$), and the maximum is $\mathbf{d}^\top \mathbf{S}_p^{-1}\mathbf{d}$ with $\mathbf{d} = \bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2$, the squared sample Mahalanobis distance.

Proof. In general, to maximize $\frac{\mathbf{u}^\top \mathbf{A}\mathbf{u}}{\mathbf{u}^\top \mathbf{B}\mathbf{u}}$ ($\mathbf{B} \succ 0$) use $L(\mathbf{u}, \lambda) = \mathbf{u}^\top \mathbf{A}\mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{B}\mathbf{u} - 1)$: $\partial L/\partial \mathbf{u} = \mathbf{0} \Rightarrow \mathbf{A}\mathbf{u} = \lambda \mathbf{B}\mathbf{u} \Rightarrow \mathbf{B}^{-1}\mathbf{A}\mathbf{u} = \lambda \mathbf{u}$, so the top eigenvector of $\mathbf{B}^{-1}\mathbf{A}$ is optimal. With $\mathbf{A} = \mathbf{d}\mathbf{d}^\top$ (rank 1) and $\mathbf{B} = \mathbf{S}_p$, $\mathbf{S}_p^{-1}\mathbf{d}\mathbf{d}^\top$ has a single nonzero eigenvalue $\mathbf{d}^\top \mathbf{S}_p^{-1}\mathbf{d}$ with unique eigenvector $\mathbf{S}_p^{-1}\mathbf{d}$ (rank 1). Directly: with $\mathbf{v} = \mathbf{S}_p^{1/2}\mathbf{u}$, Cauchy–Schwarz gives $(\mathbf{u}^\top \mathbf{d})^2/\mathbf{u}^\top \mathbf{S}_p \mathbf{u} = (\mathbf{v}^\top \mathbf{S}_p^{-1/2}\mathbf{d})^2/\mathbf{v}^\top \mathbf{v} \le \mathbf{d}^\top \mathbf{S}_p^{-1}\mathbf{d}$. ∎

Remark 10.6: Fisher's rule = LDA with equal priors

With the discriminant score $y_i = \mathbf{b}_Y^\top \mathbf{x}_i$, $\mathbf{b}_Y = \mathbf{S}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)$, and the cut-off $b_{0Y} = \mathbf{b}_Y^\top(\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2)/2$,

$$\text{“}\mathbf{b}_Y^\top \mathbf{y} - b_{0Y} < 0 \Rightarrow 1\text{”} \iff \text{“}[\mathbf{S}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)]^\top \Big(\mathbf{x} - \frac{\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2}{2}\Big) < 0 \Rightarrow 1\text{”},$$

which is sample LDA with $\pi_1 = \pi_2$. A rule derived without normality coincides with the Bayes rule under normality.

Several groups ($K \ge 3$): $\max_{\mathbf{u}} \frac{\mathbf{u}^\top \mathbf{B}\mathbf{u}}{\mathbf{u}^\top \mathbf{W}\mathbf{u}}$ ($\mathbf{B}, \mathbf{W}$ the MANOVA between/within SSP matrices) is solved by the eigenvectors of $\mathbf{W}^{-1}\mathbf{B}$; there are at most $\min(p, K-1)$ nonzero eigenvalues, the same η_i that appear in the Wilks/Pillai statistics of Ch. 6.

4 Logistic Regression

For $K = 2$, $Y \in \{0, 1\}$, model the posterior directly:

$$\operatorname{logit}(\eta(\mathbf{x})) := \log\Big(\frac{\eta(\mathbf{x})}{1 - \eta(\mathbf{x})}\Big) = b + \boldsymbol{\beta}^\top \mathbf{x}, \qquad \eta(\mathbf{x}) = P(Y = 1 \mid \mathbf{X} = \mathbf{x}).$$

Classification: “$\eta(\mathbf{x}) > \frac{1}{2} \Rightarrow 1$” $\iff$ “$f(\mathbf{x}) = b + \boldsymbol{\beta}^\top \mathbf{x} > 0 \Rightarrow 1$” (a linear boundary). The log-likelihood is

$$\begin{aligned}\ell(b, \boldsymbol{\beta}) &= \sum_{i=1}^{n} \big[y_i \log(\eta(\mathbf{x}_i; b, \boldsymbol{\beta})) + (1 - y_i)\log\{1 - \eta(\mathbf{x}_i; b, \boldsymbol{\beta})\}\big] \\ &= \sum_{i=1}^{n} \big[y_i(b + \boldsymbol{\beta}^\top \mathbf{x}_i) - \log\{1 + \exp(b + \boldsymbol{\beta}^\top \mathbf{x}_i)\}\big],\end{aligned}$$

which is concave and maximized by Newton–Raphson (IRLS). In the equal-covariance normal model $\log \frac{\pi_1 f_1}{\pi_0 f_0}$ is exactly linear in $\mathbf{x}$, so the logistic model holds; LDA models the joint distribution, logistic regression only the conditional one.

Derivation: the second equality and the derivatives

[Supplement] $\log\eta = (b+\boldsymbol{\beta}^\top\mathbf{x}) - \log(1+e^{b+\boldsymbol{\beta}^\top\mathbf{x}})$ and $\log(1-\eta) = -\log(1+e^{b+\boldsymbol{\beta}^\top\mathbf{x}})$, so $y\log\eta + (1-y)\log(1-\eta) = y(b+\boldsymbol{\beta}^\top\mathbf{x}) - \log(1+e^{b+\boldsymbol{\beta}^\top\mathbf{x}})$. The gradient is $\sum_i (y_i-\eta_i)(1,\mathbf{x}_i^\top)^\top$ and the Hessian $-\sum_i \eta_i(1-\eta_i)(1,\mathbf{x}_i^\top)^\top(1,\mathbf{x}_i^\top) \preceq 0$.

5 Simple Classifiers in High Dimensions

If $p > n$, $\mathbf{S}_p$ is singular and LDA cannot be used (e.g. the Golub data).

- **Nearest centroid**: LDA with $\boldsymbol{\Sigma} = \mathbf{I}$. $\phi(\mathbf{x}) = 1$ if $\mathbf{b}^\top(\mathbf{x} - \frac{\bar{\mathbf{x}}_1+\bar{\mathbf{x}}_2}{2}) < 0$, $\mathbf{b} = \bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1$ (assign to the closer mean).
- **Naive Bayes classifier (diagonal LDA)**: assume independent variables and replace $\mathbf{S}_p$ by $\mathbf{D}_p = \mathrm{diag}(\mathbf{S}_p)$: $\phi(\mathbf{x}) = 1$ if $\mathbf{b}^\top(\mathbf{x} - \frac{\bar{\mathbf{x}}_1+\bar{\mathbf{x}}_2}{2}) < 0$, $\mathbf{b} = \mathbf{D}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)$.
- **k-nearest neighbors**: majority vote among the k training points closest to $\mathbf{x}$; non-parametric, high variance for small k and high bias for large k.

6 Assessment of Classifiers

- True error rate $P(\phi(\mathbf{X}) \neq Y) = \sum_{k=1}^K P(\phi(\mathbf{X}) \neq k \mid Y = k)P(Y = k)$; ϕ is optimal if $P(\phi(\mathbf{X}) \neq Y) \leq P(\varphi(\mathbf{X}) \neq Y)$ for all φ.
- Empirical misclassification rate $\hat{P}(\hat{\phi}_D(\mathbf{X}) \neq Y \mid D) = \frac{1}{M}\sum_{i=1}^M \mathbf{1}_{\{\hat{\phi}(\mathbf{x}_i)\neq y_i\}}$.
- **Training and test sets**: build $\hat{\phi}$ from $\mathbf{X}_{\text{tr}}$ and estimate the misclassification rate on $\mathbf{X}_{\text{test}}$. The training error (APER) is optimistically biased.
- **Confusion matrix**: actual group by predicted group (training and testing confusion matrices).
- **K-fold cross-validation**: split into K parts, use each in turn as the test set and average. $K\uparrow \Rightarrow$ bias $\downarrow$, variance $\uparrow$; $K\downarrow \Rightarrow$ the reverse. $K = n$ is LOOCV (Lachenbruch's holdout).

7 Selected exercises

Motivation. The exercises prove the cost-weighted Bayes rule stated above, work out classification regions for non-normal densities, show how the total separation of several groups splits over Fisher's discriminants, and explain when logistic regression has no maximum likelihood estimate.

Exercise 10.1 (The minimum expected cost rule; J&W 11.3, 11.4)

For two populations with densities f_1, f_2, priors π_1, π_2 and costs $c(2 \mid 1), c(1 \mid 2) \geq 0$, a rule is a partition $\mathbb{R}^p = R_1 \cup R_2$ and

$$\mathrm{ECM}(R_1) = c(2 \mid 1)\pi_1 \int_{R_2} f_1(\mathbf{x})\,d\mathbf{x} + c(1 \mid 2)\pi_2 \int_{R_1} f_2(\mathbf{x})\,d\mathbf{x}.$$

(a) Show that ECM is minimized by $R_1^* = \{\mathbf{x} : c(2 \mid 1)\pi_1 f_1(\mathbf{x}) \geq c(1 \mid 2)\pi_2 f_2(\mathbf{x})\}$.
(b) With $c(2 \mid 1) = 50$, $c(1 \mid 2) = 100$, $\pi_2 = 0.2$, classify an item with $f_1(\mathbf{x}) = 0.3$ and $f_2(\mathbf{x}) = 0.5$.

Solution.

(a)

Step 1. Since $\int_{R_2} f_1 = 1 - \int_{R_1} f_1$,

$$\mathrm{ECM}(R_1) = c(2 \mid 1)\pi_1 + \int_{R_1} h(\mathbf{x})\, d\mathbf{x}, \qquad h(\mathbf{x}) = c(1 \mid 2)\pi_2 f_2(\mathbf{x}) - c(2 \mid 1)\pi_1 f_1(\mathbf{x}).$$

Step 2. For any measurable R_1, splitting R_1 according to the sign of h,

$$\int_{R_1} h = \int_{R_1 \cap \{h \le 0\}} h + \int_{R_1 \cap \{h > 0\}} h \;\ge\; \int_{R_1 \cap \{h \le 0\}} h \;\ge\; \int_{\{h \le 0\}} h = \int_{R_1^*} h,$$

because adding the points of $\{h \le 0\}$ outside R_1 can only decrease the integral. Hence $\mathrm{ECM}(R_1) \ge \mathrm{ECM}(R_1^*)$. Where $f_2 > 0$ this is the ratio form $f_1/f_2 \ge \{c(1 \mid 2)/c(2 \mid 1)\}(\pi_2/\pi_1)$; points with $h = 0$ may be assigned either way.

(b) The threshold is $(100/50)(0.2/0.8) = 0.5$ and $f_1/f_2 = 0.6 \ge 0.5$, so the item is assigned to population 1, although $f_2 > f_1$: the larger prior of population 1 outweighs the doubled cost. ■

Exercise 10.2 (Triangular densities; J&W 11.7)

Let $f_1(x) = 1 - |x|$ for $|x| \le 1$ and $f_2(x) = 1 - |x - 0.5|$ for $-0.5 \le x \le 1.5$ (both 0 elsewhere), with equal costs.
(a) Find the classification regions for $\pi_1 = \pi_2$ and the total probability of misclassification.
(b) Find the regions for $\pi_1 = 0.2$.

Solution. By Exercise 10.1 with equal costs, assign x to population 1 iff $\pi_1 f_1(x) \ge \pi_2 f_2(x)$. On $[-1, -0.5)$ only f_1 is positive (population 1) and on $(1, 1.5]$ only f_2 (population 2); it remains to treat $[-0.5, 1]$, where $f_2(x) = 0.5 + x$ for $x \le 0.5$ and $f_2(x) = 1.5 - x$ for $x \ge 0.5$.

(a)

Step 1. On $[-0.5, 1]$, $f_1 \ge f_2 \iff |x| \le |x - 0.5| \iff x \le 0.25$. So $R_1 = [-1, 0.25]$ and $R_2 = (0.25, 1.5]$.

Step 2. $P(2 \mid 1) = \int_{0.25}^{1} (1 - x)\, dx = 0.75^2/2 = 0.28125$ and $P(1 \mid 2) = \int_{-0.5}^{0.25} (0.5 + x)\, dx = 0.75^2/2 = 0.28125$, so the total probability of misclassification is $\frac{1}{2}(0.28125 + 0.28125) = 0.28125$.

(b) Now population 1 needs $0.2 f_1 \ge 0.8 f_2$, i.e. $f_1 \ge 4 f_2$.

Step 1. On $[-0.5, 0]$: $1 + x \ge 4(0.5 + x) \iff x \le -1/3$.

Step 2. On $[0, 0.5]$: $1 - x \ge 2 + 4x \iff x \le -0.2$, impossible. On $[0.5, 1]$: $1 - x \ge 6 - 4x \iff x \ge 5/3$, impossible.

Step 3. Hence $R_1 = [-1, -1/3]$ and $R_2 = (-1/3, 1.5]$: the smaller prior shifts the boundary from 0.25 to $-1/3$. ■

Exercise 10.3 (Total separation and Fisher's discriminants; J&W 11.22)

For g groups with means $\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_g$, common covariance $\boldsymbol{\Sigma} \succ 0$ and $\bar{\boldsymbol{\mu}} = \frac{1}{g}\sum_i \boldsymbol{\mu}_i$, let $\mathbf{B}_\mu = \sum_i (\boldsymbol{\mu}_i - \bar{\boldsymbol{\mu}})(\boldsymbol{\mu}_i - \bar{\boldsymbol{\mu}})^\top$ and define the total separation

$$\Delta_S^2 = \sum_{i=1}^{g} (\boldsymbol{\mu}_i - \bar{\boldsymbol{\mu}})^\top \boldsymbol{\Sigma}^{-1} (\boldsymbol{\mu}_i - \bar{\boldsymbol{\mu}}).$$

Let $\lambda_1 \ge \cdots \ge \lambda_s > 0$ be the nonzero eigenvalues of $\boldsymbol{\Sigma}^{-1}\mathbf{B}_\mu$ and $Y_k = \mathbf{e}_k^\top \mathbf{X}$ the discriminants, with $\boldsymbol{\Sigma}^{-1}\mathbf{B}_\mu \mathbf{e}_k = \lambda_k \mathbf{e}_k$ and $\mathbf{e}_k^\top \boldsymbol{\Sigma} \mathbf{e}_l = \delta_{kl}$. Show $\Delta_S^2 = \lambda_1 + \cdots + \lambda_s$, and that the first r

discriminants carry separation $\sum_{k\le r}\sum_i(\mathrm{E}_i Y_k-\bar\mu_{Y_k})^2=\lambda_1+\cdots+\lambda_r$.

Solution.

Step 1. $\Delta_S^2=\sum_i \operatorname{tr}\{\boldsymbol\Sigma^{-1}(\boldsymbol\mu_i-\bar{\boldsymbol\mu})(\boldsymbol\mu_i-\bar{\boldsymbol\mu})^\top\}=\operatorname{tr}(\boldsymbol\Sigma^{-1}\mathbf B_\mu)=\operatorname{tr}(\mathbf A)$, where $\mathbf A=\boldsymbol\Sigma^{-1/2}\mathbf B_\mu\boldsymbol\Sigma^{-1/2}$ is symmetric and positive semidefinite.

Step 2. $\boldsymbol\Sigma^{-1}\mathbf B_\mu=\boldsymbol\Sigma^{-1/2}\mathbf A\boldsymbol\Sigma^{1/2}$ is similar to $\mathbf A$, so they have the same eigenvalues, and tr$\mathbf A$ is the sum of the eigenvalues; the zero ones contribute nothing, so $\Delta_S^2=\lambda_1+\cdots+\lambda_s$.

Step 3. Let $\mathbf A=\sum_k\lambda_k\mathbf f_k\mathbf f_k^\top$ with orthonormal $\mathbf f_k$ and $\mathbf e_k=\boldsymbol\Sigma^{-1/2}\mathbf f_k$. Then $\boldsymbol\Sigma^{-1}\mathbf B_\mu\mathbf e_k=\boldsymbol\Sigma^{-1/2}\mathbf A\mathbf f_k=\lambda_k\mathbf e_k$ and $\mathbf e_k^\top\boldsymbol\Sigma\mathbf e_l=\mathbf f_k^\top\mathbf f_l=\delta_{kl}$, so these are the discriminants: $\operatorname{Var}(Y_k)=1$ and $\operatorname{Cov}(Y_k,Y_l)=0$.

Step 4. The group means of Y_k are $\mathbf e_k^\top\boldsymbol\mu_i$ with average $\mathbf e_k^\top\bar{\boldsymbol\mu}$, so

$$\sum_{i=1}^{g}\{\mathbf e_k^\top(\boldsymbol\mu_i-\bar{\boldsymbol\mu})\}^2=\mathbf e_k^\top\mathbf B_\mu\mathbf e_k=\mathbf f_k^\top\mathbf A\mathbf f_k=\lambda_k.$$

Summing over $k\le r$ gives $\lambda_1+\cdots+\lambda_r$; for $r=s$ this is all of Δ_S^2. Thus $(\lambda_1+\cdots+\lambda_r)/\Delta_S^2$ is the share of separation kept by the first r discriminants. ■

Exercise 10.4 (Logistic regression: linearity and separation)

(a) If $\mathbf X\mid Y=k\sim N_p(\boldsymbol\mu_k,\boldsymbol\Sigma)$, $k=0,1$, show $\operatorname{logit}\eta(\mathbf x)=b+\boldsymbol\beta^\top\mathbf x$ with $\boldsymbol\beta=\boldsymbol\Sigma^{-1}(\boldsymbol\mu_1-\boldsymbol\mu_0)$ and $b=\log(\pi_1/\pi_0)-\frac12(\boldsymbol\mu_1+\boldsymbol\mu_0)^\top\boldsymbol\Sigma^{-1}(\boldsymbol\mu_1-\boldsymbol\mu_0)$.
(b) Suppose the data are completely separated: there is $(b_0,\boldsymbol\beta_0)$ with $z_i=b_0+\boldsymbol\beta_0^\top\mathbf x_i>0$ when $y_i=1$ and $z_i<0$ when $y_i=0$. Show that the log-likelihood $\ell(b,\boldsymbol\beta)$ has no maximizer.

Solution [Supplement].

(a) $\operatorname{logit}\eta(\mathbf x)=\log\dfrac{\pi_1 f_1(\mathbf x)}{\pi_0 f_0(\mathbf x)}=\log\dfrac{\pi_1}{\pi_0}-\dfrac12\{(\mathbf x-\boldsymbol\mu_1)^\top\boldsymbol\Sigma^{-1}(\mathbf x-\boldsymbol\mu_1)-(\mathbf x-\boldsymbol\mu_0)^\top\boldsymbol\Sigma^{-1}(\mathbf x-\boldsymbol\mu_0)\}$. The terms $\mathbf x^\top\boldsymbol\Sigma^{-1}\mathbf x$ cancel, leaving $(\boldsymbol\mu_1-\boldsymbol\mu_0)^\top\boldsymbol\Sigma^{-1}\mathbf x-\frac12(\boldsymbol\mu_1^\top\boldsymbol\Sigma^{-1}\boldsymbol\mu_1-\boldsymbol\mu_0^\top\boldsymbol\Sigma^{-1}\boldsymbol\mu_0)$, and $\boldsymbol\mu_1^\top\boldsymbol\Sigma^{-1}\boldsymbol\mu_1-\boldsymbol\mu_0^\top\boldsymbol\Sigma^{-1}\boldsymbol\mu_0=(\boldsymbol\mu_1+\boldsymbol\mu_0)^\top\boldsymbol\Sigma^{-1}(\boldsymbol\mu_1-\boldsymbol\mu_0)$ by symmetry of $\boldsymbol\Sigma^{-1}$.

(b)

Step 1 ($\ell<0$). Every term $y_i\log\eta_i+(1-y_i)\log(1-\eta_i)$ is negative because $0<\eta_i<1$, so $\ell(b,\boldsymbol\beta)<0$ for every parameter.

Step 2 ($\sup\ell=0$). Along $(tb_0,t\boldsymbol\beta_0)$, $t\to\infty$: if $y_i=1$, $\eta_i=1/(1+e^{-tz_i})\to1$ since $z_i>0$; if $y_i=0$, $1-\eta_i=1/(1+e^{tz_i})\to1$ since $z_i<0$. Every term tends to 0, so $\ell(tb_0,t\boldsymbol\beta_0)\to0$.

Step 3. The supremum 0 is approached but, by Step 1, never attained, so no MLE exists. Newton–Raphson then drives $\|\boldsymbol\beta\|$ to infinity; in practice one uses a penalty (e.g. ridge) or reports the separation. LDA, which models $\mathbf X$ given Y, is unaffected. ■

Chapter 11. Cluster Analysis

K-means, K-medoids, hierarchical clustering, Gaussian mixtures and the EM algorithm, choosing K

Unlike classification (Ch. 10), clustering groups similar observations without labels (unsupervised learning). A clustering is an encoder $C : \{1, \dots, n\} \to \{1, \dots, K\}$.

1 Dissimilarity and Within-Cluster Scatter

Motivation. Clustering looks for groups without labels, so we must decide what makes a grouping good. Small within-cluster scatter is the usual criterion; because total scatter is fixed, this is the same as large between-cluster scatter.

Definition 11.1: within-cluster scatter

For a dissimilarity $d(i, j) \geq 0$,

$$W(C) = \frac{1}{2} \sum_{k=1}^{K} \frac{1}{n_k} \sum_{i:C(i)=k} \sum_{j:C(j)=k} d(i, j), \qquad n_k = \#\{i : C(i) = k\}.$$

Smaller W is better.

Theorem 11.2: with squared Euclidean distance $W(C)$ is the within-cluster sum of squares

If $d(i, j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2^2$, then

$$W(C) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|_2^2, \qquad \bar{\mathbf{x}}_{(k)} = \frac{1}{n_k} \sum_{C(i)=k} \mathbf{x}_i,$$

and $T = \sum_i \|\mathbf{x}_i - \bar{\mathbf{x}}\|^2 = W(C) + B(C)$ with $B(C) = \sum_k n_k \|\bar{\mathbf{x}}_{(k)} - \bar{\mathbf{x}}\|^2$.

Proof [Supplement]. Within cluster k, $\mathbf{x}_i - \mathbf{x}_j = (\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}) - (\mathbf{x}_j - \bar{\mathbf{x}}_{(k)})$ and $\sum_{C(i)=k} (\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}) = \mathbf{0}$, so

$$\sum_{C(i)=k} \sum_{C(j)=k} \|\mathbf{x}_i - \mathbf{x}_j\|^2 = 2n_k \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|^2;$$

multiply by $\frac{1}{2} \cdot \frac{1}{n_k}$. $T = W + B$ is the trace of the MANOVA SSP decomposition (Ch. 6). ∎

Since T is fixed, minimizing W is maximizing B.

1.1 Limits of combinatorial algorithms

The number of ways to partition n items into K non-empty clusters (Stirling number of the second kind) is

$$A(n, K) = \frac{1}{K!} \sum_{k=1}^{K} (-1)^{K-k} \binom{K}{k} k^n \sim \frac{K^n}{K!}.$$

E.g. $A(19, 4) \approx 10^{10}$: exhaustive search is impossible, so greedy iterative algorithms are used.

2 K-means and K-medoids

Motivation. Minimizing within-cluster scatter exactly is combinatorially hard. K-means alternates two easy steps, assigning points and updating centres, each of which decreases the criterion, so it converges to a local optimum.

2.1 K-means

With $d(i,j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2^2$, any point $\boldsymbol{m}$ satisfies $\sum_{C(i)=k} \|\mathbf{x}_i - \boldsymbol{m}\|^2 \geq \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|^2$, so

$$W(C) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|_2^2 = \min_{\boldsymbol{m}_1,\ldots,\boldsymbol{m}_K} L(C \mid \boldsymbol{m}), \qquad L(C \mid \boldsymbol{m}) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \boldsymbol{m}_k\|_2^2.$$

Definition 11.3: K-means algorithm

(i) Minimize over C with $\boldsymbol{m}_1, \ldots, \boldsymbol{m}_K$ fixed: update $C(i) = k$ for the closest center.

(ii) Minimize over $\boldsymbol{m}_1, \ldots, \boldsymbol{m}_K$ with C fixed: update $\boldsymbol{m}_k$ by the new average of cluster k.

(iii) Iterate (i), (ii) until $L(C \mid \boldsymbol{m})$ does not change.

Theorem 11.4: convergence

K-means stops after finitely many iterations at a local minimum.

Proof [Supplement]. Each step minimizes L in one block of variables with the other fixed, so L is non-increasing. There are finitely many encoders C and step (ii) is unique for a given C, so while L strictly decreases no C can recur; the algorithm stops in finitely many steps. ■

The result depends on the starting values, so run several starts and keep the smallest W. K-means is scale-dependent; consider standardizing.

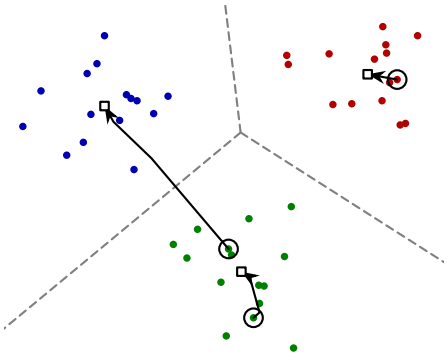


Figure. *K-means ($K = 3$) on synthetic data, 45 points from three groups. The starting centres (circles) are three observations, two of them in the bottom group. Black arrows trace the centres over the iterations until C stops changing (squares, after three updates). The dashed lines are the perpendicular bisectors of the final centres: step (i) assigns each point to the Voronoi cell of its nearest centre, so the cells are convex.*

2.2 K-medoids

Use an arbitrary dissimilarity and restrict centers to observations.

(i) Minimize over C: update $C(i) = k$ for the closest medoid $\boldsymbol{m}_k$.

(ii) Minimize over $\boldsymbol{m}_1, \dots, \boldsymbol{m}_K$: update $\boldsymbol{m}_k$ by the item in the kth cluster that minimizes the total dissimilarity to the cluster.

(iii) Iterate (i), (ii) until $L(C \mid \boldsymbol{m})$ does not change.

Using actual observations as representatives makes it robust to outliers and usable for categorical data.

3 Hierarchical Clustering

3.1 Agglomerative

Start with singletons and repeatedly merge the two closest clusters. Between-cluster distances (linkage):

Single linkage	$d(G_1, G_2) = \min_{i \in G_1, j \in G_2} d_{ij}$
Complete linkage	$d(G_1, G_2) = \max_{i \in G_1, j \in G_2} d_{ij}$
Average linkage	$d(G_1, G_2) = \text{average}_{i \in G_1, j \in G_2} d_{ij}$

Single linkage tends to produce long chained clusters, complete linkage compact round ones. The result is drawn as a dendrogram and cut at a chosen height.

Ward's method merges, at each step, the two clusters whose union increases the within-cluster sum of squares $W(C)$ the least.

Theorem 11.5: Ward's merging cost

Let $\mathrm{ESS}(G) = \sum_{i \in G} \|\mathbf{x}_i - \bar{\mathbf{x}}_G\|^2$. For disjoint clusters G, H of sizes n_G, n_H,

$$\mathrm{ESS}(G \cup H) - \mathrm{ESS}(G) - \mathrm{ESS}(H) = \frac{n_G n_H}{n_G + n_H} \|\bar{\mathbf{x}}_G - \bar{\mathbf{x}}_H\|^2.$$

Proof [Supplement].

Step 1. Applying $T = W + B$ (Theorem 11.2) to the set $G \cup H$ split into the two clusters G, H, with $N = n_G + n_H$ and $\bar{\mathbf{x}} = (n_G \bar{\mathbf{x}}_G + n_H \bar{\mathbf{x}}_H)/N$ the mean of $G \cup H$,

$$\mathrm{ESS}(G \cup H) = \mathrm{ESS}(G) + \mathrm{ESS}(H) + n_G \|\bar{\mathbf{x}}_G - \bar{\mathbf{x}}\|^2 + n_H \|\bar{\mathbf{x}}_H - \bar{\mathbf{x}}\|^2.$$

Step 2. $\bar{\mathbf{x}}_G - \bar{\mathbf{x}} = \frac{n_H}{N}(\bar{\mathbf{x}}_G - \bar{\mathbf{x}}_H)$ and $\bar{\mathbf{x}}_H - \bar{\mathbf{x}} = -\frac{n_G}{N}(\bar{\mathbf{x}}_G - \bar{\mathbf{x}}_H)$, so the last two terms equal $\frac{n_G n_H^2 + n_H n_G^2}{N^2} \|\bar{\mathbf{x}}_G - \bar{\mathbf{x}}_H\|^2 = \frac{n_G n_H}{N} \|\bar{\mathbf{x}}_G - \bar{\mathbf{x}}_H\|^2$. ■

So Ward's method is the greedy version of K-means' criterion: it is centroid distance weighted by cluster sizes, which penalizes merging two large clusters and favours compact clusters of similar size.

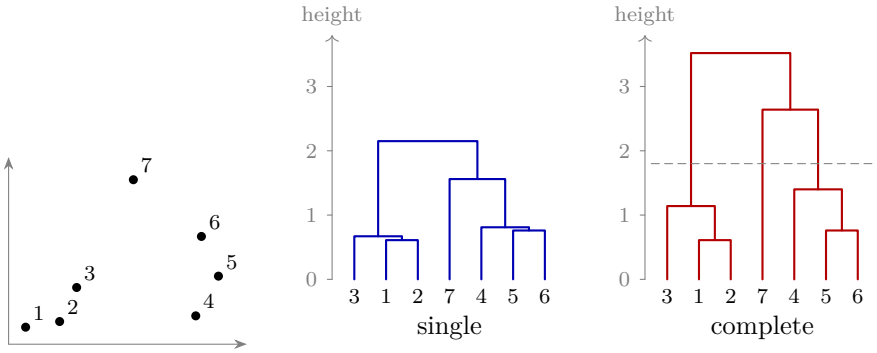


Figure. *Seven synthetic points (left) and their single- and complete-linkage dendrograms (Euclidean distance). Both merge* $\{1,2\}$ *first at height* 0.61*, but single linkage then chains the isolated point* 7 *onto* $\{4,5,6\}$ *at the smallest gap* 1.56*, while complete linkage merges* 7 *only at* 2.64*. Cutting the complete-linkage tree at the dashed height gives the three clusters* $\{1,2,3\}$*,* $\{4,5,6\}$*,* $\{7\}$*. In both trees the heights increase upwards (Exercise 11.4).*

Distances between observations:

$$p\text{-norm: } d(\mathbf{x},\mathbf{y}) = \|\mathbf{x}-\mathbf{y}\|_p = \Big(\sum_{l=1}^{d} |x_l - y_l|^p\Big)^{1/p},$$

$$\text{standardized distance: } d^2(\mathbf{x},\mathbf{y}) = \sum_{l=1}^{d} \frac{(x_l - y_l)^2}{s_l^2} = (\mathbf{x}-\mathbf{y})^\top \mathbf{D}^{-1}(\mathbf{x}-\mathbf{y}).$$

3.2 Divisive

Start from a single cluster and split repeatedly (e.g. $K = 2$ means at each step).

4 Gaussian Mixtures and the EM Algorithm

4.1 Model

$\mathbf{X}_1,\dots,\mathbf{X}_n \overset{\text{iid}}{\sim} g(\mathbf{x})$ with latent labels $\delta_1,\dots,\delta_n \overset{\text{iid}}{\sim}$, $P(\delta_i = k) = \pi_k$, and $\mathbf{X} \mid \delta = k \sim N_p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$:

$$\mathbf{X} \sim g(\mathbf{x}) = \sum_{k=1}^{K} \text{pdf}(\mathbf{x} \mid \delta = k) P(\delta = k) = \sum_{k=1}^{K} \pi_k g_k(\mathbf{x}),$$

$$L(\boldsymbol{\theta} \mid \mathbf{X}) = \prod_{i=1}^{n} g(\mathbf{x}_i) = \prod_{i=1}^{n} \Big\{ \sum_{k=1}^{K} g_k(\mathbf{x}_i)\pi_k \Big\},$$

giving $\hat{\pi}_k, \hat{\boldsymbol{\mu}}_k, \hat{\boldsymbol{\Sigma}}_k$. Then $\hat{P}(\delta = k \mid \mathbf{x}) = \hat{g}_k(\mathbf{x})\hat{P}(\delta = k)/\hat{g}(\mathbf{x})$ and $C : \mathcal{X} \to \{1,\dots,K\}$, $C(\mathbf{x}) = k$ iff $\hat{P}(\delta = k \mid \mathbf{x}) \geq \hat{P}(\delta = j \mid \mathbf{x})$ for all j (soft clustering). With all $\boldsymbol{\Sigma}_k = \sigma^2\mathbf{I}$ and $\sigma \to 0$ this becomes K-means.

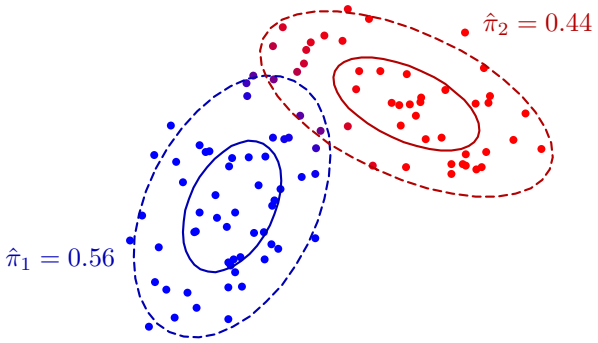


Figure. *A two-component Gaussian mixture fitted by EM to* 100 *synthetic points (*60 *and* 40 *drawn from two bivariate normals). Ellipses are the* 1- *and* 2*-Mahalanobis-distance contours of the fitted components. Each point is coloured by its responsibility* $\hat P(\delta = 1 \mid \mathbf{x})$*, from blue (*1*) to red (*0*); points in the overlap get intermediate colours, the soft assignment that* K*-means lacks.*

4.2 The EM algorithm ($K = 2$)

$\mathbf{X}_1, \dots, \mathbf{X}_n \sim \pi_1 \phi(\mathbf{x}; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) + \pi_0 \phi(\mathbf{x}; \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$, $\pi_0 = 1 - \pi_1$. The observed-data log-likelihood

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^{n} \log\{\pi_0 \phi(\mathbf{x}_i; \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) + \pi_1 \phi(\mathbf{x}_i; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)\}$$

has no closed-form maximizer (log of a sum). If $\delta_1, \dots, \delta_n \in \{0, 1\}$ were observed, $L(\boldsymbol{\theta} \mid \mathbf{X}^n, \boldsymbol{\delta}^n) = \prod_i \mathrm{pdf}(\mathbf{x}_i, \delta_i \mid \boldsymbol{\theta}) = \prod_i \mathrm{pdf}(\mathbf{x}_i \mid \delta_i, \boldsymbol{\theta}) \mathrm{pdf}(\delta_i)$ and

$$\ell(\boldsymbol{\theta} \mid \mathbf{X}^n, \boldsymbol{\delta}^n) = \sum_{i:\delta_i = 1} \log \pi_1 \phi(\mathbf{x}_i; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) + \sum_{i:\delta_i = 0} \log \pi_0 \phi(\mathbf{x}_i; \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$$

$$= \sum_{i=1}^{n} \left[\delta_i \log \pi_1 \phi_1(\mathbf{x}_i) + (1 - \delta_i) \log \pi_0 \phi_0(\mathbf{x}_i)\right],$$

which separates and gives closed forms such as $\hat\pi_1^{\mathrm{MLE}} = n_1/n$.

Definition 11.6: EM algorithm

(i) $\ell(\boldsymbol{\theta} \mid \mathbf{X}^n, \boldsymbol{\delta}^n)$ has an easy MLE, but $\delta_1, \dots, \delta_n$ are unobserved;
(ii) iteratively approximate the maximum. At $\boldsymbol{\theta}^{(m)}$:

- **E-step**: predict δ_i by $\hat\delta_i = \mathrm{E}(\delta_i \mid \mathbf{x}_i, \boldsymbol{\theta}^{(m)})$, i.e. compute $Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) := \mathrm{E}[\ell(\boldsymbol{\theta} \mid D) \mid D_{\mathrm{obs}}, \boldsymbol{\theta}^{(m)}]$. Since the complete-data log-likelihood is linear in δ_i,

$$\hat\delta_i = \frac{\pi_1^{(m)} \phi(\mathbf{x}_i; \boldsymbol{\mu}_1^{(m)}, \boldsymbol{\Sigma}_1^{(m)})}{\pi_0^{(m)} \phi(\mathbf{x}_i; \boldsymbol{\mu}_0^{(m)}, \boldsymbol{\Sigma}_0^{(m)}) + \pi_1^{(m)} \phi(\mathbf{x}_i; \boldsymbol{\mu}_1^{(m)}, \boldsymbol{\Sigma}_1^{(m)})} \quad \text{(responsibility)}.$$

- **M-step**: $\boldsymbol{\theta}^{(m+1)} = \arg\max_{\boldsymbol{\theta}} L(\boldsymbol{\theta} \mid \mathbf{X}^n, \hat{\boldsymbol{\delta}}^n)$, i.e. $\arg\max_{\boldsymbol{\theta}} Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)})$:

$$\pi_1 = \frac{\sum \hat\delta_i}{n}, \quad \boldsymbol{\mu}_1 = \frac{\sum \hat\delta_i \mathbf{x}_i}{\sum \hat\delta_i}, \quad \boldsymbol{\Sigma}_1 = \frac{\sum \hat\delta_i (\mathbf{x}_i - \boldsymbol{\mu}_1)(\mathbf{x}_i - \boldsymbol{\mu}_1)^\top}{\sum \hat\delta_i},$$

  and similarly for component 0 with weights $1 - \hat\delta_i$.

Repeat until $d(\boldsymbol{\theta}^{(m)}, \boldsymbol{\theta}^{(m+1)})$ is small.

Theorem 11.7: monotonicity of EM

$\ell(\boldsymbol{\theta}^{(m+1)}) \geq \ell(\boldsymbol{\theta}^{(m)})$.

Proof [Supplement]. $\ell(\boldsymbol{\theta}) = Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) - H(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)})$ with $H(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) = \mathrm{E}[\log p(\boldsymbol{\delta} \mid \mathbf{X}, \boldsymbol{\theta}) \mid \mathbf{X}, \boldsymbol{\theta}^{(m)}]$. By Gibbs' (Jensen's) inequality $H(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) \leq H(\boldsymbol{\theta}^{(m)} \mid \boldsymbol{\theta}^{(m)})$ (KL divergence ≥ 0). The M-step does not decrease Q, so $\ell(\boldsymbol{\theta}^{(m+1)}) - \ell(\boldsymbol{\theta}^{(m)}) = \{Q(\boldsymbol{\theta}^{(m+1)} \mid \boldsymbol{\theta}^{(m)}) - Q(\boldsymbol{\theta}^{(m)} \mid \boldsymbol{\theta}^{(m)})\} - \{H(\boldsymbol{\theta}^{(m+1)} \mid \boldsymbol{\theta}^{(m)}) - H(\boldsymbol{\theta}^{(m)} \mid \boldsymbol{\theta}^{(m)})\} \geq 0$. ■

EM may converge to a local maximum (or saddle point), and the likelihood is unbounded as some $\boldsymbol{\Sigma}_k$ becomes singular, so use several starts and regularization.

5 Choosing K and Other Topics

- **Scatter decomposition**: $W(C) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|^2$, $B(C) = \sum_{k=1}^{K} n_k \|\bar{\mathbf{x}}_{(k)} - \bar{\mathbf{x}}\|^2$. W always decreases as K grows, so it cannot choose K by itself (elbow method: where the decrease flattens).
- Indices: $\mathrm{CI}(K, c) = W(K, c)/T$ (smaller is better) and Calinski–Harabasz $\mathrm{CH}(K, c) = \dfrac{B(K, c)/(K-1)}{W(K, c)/(n-K)}$ (larger is better; the analogue of the one-way MANOVA F).
- **Gap statistic**: compare with the expected $\log W_K^*$ under a reference distribution (uniform over the data range),

$$\mathrm{Gap}(K) = \mathrm{E}^*[\log W_K^*] - \log W_K, \qquad s(K) = \text{(adjusted) s.d. of } \log W_K^*,$$

  and choose the smallest K with $\mathrm{Gap}(K) \geq \mathrm{Gap}(K+1) - s(K+1)$.
- **Clustering variables**: define $d(v_i, v_j) = 1 - r_{ij}$ or $1 - r_{ij}^2$ and apply hierarchical clustering.

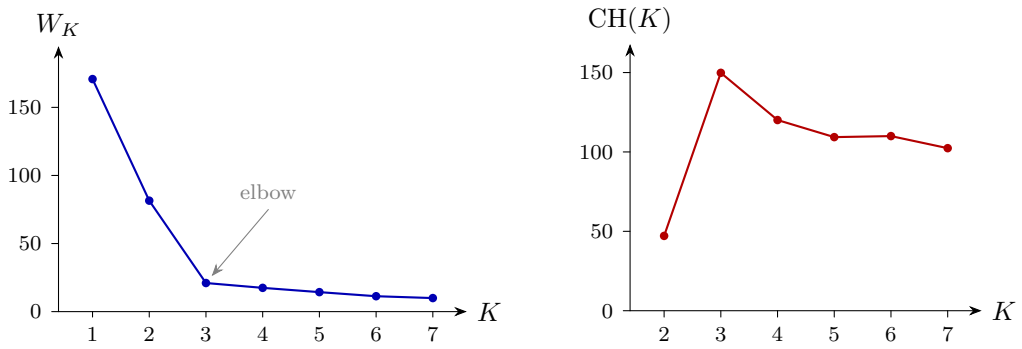


Figure. *Choosing K for the K-means data of the earlier figure (best of 20 random starts for each K). Left: W_K keeps decreasing, but the decrease flattens after $K = 3$ (elbow). Right: the Calinski–Harabasz index, which divides the between and within scatter by their degrees of freedom, has its maximum at $K = 3$.*

6 Selected exercises

Motivation. The exercises prove facts used above without proof: the count of partitions, the closed-form M-step, the unbounded mixture likelihood, and the monotone merge heights that make a dendrogram well defined.

Exercise 11.1 (Counting partitions)

Show that the number of partitions of $n \geq 1$ items into K non-empty clusters is

$$A(n,K) = \frac{1}{K!}\sum_{k=0}^{K}(-1)^{K-k}\binom{K}{k}k^n,$$

and that $A(n,K)\,K!/K^n \to 1$ as $n \to \infty$ for fixed K.

Solution [Supplement].

Step 1 (encoders). A partition into K non-empty blocks together with a labelling of the blocks by $1,\dots,K$ is the same as a surjective encoder $C : \{1,\dots,n\} \to \{1,\dots,K\}$. Each partition has $K!$ labellings, so $A(n,K) = N/K!$ with N the number of surjections.

Step 2 (inclusion–exclusion). Let E_j be the set of encoders that never use label j. For $J \subseteq \{1,\dots,K\}$ with $|J| = m$, $|\bigcap_{j\in J} E_j| = (K-m)^n$. By inclusion–exclusion,

$$N = \sum_{J}(-1)^{|J|}\Big|\bigcap_{j\in J}E_j\Big| = \sum_{m=0}^{K}(-1)^m\binom{K}{m}(K-m)^n,$$

and the substitution $k = K - m$ gives the formula. The $k = 0$ term vanishes for $n \geq 1$, which gives the form used above.

Step 3 (asymptotics). $K!\,A(n,K)/K^n = \sum_{k=0}^{K}(-1)^{K-k}\binom{K}{k}(k/K)^n \to 1$, since $(k/K)^n \to 0$ for $k < K$. For example $A(19,4) \approx 4^{19}/24 \approx 1.1 \times 10^{10}$. ■

Exercise 11.2 (Deriving the M-step)

Given responsibilities $\hat\delta_i \in (0,1)$, let

$$Q(\boldsymbol\theta) = \sum_{i=1}^{n}\big[\hat\delta_i \log\{\pi_1\phi(\mathbf{x}_i;\boldsymbol\mu_1,\boldsymbol\Sigma_1)\} + (1-\hat\delta_i)\log\{\pi_0\phi(\mathbf{x}_i;\boldsymbol\mu_0,\boldsymbol\Sigma_0)\}\big], \qquad \pi_0 = 1-\pi_1.$$

Assuming the weighted scatter matrices below are positive definite, show that Q is maximized by the M-step formulas of the EM algorithm.

Solution [Supplement].

Step 1 (separation). With $N_1 = \sum_i \hat\delta_i \in (0,n)$ and $N_0 = n - N_1$,

$$Q = \{N_1\log\pi_1 + N_0\log(1-\pi_1)\} + Q_1(\boldsymbol\mu_1,\boldsymbol\Sigma_1) + Q_0(\boldsymbol\mu_0,\boldsymbol\Sigma_0), \qquad Q_1 = \sum_i \hat\delta_i \log\phi(\mathbf{x}_i;\boldsymbol\mu_1,\boldsymbol\Sigma_1),$$

and Q_0 likewise with weights $1-\hat\delta_i$; the three parts have separate parameters.

Step 2 (π_1). $N_1\log\pi + N_0\log(1-\pi)$ is strictly concave on $(0,1)$ with derivative $N_1/\pi - N_0/(1-\pi)$, which vanishes only at $\pi = N_1/n$.

Step 3 ($\boldsymbol\mu_1$). $-2Q_1 = N_1 p\log 2\pi + N_1\log|\boldsymbol\Sigma_1| + \sum_i \hat\delta_i(\mathbf{x}_i-\boldsymbol\mu_1)^\top\boldsymbol\Sigma_1^{-1}(\mathbf{x}_i-\boldsymbol\mu_1)$. With $\boldsymbol m_1 = \sum_i \hat\delta_i\mathbf{x}_i/N_1$ we have $\sum_i\hat\delta_i(\mathbf{x}_i-\boldsymbol m_1) = \mathbf{0}$, so

$$\sum_i \hat\delta_i(\mathbf{x}_i-\boldsymbol\mu_1)^\top\boldsymbol\Sigma_1^{-1}(\mathbf{x}_i-\boldsymbol\mu_1) = \sum_i \hat\delta_i(\mathbf{x}_i-\boldsymbol m_1)^\top\boldsymbol\Sigma_1^{-1}(\mathbf{x}_i-\boldsymbol m_1) + N_1(\boldsymbol m_1-\boldsymbol\mu_1)^\top\boldsymbol\Sigma_1^{-1}(\boldsymbol m_1-\boldsymbol\mu_1),$$

minimized by $\boldsymbol\mu_1 = \boldsymbol m_1$ whatever $\boldsymbol\Sigma_1$ is (profiling, Theorem 2.25).

Step 4 ($\boldsymbol\Sigma_1$). With $\mathbf{V}_1 = \sum_i \hat\delta_i(\mathbf{x}_i-\boldsymbol m_1)(\mathbf{x}_i-\boldsymbol m_1)^\top/N_1 \succ 0$, the profiled criterion is $N_1\{\log|\boldsymbol\Sigma_1| + \operatorname{tr}(\boldsymbol\Sigma_1^{-1}\mathbf{V}_1)\} + \text{const}$. Let $\mathbf{B} = \mathbf{V}_1^{1/2}\boldsymbol\Sigma_1^{-1}\mathbf{V}_1^{1/2} \succ 0$ with eigenvalues β_j. Then

$$\log|\boldsymbol\Sigma_1| + \operatorname{tr}(\boldsymbol\Sigma_1^{-1}\mathbf{V}_1) - \log|\mathbf{V}_1| - p = \operatorname{tr}\mathbf{B} - \log|\mathbf{B}| - p = \sum_{j=1}^{p}(\beta_j - \log\beta_j - 1) \geq 0$$

by Theorem 2.14, with equality iff all $\beta_j = 1$, i.e. $\mathbf{B} = \mathbf{I}$, i.e. $\boldsymbol{\Sigma}_1 = \mathbf{V}_1$.

Step 5. The same argument with weights $1 - \hat{\delta}_i$ gives $\boldsymbol{\mu}_0$ and $\boldsymbol{\Sigma}_0$. These are the formulas of the EM algorithm: weighted versions of the MLEs of Theorem 4.7. ■

Exercise 11.3 (The mixture likelihood is unbounded)

Let $x_1, \dots, x_n$ be distinct reals and $\ell(\boldsymbol{\theta}) = \sum_i \log\{\pi\phi(x_i; \mu_1, \sigma_1^2) + (1-\pi)\phi(x_i; \mu_0, \sigma_0^2)\}$.
(a) Show $\sup_{\boldsymbol{\theta}} \ell = +\infty$.
(b) Show that if the variances are constrained to be equal, $\sigma_0 = \sigma_1 = \sigma$, and $n \geq 3$, then ℓ is bounded above.

Solution [Supplement].

(a) Fix $\pi \in (0,1)$, μ_0 and σ_0, and set $\mu_1 = x_1$. The $i = 1$ term is at least $\log\{\pi/(\sqrt{2\pi}\sigma_1)\} \to \infty$ as $\sigma_1 \downarrow 0$, while each term $i \geq 2$ is at least $\log\{(1-\pi)\phi(x_i; \mu_0, \sigma_0^2)\}$, a finite constant. Hence $\ell \to \infty$: a component collapses onto one observation, and no global MLE exists.

(b)

Step 1. Let $\delta = \min_{i \neq j} |x_i - x_j| > 0$. A mean μ_k is within distance $< \delta/2$ of at most one x_i, so for any (μ_0, μ_1) at least $n - 2 \geq 1$ observations, say x_{i_0}, satisfy $|x_{i_0} - \mu_k| \geq \delta/2$ for $k = 0, 1$.

Step 2. Every term is at most $\log\{1/(\sqrt{2\pi}\sigma)\}$, and the i_0 term is at most $\log\{1/(\sqrt{2\pi}\sigma)\} - \delta^2/(8\sigma^2)$. So

$$\ell \leq u(\sigma) := -n \log(\sqrt{2\pi}\,\sigma) - \frac{\delta^2}{8\sigma^2}.$$

Step 3. u is continuous on $(0, \infty)$ and tends to $-\infty$ as $\sigma \downarrow 0$ and as $\sigma \to \infty$, so $\sup_\sigma u(\sigma) < \infty$, which bounds ℓ. This is why EM is run with constrained or regularized covariances. ■

Exercise 11.4 (Monotone merge heights)

In agglomerative clustering let h_t be the linkage distance of the two clusters merged at step t.
(a) Show $h_1 \leq h_2 \leq \cdots$ for single, complete and average linkage.
(b) Show by example that centroid linkage, $d(G, H) = \|\bar{\mathbf{x}}_G - \bar{\mathbf{x}}_H\|$, can produce $h_2 < h_1$.

Solution [Supplement].

(a)

Step 1 (update rule). If G_1, G_2 are merged into $G = G_1 \cup G_2$, then for every other cluster H

$$d(G, H) = \min\{d(G_1, H), d(G_2, H)\}, \quad \max\{d(G_1, H), d(G_2, H)\}, \quad \frac{|G_1| d(G_1, H) + |G_2| d(G_2, H)}{|G_1| + |G_2|}$$

for single, complete and average linkage respectively, since the pairs $(i, j) \in G \times H$ are the union of $G_1 \times H$ and $G_2 \times H$.

Step 2. At step t the merged pair has the smallest distance among current clusters, so $d(G_1, H), d(G_2, H) \geq h_t$ and all distances between untouched clusters are $\geq h_t$. A minimum, maximum or weighted average of numbers $\geq h_t$ is $\geq h_t$, so every distance after the merge is $\geq h_t$, and $h_{t+1} \geq h_t$.

(b) Take the vertices of an equilateral triangle with side 1. First two vertices merge at $h_1 = 1$; their centroid is the midpoint of a side, at distance $\sqrt{3}/2$ from the third vertex, so $h_2 = \sqrt{3}/2 < h_1$ and the dendrogram has an inversion. ■